

From Platonic Solids to Regular 4-Polytopes: A Topological and Coxeter-Theoretic Classification with Lean 4 Certificate generated by LLM

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Github: <https://github.com/catskillsresearch/platonic-solids>

Abstract

We present an expository and formally verified classification of the regular convex polyhedra in $\mathbb{R}^3$ (the Platonic solids) and the regular convex polychora in $\mathbb{R}^4$, formalized in **Lean 4** against Mathlib. In dimension 3, Euler’s formula $V - E + F = 2$ together with regularity incidence $pF = 2E$, $qV = 2E$ yields the Diophantine inequality $(p-2)(q-2) < 4$ and exactly five solutions; the same identities algebraically force $E > 0$, so non-degeneracy is a theorem rather than an extra axiom. The development proves algebraic Euler–Poincaré, the radial-projection homeomorphism from the frontier of a bounded convex body to the sphere, and the cellular Betti numbers of the tetrahedron surface ($\chi = 2$) and of the 4-simplex boundary ($\chi = 0$). Singular H_0 of the metric sphere, $H_*(S^0)$, and the pair and triad exact sequences are also proved; barycentric subdivision and the small-simplex deformation retract live in `PlatonicSolids/SingularExcision/`. In dimension 4, $\chi(S^3) = 0$ yields no flag bound. Among the eleven Platonic-cell / Platonic-vertex-figure Schläfli triples, the six regular convex 4-polytopes are exactly those whose Gram determinant Δ is positive. The classification uses the rank-4 Schläfli Gram matrix and Sylvester’s criterion; it does not construct Coxeter groups as Lean objects. There is no `sorry` outside the deliberate holes in `Challenge.lean` and no project-defined axiom; the compared theorems may depend on Mathlib’s `propext`, `Classical.choice`, and `Quot.sound`. Companion artifact: `PlatonicSolids.lean` (standalone Lake package in this directory).

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1 1. Introduction: The Dimensional Hierarchy

In Euclidean geometry, a regular polytope is a figure whose symmetry group acts transitively on its flags (incident chains of vertices, edges, faces, and higher-dimensional cells).

- In **dimension 2**, there exist infinitely many regular polygons $\{p\}$ for each $p \geq 3$.
- In **dimension 3**, Euclid’s *Elements* (Book XIII) establishes that there exist exactly **5 Platonic solids**.
- In **dimension 4**, Ludwig Schläfli proved in 1852 that there exist exactly **6 regular convex 4-polytopes** (polychora).
- In **dimension $n \geq 5$** , the classification collapses permanently to **3 families**: the regular simplex, the hypercube (measure polytope), and the cross-polytope.

This distribution raises foundational mathematical and pedagogical questions: **Why does dimension 3 permit 5 shapes, dimension 4 permit 6 (admitting the exceptional self-dual 24-cell), and dimension $n \geq 5$ permit only 3? Why does topology suffice to bound the classification in 3D, while failing completely in 4D?**

This paper addresses these questions through an analytical exposition coupled with a fully verified formal certificate in Lean 4. Figure 1 is the coarsest dependency blueprint: topology bounds the 3D census, while the sign of the Schläfli determinant Δ bounds the 4D census.

2 2. Dimension 3: Topology, Incidence, and the Non-Degeneracy Insight

Figure 2 expands the 3D side of Figure 1 to the Lean lemmas that turn a convex polyhedron into a Platonic pair.

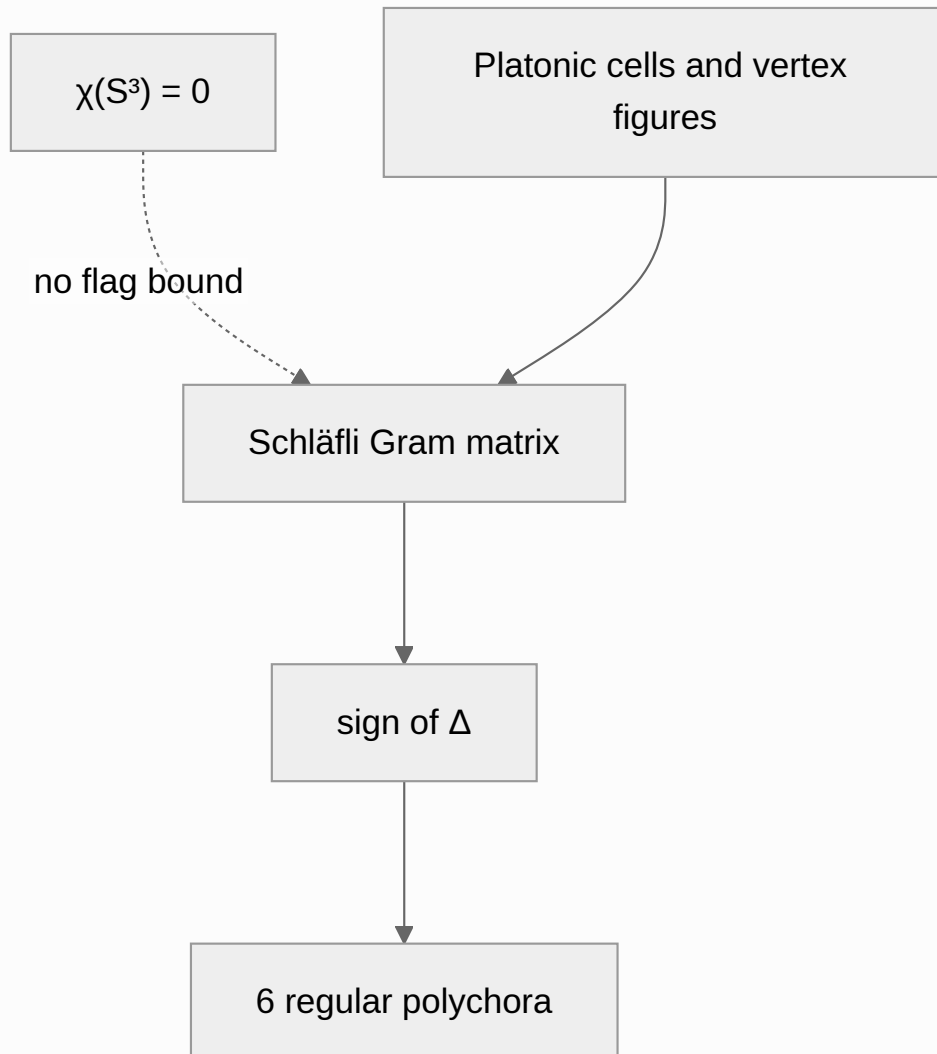
2.1 2.1. Topological Derivation of Euler’s Formula

The Lean gists are `PlatonicSolids/RadialProjection.lean` (frontier of a convex body $\simeq S^{n-1}$), `PlatonicSolids/EulerPoincare.lean` (algebraic Euler–Poincaré: $\chi(C) = \chi(H_*)$ by rank-nullity), `PlatonicSolids/SingularHomology.lean` (homotopy invariance, H_0 of path-connected spheres, and $H_*(S^0)$), `PlatonicSolids/RelativeHomology.lean` and `PlatonicSolids/MayerVietoris.lean` (pair and triad exact sequences of singular chains), `PlatonicSolids/SingularExcision.lean` (affine barycentric subdivision, the singular chain map, and the open-cover quasi-isomorphism), `PlatonicSolids/SphereSingularHomology.lean` (stereographic $U \cap V \simeq_h S^{n-1}$ and $H_n(S^n; R) \cong R$ for $n > 0$), `PlatonicSolids/Sphere2Homology.lean` (cellular $H_*(S^2; \mathbb{Q})$ of the tetrahedron surface), and `PlatonicSolids/EulerBetti.lean` (the numbers $\chi(S^2) = 2$ and $\chi(S^3) = 0$).

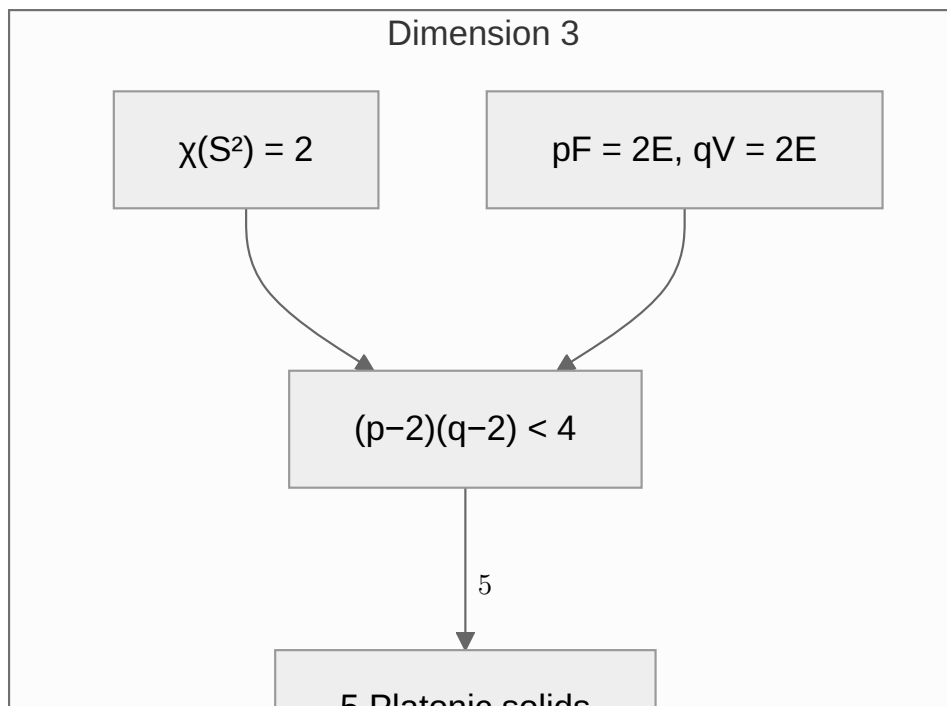
Let $P \subset \mathbb{R}^3$ be a convex polyhedron containing the origin in its interior. Radial projection $\pi : \partial P \rightarrow S^2$ onto the circumscribed 2-sphere is a homeomorphism (`frontier_convex_homeo_sphere`). This projection induces a finite CW-complex structure on S^2 with V 0-cells (vertices), E 1-cells (edges), and F 2-cells (faces).

Mathlib already has the singular homology functor and its homotopy invariance. We package the missing transport lemmas: a homotopy equivalence (or homeomorphism) of spaces induces an isomorphism of singular homology (`singularHomologyIso_of_homotopyEquiv`), a contractible space has vanishing positive homology (`isZero_singularHomology_of_contractible`), and a

Dimension 4



Dimension 3



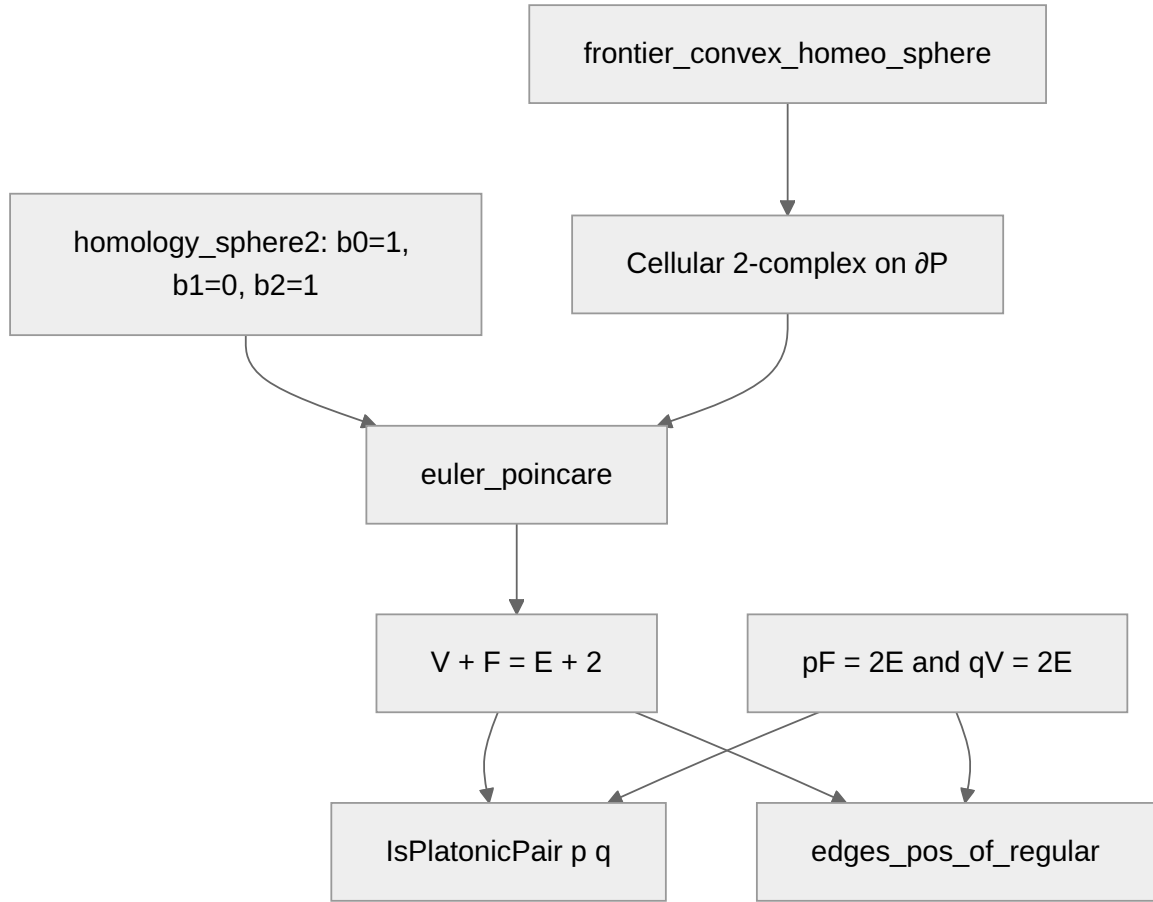


Figure 2: Dimension-3 pipeline. Radial projection identifies ∂P with S^2 ; cellular $H_*(S^2; \mathbb{Q})$ of the tetrahedron surface plus algebraic Euler–Poincaré give $V + F = E + 2$; regularity incidence $pF = 2E$, $qV = 2E$ then forces a Platonic pair and $E > 0$.

closed disk is contractible. For the metric sphere itself, path-connectedness in dimension $n \geq 1$ gives

$$H_0(S^n; R) \cong R$$

(`singularHomology0_sphere`). The 0-sphere is the two-point set $\{\pm 1\}$, which is totally disconnected, so $H_0(S^0; R) \cong R \oplus R$ and $H_n(S^0; R) = 0$ for $n > 0$.

A subspace inclusion $A \hookrightarrow X$ is a monomorphism in `TopCat`. The singular chain-complex functor preserves monomorphisms, so

$$0 \rightarrow C_*(A; R) \rightarrow C_*(X; R) \rightarrow C_*(X, A; R) \rightarrow 0$$

is short exact (`pair_shortExact`). Snake lemma supplies the long exact sequence of the pair: connecting $\delta : H_{n+1}(X, A) \rightarrow H_n(A)$ (`pairδ`) and exactness at $H_n(A)$, $H_n(X)$, and $H_n(X, A)$ (`pair_exact_at_A`, `pair_exact_at_X`, `pair_exact_at_rel`).

For a commuting square of spaces $W \rightarrow U, V \rightarrow X$, the same argument on $C_*(W) \rightarrow C_*(U) \oplus C_*(V)$ yields the triad sequence (`triad_shortExact`, `triadδ`). The square induces a comparison from the cover chains to $C_*(X)$; the square is *excisive* when that comparison is a quasi-isomorphism (`IsExcisive`). After excision, the connecting map is Mayer–Vietoris for X (`mayerVietorisδ`).

The stereographic cover of S^n is the standard pair of charts $U = S^n \setminus \{N\}$, $V = S^n \setminus \{S\}$. Each is homeomorphic to a hyperplane (`homeoPuncturedNorth`, `homeoPuncturedSouth`), hence contractible, so $H_k(U) = H_k(V) = 0$ for $k > 0$. They cover S^n (`punctured_cover`) and meet in $S^n \setminus \{N, S\}$. Stereographic projection from the north pole sends the south pole to the origin, so $U \cap V$ is homeomorphic to the punctured hyperplane and therefore homotopy-equivalent to S^{n-1} (`homotopyEquiv_puncturedInter_sphere`).

The open-cover quasi-isomorphism is proved in `PlatonicSolids/SingularExcision/`, not imported from `Mathlib`. Finite ordered affine chains carry a cone-point barycentric subdivision `subdivide` and prism `prism` with $\partial \text{Sd} = \text{Sd} \partial$ and $\partial T + T \partial = \text{id} - \text{Sd}$ (`boundary_subdivide`, `boundary_prism`). Realization of an affine simplex in Δ^q is functorial on faces (`realize_face`) and on vertex-wise maps (`realize_map`), so the generators assemble to a morphism of singular chain complexes `singularSubdivision` (`ι_singularSubdivisionHom`, `singularSubdivideGenerator_d`). The prism identity is a chain homotopy $\text{id} \simeq \text{Sd}$ (`singularPrismHomotopy`). Cover chains of a two-set open cover are identified with the small-simplex subcomplex (`coverChainsIsoSmall`). Mesh shrinking is proved: each barycentric piece has diameter at most $n/(n+1)$ times its parent (`barycentricMeshRatio`, `diam_range_subdivideStd_le`), so iterated subdivision plus the Lebesgue-number lemma (`exists_subdivideStd_iterate_image_subset_left_or_right`) puts every finite `ModuleCat` chain into the small-chain subgroup (`eventual_small_of_chain`). The algebraic deformation retract on that subgroup is instantiated (`small_cycles_and_boundaries_singular`). The small-subcomplex inclusion is a quasi-isomorphism (`smallι_quasiIso`), so `coverComparison` is a quasi-isomorphism and every two-set open cover is excisive (`IsExcisiveSubspaces_of_openCover`). The stereographic pair is therefore excisive; after the charts vanish in positive degree, Mayer–Vietoris supplies $H_{n+1}(S^{n+1}) \cong H_n(U \cap V) \cong H_n(S^n)$ for $n \geq 1$ (`singularHomology_sphere_succ_iso`). Degree-zero Mayer–Vietoris on S^1 identifies $H_1(S^1; R)$ with $\ker \varepsilon_{S^0} \cong R$ (`singularHomology_sphere1`), so $H_n(S^n; R) \cong R$ for every $n > 0$ (`singularHomology_sphere`) and vanishes in the other positive degrees (`isZero_singularHomology_sphere`). The compared 3D classification still takes $V + F = E + 2$ as a hypothesis; `platonic_solids_3d_of_sphere` is the optional bridge from those singular Betti numbers through Euler–Poincaré, and does not change Palomar types. Figure 3 records this dependency stack.

The standard simplicial 2-sphere is the surface of a tetrahedron: 4 vertices, 6 edges, 4 triangles. Oriented incidence matrices $d_1 : \mathbb{Q}^6 \rightarrow \mathbb{Q}^4$ and $d_2 : \mathbb{Q}^4 \rightarrow \mathbb{Q}^6$ satisfy $d_1 d_2 = 0$. Each has a 3×3 identity minor, so both ranks are at least 3; the product-zero rank inequality on a 6-dimensional middle term forces both ranks equal to 3. Hence

$$b_0 = 4 - 3 = 1, \quad b_1 = 3 - 3 = 0, \quad b_2 = 4 - 3 = 1$$

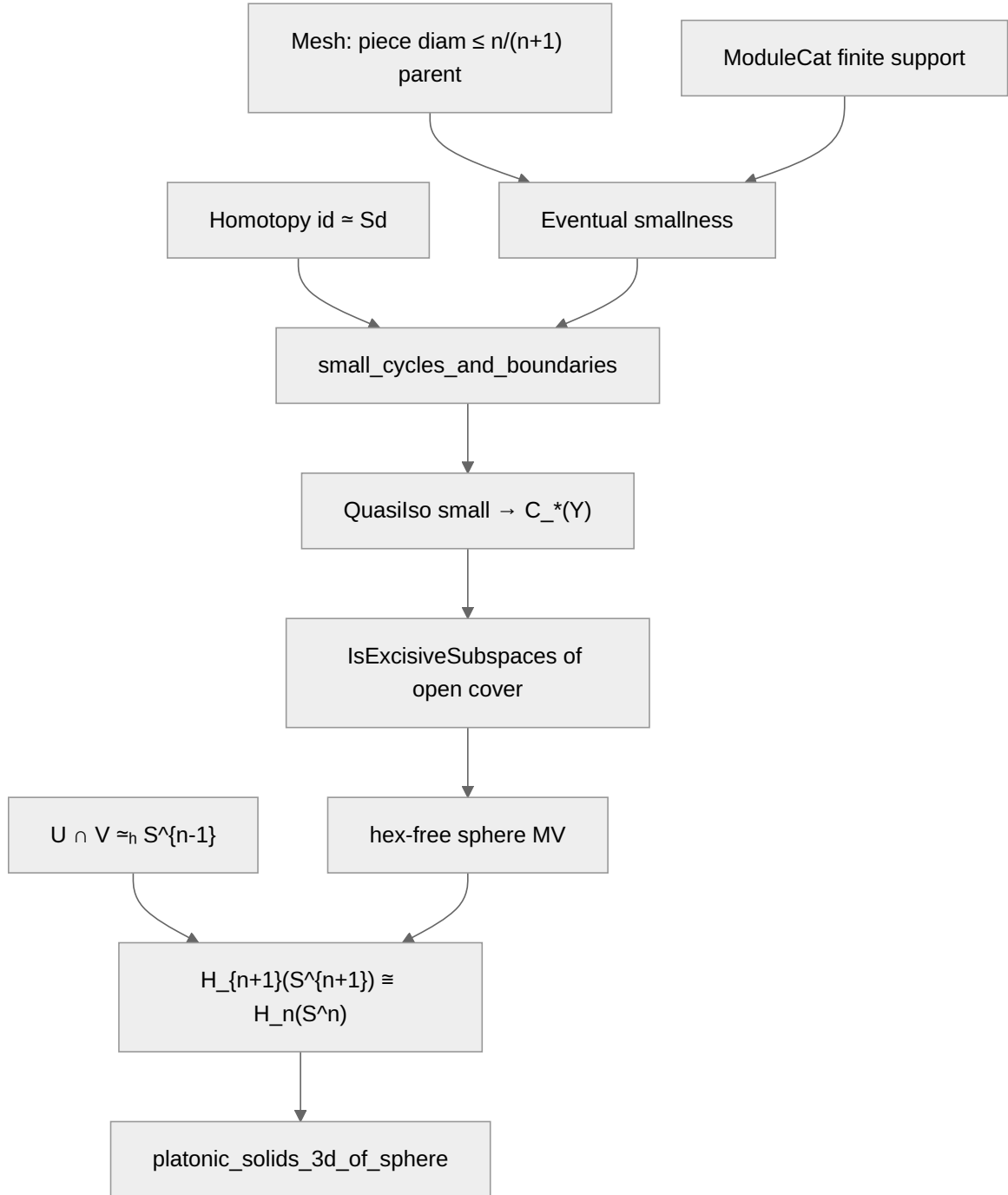


Figure 3: Singular-excision blueprint, finer than Figure 2. Mesh shrinking and the prism homotopy feed the small-chain retract; the open-cover quasi-isomorphism discharges `IsExcisiveSubspaces` and yields $H_{n+1}(S^{n+1}) \cong H_n(S^n)$.

(homology_sphere2). The Euler characteristic is the alternating sum of Betti numbers:

$$\chi(S^2) = b_0 - b_1 + b_2 = 1 - 0 + 1 = 2$$

Algebraic Euler–Poincaré (euler_poincare) is rank-nullity on any 2-complex: cell ranks equal Betti ranks. Specializing to these numbers yields Euler’s formula

$$V - E + F = 2$$

on the tetrahedron (euler_tetrahedron) and, for an arbitrary 2-complex with the same Betti numbers, the general identity euler_formula_of_sphere2. The tetrahedron incidence structure is then Platonic (tetrahedron_platonic). Combining Euler–Poincaré with regularity incidence gives the classification from homology (platonic_solids_3d_of_homology).

2.2 2.2. Incidence Combinatorics

The Lean gist is PlatonicSolids/Incidence.lean (with the pair predicate in PlatonicSolids/PlatonicPair.lean). A regular polyhedron has Schläfli symbol $\{p, q\}$, meaning: 1. Every 2D face is an identical regular p -gon ($p \geq 3$). Each face contributes p edge-face incidences. Since every edge is shared by exactly 2 faces:

$$pF = 2E \implies F = \frac{2E}{p}$$

2. Every vertex meets q faces and q edges ($q \geq 3$). Each vertex contributes q edge-vertex incidences. Since every edge connects exactly 2 vertices:

$$qV = 2E \implies V = \frac{2E}{q}$$

2.3 2.3. The Diophantine Reduction

The Lean gist is PlatonicSolids/PlatonicPair.lean; the reciprocal form $1/p + 1/q > 1/2$ is PlatonicSolids/Reciprocal.lean. Substituting these incidence relations into Euler’s formula:

$$\frac{2E}{q} - E + \frac{2E}{p} = 2$$

Multiplying by pq over $\mathbb{N}$ avoids rational arithmetic:

$$(2p + 2q)E = (pq)E + 2pq$$

Rearranging terms:

$$(2p + 2q - pq)E = 2pq$$

Because $2pq > 0$ for $p, q \geq 3$, the factor $(2p + 2q - pq)$ must be strictly positive, which yields:

$$pq < 2p + 2q \iff (p - 2)(q - 2) < 4$$

Since $p \geq 3$ and $q \geq 3$, both $(p - 2)$ and $(q - 2)$ are strictly positive integers. The only pairs of positive integers with product strictly less than 4 are:

$(p - 2)$	$(q - 2)$	$(p - 2)(q - 2)$	Pair (p, q)	Polyhedron
1	1	1	(3, 3)	Tetrahedron
1	2	2	(3, 4)	Octahedron

$(p-2)$	$(q-2)$	$(p-2)(q-2)$	Pair (p, q)	Polyhedron
2	1	2	$(4, 3)$	Cube (Hexahedron)
1	3	3	$(3, 5)$	Icosahedron
3	1	3	$(5, 3)$	Dodecahedron

2.4 2.4. Formalization Insight: Elimination of the Non-Degeneracy Axiom

The Lean gist is `edges_pos_of_regular` in `PlatonicSolids/Incidence.lean`. In informal presentations, dividing Euler’s equation by $2E$ to obtain:

$$\frac{1}{p} + \frac{1}{q} = \frac{1}{2} + \frac{1}{E} > \frac{1}{2}$$

is typically justified by appealing to physical or geometric intuition: “*a solid polyhedron clearly has edges, so $E > 0$.*”

When formalizing this theorem in Lean 4, one might initially expect to require an explicit non-degeneracy hypothesis (`hE : E > 0`). However, machine verification reveals this hypothesis to be mathematically redundant.

Recall the cleared identity over $\mathbb{N}$:

$$(2p + 2q)E = (pq)E + 2pq$$

If $E = 0$, the equation reduces to:

$$0 = 2pq$$

Because $p \geq 3$ and $q \geq 3$, we have $2pq \geq 18$, which contradicts $0 = 2pq$.

Therefore, **the topological Euler characteristic ($\chi = 2$) together with the local regularity conditions ($p, q \geq 3$) algebraically prohibits an edgeless configuration.** The condition $E \geq 1$ is an algebraic consequence of the system, not an extrinsic geometric requirement.

3 3. Dimension 4: Why Topology Fails and the Gram Matrix Takes Over

Figure 4 is the 4D counterpart of Figure 2: Platonic cells and vertex figures determine a Schläfli Gram matrix, and the sign of $\Delta = \det M$ sorts the eleven triples.

3.1 3.1. The Topological Barrier: $\chi(S^3) = 0$

The Lean gists are `PlatonicSolids/EulerPoincare.lean` (now also the 3-complex identity `euler_poincare3`), `PlatonicSolids/Sphere3Homology.lean` (cellular $H_*(S^3; \mathbb{Q})$ of the 4-simplex boundary), and `eulerChar_sphere3_betti` in `PlatonicSolids/EulerBetti.lean`.

A convex 4-polytope with nonempty interior has frontier homeomorphic to S^3 by the same radial projection as in §2.1. The standard simplicial 3-sphere is the boundary of a 4-simplex: 5 vertices, 10 edges, 10 triangles, 5 tetrahedra. Oriented incidence matrices $d_1 : \mathbb{Q}^{10} \rightarrow \mathbb{Q}^5$, $d_2 : \mathbb{Q}^{10} \rightarrow \mathbb{Q}^{10}$, $d_3 : \mathbb{Q}^5 \rightarrow \mathbb{Q}^{10}$ satisfy $d_1 d_2 = 0$ and $d_2 d_3 = 0$. Identity minors give $\text{rank } d_1 \geq 4$ and $\text{rank } d_3 \geq 4$; a 6×6 minor of d_2 has determinant 1, so $\text{rank } d_2 \geq 6$. The product-zero inequalities on the two middle dimensions of size 10 then force the ranks to be exactly 4, 6, and 4. Hence

$$b_0 = 5 - 4 = 1, \quad b_1 = (10 - 4) - 6 = 0, \quad b_2 = (10 - 6) - 4 = 0, \quad b_3 = 5 - 4 = 1$$

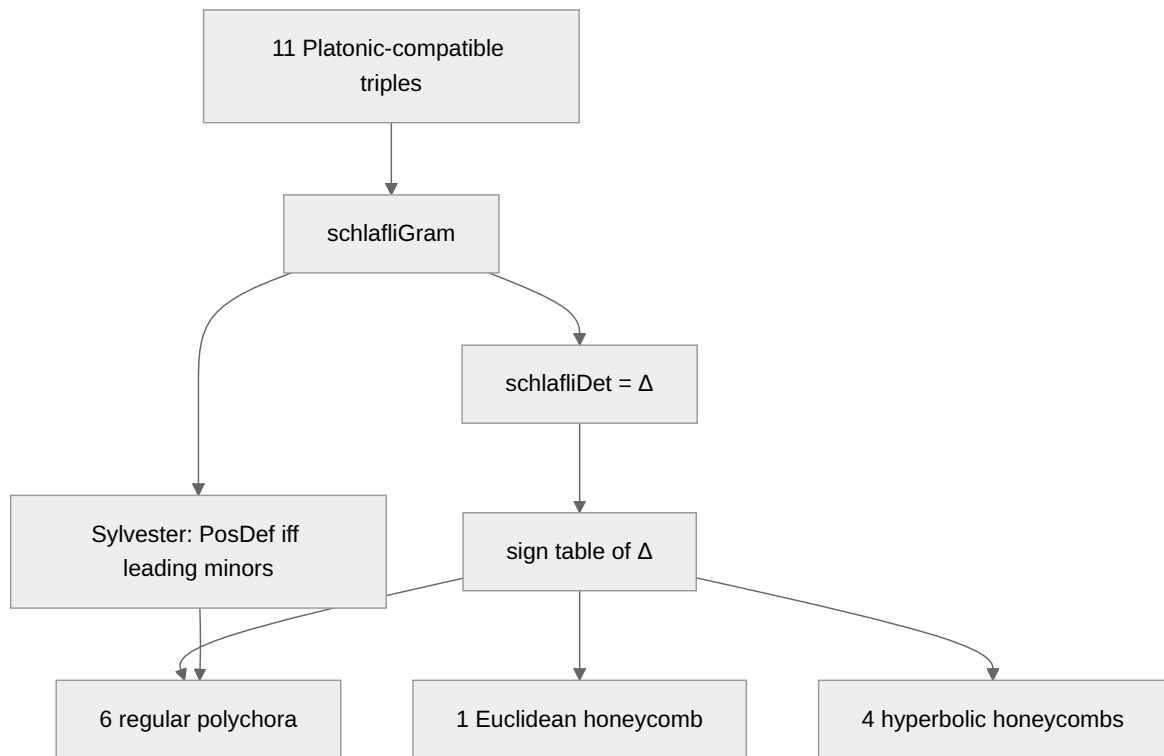


Figure 4: Dimension-4 pipeline. The eleven Platonic-compatible Schläfli triples are classified by the sign of $\Delta = \det(\text{schlafliGram})$. Positive Δ is equivalent, by Sylvester, to positive-definiteness of the Gram matrix, and picks out the six regular convex polychora.

(homology_sphere3). The Euler characteristic vanishes:

$$\chi(S^3) = b_0 - b_1 + b_2 - b_3 = 1 - 0 + 0 - 1 = 0$$

Algebraic Euler–Poincaré in dimension 3 (euler_poincare3) again equates cell ranks with Betti ranks, so any such cellulation satisfies

$$V - E + F - C = 0$$

(eulerChar_sphere3 on the 4-simplex, and euler_formula_of_sphere3 in general). Because the alternating sum is zero, dividing by any flag count yields an identity equal to zero rather than a positive reciprocal like $2/E$. **Topology alone does not impose an upper bound on p, q , or r in 4D.**

3.2 3.2. Coxeter Systems and the Schläfli Gram Matrix

The Lean gist is PlatonicSolids/GramMatrix.lean. To obtain an upper bound, informal accounts invoke **Coxeter reflection groups**. The Lean development does not construct those groups; it works with the associated **Schläfli Gram matrix** and its determinant Δ . A regular 4-polytope is described by its Schläfli symbol $\{p, q, r\}$: * The 3D cells are Platonic solids $\{p, q\}$. * The 3D vertex figures are Platonic solids $\{q, r\}$. * r cells meet around every 1D edge.

The Coxeter group W associated with $\{p, q, r\}$ is generated by reflections $\{s_1, s_2, s_3, s_4\}$ across the facet hyperplanes of a spherical fundamental simplex:

$$W = \langle s_1, s_2, s_3, s_4 \mid (s_i s_j)^{m_{ij}} = 1 \rangle$$

where the exponents are $m_{ii} = 1$, $m_{12} = p$, $m_{23} = q$, $m_{34} = r$, and $m_{ij} = 2$ for $|i - j| > 1$.

The canonical bilinear form of this system is represented by the **Schläfli Gram matrix**:

$$M = \begin{pmatrix} 1 & -\cos \frac{\pi}{p} & 0 & 0 \\ -\cos \frac{\pi}{p} & 1 & -\cos \frac{\pi}{q} & 0 \\ 0 & -\cos \frac{\pi}{q} & 1 & -\cos \frac{\pi}{r} \\ 0 & 0 & -\cos \frac{\pi}{r} & 1 \end{pmatrix}$$

3.3 3.3. Sylvester’s Criterion and Principal Minors

The Lean gists are PlatonicSolids/Angles.lean, LeadingMinors.lean, TrailingMinor.lean, D3IfPlatonic.lean, QuadForm.lean, CompletingSquare.lean, and Sylvester.lean. By the Cartan–Killing–Coxeter classification: 1. M is positive definite ($\det(M) > 0$) $\iff W$ is a finite reflection group acting on S^3 (a finite regular convex 4-polytope). 2. M is positive semidefinite with nullity 1 ($\det(M) = 0$) $\iff W$ is an affine reflection group (a Euclidean 3D space-filling honeycomb). 3. M has negative eigenvalues ($\det(M) < 0$) $\iff W$ is a hyperbolic reflection group (a honeycomb in hyperbolic space $\mathbb{H}^3$).

By Sylvester’s criterion, M is positive definite if and only if all leading principal minors $D_k = \det(M_k)$ are strictly positive: * $D_1 = 1 > 0$. * $D_2 = 1 - \cos^2 \frac{\pi}{p} = \sin^2 \frac{\pi}{p} > 0$ (satisfied for all $p \geq 3$). * $D_3 = \sin^2 \frac{\pi}{p} - \cos^2 \frac{\pi}{q}$.

$$D_3 > 0 \iff \sin \frac{\pi}{p} > \cos \frac{\pi}{q} = \sin \left(\frac{\pi}{2} - \frac{\pi}{q} \right) \iff \frac{1}{p} + \frac{1}{q} > \frac{1}{2}$$

Thus, $D_3 > 0$ requires the cell $\{p, q\}$ to be a Platonic solid. * Symmetrically, the trailing 3×3 minor requires:

$$\frac{1}{q} + \frac{1}{r} > \frac{1}{2}$$

Thus, the vertex figure $\{q, r\}$ must also be a Platonic solid.

3.4 3.4. The Critical Dimension-4 Minor: D_4

The Lean gist is `PlatonicSolids/DetFormula.lean`. Expanding $D_4 = \det(M)$ along the fourth row:

$$D_4 = D_3 - \cos^2\left(\frac{\pi}{r}\right) D_2 = \left(\sin^2\frac{\pi}{p} - \cos^2\frac{\pi}{q}\right) - \cos^2\left(\frac{\pi}{r}\right) \sin^2\left(\frac{\pi}{p}\right)$$

Factoring $\sin^2\frac{\pi}{p}$:

$$D_4 = \sin^2\left(\frac{\pi}{p}\right) \left(1 - \cos^2\frac{\pi}{r}\right) - \cos^2\left(\frac{\pi}{q}\right) = \sin^2\left(\frac{\pi}{p}\right) \sin^2\left(\frac{\pi}{r}\right) - \cos^2\left(\frac{\pi}{q}\right)$$

Therefore, the condition that $\{p, q, r\}$ forms a finite regular convex 4-polytope is:

$$\Delta(p, q, r) := \sin^2\left(\frac{\pi}{p}\right) \sin^2\left(\frac{\pi}{r}\right) - \cos^2\left(\frac{\pi}{q}\right) > 0$$

4 4. Evaluation of the 11 Candidate Triples

The Lean gists are `PlatonicSolids/CompatibleTriples.lean` (the eleven triples), `TrigValues.lean` (exact $\sin^2$ and $\cos^2$), `SignTable.lean` (sign of Δ), `Regular4.lean` (the six), and `Classification.lean` (the capstone).

Because both $\{p, q\}$ and $\{q, r\}$ must be Platonic, the indices must satisfy $p, q, r \in \{3, 4, 5\}$. Among the $5 \times 5 = 25$ combinations, only **11 triples** have a compatible middle index q .

The exact squared trigonometric values are:

$$\begin{aligned} \cos^2\frac{\pi}{3} &= \frac{1}{4}, & \sin^2\frac{\pi}{3} &= \frac{3}{4} \\ \cos^2\frac{\pi}{4} &= \frac{1}{2}, & \sin^2\frac{\pi}{4} &= \frac{1}{2} \\ \cos^2\frac{\pi}{5} &= \frac{3 + \sqrt{5}}{8}, & \sin^2\frac{\pi}{5} &= \frac{5 - \sqrt{5}}{8} \end{aligned}$$

Evaluating $\Delta(p, q, r)$ across all 11 candidate triples:

4.1 Group I: Spherical Polychora ($\Delta > 0$)

1. $\{3, 3, 3\}$ (**5-cell** / **Simplex**):

$$\Delta = \left(\frac{3}{4}\right) \left(\frac{3}{4}\right) - \frac{1}{4} = \frac{9}{16} - \frac{4}{16} = \frac{5}{16} > 0$$

2. $\{3, 3, 4\}$ (**16-cell**) & $\{4, 3, 3\}$ (**8-cell** / **Tesseract**):

$$\Delta = \left(\frac{3}{4}\right) \left(\frac{1}{2}\right) - \frac{1}{4} = \frac{3}{8} - \frac{2}{8} = \frac{1}{8} > 0$$

3. $\{3, 4, 3\}$ (**24-cell** — **Exceptional Self-Dual Polytope**):

$$\Delta = \left(\frac{3}{4}\right) \left(\frac{3}{4}\right) - \frac{1}{2} = \frac{9}{16} - \frac{8}{16} = \frac{1}{16} > 0$$

4. $\{3, 3, 5\}$ (**600-cell**) & $\{5, 3, 3\}$ (**120-cell**):

$$\Delta = \left(\frac{3}{4}\right) \left(\frac{5 - \sqrt{5}}{8}\right) - \frac{1}{4} = \frac{7 - 3\sqrt{5}}{32}$$

Since $7^2 = 49 > (3\sqrt{5})^2 = 45$, we have $7 - 3\sqrt{5} > 0$, so $\Delta > 0$.

4.2 Group II: Euclidean Honeycomb ($\Delta = 0$)

5. $\{4, 3, 4\}$ (Cubic Honeycomb of $\mathbb{R}^3$):

$$\Delta = \left(\frac{1}{2}\right) \left(\frac{1}{2}\right) - \frac{1}{4} = \frac{1}{4} - \frac{1}{4} = 0 \quad (\text{Affine / flat space-filling})$$

4.3 Group III: Hyperbolic Honeycombs ($\Delta < 0$)

6. $\{3, 5, 3\}$ (Icosacell Honeycomb):

$$\Delta = \left(\frac{3}{4}\right) \left(\frac{3}{4}\right) - \frac{3 + \sqrt{5}}{8} = \frac{3 - 2\sqrt{5}}{16} < 0 \quad (\text{since } 3^2 = 9 < 20)$$

7. $\{4, 3, 5\}$ & $\{5, 3, 4\}$:

$$\Delta = \left(\frac{1}{2}\right) \left(\frac{5 - \sqrt{5}}{8}\right) - \frac{1}{4} = \frac{1 - \sqrt{5}}{16} < 0$$

8. $\{5, 3, 5\}$:

$$\Delta = \left(\frac{5 - \sqrt{5}}{8}\right)^2 - \frac{1}{4} = \frac{14 - 10\sqrt{5}}{64} < 0 \quad (\text{since } 14^2 = 196 < 500)$$

Thus, precisely 6 triples yield positive-definite Gram matrices.

5 5. Complete Lean 4 Formal Verification

The Lake package compiles against Lean 4 and Mathlib. It contains no project-defined axiom, no `admit`, and no `sorry` outside the deliberate holes in `Challenge.lean`. Compared theorems may still depend on Mathlib's `propext`, `Classical.choice`, and `Quot.sound`. The development is split into section-sized modules under `PlatonicSolids/`, each meant to be copied as a gist next to the matching paragraph above. `PlatonicSolids.lean` is the root import. The full sources are reproduced in the appendix. Figures 5 and 6 are the module-level blueprints: first the top-level `PlatonicSolids/*.lean` graph, then the `SingularExcision/` subgraph.

The compared theorems are:

- `platonic_solids_3d` — Euler $V - E + F = 2$ plus regularity incidence $pF = 2E$, $qV = 2E$ force a Platonic pair (p, q) .
- `edges_pos_of_regular` — the same identities algebraically force $E > 0$; non-degeneracy is a theorem, not an axiom.
- `regular_polychora_classification` — Platonic cells and vertex figures with $\Delta(p, q, r) > 0$ are exactly the six regular convex 4-polytopes.

The stronger equivalence `regular_polychora_iff` identifies those six with positive-definiteness of `schlafligram` for $p, q, r \geq 3$, via Sylvester's criterion (`PlatonicSolids/Sylvester.lean`).

`Challenge.lean` restates the compared family with deliberate `sorry`s so Palomar Comparator can check types against `Solution.lean`. Supporting definitions are `IsPlatonicPair`, `IsRegular4Polytope`, `schlafligram`, and `schlafliDet`.

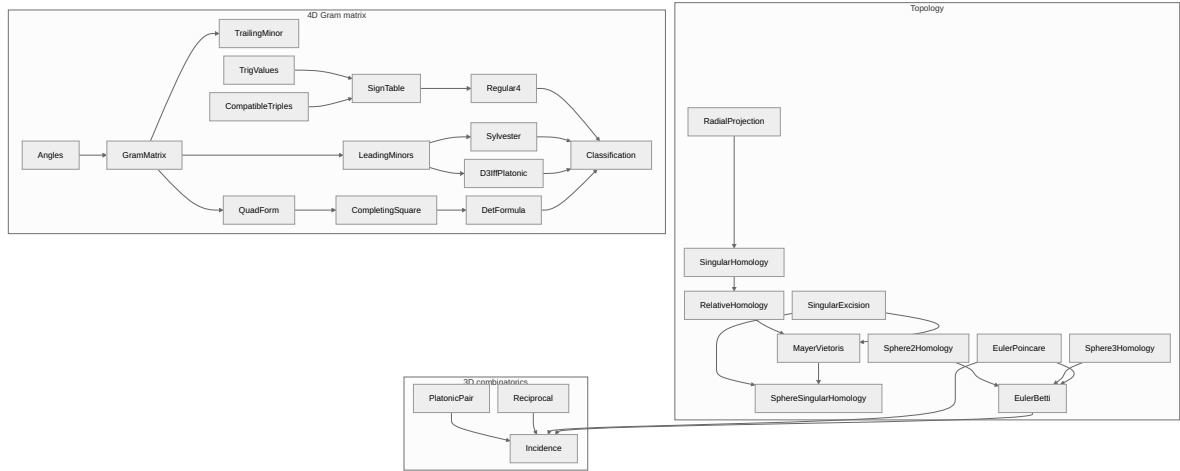


Figure 5: Top-level Lean module graph. Topology and cellular Euler–Poincaré feed the 3D incidence theorems; the Gram-matrix stack feeds the 4D classification. **SingularExcision** is the open-cover input to Mayer–Vietoris.

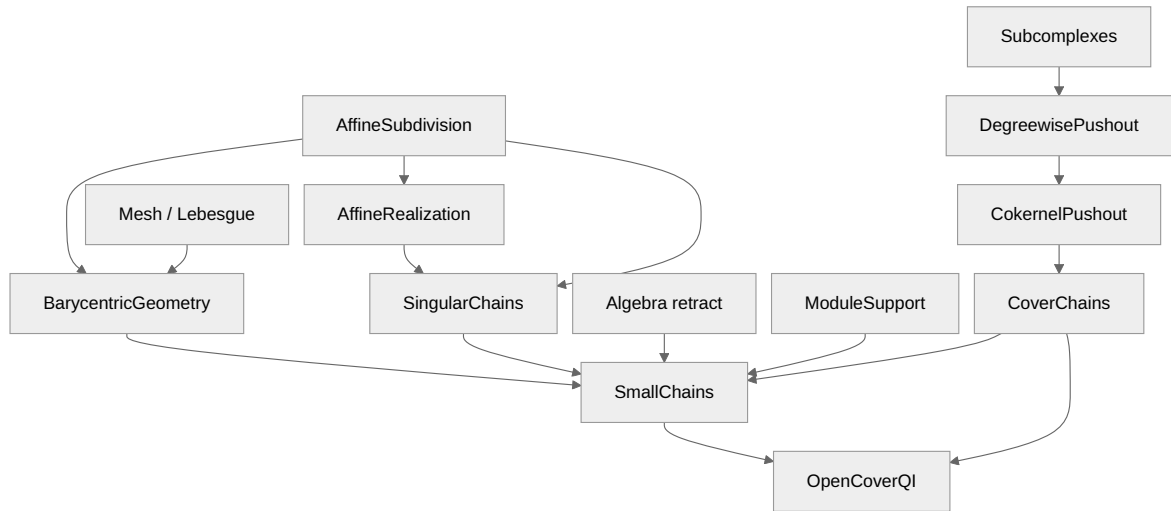


Figure 6: Finer blueprint of **PlatonicSolids/SingularExcision/**. Affine subdivision, mesh geometry, and the prism homotopy assemble the small-chain retract; cover chains are identified with small singular chains; the open-cover quasi-isomorphism discharges excision.

File	Paper	Content
PlatonicSolids/ RadialProjection.lean	§2.1	Frontier of a convex body $\simeq$ sphere
PlatonicSolids/ EulerPoincare.lean	§§2.1, 3.1	Algebraic Euler–Poincaré in dimensions 2 and 3
PlatonicSolids/ SingularHomology.lean	§2.1	Homotopy invariance; $H_0(S^n); H_*(S^0)$
PlatonicSolids/ RelativeHomology.lean	§2.1	Pair sequence $0 \rightarrow C_*(A) \rightarrow C_*(X) \rightarrow C_*(X, A) \rightarrow 0$ and LES
PlatonicSolids/ MayerVietoris.lean	§2.1	Triad SES, excision predicate, stereographic cover of S^n
PlatonicSolids/ SingularExcision.lean	§2.1	Root of the barycentric / small-chain stack
PlatonicSolids/ SingularExcision/*.lean	§2.1	Mesh shrinking, prism homotopy, small retract, cover = small, open-cover QI
PlatonicSolids/ SphereSingularHomology.lean	§2.1	$U \cap V \simeq_h S^{n-1};$ $H_n(S^n; R) \cong R$ for $n > 0$; Euler bridge
PlatonicSolids/ Sphere2Homology.lean	§2.1	Cellular $H_*(S^2; \mathbb{Q})$ of the tetrahedron surface
PlatonicSolids/ Sphere3Homology.lean	§3.1	Cellular $H_*(S^3; \mathbb{Q})$ of the 4-simplex boundary
PlatonicSolids/ EulerBetti.lean	§§2.1, 3.1	$\chi(S^2) = 2, \chi(S^3) = 0$
PlatonicSolids/ PlatonicPair.lean	§2.3	Five solutions of $(p-2)(q-2) < 4$
PlatonicSolids/ Reciprocal.lean	§§2.3, 3.3	$1/p + 1/q > 1/2$
PlatonicSolids/ Incidence.lean	§§2.2–2.4	platonic_solids_3d, edges_pos_of_regular
PlatonicSolids/ Angles.lean	§3.3	$\pi/n \in (0, \pi/2)$
PlatonicSolids/ GramMatrix.lean	§3.2	schlafligram, schlafliDet
PlatonicSolids/ LeadingMinors.lean	§3.3	D_1, D_2, D_3
PlatonicSolids/ TrailingMinor.lean	§3.3	Vertex-figure 3×3
PlatonicSolids/ D3IffPlatonic.lean	§3.3	$D_3 > 0$ iff Platonic cell
PlatonicSolids/ QuadForm.lean	§3.3	Gram quadratic form
PlatonicSolids/ CompletingSquare.lean	§3.3	LDL identity
PlatonicSolids/ DetFormula.lean	§3.4	$\det M = \Delta$
PlatonicSolids/ Sylvester.lean	§3.3	PosDef iff leading minors > 0

File	Paper	Content
PlatonicSolids/ TrigValues.lean	§4	$\sin^2(\pi/n), \cos^2(\pi/n)$ for $n \in \{3, 4, 5\}$
PlatonicSolids/ CompatibleTriples.lean	§4	Eleven compatible triples
PlatonicSolids/ SignTable.lean	§4	Sign of Δ
PlatonicSolids/ Regular4.lean	§4	The six regular 4-polytopes
PlatonicSolids/ Classification.lean	§4	Compared 4D theorem and PosDef iff

The remaining named lemmas are the computational steps of those paragraphs: `platonic_pairs_of_inequality` / `platonic_pair_iff` / `platonic_pair_iff_reciprocal` for the Dio-phantine pair; `leadingMinor_one/two/three` and `leadingMinor_three_pos_iff` / `trailing3_pos_iff` for Sylvester’s first three minors; `schlafligram_quad` / `band_quad_squares` / `schlafligram_det` for $\det M = \Delta$; `schlafliDet_*` and `schlafliDet_*_pos/neg` for the eleven signs; `regular_polychora_iff` / `posDef_of_regular` / `regular_of_posDef` for the PosDef equivalence.

6 6. Pedagogical and Formalization Significance

1. **Foundational Consistency Across Dimensions:** The mathematical transition between 3D and 4D is made explicit: 3D regular polytopes are bounded by the topology of S^2 , whereas 4D polychora are bounded by the spectral properties of Coxeter reflection groups via Sylvester’s criterion.
2. **Elimination of Redundant Hypotheses:** As demonstrated in `edges_pos_of_regular`, Lean’s linter and tactic engine show that the geometric non-degeneracy condition $E > 0$ is a theorem rather than an independent axiom.
3. **Absence of Ad-Hoc Combinatorial Filters:** Candidate 4D triples are filtered using the **Schläfli determinant** $\Delta(p, q, r) = \sin^2 \frac{\pi}{p} \sin^2 \frac{\pi}{r} - \cos^2 \frac{\pi}{q}$. Every rejection is certified by non-linear arithmetic over real algebraic numbers, with all properties of $\sqrt{5}$ derived from first principles.

7 Acknowledgements

The author expresses sincere gratitude to Professor Dana Scott (Carnegie Mellon University) for suggesting this topic and for inspiring discussions on the formalization of polyhedral geometry. Any remaining errors, misinterpretations, or omissions are solely the author’s own.

7.1 Declaration of Generative AI and AI-Assisted Technologies

In accordance with the Committee on Publication Ethics (COPE) position statements and guidelines on artificial intelligence in research and formal authorship, the author acknowledges the use of large language model tooling during the preparation of this work. Specifically, **Google Gemini 3.8**, **Cursor Grok 4.6**, and **OpenAI GPT-5.6 Sol** (accessed through Cursor)

were employed as interactive coding and drafting assistants. They assisted with exploratory formalization, initial Lean 4 proof tactics, the singular-homology and excision investigation, and the structure of the expository drafts. All mathematical claims, Lean 4 proofs, and final text were subsequently reviewed, verified, refined, and certified by the author, who assumes full responsibility for the integrity and accuracy of the content.

8 References

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A Complete Lean source

Checked by `lake build` against Lean 4 and Mathlib. The development contains no `sorry`, no `admit`, and no project-defined axiom. Palomar Comparator compares `platonic_solids_3d`, `edges_pos_of_regular`, and `regular_polychora_classification`.

A.1 PlatonicSolids.lean

Root import of the section-sized modules (39 lines).

```

/-
Copyright (c) 2026 Lars Warren Ericson. All rights reserved.
Released under Apache 2.0 license as described in the file LICENSE.
Authors: Lars Warren Ericson.
Github: https://github.com/catskillsresearch/platonic-solids/blob/main/PlatonicSolids.lean
-/
import PlatonicSolids.Angles

```

```

import PlatonicSolids.Classification
import PlatonicSolids.CompatibleTriples
import PlatonicSolids.CompletingSquare
import PlatonicSolids.D3IffPlatonic
import PlatonicSolids.DetFormula
import PlatonicSolids.EulerBetti
import PlatonicSolids.EulerPoincare
import PlatonicSolids.GramMatrix
import PlatonicSolids.Incidence
import PlatonicSolids.LeadngMinors
import PlatonicSolids.MayerVietoris
import PlatonicSolids.PlatonicPair
import PlatonicSolids.QuadForm
import PlatonicSolids.RadialProjection
import PlatonicSolids.Reciprocal
import PlatonicSolids.Regular4
import PlatonicSolids.RelativeHomology
import PlatonicSolids.SignTable
import PlatonicSolids.SingularExcision
import PlatonicSolids.SingularHomology
import PlatonicSolids.SphereSingularHomology
import PlatonicSolids.Sphere2Homology
import PlatonicSolids.Sphere3Homology
import PlatonicSolids.Sylvester
import PlatonicSolids.TrailingMinor
import PlatonicSolids.TrigValues

/-!
Root import of the section-sized modules. Each file under `PlatonicSolids/`
is meant to be readable next to the corresponding paragraph of
`PlatonicSolids.md`.
-/

```

A.2 PlatonicSolids/RadialProjection.lean

Frontier of a bounded convex body is homeomorphic to the sphere (25 lines).

```

/-
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-/
import Mathlib.Analysis.Convex.GaugeRescale

/-!
Paper §2.1. Radial projection: the frontier of a bounded convex body
with nonempty interior is homeomorphic to the unit sphere. Mathlib's
`gaugeRescaleHomeomorph` is the homeomorphism of the ambient space
that carries the frontier onto `Metric.sphere 0 1`.
-/

open Metric Set

/-- The frontier of a bounded convex set with nonempty interior is a
topological sphere. This is the radial projection of the paper, stated
for a body that need not be centred at the origin. -/
theorem frontier_convex_homeo_sphere
  {E : Type*} [NormedAddCommGroup E] [NormedSpace ℝ E] {s : Set E}
  (hc : Convex ℝ s) (hne : (interior s).Nonempty) (hb : Bornology.IsBounded s) :
  ∃ h : E  $\simeq_t$  E, h '' frontier s = sphere (0 : E) 1 :=
  let ⟨h, _, _, hf⟩ :=
    exists_homeomorph_image_interior_closure_frontier_eq_unitBall hc hne hb
  ⟨h, hf⟩

```

A.3 PlatonicSolids/EulerPoincare.lean

Algebraic Euler–Poincaré in dimensions 2 and 3 (112 lines).

```

/-
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-/
import Mathlib.LinearAlgebra.FiniteDimensional.Lemmas

/-!
Paper §2.1. Algebraic Euler–Poincaré for a 2-dimensional chain complex
of finite-dimensional vector spaces (the cellular complex of a polyhedron):

 $\chi(C) = \text{rank } C_0 - \text{rank } C_1 + \text{rank } C_2$  equals  $\chi(H) = b_0 - b_1 + b_2$ .

The identity is rank-nullity in dimensions 2 and 3. Specializing
to the Betti numbers of the tetrahedron surface
(`PlatonicSolids/Sphere2Homology.lean`) yields  $V + F = E + 2$ .
Specializing to the 4-simplex boundary
(`PlatonicSolids/Sphere3Homology.lean`) yields  $V - E + F - C = 0$ .
-/

open Module

variable {K : Type*} [Field K]
variable {C₂ C₁ C₀ : Type*}
  [AddCommGroup C₂] [Module K C₂] [FiniteDimensional K C₂]
  [AddCommGroup C₁] [Module K C₁] [FiniteDimensional K C₁]
  [AddCommGroup C₀] [Module K C₀]

/--  $b_0 = \text{rank}(C_0 / \text{im } d_1)$ . -/
noncomputable def betti0 (d₁ : C₁ →₁[K] C₀) : ℤ :=
  finrank K C₀ - finrank K (LinearMap.range d₁)

/--  $b_1 = \text{rank}(\text{ker } d_1 / \text{im } d_2)$ . -/
noncomputable def betti1 (d₂ : C₂ →₁[K] C₁) (d₁ : C₁ →₁[K] C₀) : ℤ :=
  finrank K (LinearMap.ker d₁) - finrank K (LinearMap.range d₂)

/--  $b_2 = \text{rank}(\text{ker } d_2)$ . -/
noncomputable def betti2 (d₂ : C₂ →₁[K] C₁) : ℤ :=
  finrank K (LinearMap.ker d₂)

/-- Algebraic Euler–Poincaré: cell Euler characteristic equals homological
Euler characteristic for any 2-complex with  $d_1 \cdot \text{comp } d_2 = 0$ . -/
theorem euler_poincare (d₂ : C₂ →₁[K] C₁) (d₁ : C₁ →₁[K] C₀)
  (hcomp : d₁.comp d₂ = 0) :
  (finrank K C₀ : ℤ) - finrank K C₁ + finrank K C₂ =
    betti0 d₁ - betti1 d₂ d₁ + betti2 d₂ := by
  have _hB : LinearMap.range d₂ ≤ LinearMap.ker d₁ :=
    LinearMap.range_le_ker_iff.mpr hcomp
  have hr1 : (finrank K (LinearMap.range d₁) : ℤ) + finrank K (LinearMap.ker d₁) =
    finrank K C₁ := by exact_mod_cast d₁.finite_rank_range_add_finite_rank_ker
  have hr2 : (finrank K (LinearMap.range d₂) : ℤ) + finrank K (LinearMap.ker d₂) =
    finrank K C₂ := by exact_mod_cast d₂.finite_rank_range_add_finite_rank_ker
  simp only [betti0, betti1, betti2]
  linarith

/--  $V - E + F = 2$  is the same counting identity as  $V + F = E + 2$ . -/
theorem euler_count {V E F : ℕ} (h : (V : ℤ) - E + F = 2) : V + F = E + 2 := by
  omega

/-- Cellular ranks  $V, E, F$  and the Betti numbers of  $S^2$  give Euler's formula. -/
theorem euler_formula_of_sphere2

```

```

(d₂ : C₂ →₁[K] C₁) (d₁ : C₁ →₁[K] C₀)
{V E F : ℕ}
(hV : finrank K C₀ = V) (hE : finrank K C₁ = E) (hF : finrank K C₂ = F)
(hcomp : d₁.comp d₂ = 0)
(hb0 : betti0 d₁ = 1) (hb1 : betti1 d₂ d₁ = 0) (hb2 : betti2 d₂ = 1) :
V + F = E + 2 := by
have hχ := euler_poincare d₂ d₁ hcomp
rw [hV, hE, hF, hb0, hb1, hb2] at hχ
exact euler_count hχ

variable {C₃ : Type*}
[AddCommGroup C₃] [Module K C₃] [FiniteDimensional K C₃]

/-- In a 3-complex, `b₂ = rank(ker d₂ / im d₃)`. -/
noncomputable def betti2Rel (d₃ : C₃ →₁[K] C₂) (d₂ : C₂ →₁[K] C₁) : ℤ :=
finrank K (LinearMap.ker d₂) - finrank K (LinearMap.range d₃)

/-- `b₃ = rank(ker d₃)`. -/
noncomputable def betti3 (d₃ : C₃ →₁[K] C₂) : ℤ :=
finrank K (LinearMap.ker d₃)

/-- Algebraic Euler-Poincaré in dimension 3. -/
theorem euler_poincare3 (d₃ : C₃ →₁[K] C₂) (d₂ : C₂ →₁[K] C₁) (d₁ : C₁ →₁[K] C₀)
(h32 : d₂.comp d₃ = 0) (h21 : d₁.comp d₂ = 0) :
(finrank K C₀ : ℤ) - finrank K C₁ + finrank K C₂ - finrank K C₃ =
betti0 d₁ - betti1 d₂ d₁ + betti2Rel d₃ d₂ - betti3 d₃ := by
have _hB21 : LinearMap.range d₂ ≤ LinearMap.ker d₁ :=
LinearMap.range_le_ker_iff.mpr h21
have _hB32 : LinearMap.range d₃ ≤ LinearMap.ker d₂ :=
LinearMap.range_le_ker_iff.mpr h32
have hr1 : (finrank K (LinearMap.range d₁) : ℤ) + finrank K (LinearMap.ker d₁) =
finrank K C₁ := by exact_mod_cast d₁.finrank_range_add_finrank_ker
have hr2 : (finrank K (LinearMap.range d₂) : ℤ) + finrank K (LinearMap.ker d₂) =
finrank K C₂ := by exact_mod_cast d₂.finrank_range_add_finrank_ker
have hr3 : (finrank K (LinearMap.range d₃) : ℤ) + finrank K (LinearMap.ker d₃) =
finrank K C₃ := by exact_mod_cast d₃.finrank_range_add_finrank_ker
simp only [betti0, betti1, betti2Rel, betti3]
linarith

/-- Cell counts of an `S³` complex with Betti numbers `1,0,0,1` give `χ = 0`. -/
theorem euler_formula_of_sphere3
(d₃ : C₃ →₁[K] C₂) (d₂ : C₂ →₁[K] C₁) (d₁ : C₁ →₁[K] C₀)
{V E F C : ℕ}
(hV : finrank K C₀ = V) (hE : finrank K C₁ = E)
(hF : finrank K C₂ = F) (hC : finrank K C₃ = C)
(h32 : d₂.comp d₃ = 0) (h21 : d₁.comp d₂ = 0)
(hb0 : betti0 d₁ = 1) (hb1 : betti1 d₂ d₁ = 0)
(hb2 : betti2Rel d₃ d₂ = 0) (hb3 : betti3 d₃ = 1) :
(V : ℤ) - E + F - C = 0 := by
have hχ := euler_poincare3 d₃ d₂ d₁ h32 h21
rw [hV, hE, hF, hC, hb0, hb1, hb2, hb3] at hχ
linarith

```

A.4 PlatonicSolids/SingularHomology.lean

Singular $H_0(S^n)$, $H_*(S^0)$, homotopy invariance (238 lines).

```

/-
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-/
import Mathlib.AlgebraicTopology.SingularHomology.HomologyZero
import Mathlib.AlgebraicTopology.SingularHomology.HomotopyInvariance

```

```

import Mathlib.Analysis.InnerProductSpace.PiL2
import Mathlib.Analysis.Normed.Module.Connected
import Mathlib.LinearAlgebra.Dimension.StrongRankCondition
import Mathlib.Topology.Homotopy.Contractible
import Mathlib.Topology.Homotopy.Equiv
import Mathlib.Topology.Separation.Lemmas

/-!
Singular homology lemmas that Mathlib does not package: homotopy
equivalences and homeomorphisms induce homology isomorphisms,
contractible spaces have the homology of a point,  $H_0(S^n) \cong \mathbb{R}$  for
 $n \geq 1$ , and the full calculation of  $H_*(S^0)$ .

The pair and Mayer-Vietoris sequences of singular chains are in
`PlatonicSolids/RelativeHomology.lean` and
`PlatonicSolids/MayerVietoris.lean`. Barycentric subdivision and the
open-cover quasi-isomorphism are in `PlatonicSolids/SingularExcision/`.
The calculation  $H_n(S^n) \cong \mathbb{R}$  for  $n > 0$  is in
`PlatonicSolids/SphereSingularHomology.lean`. The
paper's  $\chi(S^3) = 0$  is obtained from the cellular calculation in
`PlatonicSolids/Sphere3Homology.lean`.
-/

open AlgebraicTopology CategoryTheory Limits HomologicalComplex
open scoped ContinuousMap Topology

universe v u

variable {C : Type u} [Category.{v} C] [HasCoproducts.{0} C] [Preadditive C]
[CategoryWithHomology C]

/-- Homotopic continuous maps induce equal singular-homology maps. -/
lemma singularHomologyMap_eq_of_homotopic
  {X Y : TopCat.{0}} {f g : X → Y} (h : f.hom.Homotopic g.hom)
  (R : C) (n : ℕ) :
  homologyMap (((singularChainComplexFunctor C).obj R).map f) n =
    homologyMap (((singularChainComplexFunctor C).obj R).map g) n :=
  TopCat.Homotopy.congr_homologyMap_singularChainComplexFunctor h.some R n

/-- A homotopy equivalence of spaces induces a homotopy equivalence of
singular chain complexes. -/
noncomputable def singularChainHomotopyEquiv
  {X Y : Type} [TopologicalSpace X] [TopologicalSpace Y]
  (e : X  $\simeq_h$  Y) (R : C) :
  HomotopyEquiv
    (((singularChainComplexFunctor C).obj R).obj (.of X))
    (((singularChainComplexFunctor C).obj R).obj (.of Y)) where
  hom := ((singularChainComplexFunctor C).obj R).map (TopCat.ofHom e.toFun)
  inv := ((singularChainComplexFunctor C).obj R).map (TopCat.ofHom e.invFun)
  homotopyHomInvId := by
    have H : TopCat.Homotopy
      (TopCat.ofHom e.toFun  $\gg$  TopCat.ofHom e.invFun) ( $\mathbb{K}$  _) := by
      simpa [TopCat.ofHom_comp, TopCat.ofHom_id] using e.left_inv.some
    simpa [Functor.map_comp, Functor.map_id] using
      H.singularChainComplexFunctorObjMap R
  homotopyInvHomId := by
    have H : TopCat.Homotopy
      (TopCat.ofHom e.invFun  $\gg$  TopCat.ofHom e.toFun) ( $\mathbb{K}$  _) := by
      simpa [TopCat.ofHom_comp, TopCat.ofHom_id] using e.right_inv.some
    simpa [Functor.map_comp, Functor.map_id] using
      H.singularChainComplexFunctorObjMap R

/-- Homotopy equivalent spaces have isomorphic singular homology. -/

```

```

noncomputable def singularHomologyIso_of_homotopyEquiv
  {X Y : Type} [TopologicalSpace X] [TopologicalSpace Y]
  (e : X  $\simeq_h$  Y) (R : C) (n :  $\mathbb{N}$ ) :
  ((singularHomologyFunctor C n).obj R).obj (.of X)  $\cong$ 
    ((singularHomologyFunctor C n).obj R).obj (.of Y) :=
  (singularChainHomotopyEquiv e R).toHomologyIso n

/-- Homeomorphic spaces have isomorphic singular homology. -/
noncomputable def singularHomologyIso_of_homeomorph
  {X Y : Type} [TopologicalSpace X] [TopologicalSpace Y]
  (e : X  $\simeq_t$  Y) (R : C) (n :  $\mathbb{N}$ ) :
  ((singularHomologyFunctor C n).obj R).obj (.of X)  $\cong$ 
    ((singularHomologyFunctor C n).obj R).obj (.of Y) :=
  singularHomologyIso_of_homotopyEquiv e.toHomotopyEquiv R n

/-- A point has vanishing singular homology in positive degree. -/
lemma isZero_singularHomology_unit (R : C) {n :  $\mathbb{N}$ } (hn : n  $\neq$  0) :
  IsZero (((singularHomologyFunctor C n).obj R).obj (.of Unit)) :=
  isZero_singularHomologyFunctor_of_totallyDisconnectedSpace C n R (.of Unit) hn

/-- A contractible space has vanishing singular homology in positive degree. -/
lemma isZero_singularHomology_of_contractible
  {X : Type} [TopologicalSpace X] [ContractibleSpace X]
  (R : C) {n :  $\mathbb{N}$ } (hn : n  $\neq$  0) :
  IsZero (((singularHomologyFunctor C n).obj R).obj (.of X)) := by
  obtain ⟨e⟩ := ContractibleSpace.hequiv_unit X
  exact (isZero_singularHomology_unit R hn).of_iso
  (singularHomologyIso_of_homotopyEquiv e R n)

/-- Path-connectedness transfers along a homotopy equivalence. -/
lemma pathConnectedSpace_of_homotopyEquiv
  {X Y : Type} [TopologicalSpace X] [TopologicalSpace Y]
  [PathConnectedSpace Y] (e : ContinuousMap.HomotopyEquiv X Y) :
  PathConnectedSpace X :=
  ⟨(e.invFun (Classical.arbitrary Y)), fun x1 x2 =>
    (e.left_inv.some.evalAt x1).symm.trans
      (((PathConnectedSpace.somePath (e.toFun x1) (e.toFun x2)).map
        e.invFun.continuous).trans
        (e.left_inv.some.evalAt x2)))⟩

/-- The augmentation  $H_0(X) \rightarrow R$  is natural. -/
lemma singularHomology0 $\epsilon$ _natural {X Y : TopCat.{0}} (f : X  $\longrightarrow$  Y) (R : C) :
  homologyMap (((singularChainComplexFunctor C).obj R).map f) 0  $\gg$ 
    Y.singularHomology0 $\epsilon$  R =
  X.singularHomology0 $\epsilon$  R := by
  let SX := TopCat.toSSet.obj X
  let SY := TopCat.toSSet.obj Y
  let sf := TopCat.toSSet.map f
  let  $\varphi$  := SSet.chainComplexMap sf R
  let KX := SX.chainComplex R
  let KY := SY.chainComplex R
  apply (cancel_epi (KX.homology $\pi$  0)).1
  apply (cancel_epi (ChainComplex.cycles0Iso KX).inv).1
  apply SSet.chainComplex_hom_ext
  intro x
  have hx :
    KX.liftCycles (SX. $\iota$ ChainComplex x) 0 (by simp) (by simp) =
      SX. $\iota$ ChainComplex x  $\gg$  (ChainComplex.cycles0Iso KX).inv := by
  rw [← cancel_mono (KX.iCycles 0)]
  simp [ChainComplex.cycles0Iso]
  exact (Category.comp_id _).symm
  have hnat := HomologicalComplex.homology $\pi$ _naturality ( $\varphi$  :=  $\varphi$ ) (i := 0)
  have hcycle := HomologicalComplex.liftCycles_comp_cyclesMap

```

```

(k := SX.ιChainComplex x) (j := 0) (hj := by simp) (hk := by simp) φ
have hf : SX.ιChainComplex x >> φ.f 0 = SY.ιChainComplex (sf.app _ x) :=
  SSet.ι_chainComplexMap_f (X := SX) (Y := SY) (f := sf) (R := R) x
have hY := SSet.liftCycles_ιChainComplex_homologyπ_homology0ε SY R (sf.app _ x)
have hX := SSet.liftCycles_ιChainComplex_homologyπ_homology0ε SX R x
have lhs :
  SX.ιChainComplex x >> (ChainComplex.cyclesIso KX).inv >>
    KX.homologyπ 0 >> homologyMap φ 0 >> Y.singularHomology0ε R = ⌞ R := by
  rw [← Category.assoc, ← hx, reassoc_of% hnat, reassoc_of% hcycle]
  exact (congrArg (fun t =>
    KY.liftCycles t 0 (by simp) (by simp) >>
    KY.homologyπ 0 >> Y.singularHomology0ε R) hf).trans hY
have rhs :
  SX.ιChainComplex x >> (ChainComplex.cyclesIso KX).inv >>
    KX.homologyπ 0 >> X.singularHomology0ε R = ⌞ R := by
  rw [← Category.assoc, ← hx]
  exact hX
exact lhs.trans rhs.symm

/-- Path-connected spaces have `H₀ ≅ R` via the augmentation. -/
noncomputable def singularHomology0Iso_pathConnected
  (X : TopCat.{0}) [PathConnectedSpace X] (R : C) :
  ((singularHomologyFunctor C 0).obj R).obj X ≅ R :=
  asIso (X.singularHomology0ε R)

/-- A map of path-connected spaces induces an isomorphism on `H₀`. -/
instance singularHomology0_map_isIso
  {X Y : TopCat.{0}} [PathConnectedSpace X] [PathConnectedSpace Y]
  (f : X → Y) (R : C) :
  Iso (homologyMap (((singularChainComplexFunctor C).obj R).map f) 0) := by
  have h := singularHomology0ε_natural f R
  let eX := singularHomology0Iso_pathConnected X R
  let eY := singularHomology0Iso_pathConnected Y R
  refine ⟨eY.hom >> eX.inv, ?_, ?_⟩
  · erw [← Category.assoc, h, eX.hom_inv_id]; rfl
  · have hf : homologyMap (((singularChainComplexFunctor C).obj R).map f) 0 =
      eX.hom >> eY.inv := by
      refine (cancel_mono eY.hom).mp ?_
      erw [h, Category.assoc, eY.inv_hom_id, Category.comp_id]
      rfl
  calc
    (eY.hom >> eX.inv) >>
      homologyMap (((singularChainComplexFunctor C).obj R).map f) 0 =
      (eY.hom >> eX.inv) >> eX.hom >> eY.inv := by
      rw [hf]; rfl
    _ = eY.hom >> (eX.inv >> eX.hom) >> eY.inv := by
      simp [Category.assoc]
    _ = eY.hom >> (⌞ R) >> eY.inv := by
      rw [eX.inv_hom_id]
    _ = eY.hom >> eY.inv := by
      simp
    _ = ⌞ _ := eY.hom_inv_id

/-- The unit sphere in Euclidean dimension `n+1 ≥ 2` is path-connected. -/
instance pathConnectedSpace_sphere_euclidean
  {n : ℕ} [NeZero n] :
  PathConnectedSpace (↑(Metric.sphere (0 : EuclideanSpace ℝ (Fin (n + 1))) 1)) := by
  rw [← isPathConnected_iff_pathConnectedSpace]
  refine isPathConnected_sphere ?_ 0 (by norm_num)
  have : (1 : ℕ) < n + 1 := Nat.lt_add_of_pos_left (NeZero.pos n)
  have hfin : Module.finrank ℝ (EuclideanSpace ℝ (Fin (n + 1))) = n + 1 := by
    simp
  rw [← Module.finrank_eq_rank, hfin]

```

```

exact_mod_cast this

/--  $H_0(S^n; \mathbb{R}) \cong \mathbb{R}$  for  $n \geq 1$ . -/
noncomputable def singularHomology0_sphere
  {n : ℕ} [NeZero n] (R : C) :
    ((singularHomologyFunctor C 0).obj R).obj
      (.of ↑(Metric.sphere (0 : EuclideanSpace ℝ (Fin (n + 1))) 1)) ≅ R :=
    singularHomology0Iso_pathConnected _ R

/--  $S^0 \subset \mathbb{R}$  is the two-point set  $\{\pm 1\}$ . -/
lemma sphere_real_eq : (Metric.sphere (0 : ℝ) 1 : Set ℝ) = {1, -1} := by
  ext x
  simp [mem_sphere_iff_norm, Real.norm_eq_abs, abs_eq]

instance : TotallyDisconnectedSpace ↑(Metric.sphere (0 : ℝ) 1) := by
  refine totallyDisconnectedSpace_subtype_iff.2
    (Set.Countable.isTotallyDisconnected ?_)
  rw [sphere_real_eq]
  exact ((Set.finite_singleton (-1 : ℝ)).insert 1).countable

/--  $H_n(S^0; \mathbb{R}) = 0$  for  $n > 0$ . -/
lemma isZero_singularHomology_sphere0 (R : C) {n : ℕ} (hn : n ≠ 0) :
  IsZero (((singularHomologyFunctor C n).obj R).obj
    (.of ↑(Metric.sphere (0 : ℝ) 1))) :=
  isZero_singularHomologyFunctor_of_totallyDisconnectedSpace C n R _ hn

/--  $H_0(S^0; \mathbb{R}) \cong \mathbb{R} \oplus \mathbb{R}$ . -/
noncomputable def singularHomology0_sphere0 (R : C) :
  ((singularHomologyFunctor C 0).obj R).obj (.of ↑(Metric.sphere (0 : ℝ) 1)) ≅
    ∏ fun _ : ↑(Metric.sphere (0 : ℝ) 1) ↦ R :=
  singularHomologyFunctorZeroOfTotallyDisconnectedSpace C R
    (.of ↑(Metric.sphere (0 : ℝ) 1))

/-- The closed unit disk is contractible, hence has vanishing positive
singular homology. -/
lemma isZero_singularHomology_closedBall
  {E : Type} [NormedAddCommGroup E] [NormedSpace ℝ E]
  (R : C) {n : ℕ} (hn : n ≠ 0) :
  IsZero (((singularHomologyFunctor C n).obj R).obj
    (.of ↑(Metric.closedBall (0 : E) 1))) := by
  have : ContractibleSpace ↑(Metric.closedBall (0 : E) 1) :=
    Metric.contractibleSpace_closedBall (by norm_num)
  exact isZero_singularHomology_of_contractible R hn

```

A.5 PlatonicSolids/RelativeHomology.lean

Relative singular homology and the LES of a pair (148 lines).

```

/-
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-/
import Mathlib.Algebra.Homology.HomologicalComplexAbelian
import Mathlib.Algebra.Homology.HomologySequence
import Mathlib.AlgebraicTopology.SingularHomology.Basic
import Mathlib.Topology.Category.TopCat.EpiMono

/-!
Relative singular homology of a pair  $(X, A)$  as the cokernel of the
inclusion of singular chains. Mathlib's singular chain-complex functor
preserves monomorphisms (it is free on simplices), so a subspace
inclusion yields a short exact sequence of chain complexes and therefore
the long exact sequence of the pair.

```

The pair sequence is the first half of the path to $H_n(S^n)$. Excision of an open cover and the stereographic connecting isomorphism are in ``PlatonicSolids/SingularExcision/`` and ``PlatonicSolids/SphereSingularHomology.lean``.
 -/

```
open AlgebraicTopology CategoryTheory Limits HomologicalComplex
open scoped Topology
```

```
set_option linter.unusedSectionVars false
```

```
universe v u
```

```
variable {C : Type u} [Category.{v} C] [Abelian C]
  [HasCoproducts.{0} C] [HasPullbacks C]
```

```
variable (R : C)
```

```
/-- Inclusion of a subspace as a morphism of `TopCat`. -/
def subtypeIncl (X : Type) [TopologicalSpace X] (A : Set X) :
  TopCat.of  $\uparrow A \rightarrow$  TopCat.of X :=
  TopCat.ofHom (Subtype.val, continuous_subtype_val)
```

```
instance subtypeIncl_mono (X : Type) [TopologicalSpace X] (A : Set X) :
  Mono (subtypeIncl X A) :=
  (TopCat.mono_iff_injective _).2 Subtype.val_injective
```

```
/-- Singular chains of a subspace, included into those of the ambient space. -/
noncomputable def inclChains (X : Type) [TopologicalSpace X] (A : Set X) :
  ((singularChainComplexFunctor C).obj R).obj (.of  $\uparrow A$ )  $\rightarrow$ 
  ((singularChainComplexFunctor C).obj R).obj (.of X) :=
  ((singularChainComplexFunctor C).obj R).map (subtypeIncl X A)
```

```
instance inclChains_mono (X : Type) [TopologicalSpace X] (A : Set X) :
  Mono (inclChains (C := C) R X A) :=
  ((singularChainComplexFunctor C).obj R).map_mono (subtypeIncl X A)
```

```
/-- Relative singular chains  $C_*(X, A; R) := \text{coker}(C_*(A) \rightarrow C_*(X))$ . -/
noncomputable def relativeSingularChains (X : Type) [TopologicalSpace X] (A : Set X) :
  ChainComplex C  $\mathbb{N}$  :=
  cokernel (inclChains (C := C) R X A)
```

```
/-- The short complex  $C_*(A) \rightarrow C_*(X) \rightarrow C_*(X, A)$ . -/
noncomputable def pairShortComplex (X : Type) [TopologicalSpace X] (A : Set X) :
  ShortComplex (ChainComplex C  $\mathbb{N}$ ) :=
  ShortComplex.mk (inclChains (C := C) R X A) (cokernel. $\pi$  _) (cokernel.condition _)
```

```
instance pairShortComplex_mono_f (X : Type) [TopologicalSpace X] (A : Set X) :
  Mono (pairShortComplex (C := C) R X A).f :=
  inferInstanceAs (Mono (inclChains (C := C) R X A))
```

```
instance pairShortComplex_epi_g (X : Type) [TopologicalSpace X] (A : Set X) :
  Epi (pairShortComplex (C := C) R X A).g :=
  inferInstanceAs (Epi (cokernel. $\pi$  (inclChains (C := C) R X A)))
```

```
/-- The pair sequence of singular chains is short exact. -/
lemma pair_shortExact (X : Type) [TopologicalSpace X] (A : Set X) :
  (pairShortComplex (C := C) R X A).ShortExact :=
  ShortComplex.ShortExact.mk'
  (ShortComplex.exact_of_g_is_cokernel _
    (cokernelIsCokernel (inclChains (C := C) R X A)))
  inferInstance inferInstance
```

```

/-- Relative singular homology  $H_n(X, A; R)$ . -/
noncomputable def relativeSingularHomology
  (X : Type) [TopologicalSpace X] (A : Set X) (n : ℕ) : C :=
  (relativeSingularChains (C := C) R X A).homology n

/-- The map  $H_n(A) \rightarrow H_n(X)$  induced by the subspace inclusion. -/
noncomputable def pair_i
  (X : Type) [TopologicalSpace X] (A : Set X) (n : ℕ) :
  (((singularChainComplexFunctor C).obj R).obj (.of ↑A)).homology n →
  (((singularChainComplexFunctor C).obj R).obj (.of X)).homology n :=
  homologyMap (inclChains (C := C) R X A) n

/-- The map  $H_n(X) \rightarrow H_n(X, A)$  to relative homology. -/
noncomputable def pair_j
  (X : Type) [TopologicalSpace X] (A : Set X) (n : ℕ) :
  (((singularChainComplexFunctor C).obj R).obj (.of X)).homology n →
  relativeSingularHomology (C := C) R X A n :=
  homologyMap (cokernel.π (inclChains (C := C) R X A)) n

/-- Connecting homomorphism  $\delta : H_{n+1}(X, A) \rightarrow H_n(A)$  of the pair. -/
noncomputable def pair_δ
  (X : Type) [TopologicalSpace X] (A : Set X) (n : ℕ) :
  relativeSingularHomology (C := C) R X A (n + 1) →
  (((singularChainComplexFunctor C).obj R).obj (.of ↑A)).homology n :=
  (pair_shortExact (C := C) R X A).δ (n + 1) n (by simp)

/-- The composition  $H_n(A) \rightarrow H_n(X) \rightarrow H_n(X, A)$  is zero. -/
lemma pair_i_comp_pair_j
  (X : Type) [TopologicalSpace X] (A : Set X) (n : ℕ) :
  pair_i (C := C) R X A n >> pair_j (C := C) R X A n = 0 := by
  simpa [pair_i, pair_j] using
    (by
      rw [← homologyMap_comp, cokernel.condition, homologyMap_zero] :
      homologyMap (inclChains (C := C) R X A) n >>
      homologyMap (cokernel.π (inclChains (C := C) R X A)) n = 0)

/-- The composition  $H_n(X) \rightarrow H_n(X, A) \rightarrow H_{n-1}(A)$  is zero. -/
lemma pair_j_comp_pair_δ
  (X : Type) [TopologicalSpace X] (A : Set X) (n : ℕ) :
  pair_j (C := C) R X A (n + 1) >> pair_δ (C := C) R X A n = 0 :=
  (pair_shortExact (C := C) R X A).comp_δ (n + 1) n (by simp)

/-- The composition  $H_{n+1}(X, A) \rightarrow H_n(A) \rightarrow H_n(X)$  is zero. -/
lemma pair_δ_comp_pair_i
  (X : Type) [TopologicalSpace X] (A : Set X) (n : ℕ) :
  pair_δ (C := C) R X A n >> pair_i (C := C) R X A n = 0 :=
  (pair_shortExact (C := C) R X A).δ_comp (n + 1) n (by simp)

/-- Exactness of  $H_n(A) \rightarrow H_n(X) \rightarrow H_n(X, A)$ . -/
lemma pair_exact_at_X
  (X : Type) [TopologicalSpace X] (A : Set X) (n : ℕ) :
  (ShortComplex.mk (pair_i (C := C) R X A n) (pair_j (C := C) R X A n)
    (pair_i_comp_pair_j (C := C) R X A n)).Exact := by
  simpa [pair_i, pair_j, pairShortComplex] using
    (pair_shortExact (C := C) R X A).homology_exact₂ n

/-- Exactness of  $H_n(X) \rightarrow H_n(X, A) \rightarrow H_{n-1}(A)$ . -/
lemma pair_exact_at_rel
  (X : Type) [TopologicalSpace X] (A : Set X) (n : ℕ) :
  (ShortComplex.mk (pair_j (C := C) R X A (n + 1)) (pair_δ (C := C) R X A n)
    (pair_j_comp_pair_δ (C := C) R X A n)).Exact :=
  (pair_shortExact (C := C) R X A).homology_exact₃ (n + 1) n (by simp)

```

```

/-- Exactness of  $H_{n+1}(X, A) \rightarrow H_n(A) \rightarrow H_n(X)$ . -/
lemma pair_exact_at_A
  (X : Type) [TopologicalSpace X] (A : Set X) (n : ℕ) :
    (ShortComplex.mk (pairδ (C := C) R X A n) (pair_i (C := C) R X A n)
      (pairδ_comp_pair_i (C := C) R X A n)).Exact :=
    (pair_shortExact (C := C) R X A).homology_exact₁ (n + 1) n (by simp)

```

A.6 PlatonicSolids/MayerVietoris.lean

Mayer–Vietoris of an excisive triad; stereographic cover of S^n (303 lines).

```

/-
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-/

import Mathlib.Algebra.Homology.HomologicalComplexBiprod
import Mathlib.Algebra.Homology.QuasiIso
import Mathlib.Analysis.Convex.Contractible
import Mathlib.Analysis.InnerProductSpace.PiL2
import Mathlib.Geometry.Manifold.Instances.Sphere
import Mathlib.Topology.Homeomorph.Lemmas
import PlatonicSolids.RelativeHomology
import PlatonicSolids.SingularHomology

/-!
Mayer-Vietoris for a commuting square of spaces
...
  W --iU--> U
  |         |
  iV        jU
  |         |
  v         v
  V --jV--> X
...

The map  $C_*(W) \rightarrow C_*(U) \oplus C_*(V)$ ,  $\sigma \mapsto (iU_\# \sigma, -iV_\# \sigma)$  is a monomorphism
as soon as  $iU$  is. Its cokernel is the chain complex of the cover; the
commuting square induces a comparison  $\text{coker} \rightarrow C_*(X)$ . The square is
*excisive* when that comparison is a quasi-isomorphism. In that case the
homology sequence of the cover is the Mayer-Vietoris sequence of  $X$ .

The stereographic cover of  $S^n$  (complements of the poles) is the
classical excisive pair: each chart is contractible. Barycentric
subdivision, mesh shrinking, and the open-cover quasi-isomorphism live
in PlatonicSolids/SingularExcision/. For ModuleCat coefficients the
pair is excisive (IsExcisiveSubspaces_of_openCover); the hex-free
connecting map and  $H_n(S^n) \cong \mathbb{R}$  for  $n > 0$  are in
PlatonicSolids/SphereSingularHomology.lean.
-/

open AlgebraicTopology CategoryTheory Limits HomologicalComplex Metric Set
open scoped Topology ContinuousMap

set_option linter.unusedSectionVars false

universe v u

variable {C : Type u} [Category.{v} C] [Abelian C]
  [HasCoproducts.{0} C] [HasPullbacks C]

variable (R : C)

```

```

/--! ## Abstract triad -/

variable {W U V X : TopCat.{0}}
variable (iU : W → U) (iV : W → V) (jU : U → X) (jV : V → X)

/-- `C_*(W) → C_*(U) ⊕ C_*(V)`, the Mayer-Vietoris inclusion of the intersection. -/
noncomputable def mvIncl :
  ((singularChainComplexFunctor C).obj R).obj W →
  ((singularChainComplexFunctor C).obj R).obj U ⊕
  ((singularChainComplexFunctor C).obj R).obj V :=
  biprod.lift
  (((singularChainComplexFunctor C).obj R).map iU)
  (-(((singularChainComplexFunctor C).obj R).map iV))

instance mvIncl_mono [Mono iU] : Mono (mvIncl (C := C) R iU iV) :=
  haveI : Mono (((singularChainComplexFunctor C).obj R).map iU) :=
    ((singularChainComplexFunctor C).obj R).map_mono iU
  mono_of_mono_fac (biprod.lift_fst
    (((singularChainComplexFunctor C).obj R).map iU)
    (-(((singularChainComplexFunctor C).obj R).map iV)))

/-- Chains of the cover: `coker(C_*(W) → C_*(U) ⊕ C_*(V))`. -/
noncomputable def coverChains : ChainComplex C ℕ :=
  cokernel (mvIncl (C := C) R iU iV)

/-- The short complex of the triad. -/
noncomputable def triadShortComplex [Mono iU] :
  ShortComplex (ChainComplex C ℕ) :=
  ShortComplex.mk (mvIncl (C := C) R iU iV) (cokernel.π _) (cokernel.condition _)

instance triadShortComplex_mono_f [Mono iU] :
  Mono (triadShortComplex (C := C) R iU iV).f :=
  inferInstanceAs (Mono (mvIncl (C := C) R iU iV))

instance triadShortComplex_epi_g [Mono iU] :
  Epi (triadShortComplex (C := C) R iU iV).g :=
  inferInstanceAs (Epi (cokernel.π (mvIncl (C := C) R iU iV)))

/-- The triad sequence of singular chains is short exact. -/
lemma triad_shortExact [Mono iU] :
  (triadShortComplex (C := C) R iU iV).ShortExact :=
  ShortComplex.ShortExact.mk'
  (ShortComplex.exact_of_g_is_cokernel _
    (cokernelIsCokernel (mvIncl (C := C) R iU iV)))
  inferInstance inferInstance

/-- `C_*(U) ⊕ C_*(V) → C_*(X)` induced by the two inclusions into `X`. -/
noncomputable def coverToX (_hcomm : iU ≫ jU = iV ≫ jV) :
  ((singularChainComplexFunctor C).obj R).obj U ⊕
  ((singularChainComplexFunctor C).obj R).obj V →
  ((singularChainComplexFunctor C).obj R).obj X :=
  biprod.desc
  (((singularChainComplexFunctor C).obj R).map jU)
  (((singularChainComplexFunctor C).obj R).map jV)

lemma mvIncl_comp_coverToX (hcomm : iU ≫ jU = iV ≫ jV) :
  mvIncl (C := C) R iU iV ≫ coverToX (C := C) R iU iV jU jV hcomm = 0 := by
  dsimp [mvIncl, coverToX]
  ext
  simp [← Functor.map_comp, hcomm]

/-- Comparison of cover chains with ambient chains. -/

```

```

noncomputable def coverComparison (hcomm : iU >> jU = iV >> jV) :
  coverChains (C := C) R iU iV →
    ((singularChainComplexFunctor C).obj R).obj X :=
  cokernel.desc _ (coverToX (C := C) R iU iV jU jV hcomm)
  (mvIncl_comp_coverToX (C := C) R iU iV jU jV hcomm)

/-- A commuting square of spaces is excisive when cover chains compute  $H_*(X)$ . -/
def IsExcisive (hcomm : iU >> jU = iV >> jV) : Prop :=
  QuasiIso (coverComparison (C := C) R iU iV jU jV hcomm)

/-- Connecting map of the triad:  $H_{n+1}(U+V) \rightarrow H_n(W)$ . -/
noncomputable def triad $\delta$  [Mono iU] (n :  $\mathbb{N}$ ) :
  (coverChains (C := C) R iU iV).homology (n + 1) →
    (((singularChainComplexFunctor C).obj R).obj W).homology n :=
  (triad_shortExact (C := C) R iU iV). $\delta$  (n + 1) n (by simp)

/-- If the square is excisive, homology of the cover is homology of  $X$ . -/
noncomputable def excisiveHomologyIso
  (hcomm : iU >> jU = iV >> jV)
  (h : IsExcisive (C := C) R iU iV jU jV hcomm) (n :  $\mathbb{N}$ ) :
  (coverChains (C := C) R iU iV).homology n  $\cong$ 
    (((singularChainComplexFunctor C).obj R).obj X).homology n :=
  have : QuasiIso (coverComparison (C := C) R iU iV jU jV hcomm) := h
  asIso (homologyMap (coverComparison (C := C) R iU iV jU jV hcomm) n)

/-- Mayer-Vietoris connecting map, after excision. -/
noncomputable def mayerVietoris $\delta$  [Mono iU]
  (hcomm : iU >> jU = iV >> jV)
  (h : IsExcisive (C := C) R iU iV jU jV hcomm) (n :  $\mathbb{N}$ ) :
  (((singularChainComplexFunctor C).obj R).obj X).homology (n + 1) →
    (((singularChainComplexFunctor C).obj R).obj W).homology n :=
  (excisiveHomologyIso (C := C) R iU iV jU jV hcomm h (n + 1)).inv >>
  triad $\delta$  (C := C) R iU iV n

/-- Exactness of  $H_n(W) \rightarrow H_n(U) \oplus H_n(V) \rightarrow H_n(U+V)$ . -/
lemma triad_exact_at_sum [Mono iU] (n :  $\mathbb{N}$ ) :
  (ShortComplex.mk
    (homologyMap (mvIncl (C := C) R iU iV) n)
    (homologyMap (cokernel. $\pi$  (mvIncl (C := C) R iU iV)) n)
    (by rw [← homologyMap_comp, cokernel.condition, homologyMap_zero])).Exact :=
  (triad_shortExact (C := C) R iU iV).homology_exact $_2$  n

/-- Exactness of  $H_{n+1}(U+V) \rightarrow H_n(W) \rightarrow H_n(U) \oplus H_n(V)$ . -/
lemma triad_exact_at_W [Mono iU] (n :  $\mathbb{N}$ ) :
  (ShortComplex.mk (triad $\delta$  (C := C) R iU iV n)
    (homologyMap (mvIncl (C := C) R iU iV) n)
    (by
      simpa [triad $\delta$ , triadShortComplex] using
        (triad_shortExact (C := C) R iU iV). $\delta$ _comp (n + 1) n (by simp))).Exact :=
  (triad_shortExact (C := C) R iU iV).homology_exact $_1$  (n + 1) n (by simp)

/-! ## Subspace covers -/

variable {Y : Type} [TopologicalSpace Y]

/-- Inclusion of an intersection as a subspace of the first set. -/
def interInclLeft (A B : Set Y) : TopCat.of  $\uparrow(A \cap B) \rightarrow$  TopCat.of  $\uparrow A :=$ 
  TopCat.ofHom (ContinuousMap.inclusion (fun _ hx => hx.1))

/-- Inclusion of an intersection as a subspace of the second set. -/
def interInclRight (A B : Set Y) : TopCat.of  $\uparrow(A \cap B) \rightarrow$  TopCat.of  $\uparrow B :=$ 
  TopCat.ofHom (ContinuousMap.inclusion (fun _ hx => hx.2))

```

```

instance interInclLeft_mono (A B : Set Y) : Mono (interInclLeft A B) :=
  (TopCat.mono_iff_injective _).2 fun _ _ h =>
    Subtype.ext (congrArg (Subtype.val : ↑A → Y) h)

instance interInclRight_mono (A B : Set Y) : Mono (interInclRight A B) :=
  (TopCat.mono_iff_injective _).2 fun _ _ h =>
    Subtype.ext (congrArg (Subtype.val : ↑B → Y) h)

lemma subset_square_comm (A B : Set Y) :
  interInclLeft A B >> subtypeIncl Y A =
    interInclRight A B >> subtypeIncl Y B := by
  ext x
  rfl

/-- A pair of subspaces is excisive when their cover computes `H_*(Y)`. -/
def IsExcisiveSubspaces (A B : Set Y) : Prop :=
  IsExcisive (C := C) R (interInclLeft A B) (interInclRight A B)
  (subtypeIncl Y A) (subtypeIncl Y B) (subset_square_comm A B)

/-! ## Stereographic cover of the metric sphere -/

/-- The unit `n`-sphere in Euclidean space of dimension `n+1`. -/
abbrev MetricSphere (n : ℕ) : Type :=
  ↑(sphere (0 : EuclideanSpace ℝ (Fin (n + 1))) 1)

/-- North pole `e_0` of `S^n`. -/
noncomputable def northPole (n : ℕ) : MetricSphere n :=
  ⟨EuclideanSpace.single 0 1, by simp [mem_sphere_iff_norm]⟩

/-- South pole `-e_0` of `S^n`. -/
noncomputable def southPole (n : ℕ) : MetricSphere n :=
  ⟨-EuclideanSpace.single 0 1, by simp [mem_sphere_iff_norm]⟩

lemma northPole_ne_southPole (n : ℕ) : northPole n ≠ southPole n := by
  intro h
  have h1 : (1 : ℝ) = -1 := by
    simpa [northPole, southPole] using
      congrArg (fun p : MetricSphere n =>
        (p : EuclideanSpace ℝ (Fin (n + 1))) 0) h
  norm_num at h1

/-- Open chart `S^n \ {N}`. -/
def puncturedNorth (n : ℕ) : Set (MetricSphere n) :=
  {northPole n}^c

/-- Open chart `S^n \ {S}`. -/
def puncturedSouth (n : ℕ) : Set (MetricSphere n) :=
  {southPole n}^c

lemma isOpen_puncturedNorth (n : ℕ) : IsOpen (puncturedNorth n) :=
  isOpen_compl_singleton

lemma isOpen_puncturedSouth (n : ℕ) : IsOpen (puncturedSouth n) :=
  isOpen_compl_singleton

lemma punctured_cover (n : ℕ) :
  puncturedNorth n ∪ puncturedSouth n = univ := by
  ext x
  simp [puncturedNorth, puncturedSouth]
  exact or_iff_not_imp_left.2 fun hx => by
    rintro rfl
    exact hx (northPole_ne_southPole n).symm

```

```

lemma punctured_inter (n : ℕ) :
  puncturedNorth n ∩ puncturedSouth n = {northPole n, southPole n}c := by
  ext x
  simp [puncturedNorth, puncturedSouth]

lemma norm_northPole (n : ℕ) :
  ||(northPole n : EuclideanSpace ℝ (Fin (n + 1)))|| = 1 := by
  simp [northPole]

lemma norm_southPole (n : ℕ) :
  ||(southPole n : EuclideanSpace ℝ (Fin (n + 1)))|| = 1 := by
  simp [southPole]

/-- Stereographic projection from the north pole is a homeomorphism
`S^n \ {N}` homeomorphic to the orthogonal complement of `ℝ · N`. -/
noncomputable def homeoPuncturedNorth (n : ℕ) :
  ↑(puncturedNorth n) ≃t
  ((ℝ · (northPole n : EuclideanSpace ℝ (Fin (n + 1))))⊥) :=
  have hsrc : (stereographic (norm_northPole n)).source = puncturedNorth n := by
  ext x
  simp [puncturedNorth, northPole, stereographic_source]
  (Homeomorph.setCongr hsrc).trans
  ((stereographic (norm_northPole n)).toHomeomorphSourceTarget.trans
    (Homeomorph.Set.univ _))

instance contractible_puncturedNorth (n : ℕ) :
  ContractibleSpace ↑(puncturedNorth n) :=
  (homeoPuncturedNorth n).contractibleSpace

/-- Stereographic projection from the south pole is a homeomorphism
`S^n \ {S}` homeomorphic to the orthogonal complement of `ℝ · S`. -/
noncomputable def homeoPuncturedSouth (n : ℕ) :
  ↑(puncturedSouth n) ≃t
  ((ℝ · (southPole n : EuclideanSpace ℝ (Fin (n + 1))))⊥) :=
  have hsrc : (stereographic (norm_southPole n)).source = puncturedSouth n := by
  ext x
  simp [puncturedSouth, southPole, stereographic_source]
  (Homeomorph.setCongr hsrc).trans
  ((stereographic (norm_southPole n)).toHomeomorphSourceTarget.trans
    (Homeomorph.Set.univ _))

instance contractible_puncturedSouth (n : ℕ) :
  ContractibleSpace ↑(puncturedSouth n) :=
  (homeoPuncturedSouth n).contractibleSpace

/-- Positive singular homology of the northern stereographic chart vanishes. -/
lemma isZero_singularHomology_puncturedNorth
  (n : ℕ) {k : ℕ} (hk : k ≠ 0) :
  IsZero (((singularHomologyFunctor C k).obj R).obj
    (.of ↑(puncturedNorth n))) :=
  isZero_singularHomology_of_contractible R hk

/-- Positive singular homology of the southern stereographic chart vanishes. -/
lemma isZero_singularHomology_puncturedSouth
  (n : ℕ) {k : ℕ} (hk : k ≠ 0) :
  IsZero (((singularHomologyFunctor C k).obj R).obj
    (.of ↑(puncturedSouth n))) :=
  isZero_singularHomology_of_contractible R hk

-- The hex-free stereographic connecting map is
-- `singularHomology_sphere_mayerVietorisδ` in
-- `PlatonicSolids/SphereSingularHomology.lean` (ModuleCat coefficients).

```

A.7 PlatonicSolids/SingularExcision.lean

Root import of barycentric subdivision and small-chain excision (22 lines).

```
/-
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-/
import PlatonicSolids.SingularExcision.AffineSubdivision
import PlatonicSolids.SingularExcision.AffineRealization
import PlatonicSolids.SingularExcision.Algebra
import PlatonicSolids.SingularExcision.BarycentricGeometry
import PlatonicSolids.SingularExcision.CokernelPushout
import PlatonicSolids.SingularExcision.CoverChains
import PlatonicSolids.SingularExcision.DegreePushout
import PlatonicSolids.SingularExcision.Mesh
import PlatonicSolids.SingularExcision.ModuleSupport
import PlatonicSolids.SingularExcision.OpenCoverQI
import PlatonicSolids.SingularExcision.SingularChains
import PlatonicSolids.SingularExcision.SmallChains
import PlatonicSolids.SingularExcision.Subcomplexes

/-!
Root import for the barycentric-subdivision and small-chain foundations
used by the singular-chain excision proof.
-/
```

A.8 PlatonicSolids/SingularExcision/Algebra.lean

Ungraded subdivision retract on an abelian group (146 lines).

```
/-
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-/
import Mathlib.Algebra.Group.Subgroup.Defs
import Mathlib.Algebra.GroupWithZero.Divisibility
import Mathlib.Logic.Function.Iterate
import Mathlib.Tactic.Abel

/-!
The elementwise algebraic core of the small-simplices proof of singular
excision. It is deliberately independent of singular simplices and is
applied to the direct sum of the chain groups of a chain complex of
modules.
-/

open Function

namespace SingularExcision

/-- Ungraded data underlying subdivision: the differential `d`, subdivision
`S`, and prism `H`, with `id - S = dH + Hd`. -/
structure SubdivisionData (A : Type*) [AddCommGroup A] where
  d : A →+ A
  S : A →+ A
  H : A →+ A
  S_comm : ∀ x, d (S x) = S (d x)
  homotopy : ∀ x, x - S x = d (H x) + H (d x)

namespace SubdivisionData

variable {A : Type*} [AddCommGroup A] (D : SubdivisionData A)
```

```

def Cycle (x : A) : Prop :=
  D.d x = 0

def Homologous (x y : A) : Prop :=
  ∃ w : A, x - y = D.d w

lemma cycle_S {x : A} (hx : D.Cycle x) :
  D.Cycle (D.S x) := by
  rw [Cycle, D.S_comm, hx, map_zero]

lemma cycle_iterate {x : A} (hx : D.Cycle x) (N : ℕ) :
  D.Cycle (D.S^[N] x) := by
  induction N with
  | zero => simp
  | succ N ih =>
    rw [iterate_succ_apply']
    exact D.cycle_S ih

lemma homologous_refl (x : A) :
  D.Homologous x x :=
  ⟨0, by simp⟩

lemma homologous_S_of_cycle {x : A} (hx : D.Cycle x) :
  D.Homologous x (D.S x) := by
  refine ⟨D.H x, ?_⟩
  rw [D.homotopy, show D.d x = 0 from hx, map_zero, add_zero]

lemma homologous_trans {x y z : A}
  (hxy : D.Homologous x y) (hyz : D.Homologous y z) :
  D.Homologous x z := by
  obtain ⟨u, hu⟩ := hxy
  obtain ⟨v, hv⟩ := hyz
  refine ⟨u + v, ?_⟩
  rw [map_add, ← hu, ← hv]
  abel

lemma homologous_iterate_of_cycle {x : A} (hx : D.Cycle x) (N : ℕ) :
  D.Homologous x (D.S^[N] x) := by
  induction N with
  | zero => simp using D.homologous_refl x
  | succ N ih =>
    rw [iterate_succ_apply']
    exact D.homologous_trans ih
    (D.homologous_S_of_cycle (D.cycle_iterate hx N))

variable (B : AddSubgroup A)

/-- The small chains are closed under boundary, subdivision, and prism. -/
structure SmallInvariant : Prop where
  d_mem : ∀ {x}, x ∈ B → D.d x ∈ B
  S_mem : ∀ {x}, x ∈ B → D.S x ∈ B
  H_mem : ∀ {x}, x ∈ B → D.H x ∈ B

lemma iterate_mem (hB : D.SmallInvariant B) {x : A}
  (hx : x ∈ B) (N : ℕ) :
  D.S^[N] x ∈ B := by
  induction N with
  | zero => simp
  | succ N ih =>
    rw [iterate_succ_apply']
    exact hB.S_mem ih

```

```

lemma small_boundary_of_small_homologous
  (hB : D.SmallInvariant B) {x : A}
  (hx : x ∈ B) (hxcycle : D.Cycle x) (N : ℕ) :
  ∃ w ∈ B, x - D.S^[N] x = D.d w := by
induction N with
| zero => exact ⟨0, B.zero_mem, by simp⟩
| succ N ih =>
  obtain ⟨w, hwB, hw⟩ := ih
  let xN := D.S^[N] x
  refine ⟨w + D.H xN,
    B.add_mem hwB (hB.H_mem (D.iterate_mem B hB hx N)), ?_⟩
  rw [map_add, ← hw]
  have hxNcycle := D.cycle_iterate hxcycle N
  have hstep : xN - D.S xN = D.d (D.H xN) := by
    rw [D.homotopy, show D.d xN = 0 from hxNcycle, map_zero, add_zero]
  rw [← hstep, iterate_succ_apply']
  abel

/-- If subdivision eventually puts every finite chain in the small-chain
subgroup, every homology class has a small representative and every small
boundary bounds a small chain. -/
theorem small_cycles_and_boundaries
  (hB : D.SmallInvariant B)
  (eventual : ∀ x : A, ∃ N, D.S^[N] x ∈ B) :
  (∀ z : A, D.Cycle z →
    ∃ b ∈ B, D.Cycle b ∧ D.Homologous z b) ∧
  (∀ b : A, b ∈ B → D.Cycle b →
    (∃ x : A, b = D.d x) → ∃ y ∈ B, b = D.d y) := by
constructor
· intro z hz
  obtain ⟨N, hN⟩ := eventual z
  exact ⟨D.S^[N] z, hN, D.cycle_iterate hz N,
    D.homologous_iterate_of_cycle hz N⟩
· intro b hb hbcycle hboundary
  obtain ⟨x, rfl⟩ := hboundary
  obtain ⟨N, hNx⟩ := eventual x
  obtain ⟨w, hwB, hw⟩ :=
    D.small_boundary_of_small_homologous B hB hb hbcycle N
  refine ⟨D.S^[N] x + w, B.add_mem hNx hwB, ?_⟩
  rw [map_add, ← hw]
  have hcomm : ∀ M : ℕ, D.S^[M] (D.d x) = D.d (D.S^[M] x) := by
    intro M
    induction M with
    | zero => simp
    | succ M ih =>
      rw [iterate_succ_apply', iterate_succ_apply', D.S_comm, ih]
  rw [← hcomm N]
  abel

end SubdivisionData
end SingularExcision

```

A.9 PlatonicSolids/SingularExcision/AffineSubdivision.lean

Affine barycentric subdivision and prism (823 lines).

```

/-
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-/
import Mathlib.Data.Finsupp.BigOperators
import Mathlib.Data.Finsupp.Ext
import Mathlib.LinearAlgebra.AffineSpace.Simplex.Centroid

```

```

import Mathlib.Tactic.Linarith
import Mathlib.Tactic.Module

/ -!
# Finite affine chains for singular excision

This file isolates the integral signs in barycentric subdivision from
the categorical singular-chain construction. An `AffineSimplex V n`
is an ordered list of `n + 1` vertices in `V`.

The cone convention is chosen so that
`∂ (cone v c) = c - cone v (∂ c)`. In dimension one this produces the
usual permutation formula `[a,m] - [b,m]` for subdivision of `[a,b]`.
- /

open scoped BigOperators

namespace PlatonicSolids.SingularExcision

noncomputable section

set_option maxHeartbeats 800000

/ - - An ordered affine `n`-simplex, represented by its vertices. - /
abbrev AffineSimplex (V : Type*) (n : ℕ) := Fin (n + 1) → V

/ - - Finite integral chains of ordered affine simplices. - /
abbrev AffineChain (V : Type*) (n : ℕ) := AffineSimplex V n →₀ ℤ

lemma AffineChain.eq_of_single {V : Type*} {n : ℕ} {M : Type*} [AddCommGroup M]
  (f g : AffineChain V n →+ M) (c : AffineChain V n)
  (h : ∀ σ z, f (Finsupp.single σ z) = g (Finsupp.single σ z)) :
  f c = g c :=
  DFunLike.congr_fun (Finsupp.addHom_ext h) c

/ - - Delete vertex `i` from an ordered simplex. - /
def AffineSimplex.face {V : Type*} {n : ℕ}
  (σ : AffineSimplex V (n + 1)) (i : Fin (n + 2)) :
  AffineSimplex V n :=
  fun j ↦ σ (i.succAbove j)

/ - - The chain consisting of one oriented simplex. - /
def simplex {V : Type*} {n : ℕ} (σ : AffineSimplex V n) :
  AffineChain V n :=
  Finsupp.single σ 1

/ - - Boundary of one ordered affine simplex. - /
def simplexBoundary {V : Type*} {n : ℕ}
  (σ : AffineSimplex V (n + 1)) : AffineChain V n :=
  ∑ i : Fin (n + 2), (-1 : ℤ) ^ (i : ℕ) • simplex (σ.face i)

/ - - The integral boundary map on finite affine chains. - /
def boundary {V : Type*} {n : ℕ} :
  AffineChain V (n + 1) →+ AffineChain V n :=
  Finsupp.liftAddHom fun σ ↦
    (smulAddHom ℤ (AffineChain V n)).flip (simplexBoundary σ)

@[simp]
theorem boundary_simplex {V : Type*} {n : ℕ} (σ : AffineSimplex V (n + 1)) :
  boundary (simplex σ) = simplexBoundary σ := by
  simp [boundary, simplex]

/ - - A vertex as a zero-chain. - /

```

```

def vertex {V : Type*} (v : V) : AffineChain V 0 :=
  simplex (fun _ ↦ v)

/-- The ordered edge `[a,b]`. -/
def edge {V : Type*} (a b : V) : AffineChain V 1 :=
  simplex (Fin.cases a fun _ ↦ b)

/-- The ordered triangle `[a,b,c]`. -/
def triangle {V : Type*} (a b c : V) : AffineChain V 2 :=
  simplex (Fin.cases a (Fin.cases b fun _ ↦ c))

/-- The boundary of an edge is its endpoint minus its startpoint. -/
theorem boundary_edge {V : Type*} (a b : V) :
  boundary (edge a b) = vertex b - vertex a := by
  classical
  let  $\sigma$  : AffineSimplex V 1 := Fin.cases a fun _ ↦ b
  have h0 :  $\sigma$ .face (0 : Fin 2) = (fun _ ↦ b) := by
    funext i
    fin_cases i
    rfl
  have h1 :  $\sigma$ .face (1 : Fin 2) = (fun _ ↦ a) := by
    funext i
    fin_cases i
    rfl
  rw [edge, boundary_simplex]
  simp [simplexBoundary, Fin.sum_univ_succ, h0, h1,  $\sigma$ , vertex]
  abel

/-- The low-dimensional oriented boundary formula for a triangle. -/
theorem boundary_triangle {V : Type*} (a b c : V) :
  boundary (triangle a b c) = edge b c - edge a c + edge a b := by
  classical
  let  $\sigma$  : AffineSimplex V 2 := Fin.cases a (Fin.cases b fun _ ↦ c)
  have h0 :  $\sigma$ .face (0 : Fin 3) = (Fin.cases b fun _ ↦ c) := by
    funext i
    fin_cases i <|> rfl
  have h1 :  $\sigma$ .face (1 : Fin 3) = (Fin.cases a fun _ ↦ c) := by
    funext i
    fin_cases i <|> rfl
  have h2 :  $\sigma$ .face (2 : Fin 3) = (Fin.cases a fun _ ↦ b) := by
    funext i
    fin_cases i <|> rfl
  rw [triangle, boundary_simplex]
  simp [simplexBoundary, Fin.sum_univ_succ, h0, h1, h2,  $\sigma$ , edge]
  abel

/-- Append an apex to the ordered vertices of a simplex. -/
def AffineSimplex.append {V : Type*} {n : ℕ}
  ( $\sigma$  : AffineSimplex V n) (v : V) : AffineSimplex V (n + 1) :=
  Fin.lastCases v  $\sigma$ 

/-- Deleting the appended last vertex recovers the original simplex. -/
@[simp]
lemma AffineSimplex.append_face_last {V : Type*} {n : ℕ}
  ( $\sigma$  : AffineSimplex V n) (v : V) :
  ( $\sigma$ .append v).face (Fin.last (n + 1)) =  $\sigma$  := by
  funext i
  simp [AffineSimplex.face, AffineSimplex.append,
    Fin.succAbove_last_apply]

/-- Deleting an original vertex commutes with appending a last vertex. -/
@[simp]
lemma AffineSimplex.append_face_castSucc {V : Type*} {n : ℕ}

```

```

    (σ : AffineSimplex V (n + 1)) (v : V) (i : Fin (n + 2)) :
    (σ.append v).face i.castSucc = (σ.face i).append v := by
  funext j
  refine Fin.lastCases ?_ (fun k ↦ ?_) j
  · simp [AffineSimplex.face, AffineSimplex.append]
  · simp [AffineSimplex.face, AffineSimplex.append,
    Fin.castSucc_succAbove_castSucc]

/-- Cone on one simplex, with the dimension-dependent orientation sign. -/
def simplexCone {V : Type*} {n : ℕ} (v : V) (σ : AffineSimplex V n) :
  AffineChain V (n + 1) :=
  (-1 : ℤ) ^ (n + 1) • simplex (σ.append v)

/-- Extend coning linearly to finite chains. -/
def cone {V : Type*} {n : ℕ} (v : V) :
  AffineChain V n →+ AffineChain V (n + 1) :=
  Finsupp.liftAddHom fun σ ↦
    (smulAddHom ℤ (AffineChain V (n + 1))).flip (simplexCone v σ)

@[simp]
theorem cone_simplex {V : Type*} {n : ℕ} (v : V) (σ : AffineSimplex V n) :
  cone v (simplex σ) = simplexCone v σ := by
  simp [cone, simplex]

theorem simplexCone_vertex {V : Type*} (v a : V) :
  simplexCone v (fun _ : Fin 1 ↦ a) = -edge a v := by
  unfold simplexCone edge
  norm_num
  congr 1
  funext i
  fin_cases i <,> rfl

theorem simplexCone_edge {V : Type*} (v a b : V) :
  simplexCone v (Fin.cases a fun _ ↦ b) = triangle a b v := by
  unfold simplexCone triangle
  norm_num
  congr 1
  funext i
  fin_cases i <,> rfl

/-- Boundary of a cone on a positive-dimensional simplex. -/
theorem boundary_simplexCone {V : Type*} {n : ℕ}
  (v : V) (σ : AffineSimplex V (n + 1)) :
  boundary (simplexCone v σ) =
    simplex σ - cone v (simplexBoundary σ) := by
  classical
  rw [simplexCone, map_zsmul, boundary_simplex, simplexBoundary]
  rw [Fin.sum_univ_castSucc]
  simp only [AffineSimplex.append_face_castSucc,
    AffineSimplex.append_face_last, Fin.val_castSucc, Fin.val_last]
  simp only [simplexBoundary]
  rw [map_sum]
  simp_rw [map_zsmul, cone_simplex, simplexCone]
  simp only [Nat.add_assoc, Nat.reduceAdd, smul_add, Finset.smul_sum]
  simp_rw [smul_smul]
  have hsign : (-1 : ℤ) ^ (n + 2) * (-1 : ℤ) ^ (n + 2) = 1 := by
    rw [← pow_add, show n + 2 + (n + 2) = 2 * (n + 2) by omega, pow_mul]
    norm_num
  rw [hsign, one_smul]
  rw [sub_eq_add_neg]
  rw [add_comm (∑ x : Fin (n + 2),
    ((-1 : ℤ) ^ (n + 2) * (-1 : ℤ) ^ (x : ℕ)) •
    simplex ((σ.face x).append v)) (simplex σ))]

```

```

congr 1
rw [← Finset.sum_neg_distrib]
apply Finset.sum_congr rfl
intro i _
rw [← neg_smul]
congr 1
have hnext :  $(-1 : \mathbb{Z}) \wedge (n + 2) = -((-1 : \mathbb{Z}) \wedge (n + 1))$  := by
  rw [show  $n + 2 = \text{Nat.succ } (n + 1)$  by omega, pow_succ]
  ring
rw [hnext]
ring

/-- Boundary of a cone on any positive-dimensional affine chain. -/
theorem boundary_cone {V : Type*} {n : ℕ}
  (v : V) (c : AffineChain V (n + 1)) :
  boundary (cone v c) = c - cone v (boundary c) := by
  classical
  refine AffineChain.eq_of_single
  (boundary.comp (cone v))
  (AddMonoidHom.id _ - (cone v).comp boundary) c ?_
intro σ z
change boundary (cone v (Finsupp.single σ z)) =
  Finsupp.single σ z - cone v (boundary (Finsupp.single σ z))
rw [show Finsupp.single σ z = z • simplex σ by simp [simplex]]
simp [map_zsmul, cone_simplex, boundary_simplexCone, boundary_simplex, smul_sub]

@[simp]
theorem cone_vertex {V : Type*} (v a : V) :
  cone v (vertex a) = -edge a v := by
  rw [vertex, cone_simplex, simplexCone_vertex]

@[simp]
theorem cone_edge {V : Type*} (v a b : V) :
  cone v (edge a b) = triangle a b v := by
  rw [edge, cone_simplex, simplexCone_edge]

/-- The cone identity on zero-simplices. -/
theorem boundary_cone_vertex {V : Type*} (v a : V) :
  boundary (cone v (vertex a)) = vertex a - vertex v := by
  classical
  rw [cone_vertex]
  rw [map_neg, boundary_edge]
  abel

/-- The cone identity on edges. -/
theorem boundary_cone_edge {V : Type*} (v a b : V) :
  boundary (cone v (edge a b)) =
    edge a b - cone v (vertex b - vertex a) := by
  classical
  rw [cone_edge, map_sub, cone_vertex, cone_vertex]
  rw [boundary_triangle]
  abel

/-- The affine barycenter of the vertices of a simplex in a vector space. -/
def AffineSimplex.barycenter {V : Type*} [AddCommGroup V] [Module ℚ V]
  {n : ℕ} (σ : AffineSimplex V n) : V :=
  (n + 1 : ℚ)-1 • ∑ i, σ i

/-- The midpoint used by one-dimensional barycentric subdivision. -/
def midpoint {V : Type*} [AddCommGroup V] [Module ℚ V] (a b : V) : V :=
  (2 : ℚ)-1 • (a + b)

theorem barycenter_edge {V : Type*} [AddCommGroup V] [Module ℚ V] (a b : V) :

```

```

    AffineSimplex.barycenter
    ((Fin.cases a fun _ ↦ b) : AffineSimplex V 1) = midpoint a b := by
norm_num [AffineSimplex.barycenter, midpoint, Fin.sum_univ_succ]

/-- The first simplicial identity, as an equality of ordered faces. -/
lemma AffineSimplex.face_face {V : Type*} {n : ℕ}
  (σ : AffineSimplex V (n + 2)) {i j : Fin (n + 2)} (hij : i ≤ j) :
  (σ.face j.succ).face i = (σ.face i.castSucc).face j := by
  funext k
  simp only [AffineSimplex.face]
  congr 1
  change j.succ.succAbove (i.succAbove k) =
    i.castSucc.succAbove (j.succAbove k)
  dsimp [Fin.succAbove]
  rcases i with ⟨i, _⟩
  rcases j with ⟨j, _⟩
  rcases k with ⟨k, _⟩
  split_ifs <;> · simp at * <;> lia

/-- The signed sum of coefficients of a finite affine chain. -/
def coeffSum {V : Type*} {n : ℕ} : AffineChain V n →+ ℤ :=
  Finsupp.liftAddHom fun _ => AddMonoidHom.id ℤ

@[simp]
lemma coeffSum_single {V : Type*} {n : ℕ} (σ : AffineSimplex V n) (z : ℤ) :
  coeffSum (Finsupp.single σ z) = z := by
  simp [coeffSum]

@[simp]
lemma coeffSum_simplex {V : Type*} {n : ℕ} (σ : AffineSimplex V n) :
  coeffSum (simplex σ) = 1 := by
  simp [simplex]

@[simp]
lemma coeffSum_vertex {V : Type*} (v : V) :
  coeffSum (vertex v) = 1 := by
  simp [vertex]

lemma coeffSum_simplexBoundary_one {V : Type*} (σ : AffineSimplex V 1) :
  coeffSum (simplexBoundary σ) = 0 := by
  rw [simplexBoundary, Fin.sum_univ_two]
  simp [coeffSum_simplex]

/-- Cone identity in dimension zero: the extra term is the augmentation. -/
theorem boundary_cone_zero {V : Type*} (v : V) (c : AffineChain V 0) :
  boundary (cone v c) = c - coeffSum c • vertex v := by
  classical
  refine AffineChain.eq_of_single
  (boundary.comp (cone v))
  (AddMonoidHom.id _ -
    ((smulAddHom ℤ (AffineChain V 0)).flip (vertex v)).comp coeffSum) c ?_
  intro σ z
  have hσ : σ = fun _ : Fin 1 ↦ σ 0 := by
    funext i
    fin_cases i
    rfl
  change boundary (cone v (Finsupp.single σ z)) =
    Finsupp.single σ z - coeffSum (Finsupp.single σ z) • vertex v
  have hsingle : Finsupp.single σ z = z • vertex (σ 0) := by
    rw [hσ, vertex, simplex, Finsupp.smul_single, smul_eq_mul, mul_one]
  rw [hsingle, map_zsmul, cone_vertex, map_zsmul, map_neg, boundary_edge,
    map_zsmul, coeffSum_vertex]
  module

```

```

/-- Double faces cancel:  $\partial \circ \partial = 0$  on a generator. -/
theorem boundary_simplexBoundary {V : Type*} {n : ℕ}
  (σ : AffineSimplex V (n + 2)) :
  boundary (simplexBoundary σ) = 0 := by
  classical
  simp only [simplexBoundary, map_sum, map_zsmul, boundary_simplex]
  simp only [Finset.smul_sum, smul_smul]
  rw [Finset.sum_comm, ← Finset.sum_product']
  let P := Fin (n + 2) × Fin (n + 3)
  let S : Finset P := {ij : P | (ij.2 : ℕ) ≤ (ij.1 : ℕ)}
  rw [Finset.univ_product_univ, ← Finset.sum_add_sum_compl S,
    ← eq_neg_iff_add_eq_zero, ← Finset.sum_neg_distrib]
  let φ : ∀ ij : P, ij ∈ S → P := fun ij hij =>
    (Fin.castLT ij.2 (lt_of_le_of_lt (Finset.mem_filter.mp hij).right (Fin.is_lt ij.1)),
    ij.1.succ)
  apply Finset.sum_bij φ
  · intro ij hij
    simp_rw [S, φ, Finset.compl_filter, Finset.mem_filter_univ, Fin.val_succ,
      Fin.val_castLT] at hij ⊢
    omega
  · rintro ⟨j, i⟩ hij ⟨j', i'⟩ hij' h
    rw [Prod.mk_inj]
    exact ⟨by simpa [φ] using congr_arg Prod.snd h,
      by simpa [φ, Fin.castSucc_castLT] using
        congr_arg Fin.castSucc (congr_arg Prod.fst h)⟩
  · rintro ⟨j', i'⟩ hij'
    simp_rw [S, Finset.compl_filter, Finset.mem_filter_univ, not_le] at hij'
    refine ⟨(i'.pred ?_, Fin.castSucc j'), ?_, ?_⟩
    · exact_mod_cast hij'.ne_zero
    · simpa [S] using Nat.le_sub_one_of_lt hij'
    · simp [φ, Fin.castLT_castSucc, Fin.succ_pred]
  · rintro ⟨j, i⟩ hij
    simp only [← neg_smul]
    congr 1
    · simp [φ, Fin.val_succ, pow_add, pow_one, mul_neg, neg_neg, mul_comm]
    · have hij' : i ≤ j.castSucc := by
        simpa [S, Fin.le_iff_val_le_val] using hij
      have hlt : (i : ℕ) < n + 2 :=
        Nat.lt_of_le_of_lt (Fin.le_iff_val_le_val.mp hij') j.isLt
      have hcast : (i.castLT hlt).castSucc = i := by
        ext
        simp
      have hface :
        (σ.face j.succ).face (i.castLT hlt) = (σ.face i).face j := by
        have := AffineSimplex.face_face (σ := σ)
        (i := i.castLT hlt) (j := j) (by
          simpa [Fin.le_iff_val_le_val, Fin.val_castLT] using hij')
        rwa [hcast] at this
      congr 1
      simpa [φ] using hface.symm

theorem boundary_boundary {V : Type*} {n : ℕ} (c : AffineChain V (n + 2)) :
  boundary (boundary c) = 0 := by
  classical
  refine AffineChain.eq_of_single
  (boundary.comp boundary) 0 c ?_
  intro σ z
  change boundary (boundary (Finsupp.single σ z)) = 0
  rw [show Finsupp.single σ z = z • simplex σ by simp [simplex]]
  simp [map_zsmul, boundary_simplex, boundary_simplexBoundary]

/-- Barycentric subdivision of finite affine chains, coning on a

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```

chosen point of each simplex (classically the barycenter). -/
def subdivide {V : Type*} (b : {n : ℕ} → AffineSimplex V n → V) :
  (n : ℕ) → AffineChain V n →+ AffineChain V n
| 0 => AddMonoidHom.id _
| n + 1 =>
  Finsupp.liftAddHom fun σ =>
    (smulAddHom ℤ (AffineChain V (n + 1))).flip
    (cone (b σ) (subdivide b n (simplexBoundary σ)))

@[simp]
theorem subdivide_zero {V : Type*} (b : {n : ℕ} → AffineSimplex V n → V)
  (c : AffineChain V 0) :
  subdivide b 0 c = c :=
  rfl

theorem subdivide_single {V : Type*} (b : {n : ℕ} → AffineSimplex V n → V)
  {n : ℕ} (σ : AffineSimplex V (n + 1)) (z : ℤ) :
  subdivide b (n + 1) (Finsupp.single σ z) =
  z • cone (b σ) (subdivide b n (simplexBoundary σ)) := by
  simp [subdivide]

theorem subdivide_simplex {V : Type*} (b : {n : ℕ} → AffineSimplex V n → V)
  {n : ℕ} (σ : AffineSimplex V (n + 1)) :
  subdivide b (n + 1) (simplex σ) =
  cone (b σ) (subdivide b n (simplexBoundary σ)) := by
  simpa [simplex] using subdivide_single b σ 1

/-- Subdivision commutes with the boundary. -/
theorem boundary_subdivide {V : Type*} (b : {n : ℕ} → AffineSimplex V n → V)
  (n : ℕ) (c : AffineChain V (n + 1)) :
  boundary (subdivide b (n + 1) c) = subdivide b n (boundary c) := by
  induction n with
  | zero =>
    induction c using Finsupp.induction with
    | zero => simp
    | single_add σ z c _ _ ih =>
      rw [map_add, map_add, map_add, ih]
      rw [subdivide_single b, map_zsmul, subdivide_zero]
      rw [boundary_cone_zero, coeffSum_simplexBoundary_one, zero_smul,
        sub_zero]
      rw [show Finsupp.single σ z = z • simplex σ by simp [simplex]]
      simp [map_zsmul, boundary_simplex]
  | succ n ih =>
    induction c using Finsupp.induction with
    | zero => simp
    | single_add σ z c _ _ ihc =>
      rw [map_add, map_add, map_add, ihc]
      rw [subdivide_single b, map_zsmul, boundary_cone, ih,
        boundary_simplexBoundary]
      simp only [map_zero, sub_zero]
      rw [show Finsupp.single σ z = z • simplex σ by simp [simplex]]
      simp [map_zsmul, boundary_simplex]

/-- The prism operator: a chain homotopy from the identity to subdivision. -/
def prism {V : Type*} (b : {n : ℕ} → AffineSimplex V n → V) :
  (n : ℕ) → AffineChain V n →+ AffineChain V (n + 1)
| 0 => 0
| n + 1 =>
  Finsupp.liftAddHom fun σ =>
    (smulAddHom ℤ (AffineChain V (n + 2))).flip
    (cone (b σ)
      (simplex σ - subdivide b (n + 1) (simplex σ) -
        prism b n (simplexBoundary σ)))

```

```

@[simp]
theorem prism_zero {V : Type*} (b : {n : ℕ} → AffineSimplex V n → V)
  (c : AffineChain V 0) :
  prism b 0 c = 0 :=
  rfl

theorem prism_single {V : Type*} (b : {n : ℕ} → AffineSimplex V n → V)
  {n : ℕ} (σ : AffineSimplex V (n + 1)) (z : ℤ) :
  prism b (n + 1) (Finsupp.single σ z) =
    z • cone (b σ)
    (simplex σ - subdivide b (n + 1) (simplex σ) -
     prism b n (simplexBoundary σ)) := by
  simp [prism, smul_sub]

/-- The prism identity  $\partial T + T\partial = \text{id} - Sd$ . -/
theorem boundary_prism {V : Type*} (b : {n : ℕ} → AffineSimplex V n → V)
  (n : ℕ) (c : AffineChain V (n + 1)) :
  boundary (prism b (n + 1) c) + prism b n (boundary c) =
    c - subdivide b (n + 1) c := by
  induction n with
  | zero =>
    induction c using Finsupp.induction with
    | zero => simp
    | single_add σ z c _ _ ih =>
      have hsingle :
        boundary (prism b 1 (Finsupp.single σ z)) +
        prism b 0 (boundary (Finsupp.single σ z)) =
        Finsupp.single σ z - subdivide b 1 (Finsupp.single σ z) := by
        rw [prism_single, map_zsmul]
        simp only [prism_zero, sub_zero, add_zero]
      have hw :
        boundary (simplex σ - subdivide b 1 (simplex σ)) = 0 := by
        rw [map_sub, boundary_subdivide, boundary_simplex,
            subdivide_zero]
      abel
      rw [boundary_cone, hw]
      simp only [map_zero, sub_zero]
      rw [show Finsupp.single σ z = z • simplex σ by
          rw [simplex, Finsupp.smul_single, smul_eq_mul, mul_one]]
      simp [map_zsmul]
      module
      simp only [map_add]
      calc
      _ = (boundary (prism b 1 (Finsupp.single σ z)) +
          prism b 0 (boundary (Finsupp.single σ z))) +
          (boundary (prism b 1 c) + prism b 0 (boundary c)) := by abel
      _ = (Finsupp.single σ z - subdivide b 1 (Finsupp.single σ z)) +
          (c - subdivide b 1 c) := by rw [hsingle, ih]
      _ = _ := by abel
  | succ n ih =>
    induction c using Finsupp.induction with
    | zero => simp
    | single_add σ z c _ _ ihc =>
      have hsingle :
        boundary (prism b (n + 2) (Finsupp.single σ z)) +
        prism b (n + 1) (boundary (Finsupp.single σ z)) =
        Finsupp.single σ z - subdivide b (n + 2) (Finsupp.single σ z) := by
        rw [prism_single, map_zsmul]
      have hw :
        boundary (simplex σ - subdivide b (n + 2) (simplex σ) -
            prism b (n + 1) (simplexBoundary σ)) = 0 := by
        have hih := ih (simplexBoundary σ)

```

```

    have hsd := boundary_subdivide b (n + 1) (simplex  $\sigma$ )
    rw [map_sub, map_sub, hsd, boundary_simplex]
    simp [boundary_simplexBoundary] at hih
    rw [hih]
    abel
  rw [boundary_cone, hw]
  simp only [map_zero, sub_zero]
  rw [show Finsupp.single  $\sigma$  z = z • simplex  $\sigma$  by
    rw [simplex, Finsupp.smul_single, smul_eq_mul, mul_one]]
  simp [map_zsmul, boundary_simplex]
  module
  simp only [map_add]
  calc
    _ = (boundary (prism b (n + 2) (Finsupp.single  $\sigma$  z)) +
      prism b (n + 1) (boundary (Finsupp.single  $\sigma$  z))) +
      (boundary (prism b (n + 2) c) +
      prism b (n + 1) (boundary c)) := by abel
    _ = (Finsupp.single  $\sigma$  z - subdivide b (n + 2) (Finsupp.single  $\sigma$  z)) +
      (c - subdivide b (n + 2) c) := by rw [hsingle, ihc]
    _ = _ := by abel

/-- Barycentric subdivision of an oriented affine interval.

The expression `[a,m] - [b,m]` is the signed-permutation form of
`[a,m] + [m,b]`; no quotient by affine reparametrization is taken here.
-/
def subdivideInterval {V : Type*} [AddCommGroup V] [Module  $\mathbb{Q}$  V]
  (a b : V) : AffineChain V 1 :=
  edge a (midpoint a b) - edge b (midpoint a b)

/-- Subdivision is a chain map on the one-dimensional checkpoint. -/
theorem boundary_subdivideInterval {V : Type*} [AddCommGroup V] [Module  $\mathbb{Q}$  V]
  (a b : V) :
  boundary (subdivideInterval a b) = vertex b - vertex a := by
  simp [subdivideInterval, boundary_edge]

/-- The universal one-dimensional prism between an edge and its subdivision. -/
def intervalPrism {V : Type*} [AddCommGroup V] [Module  $\mathbb{Q}$  V]
  (a b : V) : AffineChain V 2 :=
  cone (midpoint a b) (edge a b - subdivideInterval a b)

/-- The prism identity ` $\partial P = \text{id} - S_d$ ` for an affine interval. -/
theorem boundary_intervalPrism {V : Type*} [AddCommGroup V] [Module  $\mathbb{Q}$  V]
  (a b : V) :
  boundary (intervalPrism a b) = edge a b - subdivideInterval a b := by
  classical
  unfold intervalPrism subdivideInterval
  simp only [map_sub]
  rw [boundary_cone_edge, boundary_cone_edge, boundary_cone_edge]
  simp only [map_sub, cone_vertex]
  abel

/-- Push vertices of an ordered simplex forward along a function. -/
def AffineSimplex.map {V W : Type*} {n :  $\mathbb{N}$ } (f : V  $\rightarrow$  W)
  ( $\sigma$  : AffineSimplex V n) : AffineSimplex W n :=
  f  $\circ$   $\sigma$ 

@[simp]
theorem AffineSimplex.map_apply {V W : Type*} {n :  $\mathbb{N}$ } (f : V  $\rightarrow$  W)
  ( $\sigma$  : AffineSimplex V n) (i : Fin (n + 1)) :
  AffineSimplex.map f  $\sigma$  i = f ( $\sigma$  i) :=
  rfl

```

```

@[simp]
theorem AffineSimplex.map_id {V : Type*} {n : ℕ} (σ : AffineSimplex V n) :
  AffineSimplex.map id σ = σ :=
  rfl

@[simp]
theorem AffineSimplex.map_comp {U V W : Type*} {n : ℕ}
  (g : V → W) (f : U → V) (σ : AffineSimplex U n) :
  AffineSimplex.map g (AffineSimplex.map f σ) =
  AffineSimplex.map (g ∘ f) σ :=
  rfl

@[simp]
theorem AffineSimplex.map_face {V W : Type*} {n : ℕ} (f : V → W)
  (σ : AffineSimplex V (n + 1)) (i : Fin (n + 2)) :
  AffineSimplex.map f (σ.face i) = (AffineSimplex.map f σ).face i :=
  rfl

@[simp]
theorem AffineSimplex.map_append {V W : Type*} {n : ℕ} (f : V → W)
  (σ : AffineSimplex V n) (v : V) :
  AffineSimplex.map f (σ.append v) = (AffineSimplex.map f σ).append (f v) := by
  funext i
  refine Fin.lastCases ?_ (fun _ ↦ ?_) i
  · simp [AffineSimplex.map, AffineSimplex.append]
  · simp [AffineSimplex.map, AffineSimplex.append]

/-- Push a finite affine chain forward along a function of vertices. -/
def AffineChain.map {V W : Type*} (f : V → W) {n : ℕ} :
  AffineChain V n →+ AffineChain W n :=
  Finsupp.mapDomain.addMonoidHom (AffineSimplex.map f)

@[simp]
theorem AffineChain.map_single {V W : Type*} (f : V → W) {n : ℕ}
  (σ : AffineSimplex V n) (z : ℤ) :
  AffineChain.map f (Finsupp.single σ z) =
  Finsupp.single (AffineSimplex.map f σ) z := by
  simp [AffineChain.map, Finsupp.mapDomain_single]

@[simp]
theorem AffineChain.map_simplex {V W : Type*} (f : V → W) {n : ℕ}
  (σ : AffineSimplex V n) :
  AffineChain.map f (simplex σ) = simplex (AffineSimplex.map f σ) := by
  simp [simplex]

@[simp]
theorem AffineChain.map_vertex {V W : Type*} (f : V → W) (v : V) :
  AffineChain.map f (vertex v) = vertex (f v) := by
  simp [vertex, AffineSimplex.map]
  rfl

theorem AffineChain.map_simplexBoundary {V W : Type*} (f : V → W) {n : ℕ}
  (σ : AffineSimplex V (n + 1)) :
  AffineChain.map f (simplexBoundary σ) =
  simplexBoundary (AffineSimplex.map f σ) := by
  simp [simplexBoundary, map_sum, map_zsmul]

theorem AffineChain.map_boundary {V W : Type*} (f : V → W) {n : ℕ}
  (c : AffineChain V (n + 1)) :
  AffineChain.map f (boundary c) = boundary (AffineChain.map f c) := by
  refine AffineChain.eq_of_single
  ((AffineChain.map f).comp boundary)
  (boundary.comp (AffineChain.map f)) c ?_

```

```

intro  $\sigma$  z
change AffineChain.map f (boundary (Finsupp.single  $\sigma$  z)) =
  boundary (AffineChain.map f (Finsupp.single  $\sigma$  z))
rw [show Finsupp.single  $\sigma$  z = z • simplex  $\sigma$  by simp [simplex]]
simp [map_zsmul, boundary_simplex, AffineChain.map_simplexBoundary]

theorem AffineChain.map_simplexCone {V W : Type*} (f : V → W) {n : ℕ}
  (v : V) ( $\sigma$  : AffineSimplex V n) :
  AffineChain.map f (simplexCone v  $\sigma$ ) =
    simplexCone (f v) (AffineSimplex.map f  $\sigma$ ) := by
  simp [simplexCone, map_zsmul, AffineSimplex.map_append]

theorem AffineChain.map_cone {V W : Type*} (f : V → W) {n : ℕ}
  (v : V) (c : AffineChain V n) :
  AffineChain.map f (cone v c) = cone (f v) (AffineChain.map f c) := by
  refine AffineChain.eq_of_single
    ((AffineChain.map f).comp (cone v))
    ((cone (f v)).comp (AffineChain.map f)) c ?_
  intro  $\sigma$  z
  change AffineChain.map f (cone v (Finsupp.single  $\sigma$  z)) =
    cone (f v) (AffineChain.map f (Finsupp.single  $\sigma$  z))
  rw [show Finsupp.single  $\sigma$  z = z • simplex  $\sigma$  by simp [simplex]]
  simp [map_zsmul, cone_simplex, AffineChain.map_simplexCone]

/-- Subdivision is natural for maps that send chosen cone points to
chosen cone points. -/
theorem AffineChain.map_subdivide {V W : Type*}
  (bV : {n : ℕ} → AffineSimplex V n → V)
  (bW : {n : ℕ} → AffineSimplex W n → W) (f : V → W)
  (hb : ∀ {n : ℕ} ( $\sigma$  : AffineSimplex V n),
    f (bV  $\sigma$ ) = bW (AffineSimplex.map f  $\sigma$ ))
  (n : ℕ) (c : AffineChain V n) :
  AffineChain.map f (subdivide bV n c) =
    subdivide bW n (AffineChain.map f c) := by
  induction n with
  | zero => simp
  | succ n ih =>
    refine AffineChain.eq_of_single
      ((AffineChain.map f).comp (subdivide bV (n + 1)))
      ((subdivide bW (n + 1)).comp (AffineChain.map f)) c ?_
    intro  $\sigma$  z
    change AffineChain.map f (subdivide bV (n + 1) (Finsupp.single  $\sigma$  z)) =
      subdivide bW (n + 1) (AffineChain.map f (Finsupp.single  $\sigma$  z))
    rw [subdivide_single, AffineChain.map_single, subdivide_single, map_zsmul,
      AffineChain.map_cone, ih, AffineChain.map_simplexBoundary, hb]

/-- The prism is natural for the same cone-point maps. -/
theorem AffineChain.map_prism {V W : Type*}
  (bV : {n : ℕ} → AffineSimplex V n → V)
  (bW : {n : ℕ} → AffineSimplex W n → W) (f : V → W)
  (hb : ∀ {n : ℕ} ( $\sigma$  : AffineSimplex V n),
    f (bV  $\sigma$ ) = bW (AffineSimplex.map f  $\sigma$ ))
  (n : ℕ) (c : AffineChain V n) :
  AffineChain.map f (prism bV n c) =
    prism bW n (AffineChain.map f c) := by
  induction n with
  | zero => simp
  | succ n ih =>
    refine AffineChain.eq_of_single
      ((AffineChain.map f).comp (prism bV (n + 1)))
      ((prism bW (n + 1)).comp (AffineChain.map f)) c ?_
    intro  $\sigma$  z
    change AffineChain.map f (prism bV (n + 1) (Finsupp.single  $\sigma$  z)) =

```

```

    prism bW (n + 1) (AffineChain.map f (Finsupp.single σ z))
  rw [prism_single, AffineChain.map_single, prism_single, map_zsmul,
    AffineChain.map_cone]
  simp [map_sub, AffineChain.map_subdivide bV bW f hb,
    ih, AffineChain.map_simplexBoundary, hb]

lemma mem_support_simplexCone {V : Type*} {n : ℕ} (v : V)
  (σ : AffineSimplex V n) {τ : AffineSimplex V (n + 1)}
  (hτ : τ ∈ (simplexCone v σ).support) :
  τ = σ.append v := by
  have hz : (-1 : ℤ) ^ (n + 1) ≠ 0 := pow_ne_zero _ (by decide)
  simpa [simplexCone, simplex, Finsupp.support_smul_eq hz,
    Finsupp.support_single_ne_zero _ (one_ne_zero (α := ℤ))] using hτ

lemma cone_single {V : Type*} {n : ℕ} (v : V)
  (σ : AffineSimplex V n) (z : ℤ) :
  cone v (Finsupp.single σ z) = z • simplexCone v σ := by
  rw [show Finsupp.single σ z = z • simplex σ by simp [simplex]]
  simp [map_zsmul, cone_simplex]

lemma mem_support_cone {V : Type*} {n : ℕ} (v : V)
  (c : AffineChain V n) {τ : AffineSimplex V (n + 1)}
  (hτ : τ ∈ (cone v c).support) :
  ∃ σ ∈ c.support, τ = σ.append v := by
  classical
  induction c using Finsupp.induction with
  | zero =>
    simp at hτ
  | single_add σ z c hmem hn ih =>
    rw [map_add, cone_single] at hτ
    have hτ' :
      τ ∈ (z • simplexCone v σ).support ∪ (cone v c).support :=
      Finset.mem_of_subset Finsupp.support_add hτ
    rw [Finset.mem_union] at hτ'
    rcases hτ' with hσ | hc
    · refine ⟨σ, ?_,
      mem_support_simplexCone v σ (Finset.mem_of_subset Finsupp.support_smul hσ))
      have hcσ : c σ = 0 := Finsupp.notMem_support_iff.mp hmem
      simp [Finsupp.mem_support_iff, hcσ, hn]
    · obtain ⟨ρ, hρ, hτρ⟩ := ih hc
      refine ⟨ρ, ?_, hτρ⟩
      simp [Finsupp.mem_support_iff] at hρ ⊢
      have hρσ : ρ ≠ σ := fun h => hmem (h ► (Finsupp.mem_support_iff.mpr hρ))
      simpa [hρσ] using hρ

lemma mem_support_subdivide_simplexBoundary {V : Type*}
  (b : {n : ℕ} → AffineSimplex V n → V) {n : ℕ}
  (σ : AffineSimplex V (n + 1)) {τ : AffineSimplex V n}
  (hτ : τ ∈ (subdivide b n (simplexBoundary σ)).support) :
  ∃ i : Fin (n + 2),
  τ ∈ (subdivide b n (simplex (σ.face i))).support := by
  classical
  simp only [simplexBoundary, map_sum, map_zsmul] at hτ
  have hsum (s : Finset (Fin (n + 2))) :
    τ ∈ (∑ i ∈ s,
      (-1 : ℤ) ^ (i : ℕ) •
        subdivide b n (simplex (σ.face i))).support →
    ∃ i ∈ s, τ ∈ (subdivide b n (simplex (σ.face i))).support := by
  induction s using Finset.induction with
  | empty => simp
  | insert i s his ih =>
    intro h
    rw [Finset.sum_insert his] at h

```

```

    have h' := Finset.mem_of_subset Finsupp.support_add h
    rw [Finset.mem_union] at h'
    rcases h' with hi | hs
    · exact ⟨i, Finset.mem_insert_self _ _,
      Finset.mem_of_subset Finsupp.support_smul hi⟩
    · obtain ⟨j, hj, hjτ⟩ := ih hs
      exact ⟨j, Finset.mem_insert_of_mem hj, hjτ⟩
  obtain ⟨i, _, hi⟩ := hsum _ hτ
  exact ⟨i, hi⟩

lemma mem_support_subdivide_simplex {V : Type*}
  (b : {n : ℕ} → AffineSimplex V n → V) {n : ℕ}
  (σ : AffineSimplex V (n + 1)) {τ : AffineSimplex V (n + 1)}
  (hτ : τ ∈ (subdivide b (n + 1) (simplex σ)).support) :
  ∃ i : Fin (n + 2),
    ∃ ρ ∈ (subdivide b n (simplex (σ.face i))).support,
      τ = ρ.append (b σ) := by
  rw [subdivide_simplex] at hτ
  obtain ⟨ρ, hρ, hτρ⟩ := mem_support_cone (b σ) _ hτ
  obtain ⟨i, hi⟩ := mem_support_subdivide_simplexBoundary b σ hρ
  exact ⟨i, ρ, hi, hτρ⟩

/-- Every simplex in the support of a subdivided chain comes from
subdividing one of the original generators. -/
lemma mem_support_subdivide {V : Type*}
  (b : {n : ℕ} → AffineSimplex V n → V) (n : ℕ)
  {c : AffineChain V n} {τ : AffineSimplex V n}
  (hτ : τ ∈ (subdivide b n c).support) :
  ∃ σ ∈ c.support, τ ∈ (subdivide b n (simplex σ)).support := by
  classical
  induction c using Finsupp.induction with
  | zero =>
    simp at hτ
  | single_add σ z c hmem hn ih =>
    rw [map_add] at hτ
    have hτ' :
      τ ∈ (subdivide b n (Finsupp.single σ z)).support ∪
      (subdivide b n c).support :=
      Finset.mem_of_subset Finsupp.support_add hτ
    rw [Finset.mem_union] at hτ'
    rcases hτ' with hσ | hc
    · refine ⟨σ, ?, ?⟩
      · have hcσ : c σ = 0 := Finsupp.notMem_support_iff.mp hmem
        simp [Finsupp.mem_support_iff, hcσ, hn]
      · have hsmul :
          subdivide b n (Finsupp.single σ z) =
            z • subdivide b n (simplex σ) := by
            rw [show Finsupp.single σ z = z • simplex σ by simp [simplex]]
            simp
        rw [hsmul] at hσ
        exact Finset.mem_of_subset Finsupp.support_smul hσ
    · obtain ⟨ρ, hρ, hτρ⟩ := ih hc
      refine ⟨ρ, ?, hτρ⟩
      simp [Finsupp.mem_support_iff] at hρ
      have hρσ : ρ ≠ σ := fun h => hmem (h ▶ (Finsupp.mem_support_iff.mpr hρ))
      simpa [hρσ] using hρ

end

end PlatonicSolids.SingularExcision

```

A.10 PlatonicSolids/SingularExcision/AffineRealization.lean

Realization of affine simplices in a space (322 lines).

```

/-
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-/
import Mathlib.AlgebraicTopology.SingularSet
import Mathlib.AlgebraicTopology.SimplicialSet.TopAdj
import Mathlib.LinearAlgebra.Finsupp.Pi
import PlatonicSolids.SingularExcision.AffineSubdivision

/-!
# Realization of finite affine simplices

Finite affine chains become singular chains by realizing a list of vertices
in a standard simplex as its barycentric affine map.
-/

open CategoryTheory
open scoped BigOperators Simplicial

namespace PlatonicSolids.SingularExcision

noncomputable section

/-- Barycentric realization of ordered vertices in a standard simplex. -/
def AffineSimplex.realize {p q : ℕ}
  (σ : AffineSimplex (stdSimplex ℝ (Fin (q + 1))) p) :
  C(stdSimplex ℝ (Fin (p + 1)), stdSimplex ℝ (Fin (q + 1))) where
toFun x :=
  ⟨fun j ↦ ∑ i, x.1 i * (σ i).1 j, by
    constructor
    · intro j
      exact Finset.sum_nonneg fun i _ ↦
        mul_nonneg (x.2.1 i) ((σ i).2.1 j)
    · rw [Finset.sum_comm]
      calc
        ∑ i, ∑ j, x.1 i * (σ i).1 j =
          ∑ i, x.1 i * ∑ j, (σ i).1 j := by
            congr with i
            rw [Finset.mul_sum]
        _ = ∑ i, x.1 i := by
            congr with i
            rw [(σ i).2.2, mul_one]
        _ = 1 := x.2.2⟩
continuous_toFun := by
  apply Continuous.subtype_mk
  apply continuous_pi
  intro j
  apply continuous_finsetSum
  intro i _
  exact ((continuous_apply i).comp continuous_subtype_val).mul continuous_const

@[simp]
theorem AffineSimplex.realize_apply {p q : ℕ}
  (σ : AffineSimplex (stdSimplex ℝ (Fin (q + 1))) p)
  (x : stdSimplex ℝ (Fin (p + 1))) (j : Fin (q + 1)) :
  (σ.realize x : Fin (q + 1) → ℝ) j =
  ∑ i, x.1 i * (σ i).1 j :=
  rfl

/-- Realization sends each standard vertex to the corresponding listed vertex. -/

```

```

@[simp]
theorem AffineSimplex.realize_vertex {p q : ℕ}
  (σ : AffineSimplex (stdSimplex ℝ (Fin (q + 1))) p)
  (i : Fin (p + 1)) :
  σ.realize (stdSimplex.vertex i) = σ i := by
  classical
  apply Subtype.ext
  funext j
  change ∑ k, (Pi.single i (1 : ℝ) : Fin (p + 1) → ℝ) k * (σ k).1 j =
    (σ i).1 j
  rw [Finset.sum_eq_single i]
  · simp
  · intro k _ hki
    simp [Pi.single_eq_of_ne hki]
  · simp

/-- Pairing against a `FunOnFinite` pushforward is the original pairing. -/
lemma sum_linearMap_mul {α β : Type*} [Fintype α] [Fintype β] [DecidableEq β]
  (f : α → β) (s : α → ℝ) (t : β → ℝ) :
  ∑ y : β, FunOnFinite.linearMap ℝ ℝ f s y * t y = ∑ x : α, s x * t (f x) := by
  simp_rw [FunOnFinite.linearMap_apply_apply]
  calc
  ∑ y : β, (∑ x ∈ Finset.univ.filter (fun x ↦ f x = y), s x) * t y =
    ∑ y : β, ∑ x ∈ Finset.univ.filter (fun x ↦ f x = y), s x * t y := by
    simp_rw [Finset.sum_mul]
  = ∑ y : β, ∑ x ∈ Finset.univ.filter (fun x ↦ f x = y), s x * t (f x) := by
    refine Finset.sum_congr rfl fun y _ ↦ Finset.sum_congr rfl fun x hx ↦ ?_
    rw [(Finset.mem_filter.mp hx).2]
  = ∑ x : α, s x * t (f x) :=
    Finset.sum_fiberwise Finset.univ f (fun x ↦ s x * t (f x))

/-- Realization commutes with deleting a vertex. -/
theorem AffineSimplex.realize_face {p q : ℕ}
  (σ : AffineSimplex (stdSimplex ℝ (Fin (q + 1))) (p + 1))
  (i : Fin (p + 2)) (x : stdSimplex ℝ (Fin (p + 1))) :
  σ.realize (stdSimplex.map i.succAbove x) = (σ.face i).realize x := by
  classical
  apply Subtype.ext
  funext j
  simp [stdSimplex.map_coe] using
    sum_linearMap_mul i.succAbove x.1 (fun k ↦ (σ k).1 j)

/-- Reparametrize a singular simplex by a realized finite affine simplex. -/
noncomputable def reparametrizeSingularSimplex {X : TopCat.{0}} {p q : ℕ}
  (s : (TopCat.toSSet.obj X) _[q])
  (σ : AffineSimplex (stdSimplex ℝ (Fin (q + 1))) p) :
  (TopCat.toSSet.obj X) _[p] :=
  (X.toSSetObjEquiv _).symm ((X.toSSetObjEquiv _ s).comp σ.realize)

@[simp]
theorem reparametrizeSingularSimplex_toContinuousMap
  {X : TopCat.{0}} {p q : ℕ}
  (s : (TopCat.toSSet.obj X) _[q])
  (σ : AffineSimplex (stdSimplex ℝ (Fin (q + 1))) p) :
  X.toSSetObjEquiv _ (reparametrizeSingularSimplex s σ) =
  (X.toSSetObjEquiv _ s).comp σ.realize := by
  simp [reparametrizeSingularSimplex]

/-- Faces of a reparametrized simplex are reparametrizations of the faces. -/
theorem reparametrizeSingularSimplex_face {X : TopCat.{0}} {p q : ℕ}
  (s : (TopCat.toSSet.obj X) _[q])
  (σ : AffineSimplex (stdSimplex ℝ (Fin (q + 1))) (p + 1))
  (i : Fin (p + 2)) :

```

```

    reparametrizeSingularSimplex s (σ.face i) =
      (TopCat.toSSet.obj X).δ i (reparametrizeSingularSimplex s σ) := by
    apply (X.toSSetObjEquiv _).injective
  ext x
  rw [reparametrizeSingularSimplex_toContinuousMap,
      TopCat.toSSetObjEquiv_δ_apply,
      reparametrizeSingularSimplex_toContinuousMap]
  simp [AffineSimplex.realize_face]

/-- The identity affine simplex on a standard simplex. -/
def idSimplex (n : ℕ) :
  AffineSimplex (stdSimplex ℝ (Fin (n + 1))) n :=
  stdSimplex.vertex

/-- The identity affine simplex realizes as the identity map. -/
@[simp]
theorem realize_idSimplex (n : ℕ) :
  (idSimplex n).realize = ContinuousMap.id _ := by
  classical
  ext x j
  change ∑ i, x.1 i * (stdSimplex.vertex i : Fin (n + 1) → ℝ) j = x.1 j
  simp_rw [stdSimplex.vertex_coe]
  rw [Finset.sum_eq_single j]
  · simp
  · intro i _ hij
    rw [Pi.single_eq_of_ne hij.symm, mul_zero]
  · simp

/-- Reparametrizing by the identity simplex is the identity. -/
@[simp]
theorem reparametrizeSingularSimplex_id {X : TopCat.{0}} {n : ℕ}
  (s : (TopCat.toSSet.obj X) _[n]) :
  reparametrizeSingularSimplex s (idSimplex n) = s := by
  apply (X.toSSetObjEquiv _).injective
  rw [reparametrizeSingularSimplex_toContinuousMap, realize_idSimplex]
  ext x
  simp

/-- Reparametrizing by a face of the identity simplex is an ordinary face. -/
theorem reparametrizeSingularSimplex_id_face {X : TopCat.{0}} {n : ℕ}
  (s : (TopCat.toSSet.obj X) _[n + 1]) (i : Fin (n + 2)) :
  reparametrizeSingularSimplex s ((idSimplex (n + 1)).face i) =
  (TopCat.toSSet.obj X).δ i s := by
  rw [reparametrizeSingularSimplex_face, reparametrizeSingularSimplex_id]

/-- Geometric barycenter of an affine simplex inside a standard simplex. -/
def AffineSimplex.stdBarycenter {p q : ℕ}
  (σ : AffineSimplex (stdSimplex ℝ (Fin (q + 1))) p) :
  stdSimplex ℝ (Fin (q + 1)) :=
  σ.realize stdSimplex.barycenter

/-- Realization is functorial for vertex-wise maps coming from an
ambient realized simplex. -/
theorem AffineSimplex.realize_map {p q k : ℕ}
  (σ : AffineSimplex (stdSimplex ℝ (Fin (q + 1))) p)
  (ρ : AffineSimplex (stdSimplex ℝ (Fin (p + 1))) k) :
  (AffineSimplex.map σ.realize ρ).realize = σ.realize.comp ρ.realize := by
  classical
  apply ContinuousMap.ext
  intro x
  apply Subtype.ext
  ext j
  change (∑ i, x.1 i * ∑ m, (ρ i).1 m * (σ m).1 j) =

```

```


$$\sum m, (\sum i, x.1 i * (\rho i).1 m) * (\sigma m).1 j$$

simp_rw [Finset.mul_sum]
rw [Finset.sum_comm]
refine Finset.sum_congr rfl fun m _ => ?_
rw [Finset.sum_mul]
refine Finset.sum_congr rfl fun i _ => ?_
ring

theorem AffineSimplex.stdBarycenter_map {p q k : ℕ}
  (σ : AffineSimplex (stdSimplex ℝ (Fin (q + 1))) p)
  (ρ : AffineSimplex (stdSimplex ℝ (Fin (p + 1))) k) :
  AffineSimplex.stdBarycenter (AffineSimplex.map σ.realize ρ) =
    σ.realize (AffineSimplex.stdBarycenter ρ) := by
  simp [AffineSimplex.stdBarycenter, AffineSimplex.realize_map]

@[simp]
theorem AffineSimplex.map_realize_idSimplex {p q : ℕ}
  (σ : AffineSimplex (stdSimplex ℝ (Fin (q + 1))) p) :
  AffineSimplex.map σ.realize (idSimplex p) = σ := by
  funext i
  simp [AffineSimplex.map, idSimplex]

theorem reparametrizeSingularSimplex_map {X : TopCat.{0}} {p q k : ℕ}
  (s : (TopCat.toSSet.obj X) _[q])
  (σ : AffineSimplex (stdSimplex ℝ (Fin (q + 1))) p)
  (ρ : AffineSimplex (stdSimplex ℝ (Fin (p + 1))) k) :
  reparametrizeSingularSimplex s (AffineSimplex.map σ.realize ρ) =
    reparametrizeSingularSimplex (reparametrizeSingularSimplex s σ) ρ := by
  apply (X.toSSetObjEquiv _).injective
  rw [reparametrizeSingularSimplex_toContinuousMap,
    reparametrizeSingularSimplex_toContinuousMap,
    reparametrizeSingularSimplex_toContinuousMap,
    AffineSimplex.realize_map]
  ext x
  simp [ContinuousMap.comp_assoc]

/-- Barycentric subdivision of affine chains in a standard simplex. -/
def subdivideStd {q : ℕ} :
  (n : ℕ) → AffineChain (stdSimplex ℝ (Fin (q + 1))) n →+
    AffineChain (stdSimplex ℝ (Fin (q + 1))) n :=
  subdivide AffineSimplex.stdBarycenter

/-- Prism operator for affine chains in a standard simplex. -/
def prismStd {q : ℕ} :
  (n : ℕ) → AffineChain (stdSimplex ℝ (Fin (q + 1))) n →+
    AffineChain (stdSimplex ℝ (Fin (q + 1))) (n + 1) :=
  prism AffineSimplex.stdBarycenter

theorem boundary_subdivideStd {q n : ℕ}
  (c : AffineChain (stdSimplex ℝ (Fin (q + 1))) (n + 1)) :
  boundary (subdivideStd (q := q) (n + 1) c) =
    subdivideStd (q := q) n (boundary c) :=
  boundary_subdivide AffineSimplex.stdBarycenter n c

theorem boundary_prismStd {q n : ℕ}
  (c : AffineChain (stdSimplex ℝ (Fin (q + 1))) (n + 1)) :
  boundary (prismStd (q := q) (n + 1) c) +
    prismStd (q := q) n (boundary c) =
    c - subdivideStd (q := q) (n + 1) c :=
  boundary_prism AffineSimplex.stdBarycenter n c

/-- Subdivision of the identity simplex. -/
def standardSubdivision (n : ℕ) :

```

```

    AffineChain (stdSimplex  $\mathbb{R}$  (Fin (n + 1))) n :=
    subdivideStd (q := n) n (simplex (idSimplex n))

/-- Prism of the identity simplex. -/
def standardPrism (n :  $\mathbb{N}$ ) :
    AffineChain (stdSimplex  $\mathbb{R}$  (Fin (n + 1))) (n + 1) :=
    prismStd (q := n) n (simplex (idSimplex n))

theorem AffineChain.map_subdivideStd {p q :  $\mathbb{N}$ }
    ( $\sigma$  : AffineSimplex (stdSimplex  $\mathbb{R}$  (Fin (q + 1))) p)
    (n :  $\mathbb{N}$ ) (c : AffineChain (stdSimplex  $\mathbb{R}$  (Fin (p + 1))) n) :
    AffineChain.map  $\sigma$ .realize (subdivideStd (q := p) n c) =
    subdivideStd (q := q) n (AffineChain.map  $\sigma$ .realize c) :=
    AffineChain.map_subdivide
    AffineSimplex.stdBarycenter AffineSimplex.stdBarycenter  $\sigma$ .realize
    (fun  $\rho \mapsto$  (AffineSimplex.stdBarycenter_map  $\sigma$   $\rho$ ).symm) n c

theorem AffineChain.map_prismStd {p q :  $\mathbb{N}$ }
    ( $\sigma$  : AffineSimplex (stdSimplex  $\mathbb{R}$  (Fin (q + 1))) p)
    (n :  $\mathbb{N}$ ) (c : AffineChain (stdSimplex  $\mathbb{R}$  (Fin (p + 1))) n) :
    AffineChain.map  $\sigma$ .realize (prismStd (q := p) n c) =
    prismStd (q := q) n (AffineChain.map  $\sigma$ .realize c) :=
    AffineChain.map_prism
    AffineSimplex.stdBarycenter AffineSimplex.stdBarycenter  $\sigma$ .realize
    (fun  $\rho \mapsto$  (AffineSimplex.stdBarycenter_map  $\sigma$   $\rho$ ).symm) n c

theorem AffineChain.map_standardSubdivision {p q :  $\mathbb{N}$ }
    ( $\sigma$  : AffineSimplex (stdSimplex  $\mathbb{R}$  (Fin (q + 1))) p) :
    AffineChain.map  $\sigma$ .realize (standardSubdivision p) =
    subdivideStd (q := q) p (simplex  $\sigma$ ) := by
    simp [standardSubdivision, AffineChain.map_subdivideStd]

theorem AffineChain.map_standardPrism {p q :  $\mathbb{N}$ }
    ( $\sigma$  : AffineSimplex (stdSimplex  $\mathbb{R}$  (Fin (q + 1))) p) :
    AffineChain.map  $\sigma$ .realize (standardPrism p) =
    prismStd (q := q) p (simplex  $\sigma$ ) := by
    simp [standardPrism, AffineChain.map_prismStd]

/-- The signed barycentric subdivision chain of the standard interval. -/
def standardIntervalSubdivision :
    AffineChain (stdSimplex  $\mathbb{R}$  (Fin 2)) 1 :=
    edge (stdSimplex.vertex 0) stdSimplex.barycenter -
    edge (stdSimplex.vertex 1) stdSimplex.barycenter

/-- The standard interval subdivision has the original endpoint boundary. -/
theorem boundary_standardIntervalSubdivision :
    boundary standardIntervalSubdivision =
    vertex (stdSimplex.vertex (1 : Fin 2)) -
    vertex (stdSimplex.vertex (0 : Fin 2)) := by
    simp [standardIntervalSubdivision, boundary_edge]

/-- Universal affine prism for subdivision of the standard interval. -/
def standardIntervalPrism :
    AffineChain (stdSimplex  $\mathbb{R}$  (Fin 2)) 2 :=
    cone stdSimplex.barycenter
    (edge (stdSimplex.vertex 0) (stdSimplex.vertex 1) -
    standardIntervalSubdivision)

/-- The standard one-dimensional prism identity. -/
theorem boundary_standardIntervalPrism :
    boundary standardIntervalPrism =
    edge (stdSimplex.vertex 0) (stdSimplex.vertex 1) -
    standardIntervalSubdivision := by

```

```

classical
unfold standardIntervalPrism standardIntervalSubdivision
simp only [map_sub]
rw [boundary_cone_edge, boundary_cone_edge, boundary_cone_edge]
simp only [map_sub, cone_vertex]
abel

end

end PlatonicSolids.SingularExcision

```

A.11 PlatonicSolids/SingularExcision/Mesh.lean

Lebesgue number and mesh-ratio $n/(n+1)$ (117 lines).

```

/-
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-/
import Mathlib.Analysis.Convex.StdSimplex
import Mathlib.Topology.MetricSpace.Pseudo.Lemmas

/-!
Metric input for the small-simplices theorem.

The Lebesgue-number lemma is most useful for subdivision in the following
form: every nonempty subset of the compact domain whose diameter is smaller
than one fixed positive number lies in one member of the open cover.
-/

open Metric Set

universe u v

variable {α : Type u} {ι : Sort v} [PseudoMetricSpace α]

/-- The classical mesh contraction factor for barycentric subdivision
of an `n`-simplex. -/
noncomputable def barycentricMeshRatio (n : ℕ) : ℝ :=
  n / (n + 1)

lemma barycentricMeshRatio_nonneg (n : ℕ) :
  0 ≤ barycentricMeshRatio n := by
  rw [barycentricMeshRatio]
  exact div_nonneg (Nat.cast_nonneg n) (by positivity)

lemma barycentricMeshRatio_lt_one (n : ℕ) :
  barycentricMeshRatio n < 1 := by
  rw [barycentricMeshRatio, div_lt_one]
  · exact_mod_cast Nat.lt_succ_self n
  · positivity

/-- Iterated barycentric subdivision eventually has mesh below every
positive threshold. -/
lemma exists_barycentricMeshRatio_pow_lt (n : ℕ) {ε : ℝ} (hε : 0 < ε) :
  ∃ k : ℕ, barycentricMeshRatio n ^ k < ε :=
  exists_pow_lt_of_lt_one hε (barycentricMeshRatio_lt_one n)

/-- Diameter form of the Lebesgue-number lemma. -/
theorem exists_diam_lt_forall_subset_openCover
  {K : Set α} {U : ι → Set α}
  (hK : IsCompact K) (hU : ∀ i, IsOpen (U i))
  (hcover : K ⊆ ⋃ i, U i) :

```

```

    ∃ δ > 0, ∀ {{s : Set α}}, s ⊆ K → s.Nonempty →
      Metric.diam s < δ → ∃ i, s ⊆ U i := by
obtain ⟨δ, hδ, hball⟩ :=
  lebesgue_number_lemma_of_metric hK hU hcover
refine ⟨δ, hδ, ?_⟩
intro s hsK hsne hsdiam
obtain ⟨x, hxs⟩ := hsne
obtain ⟨i, hi⟩ := hball x (hsK hxs)
refine ⟨i, fun y hys ↦ hi ?_⟩
rw [mem_ball, dist_comm]
exact (Metric.dist_le_diam_of_mem (hK.isBounded.subset hsK) hxs hys).trans_lt hsdiam

/-- A continuous simplex pulls a two-set open cover back to an open cover
of its compact standard-simplex domain, with a uniform diameter bound. -/
theorem exists_diam_lt_forall_image_subset_left_or_right
  {n : ℕ} {Y : Type u} [TopologicalSpace Y]
  (σ : C(stdSimplex ℝ (Fin (n + 1)), Y))
  {A B : Set Y} (hA : IsOpen A) (hB : IsOpen B)
  (hAB : A ∪ B = univ) :
  ∃ δ > 0, ∀ {{s : Set (stdSimplex ℝ (Fin (n + 1)))}},
    s.Nonempty → Metric.diam s < δ →
      σ '' s ⊆ A ∨ σ '' s ⊆ B := by
let U : Bool → Set (stdSimplex ℝ (Fin (n + 1))) :=
  fun b ↦ if b then σ-1 A else σ-1 B
have hU : ∀ b, IsOpen (U b) := by
  intro b
  cases b
  · simpa [U] using hB.preimage σ.continuous
  · simpa [U] using hA.preimage σ.continuous
have hcover : (univ : Set (stdSimplex ℝ (Fin (n + 1)))) ⊆ ⋃ b, U b := by
  intro x hx
  have hxAB : σ x ∈ A ∪ B := by rw [hAB]; exact mem_univ _
  rcases hxAB with hxA | hxB
  · exact mem_iUnion.2 ⟨true, by simpa [U]⟩
  · exact mem_iUnion.2 ⟨false, by simpa [U]⟩
obtain ⟨δ, hδ, hsmall⟩ :=
  exists_diam_lt_forall_subset_openCover
  (U := U) isCompact_univ hU hcover
refine ⟨δ, hδ, ?_⟩
intro s hsne hsdiam
obtain ⟨b, hb⟩ := hsmall (subset_univ s) hsne hsdiam
cases b
· right
  rintro _ ⟨x, hx, rfl⟩
  exact hb hx
· left
  rintro _ ⟨x, hx, rfl⟩
  exact hb hx

/-- A geometric mesh bound times an iterated contraction factor
eventually falls below every positive threshold. The remaining
geometric input is that each barycentric piece of an `n`-simplex
has diameter at most `barycentricMeshRatio n` times the parent. -/
lemma exists_barycentricMeshRatio_pow_mul_lt (n : ℕ) (C : ℝ) {ε : ℝ}
  (hε : 0 < ε) :
  ∃ k : ℕ, barycentricMeshRatio n ^ k * C < ε := by
  rcases le_or_gt C 0 with hC | hC
  · exact ⟨0, by simp; linarith⟩
  · obtain ⟨k, hk⟩ :=
      exists_barycentricMeshRatio_pow_lt n (div_pos hε hC)
    exact ⟨k, (lt_div_iff₀ hC).mp hk⟩

lemma exists_barycentricMeshRatio_pow_mul_diam_stdSimplex_lt

```

```

(n : ℕ) {ε : ℝ} (hε : 0 < ε) :
  ∃ k : ℕ,
    barycentricMeshRatio n ^ k *
      Metric.diam
        (Set.univ : Set (stdSimplex ℝ (Fin (n + 1)))) < ε :=
exists_barycentricMeshRatio_pow_mul_lt n _ hε

```

A.12 PlatonicSolids/SingularExcision/BarycentricGeometry.lean

Diameter of barycentric pieces and eventual smallness (380 lines).

```

/-
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-/
import Mathlib.Analysis.Normed.Module.Convex
import PlatonicSolids.SingularExcision.AffineRealization
import PlatonicSolids.SingularExcision.Mesh

/-!
Diameter of barycentric pieces. Each cone-on-face piece of an
`n`-simplex has diameter at most `n/(n+1)` times the parent, so
iterated subdivision has mesh going to zero.
-/

open Metric Set Bornology
open scoped BigOperators

namespace PlatonicSolids.SingularExcision

noncomputable section

lemma barycentricMeshRatio_le_succ (n : ℕ) :
  barycentricMeshRatio n ≤ barycentricMeshRatio (n + 1) := by
  simp only [barycentricMeshRatio]
  rw [div_le_div_iff₀ (by positivity) (by positivity)]
  norm_cast
  nlinarith

lemma AffineSimplex.range_face_subset {V : Type*} {n : ℕ}
  (σ : AffineSimplex V (n + 1)) (i : Fin (n + 2)) :
  Set.range (σ.face i) ⊆ Set.range σ :=
  fun _ ⟨j, hj⟩ => ⟨i.succAbove j, hj⟩

lemma AffineSimplex.range_append {V : Type*} {n : ℕ}
  (ρ : AffineSimplex V n) (v : V) :
  Set.range (ρ.append v) = insert v (Set.range ρ) := by
  ext x
  constructor
  · rintro ⟨i, rfl⟩
    induction i using Fin.lastCases with
    | last =>
      left
      simp [AffineSimplex.append]
    | cast j =>
      right
      exact ⟨j, by simp [AffineSimplex.append]⟩
  · rintro (rfl | ⟨j, rfl⟩)
  · exact ⟨Fin.last _, by simp [AffineSimplex.append]⟩
  · exact ⟨j.castSucc, by simp [AffineSimplex.append]⟩

lemma AffineSimplex.realize_val {p q : ℕ}
  (σ : AffineSimplex (stdSimplex ℝ (Fin (q + 1))) p)

```

```

    (x : stdSimplex ℝ (Fin (p + 1))) :
    (σ.realize x : Fin (q + 1) → ℝ) =
      ∑ i, x.1 i • (σ i : Fin (q + 1) → ℝ) := by
  funext j
  rw [realize_apply, Finset.sum_apply]
  refine Finset.sum_congr rfl fun i _ => ?_
  rw [Pi.smul_apply, smul_eq_mul]
  rfl

lemma AffineSimplex.realize_mem_convexHull {p q : ℕ}
  (σ : AffineSimplex (stdSimplex ℝ (Fin (q + 1))) p)
  (x : stdSimplex ℝ (Fin (p + 1))) :
  (σ.realize x : Fin (q + 1) → ℝ) ∈
    convexHull ℝ ((↑) '' Set.range σ) := by
  rw [AffineSimplex.realize_val]
  exact (convex_convexHull ℝ _).sum_mem
    (fun i _ => x.2.1 i) x.2.2
    (fun i _ => subset_convexHull ℝ _ ⟨σ i, ⟨i, rfl⟩, rfl⟩)

lemma AffineSimplex.stdBarycenter_mem_convexHull {p q : ℕ}
  (σ : AffineSimplex (stdSimplex ℝ (Fin (q + 1))) p) :
  (σ.stdBarycenter : Fin (q + 1) → ℝ) ∈
    convexHull ℝ ((↑) '' Set.range σ) :=
  σ.realize_mem_convexHull stdSimplex.barycenter

lemma AffineSimplex.range_subset_range_realize {p q : ℕ}
  (σ : AffineSimplex (stdSimplex ℝ (Fin (q + 1))) p) :
  Set.range σ ⊆ Set.range σ.realize := by
  rintro _ ⟨i, rfl⟩
  exact ⟨stdSimplex.vertex i, σ.realize_vertex i⟩

lemma isBounded_range_affineSimplex {p q : ℕ}
  (σ : AffineSimplex (stdSimplex ℝ (Fin (q + 1))) p) :
  IsBounded (Set.range σ) :=
  (Set.finite_range σ).isBounded

lemma isBounded_range_realize {p q : ℕ}
  (σ : AffineSimplex (stdSimplex ℝ (Fin (q + 1))) p) :
  IsBounded (Set.range σ.realize) :=
  (isCompact_range σ.realize.continuous).isBounded

lemma diam_coe_image_of_bounded {ι : Type*} [Fintype ι]
  {s : Set (stdSimplex ℝ ι)} (hs : IsBounded s)
  (hs' : IsBounded ((↑) '' s : Set (ι → ℝ))) :
  diam ((↑) '' s : Set (ι → ℝ)) = diam s := by
  refine le_antisymm ?_ ?_
  · refine diam_le_of_forall_dist_le diam_nonneg ?_
    rintro _ ⟨x, hx, rfl⟩ _ ⟨y, hy, rfl⟩
    simpa [Subtype.dist_eq] using dist_le_diam_of_mem hs hx hy
  · refine diam_le_of_forall_dist_le diam_nonneg ?_
    intro x hx y hy
    simpa [Subtype.dist_eq] using
      dist_le_diam_of_mem hs' ⟨x, hx, rfl⟩ ⟨y, hy, rfl⟩

lemma diam_coe_image_finite {ι : Type*} [Fintype ι]
  (s : Set (stdSimplex ℝ ι)) (hs : s.Finite) :
  diam ((↑) '' s : Set (ι → ℝ)) = diam s :=
  diam_coe_image_of_bounded hs.isBounded (hs.image _).isBounded

/-- Realization cannot increase diameter: it is the convex hull of the
listed vertices in the ambient coordinate space. -/
lemma diam_range_realize_le {p q : ℕ}
  (σ : AffineSimplex (stdSimplex ℝ (Fin (q + 1))) p) :

```

```

    diam (Set.range  $\sigma$ .realize)  $\leq$  diam (Set.range  $\sigma$ ) := by
  refine diam_le_of_forall_dist_le diam_nonneg ?_
  intro x hx y hy
  obtain ⟨a, rfl⟩ := hx
  obtain ⟨b, rfl⟩ := hy
  have ha :=  $\sigma$ .realize_mem_convexHull a
  have hb :=  $\sigma$ .realize_mem_convexHull b
  have hB : IsBounded (( $\uparrow$ ) '' Set.range  $\sigma$  : Set (Fin (q + 1)  $\rightarrow$   $\mathbb{R}$ )) :=
    ((Set.finite_range  $\sigma$ ).image _).isBounded
  have hle : dist ( $\sigma$ .realize a : Fin (q + 1)  $\rightarrow$   $\mathbb{R}$ )
    ( $\sigma$ .realize b : Fin (q + 1)  $\rightarrow$   $\mathbb{R}$ )  $\leq$ 
    diam (( $\uparrow$ ) '' Set.range  $\sigma$  : Set (Fin (q + 1)  $\rightarrow$   $\mathbb{R}$ )) :=
    (dist_le_diam_of_mem (isBounded_convexHull.mpr hB) ha hb).trans
    (by rw [convexHull_diam])
  simp [Subtype.dist_eq, diam_coe_image_finite _ (Set.finite_range  $\sigma$ )] using hle

lemma diam_range_realize_eq {p q :  $\mathbb{N}$ }
  ( $\sigma$  : AffineSimplex (stdSimplex  $\mathbb{R}$  (Fin (q + 1))) p) :
  diam (Set.range  $\sigma$ .realize) = diam (Set.range  $\sigma$ ) :=
  le_antisymm (diam_range_realize_le  $\sigma$ )
  (diam_mono  $\sigma$ .range_subset_range_realize (isBounded_range_realize  $\sigma$ ))

lemma diam_range_affineSimplex_zero {q :  $\mathbb{N}$ }
  ( $\sigma$  : AffineSimplex (stdSimplex  $\mathbb{R}$  (Fin (q + 1))) 0) :
  diam (Set.range  $\sigma$ ) = 0 := by
  refine diam_subsingleton ?_
  intro x hx y hy
  obtain ⟨i, rfl⟩ := hx
  obtain ⟨j, rfl⟩ := hy
  rw [Fin.eq_zero i, Fin.eq_zero j]

/-- Distance from the barycenter to a vertex is at most the mesh ratio
times the vertex diameter. -/
lemma dist_stdBarycenter_vertex {p q :  $\mathbb{N}$ }
  ( $\sigma$  : AffineSimplex (stdSimplex  $\mathbb{R}$  (Fin (q + 1))) p)
  (i : Fin (p + 1)) :
  dist  $\sigma$ .stdBarycenter ( $\sigma$  i)  $\leq$ 
  barycentricMeshRatio p * diam (Set.range  $\sigma$ ) := by
  classical
  have hne : (p + 1 :  $\mathbb{R}$ )  $\neq$  0 := by positivity
  have hval : ( $\sigma$ .stdBarycenter : Fin (q + 1)  $\rightarrow$   $\mathbb{R}$ ) =
     $\sum j : \text{Fin } (p + 1), (p + 1 : \mathbb{R})^{-1} \bullet (\sigma j : \text{Fin } (q + 1) \rightarrow \mathbb{R})$  := by
    rw [AffineSimplex.stdBarycenter, AffineSimplex.realize_val]
    refine Finset.sum_congr rfl fun j _ => ?_
    simp [stdSimplex.barycenter, Fintype.card_fin]
  have hsum :  $\sum j : \text{Fin } (p + 1), (p + 1 : \mathbb{R})^{-1} = (1 : \mathbb{R})$  := by
    simp [Finset.card_univ, Fintype.card_fin, nsmul_eq_mul,
      mul_inv_cancel₀ hne]
  have hi : ( $\sigma$  i : Fin (q + 1)  $\rightarrow$   $\mathbb{R}$ ) =
     $\sum j : \text{Fin } (p + 1), (p + 1 : \mathbb{R})^{-1} \bullet (\sigma i : \text{Fin } (q + 1) \rightarrow \mathbb{R})$  := by
    rw [← Finset.sum_smul, hsum, one_smul]
  have hdiff :
    ( $\sigma$ .stdBarycenter : Fin (q + 1)  $\rightarrow$   $\mathbb{R}$ ) - ( $\sigma$  i : Fin (q + 1)  $\rightarrow$   $\mathbb{R}$ ) =
     $\sum j : \text{Fin } (p + 1),$ 
     $(p + 1 : \mathbb{R})^{-1} \bullet$ 
     $((\sigma j : \text{Fin } (q + 1) \rightarrow \mathbb{R}) - (\sigma i : \text{Fin } (q + 1) \rightarrow \mathbb{R}))$  := by
    rw [hval]
    conv_lhs => rw [hi]
    rw [← Finset.sum_sub_distrib]
    exact Finset.sum_congr rfl fun j _ => (smul_sub _ _ _).symm
  have hsub :
    ( $\sigma$ .stdBarycenter : Fin (q + 1)  $\rightarrow$   $\mathbb{R}$ ) - ( $\sigma$  i : Fin (q + 1)  $\rightarrow$   $\mathbb{R}$ ) =
     $\sum j \in \text{Finset.univ.erase } i,$ 

```

```

      (p + 1 : ℝ)-1 •
      ((σ j : Fin (q + 1) → ℝ) - (σ i : Fin (q + 1) → ℝ)) := by
rw [hdiff, ← Finset.sum_erase_add (f := fun j =>
      (p + 1 : ℝ)-1 •
      ((σ j : Fin (q + 1) → ℝ) - (σ i : Fin (q + 1) → ℝ)))
      (h := Finset.mem_univ i)]
simp [sub_self]
rw [Subtype.dist_eq, dist_eq_norm]
change ||(σ.stdBarycenter : Fin (q + 1) → ℝ) - (σ i : Fin (q + 1) → ℝ)|| ≤ _
rw [hsub]
refine (norm_sum_le _ _).trans ?_
simp only [norm_smul, Real.norm_eq_abs,
      abs_of_nonneg (inv_nonneg.2 (by positivity : (0 : ℝ) ≤ p + 1))]
have hterm (j : Fin (p + 1)) :
      (p + 1 : ℝ)-1 *
      ||(σ j : Fin (q + 1) → ℝ) - (σ i : Fin (q + 1) → ℝ)|| ≤
      (p + 1 : ℝ)-1 * diam (Set.range σ) := by
      refine mul_le_mul_of_nonneg_left ?_ (inv_nonneg.2 (by positivity))
      simp [dist_eq_norm, Subtype.dist_eq] using
      dist_le_diam_of_mem (isBounded_range_affineSimplex σ)
      ⟨j, rfl⟩ ⟨i, rfl⟩
refine (Finset.sum_le_sum fun j _ => hterm j).trans ?_
simp only [Finset.sum_const, nsmul_eq_mul,
      Finset.card_erase_of_mem (Finset.mem_univ i), Finset.card_univ,
      Fintype.card_fin]
have hcard : ((p + 1 : ℕ) - 1 : ℕ) : ℝ = (p : ℝ) := by
      norm_cast
rw [hcard, barycentricMeshRatio, div_eq_mul_inv, mul_assoc]

/-- Distance from the barycenter to any point of the convex hull of the
vertices is controlled by the same mesh ratio. -/
lemma dist_stdBarycenter_mem_convexHull {p q : ℕ}
      (σ : AffineSimplex (stdSimplex ℝ (Fin (q + 1))) p)
      {x : Fin (q + 1) → ℝ}
      (hx : x ∈ convexHull ℝ ((↑) '' Set.range σ)) :
      dist (σ.stdBarycenter : Fin (q + 1) → ℝ) x ≤
      barycentricMeshRatio p * diam (Set.range σ) := by
obtain ⟨y, hy, hle⟩ :=
      convexHull_exists_dist_ge hx (σ.stdBarycenter : Fin (q + 1) → ℝ)
obtain ⟨v, ⟨i, rfl⟩, rfl⟩ := hy
rw [dist_comm]
exact hle.trans (by
      simp [dist_comm, Subtype.dist_eq] using dist_stdBarycenter_vertex σ i)

/-- Vertices of every barycentric piece lie in the convex hull of the
parent vertices. -/
lemma vertices_subdivideStd_mem_convexHull {q : ℕ} (n : ℕ) :
      ∀ (σ : AffineSimplex (stdSimplex ℝ (Fin (q + 1))) n)
      {τ : AffineSimplex (stdSimplex ℝ (Fin (q + 1))) n},
      τ ∈ (subdivideStd (q := q) n (simplex σ)).support →
      ∀ i : Fin (n + 1),
      (τ i : Fin (q + 1) → ℝ) ∈ convexHull ℝ ((↑) '' Set.range σ) := by
induction n with
| zero =>
      intro σ τ hτ i
      classical
      simp [subdivideStd, subdivide_zero, simplex,
      Finsupp.mem_support_single] at hτ
      rcases hτ with ⟨rfl, _⟩
      exact subset_convexHull ℝ _ ⟨σ i, ⟨i, rfl⟩, rfl⟩
| succ n ih =>
      intro σ τ hτ i
      obtain ⟨j, ρ, hρ, rfl⟩ :=

```

```

    mem_support_subdivide_simplex AffineSimplex.stdBarycenter  $\sigma$  h $\tau$ 
induction i using Fin.lastCases with
| last =>
    simp [AffineSimplex.append] using  $\sigma$ .stdBarycenter_mem_convexHull
| cast k =>
    exact convexHull_mono (image_mono ( $\sigma$ .range_face_subset j))
    (by simp [AffineSimplex.append] using ih ( $\sigma$ .face j) h $\rho$  k)

/-- Vertex-diameter of a barycentric piece is at most the mesh ratio
times the parent vertex-diameter. -/
lemma diam_range_subdivideStd_le {q :  $\mathbb{N}$ } (n :  $\mathbb{N}$ ) :
   $\forall$  ( $\sigma$  : AffineSimplex (stdSimplex  $\mathbb{R}$  (Fin (q + 1))) n)
  ( $\tau$  : AffineSimplex (stdSimplex  $\mathbb{R}$  (Fin (q + 1))) n),
   $\tau \in$  (subdivideStd (q := q) n (simplex  $\sigma$ )).support  $\rightarrow$ 
  diam (Set.range  $\tau$ )  $\leq$ 
    barycentricMeshRatio n * diam (Set.range  $\sigma$ ) := by
induction n with
| zero =>
    intro  $\sigma$   $\tau$  h $\tau$ 
    classical
    simp [subdivideStd, subdivide_zero, simplex,
      Finsupp.mem_support_single] at h $\tau$ 
    rcases h $\tau$  with (rfl, _)
    simp [diam_range_affineSimplex_zero]
| succ n ih =>
    intro  $\sigma$   $\tau$  h $\tau$ 
    obtain (j,  $\rho$ , h $\rho$ , rfl) :=
      mem_support_subdivide_simplex AffineSimplex.stdBarycenter  $\sigma$  h $\tau$ 
    set C := barycentricMeshRatio (n + 1) * diam (Set.range  $\sigma$ )
    have hC : 0  $\leq$  C :=
      mul_nonneg (barycentricMeshRatio_nonneg _) diam_nonneg
    have h $\rho$ diam : diam (Set.range  $\rho$ )  $\leq$  C := by
      have h1 := ih ( $\sigma$ .face j) ( $\tau$  :=  $\rho$ ) h $\rho$ 
      have h2 : diam (Set.range ( $\sigma$ .face j))  $\leq$  diam (Set.range  $\sigma$ ) :=
        diam_mono ( $\sigma$ .range_face_subset j) (isBounded_range_affineSimplex  $\sigma$ )
      have h3 : barycentricMeshRatio n * diam (Set.range  $\sigma$ )  $\leq$  C :=
        mul_le_mul_of_nonneg_right (barycentricMeshRatio_le_succ n)
        diam_nonneg
      exact (h1.trans (mul_le_mul_of_nonneg_left h2
        (barycentricMeshRatio_nonneg n))).trans h3
    have hdist {x : stdSimplex  $\mathbb{R}$  (Fin (q + 1))}
      (hx : x  $\in$  Set.range  $\rho$ ) :
      dist  $\sigma$ .stdBarycenter x  $\leq$  C := by
      obtain (k, rfl) := hx
      have hx' : ( $\rho$  k : Fin (q + 1)  $\rightarrow$   $\mathbb{R}$ )  $\in$ 
        convexHull  $\mathbb{R}$  (( $\uparrow$ ) '' Set.range  $\sigma$ ) :=
        convexHull_mono (image_mono ( $\sigma$ .range_face_subset j))
        (vertices_subdivideStd_mem_convexHull n ( $\sigma$ .face j) h $\rho$  k)
      simp [C, Subtype.dist_eq] using
        dist_stdBarycenter_mem_convexHull  $\sigma$  hx'
    rw [AffineSimplex.range_append]
    refine diam_le_of_forall_dist_le hC ?_
    intro x hx y hy
    simp only [mem_insert_iff] at hx hy
    rcases hx with rfl | hx <|> rcases hy with rfl | hy
    · simp [dist_self] using hC
    · exact hdist hy
    · rw [dist_comm]
      exact hdist hx
    · exact (dist_le_diam_of_mem (Set.finite_range  $\rho$ ).isBounded hx hy).trans
      h $\rho$ diam

lemma diam_range_idSimplex_le (n :  $\mathbb{N}$ ) :

```

```

    diam (Set.range (idSimplex n)) ≤
      diam (univ : Set (stdSimplex ℝ (Fin (n + 1)))) :=
    diam_mono (subset_univ _) isBounded_of_compactSpace

/-- After `k` barycentric subdivisions of the identity `n`-simplex,
every remaining piece has vertex-diameter at most the contracted mesh. -/
lemma diam_range_subdivideStd_iterate_le (n k : ℕ)
  {τ : AffineSimplex (stdSimplex ℝ (Fin (n + 1))) n}
  (hτ : τ ∈ ((subdivideStd (q := n) n)^[k]
    (simplex (idSimplex n))).support) :
  diam (Set.range τ) ≤
    barycentricMeshRatio n ^ k * diam (Set.range (idSimplex n)) := by
  induction k generalizing τ with
  | zero =>
    classical
    simp [Function.iterate_zero, simplex, Finsupp.mem_support_single] at hτ
    rcases hτ with ⟨rfl, _⟩
    simp
  | succ k ih =>
    rw [Function.iterate_succ_apply'] at hτ
    obtain ⟨ρ, hρ, hτρ⟩ :=
      mem_support_subdivide AffineSimplex.stdBarycenter n hτ
    have h1 := diam_range_subdivideStd_le (q := n) n ρ hτρ
    have h2 := ih hρ
    have h3 :
      barycentricMeshRatio n *
        (barycentricMeshRatio n ^ k * diam (Set.range (idSimplex n))) =
        barycentricMeshRatio n ^ (k + 1) *
        diam (Set.range (idSimplex n)) := by
      rw [← mul_assoc, mul_comm (barycentricMeshRatio n), ← pow_succ]
    exact (h1.trans (mul_le_mul_of_nonneg_left h2
      (barycentricMeshRatio_nonneg n))).trans_eq h3

lemma diam_range_realize_subdivideStd_iterate_le (n k : ℕ)
  {τ : AffineSimplex (stdSimplex ℝ (Fin (n + 1))) n}
  (hτ : τ ∈ ((subdivideStd (q := n) n)^[k]
    (simplex (idSimplex n))).support) :
  diam (Set.range τ.realize) ≤
    barycentricMeshRatio n ^ k *
    diam (univ : Set (stdSimplex ℝ (Fin (n + 1)))) :=
  (diam_range_realize_le τ).trans <|
  (diam_range_subdivideStd_iterate_le n k hτ).trans <|
  mul_le_mul_of_nonneg_left (diam_range_idSimplex_le n)
  (pow_nonneg (barycentricMeshRatio_nonneg n) k)

/-- Iterated barycentric pieces of `Δ^n` become arbitrarily small. -/
lemma exists_subdivideStd_iterate_diam_lt (n : ℕ) {ε : ℝ} (hε : 0 < ε) :
  ∃ k : ℕ, ∀ τ ∈ ((subdivideStd (q := n) n)^[k]
    (simplex (idSimplex n))).support,
    diam (Set.range τ.realize) < ε := by
  obtain ⟨k, hk⟩ :=
    exists_barycentricMeshRatio_pow_mul_diam_stdSimplex_lt n hε
  refine ⟨k, fun τ hτ => ?_⟩
  exact (diam_range_realize_subdivideStd_iterate_le n k hτ).trans_lt hk

/-- Every piece of a sufficiently iterated subdivision of a singular
simplex lands in one member of a two-set open cover. -/
lemma exists_subdivideStd_iterate_image_subset_left_or_right
  {n : ℕ} {Y : Type*} [TopologicalSpace Y]
  (σ : C(stdSimplex ℝ (Fin (n + 1)), Y))
  {A B : Set Y} (hA : IsOpen A) (hB : IsOpen B)
  (hAB : A ∪ B = univ) :
  ∃ N : ℕ, ∀ τ ∈ ((subdivideStd (q := n) n)^[N]

```

```

      (simplex (idSimplex n))).support,
     $\sigma$  '' Set.range  $\tau$ .realize  $\subseteq A \vee$ 
     $\sigma$  '' Set.range  $\tau$ .realize  $\subseteq B :=$  by
  obtain  $\langle \delta, h\delta, hsmall \rangle :=$ 
    exists_diam_lt_forall_image_subset_left_or_right  $\sigma$  hA hB hAB
  obtain  $\langle N, hN \rangle :=$  exists_subdivideStd_iterate_diam_lt n h $\delta$ 
  refine  $\langle N, \text{fun } \tau \text{ h}\tau \Rightarrow ?_ \rangle$ 
  exact hsmall  $\langle \tau$ .realize (stdSimplex.vertex 0),  $\langle$ stdSimplex.vertex 0, rfl $\rangle \rangle$ 
    (hN  $\tau$  h $\tau$ )

end

end PlatonicSolids.SingularExcision

```

A.13 PlatonicSolids/SingularExcision/SingularChains.lean

Singular Sd as a chain map and the prism homotopy (472 lines).

```

/-
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-/
import Mathlib.Algebra.Homology.Homotopy
import Mathlib.AlgebraicTopology.SingularHomology.Basic
import PlatonicSolids.SingularExcision.AffineRealization

/-!
# Interpreting finite affine chains as singular chains

A finite integral affine chain in the standard `q`-simplex is composed
with a singular `q`-simplex, then interpreted in Mathlib's coproduct of
copies of the coefficient object. Face-compatibility of realization
makes the barycentric generators assemble to a chain map, with the
prism as a chain homotopy to the identity.
-/

open AlgebraicTopology CategoryTheory Limits HomologicalComplex
open scoped Simplicial

namespace PlatonicSolids.SingularExcision

noncomputable section

universe v u

variable {C : Type u} [Category.{v} C] [HasCoproducts.{0} C] [Preadditive C]

/-- The singular-chain generator obtained by affine reparametrization. -/
noncomputable def realizeAffineSimplex
  (R : C) {X : TopCat.{0}} {p q : ℕ}
  (s : (TopCat.toSSet.obj X) _[q])
  ( $\sigma$  : AffineSimplex (stdSimplex  $\mathbb{R}$  (Fin (q + 1))) p) :
  R  $\longrightarrow$  (((singularChainComplexFunctor C).obj R).obj X).X p :=
  (TopCat.toSSet.obj X).ιChainComplex
  (reparametrizeSingularSimplex s  $\sigma$ )

/-- Interpret a finite integral affine chain as a morphism into singular chains. -/
noncomputable def realizeAffineChain
  (R : C) {X : TopCat.{0}} {p q : ℕ}
  (s : (TopCat.toSSet.obj X) _[q]) :
  AffineChain (stdSimplex  $\mathbb{R}$  (Fin (q + 1))) p  $\rightarrow$ 
  (R  $\longrightarrow$  (((singularChainComplexFunctor C).obj R).obj X).X p) :=
  Finsupp.liftAddHom fun  $\sigma \Rightarrow$ 

```

```

(smulAddHom  $\mathbb{Z}$ 
  (R  $\longrightarrow$  (((singularChainComplexFunctor C).obj R).obj X).X p)).flip
  (realizeAffineSimplex R s  $\sigma$ )

@[simp]
theorem realizeAffineChain_zero
  (R : C) {X : TopCat.{0}} {p q :  $\mathbb{N}$ }
  (s : (TopCat.toSSet.obj X)  $_{[q]}$ ) :
  realizeAffineChain R s
    (0 : AffineChain (stdSimplex  $\mathbb{R}$  (Fin (q + 1))) p) = 0 := by
  simp

@[simp]
theorem realizeAffineChain_single
  (R : C) {X : TopCat.{0}} {p q :  $\mathbb{N}$ }
  (s : (TopCat.toSSet.obj X)  $_{[q]}$ )
  ( $\sigma$  : AffineSimplex (stdSimplex  $\mathbb{R}$  (Fin (q + 1))) p) (z :  $\mathbb{Z}$ ) :
  realizeAffineChain R s (Finsupp.single  $\sigma$  z) =
    z • realizeAffineSimplex R s  $\sigma$  := by
  simp [realizeAffineChain]

@[simp]
theorem realizeAffineChain_simplex
  (R : C) {X : TopCat.{0}} {p q :  $\mathbb{N}$ }
  (s : (TopCat.toSSet.obj X)  $_{[q]}$ )
  ( $\sigma$  : AffineSimplex (stdSimplex  $\mathbb{R}$  (Fin (q + 1))) p) :
  realizeAffineChain R s (simplex  $\sigma$ ) = realizeAffineSimplex R s  $\sigma$  := by
  simp [simplex]

theorem realizeAffineSimplex_map
  (R : C) {X : TopCat.{0}} {p q k :  $\mathbb{N}$ }
  (s : (TopCat.toSSet.obj X)  $_{[q]}$ )
  ( $\sigma$  : AffineSimplex (stdSimplex  $\mathbb{R}$  (Fin (q + 1))) p)
  ( $\rho$  : AffineSimplex (stdSimplex  $\mathbb{R}$  (Fin (p + 1))) k) :
  realizeAffineSimplex R s (AffineSimplex.map  $\sigma$ .realize  $\rho$ ) =
    realizeAffineSimplex R (reparametrizeSingularSimplex s  $\sigma$ )  $\rho$  := by
  simp [realizeAffineSimplex, reparametrizeSingularSimplex_map]

theorem realizeAffineChain_map
  (R : C) {X : TopCat.{0}} {p q k :  $\mathbb{N}$ }
  (s : (TopCat.toSSet.obj X)  $_{[q]}$ )
  ( $\sigma$  : AffineSimplex (stdSimplex  $\mathbb{R}$  (Fin (q + 1))) p)
  (c : AffineChain (stdSimplex  $\mathbb{R}$  (Fin (p + 1))) k) :
  realizeAffineChain R s (AffineChain.map  $\sigma$ .realize c) =
    realizeAffineChain R (reparametrizeSingularSimplex s  $\sigma$ ) c := by
  induction c using Finsupp.induction with
  | zero => simp
  | single_add  $\tau$  z c _ _ ih =>
    simp [map_add, ih, AffineChain.map_single, realizeAffineSimplex_map]

theorem realizeAffineSimplex_d
  (R : C) {X : TopCat.{0}} {p q :  $\mathbb{N}$ }
  (s : (TopCat.toSSet.obj X)  $_{[q]}$ )
  ( $\sigma$  : AffineSimplex (stdSimplex  $\mathbb{R}$  (Fin (q + 1))) (p + 1)) :
  realizeAffineSimplex R s  $\sigma \gg$ 
    (((singularChainComplexFunctor C).obj R).obj X).d (p + 1) p =
     $\sum i : \text{Fin } (p + 2),$ 
    (-1 :  $\mathbb{Z}$ )  $^{\sim (i : \mathbb{N})}$  • realizeAffineSimplex R s ( $\sigma$ .face i) := by
  simp only [realizeAffineSimplex]
  change _  $\gg$  ((TopCat.toSSet.obj X).chainComplex R).d (p + 1) p = _
  rw [SSet.ChainComplex_d]
  refine Finset.sum_congr rfl fun i _  $\mapsto$  ?_
  congr 1

```

```

rw [reparametrizeSingularSimplex_face]

theorem realizeAffineChain_simplexBoundary
  (R : C) {X : TopCat.{0}} {p q : ℕ}
  (s : (TopCat.toSSet.obj X) _[q])
  (σ : AffineSimplex (stdSimplex ℝ (Fin (q + 1))) (p + 1)) :
  realizeAffineChain R s (simplexBoundary σ) =
    ∑ i : Fin (p + 2),
      (-1 : ℤ) ^ (i : ℕ) • realizeAffineSimplex R s (σ.face i) := by
  simp [simplexBoundary, map_sum, map_zsmul]

theorem realizeAffineChain_boundary
  (R : C) {X : TopCat.{0}} {p q : ℕ}
  (s : (TopCat.toSSet.obj X) _[q])
  (c : AffineChain (stdSimplex ℝ (Fin (q + 1))) (p + 1)) :
  realizeAffineChain R s (boundary c) =
    realizeAffineChain R s c >>
      (((singularChainComplexFunctor C).obj R).obj X).d (p + 1) p := by
  induction c using Finsupp.induction with
  | zero => simp
  | single_add σ z c _ _ ih =>
    simp [map_add, ih, Preadditive.add_comp]
    rw [show Finsupp.single σ z = z • simplex σ by simp [simplex]]
    simp [map_zsmul, boundary_simplex,
      realizeAffineChain_simplexBoundary, realizeAffineSimplex_d]

theorem realizeAffineChain_subdivideStd_simplex
  (R : C) {X : TopCat.{0}} {p q : ℕ}
  (s : (TopCat.toSSet.obj X) _[q])
  (σ : AffineSimplex (stdSimplex ℝ (Fin (q + 1))) p) :
  realizeAffineChain R s (subdivideStd (q := q) p (simplex σ)) =
    realizeAffineChain R (reparametrizeSingularSimplex s σ)
      (standardSubdivision p) := by
  rw [← AffineChain.map_standardSubdivision, realizeAffineChain_map]

theorem realizeAffineChain_prismStd_simplex
  (R : C) {X : TopCat.{0}} {p q : ℕ}
  (s : (TopCat.toSSet.obj X) _[q])
  (σ : AffineSimplex (stdSimplex ℝ (Fin (q + 1))) p) :
  realizeAffineChain R s (prismStd (q := q) p (simplex σ)) =
    realizeAffineChain R (reparametrizeSingularSimplex s σ)
      (standardPrism p) := by
  rw [← AffineChain.map_standardPrism, realizeAffineChain_map]

@[simp]
theorem realizeAffineSimplex_id
  (R : C) {X : TopCat.{0}} {q : ℕ}
  (s : (TopCat.toSSet.obj X) _[q]) :
  realizeAffineSimplex R s (idSimplex q) =
    (TopCat.toSSet.obj X).ιChainComplex s := by
  simp [realizeAffineSimplex]

/-- Subdivision of one singular simplex, on its coproduct generator. -/
noncomputable def singularSubdivideGenerator
  (R : C) {X : TopCat.{0}} {n : ℕ}
  (s : (TopCat.toSSet.obj X) _[n]) :
  R → (((singularChainComplexFunctor C).obj R).obj X).X n :=
  realizeAffineChain R s (standardSubdivision n)

/-- Prism of one singular simplex, on its coproduct generator. -/
noncomputable def singularPrismGenerator
  (R : C) {X : TopCat.{0}} {n : ℕ}
  (s : (TopCat.toSSet.obj X) _[n]) :

```

```

R → (((singularChainComplexFunctor C).obj R).obj X).X (n + 1) :=
realizeAffineChain R s (standardPrism n)

theorem singularPrismGenerator_homotopy
(R : C) {X : TopCat.{0}} {n : ℕ}
(s : (TopCat.toSSet.obj X) _[n + 1]) :
singularPrismGenerator R s >>
(((singularChainComplexFunctor C).obj R).obj X).d (n + 2) (n + 1) +
realizeAffineChain R s
(prismStd (q := n + 1) n (simplexBoundary (idSimplex (n + 1)))) =
(TopCat.toSSet.obj X).ιChainComplex s -
singularSubdivideGenerator R s := by
have hprism :=
boundary_prismStd (q := n + 1) (simplex (idSimplex (n + 1)))
simp only [singularPrismGenerator, singularSubdivideGenerator,
standardPrism, standardSubdivision]
rw [← realizeAffineChain_boundary]
have hreal := congrArg (realizeAffineChain R s) hprism
simp [map_add, map_sub, boundary_simplex, realizeAffineSimplex_id] at hreal
exact hreal

/-- Subdivision of one singular one-simplex, on its coproduct generator. -/
noncomputable def singularIntervalSubdivision
(R : C) {X : TopCat.{0}}
(s : (TopCat.toSSet.obj X) _[1]) :
R → (((singularChainComplexFunctor C).obj R).obj X).X 1 :=
realizeAffineChain R s standardIntervalSubdivision

/-- The one-dimensional prism, interpreted as a singular two-chain. -/
noncomputable def singularIntervalPrism
(R : C) {X : TopCat.{0}}
(s : (TopCat.toSSet.obj X) _[1]) :
R → (((singularChainComplexFunctor C).obj R).obj X).X 2 :=
realizeAffineChain R s standardIntervalPrism

/-- Degreewise barycentric subdivision of singular chains. -/
noncomputable def singularSubdivisionHom
(R : C) (X : TopCat.{0}) (n : ℕ) :
(((singularChainComplexFunctor C).obj R).obj X).X n →
(((singularChainComplexFunctor C).obj R).obj X).X n :=
((TopCat.toSSet.obj X).isColimitChainComplexXCofan R n).desc
(Cofan.mk _ fun s => singularSubdivideGenerator R s)

@[reassoc (attr := simp)]
theorem ι_singularSubdivisionHom
(R : C) (X : TopCat.{0}) {n : ℕ}
(s : (TopCat.toSSet.obj X) _[n]) :
(TopCat.toSSet.obj X).ιChainComplex s >> singularSubdivisionHom R X n =
singularSubdivideGenerator R s :=
((TopCat.toSSet.obj X).isColimitChainComplexXCofan R n).fac _ ⟨s⟩

/-- Degreewise prism operator on singular chains. -/
noncomputable def singularPrismHom
(R : C) (X : TopCat.{0}) (n : ℕ) :
(((singularChainComplexFunctor C).obj R).obj X).X n →
(((singularChainComplexFunctor C).obj R).obj X).X (n + 1) :=
((TopCat.toSSet.obj X).isColimitChainComplexXCofan R n).desc
(Cofan.mk _ fun s => singularPrismGenerator R s)

@[reassoc (attr := simp)]
theorem ι_singularPrismHom
(R : C) (X : TopCat.{0}) {n : ℕ}
(s : (TopCat.toSSet.obj X) _[n]) :

```

```

    (TopCat.toSSet.obj X).ιChainComplex s >> singularPrismHom R X n =
      singularPrismGenerator R s :=
    ((TopCat.toSSet.obj X).isColimitChainComplexXCofan R n).fac _ ⟨s⟩

theorem singularSubdivideGenerator_d
  (R : C) {X : TopCat.{0}} {n : ℕ}
  (s : (TopCat.toSSet.obj X) _[n + 1]) :
  singularSubdivideGenerator R s >>
    (((singularChainComplexFunctor C).obj R).obj X).d (n + 1) n =
    ∑ i : Fin (n + 2),
      (-1 : ℤ) ^ (i : ℕ) •
        singularSubdivideGenerator R ((TopCat.toSSet.obj X).δ i s) := by
have hbound :
  boundary (standardSubdivision (n + 1)) =
    subdivideStd (q := n + 1) n
      (simplexBoundary (idSimplex (n + 1))) := by
simp [standardSubdivision, boundary_simplex] using
  boundary_subdivideStd (q := n + 1) (simplex (idSimplex (n + 1)))
simp only [singularSubdivideGenerator]
rw [← realizeAffineChain_boundary, hbound]
simp only [simplexBoundary, map_sum, map_zsmul]
refine Finset.sum_congr rfl fun i _ => ?_
rw [realizeAffineChain_subdivideStd_simplex, reparametrizeSingularSimplex_id_face]

/-- Barycentric subdivision of singular chains, as a chain map. -/
noncomputable def singularSubdivision
  (R : C) (X : TopCat.{0}) :
  (((singularChainComplexFunctor C).obj R).obj X →
    ((singularChainComplexFunctor C).obj R).obj X where
f := singularSubdivisionHom R X
comm' i j hij := by
  rw [ComplexShape.down_Rel] at hij
  subst hij
  apply (TopCat.toSSet.obj X).chainComplex_hom_ext
  intro s
  calc
    (TopCat.toSSet.obj X).ιChainComplex s >>
      singularSubdivisionHom R X (j + 1) >>
        (((singularChainComplexFunctor C).obj R).obj X).d (j + 1) j =
        singularSubdivideGenerator R s >>
          (((singularChainComplexFunctor C).obj R).obj X).d (j + 1) j := by
  rw [← Category.assoc, ι_singularSubdivisionHom]
  _ = ∑ i : Fin (j + 2),
    (-1 : ℤ) ^ (i : ℕ) •
      singularSubdivideGenerator R ((TopCat.toSSet.obj X).δ i s) :=
    singularSubdivideGenerator_d R s
  _ = (TopCat.toSSet.obj X).ιChainComplex s >>
    (((singularChainComplexFunctor C).obj R).obj X).d (j + 1) j >>
      singularSubdivisionHom R X j := by
have hd :
  (TopCat.toSSet.obj X).ιChainComplex s >>
    (((singularChainComplexFunctor C).obj R).obj X).d (j + 1) j =
    ∑ i : Fin (j + 2),
      (-1 : ℤ) ^ (i : ℕ) •
        (TopCat.toSSet.obj X).ιChainComplex
          ((TopCat.toSSet.obj X).δ i s) := by
  change _ >> ((TopCat.toSSet.obj X).chainComplex R).d (j + 1) j = _
  rw [SSet.ιChainComplex_d]
  rw [← Category.assoc, hd]
  refine Eq.trans ?_
  (Preadditive.sum_comp (s := Finset.univ)
    (fun i : Fin (j + 2) =>
      (-1 : ℤ) ^ (i : ℕ) •

```

```

      (TopCat.toSSet.obj X).ιChainComplex
      ((TopCat.toSSet.obj X).δ i s))
      (singularSubdivisionHom R X j)).symm
    refine Finset.sum_congr rfl fun i _ => ?_
    rw [← ι_singularSubdivisionHom, ← Preadditive.zsmul_comp]
    rfl

lemma singularSubdivisionHom_zero
  (R : C) (X : TopCat.{0}) :
    singularSubdivisionHom R X 0 = 0 := by
  apply (TopCat.toSSet.obj X).chainComplex_hom_ext
  intro s
  rw [ι_singularSubdivisionHom]
  simp [singularSubdivideGenerator, standardSubdivision, subdivideStd,
    realizeAffineSimplex_id]
  exact (Category.comp_id _).symm

lemma singularPrismHom_zero
  (R : C) (X : TopCat.{0}) :
    singularPrismHom R X 0 = 0 := by
  apply (TopCat.toSSet.obj X).chainComplex_hom_ext
  intro s
  rw [ι_singularPrismHom]
  simp [singularPrismGenerator, standardPrism, prismStd]
  exact (Limits.comp_zero (f := (TopCat.toSSet.obj X).ιChainComplex s)).symm

theorem singularPrismGenerator_d
  (R : C) {X : TopCat.{0}} {n : ℕ}
  (s : (TopCat.toSSet.obj X) _[n + 1]) :
    realizeAffineChain R s
    (prismStd (q := n + 1) n (simplexBoundary (idSimplex (n + 1)))) =
    (TopCat.toSSet.obj X).ιChainComplex s >>
    (((singularChainComplexFunctor C).obj R).obj X).d (n + 1) n >>
    singularPrismHom R X n := by
  have hd :
    (TopCat.toSSet.obj X).ιChainComplex s >>
    (((singularChainComplexFunctor C).obj R).obj X).d (n + 1) n =
    ∑ i : Fin (n + 2),
    (-1 : ℤ) ^ (i : ℕ) •
    (TopCat.toSSet.obj X).ιChainComplex
    ((TopCat.toSSet.obj X).δ i s) := by
  change _ >> ((TopCat.toSSet.obj X).chainComplex R).d (n + 1) n = _
  rw [SSet.ιChainComplex_d]
  simp only [simplexBoundary, map_sum, map_zsmul]
  rw [← Category.assoc, hd]
  refine Eq.trans ?_
  (Preadditive.sum_comp (s := Finset.univ)
    (fun i : Fin (n + 2) =>
      (-1 : ℤ) ^ (i : ℕ) •
      (TopCat.toSSet.obj X).ιChainComplex
      ((TopCat.toSSet.obj X).δ i s))
    (singularPrismHom R X n)).symm
  refine Finset.sum_congr rfl fun i _ => ?_
  have hpiece :
    realizeAffineChain R s
    (prismStd (q := n + 1) n (simplex ((idSimplex (n + 1)).face i))) =
    (TopCat.toSSet.obj X).ιChainComplex
    ((TopCat.toSSet.obj X).δ i s) >>
    singularPrismHom R X n := by
  rw [realizeAffineChain_prismStd_simplex, reparametrizeSingularSimplex_id_face]
  exact (ι_singularPrismHom R X _).symm
  rw [hpiece]
  exact (Preadditive.zsmul_comp

```

```

((TopCat.toSSet.obj X).ιChainComplex
  ((TopCat.toSSet.obj X).δ i s))
(singularPrismHom R X n)
((-1 : ℤ) ^ (i : ℕ)).symm

/-- The prism operator is a chain homotopy from the identity to
barycentric subdivision. -/
noncomputable def singularPrismHomotopy
  (R : C) (X : TopCat.{0}) :
    Homotopy
      (⊥ (((singularChainComplexFunctor C).obj R).obj X))
        (singularSubdivision R X) where
hom i j :=
  if h : i + 1 = j then
    singularPrismHom R X i >> eqToHom (by rw [h])
  else
    0
zero i j hij := by
  refine dif_neg ?_
  intro h
  exact hij (by rw [ComplexShape.down_Rel, h])
comm i := by
  rw [Homotopy.prevD_chainComplex]
  cases i with
  | zero =>
    rw [Homotopy.dNext_zero_chainComplex, zero_add]
    simp [dif_pos, singularPrismHom_zero]
    exact (singularSubdivisionHom_zero R X).symm
  | succ n =>
    rw [Homotopy.dNext_succ_chainComplex]
    apply (TopCat.toSSet.obj X).chainComplex_hom_ext
    intro s
    simp only [dif_pos, eqToHom_refl]
    have h := singularPrismGenerator_homotopy R s
    have hd := singularPrismGenerator_d R s
    change
      (TopCat.toSSet.obj X).ιChainComplex s >>
        ⊥ (((TopCat.toSSet.obj X).chainComplex R).X (n + 1)) =
      -
    simp [Category.comp_id, eqToHom_refl]
    have hadd :=
      (Preadditive.comp_add _ _ _
        ((TopCat.toSSet.obj X).ιChainComplex s)
        (((singularChainComplexFunctor C).obj R).obj X).d
          (n + 1) n >>
          singularPrismHom R X n +
          singularPrismHom R X (n + 1) >>
          (((singularChainComplexFunctor C).obj R).obj X).d
            (n + 2) (n + 1))
        ((singularSubdivision R X).f (n + 1))).trans
      (congrArg
        (fun t => t +
          (TopCat.toSSet.obj X).ιChainComplex s >>
          (singularSubdivision R X).f (n + 1))
        (Preadditive.comp_add _ _ _
          ((TopCat.toSSet.obj X).ιChainComplex s)
          (((singularChainComplexFunctor C).obj R).obj X).d
            (n + 1) n >>
            singularPrismHom R X n)
          (singularPrismHom R X (n + 1) >>
            (((singularChainComplexFunctor C).obj R).obj X).d
              (n + 2) (n + 1))))
    erw [hadd]

```

```

rw [← Category.assoc
  ((TopCat.toSSet.obj X).ιChainComplex s)
  (singularPrismHom R X (n + 1)),
  ι_singularPrismHom]
change _ =
  (TopCat.toSSet.obj X).ιChainComplex s >>
    (((singularChainComplexFunctor C).obj R).obj X).d
      (n + 1) n >>
        singularPrismHom R X n +
        singularPrismGenerator R s >>
          ((singularChainComplexFunctor C).obj R).obj X).d
            (n + 2) (n + 1) +
            (TopCat.toSSet.obj X).ιChainComplex s >>
              singularSubdivisionHom R X (n + 1)
rw [ι_singularSubdivisionHom, ← hd,
  add_comm (realizeAffineChain R s _), h]
exact (sub_add_cancel
  ((TopCat.toSSet.obj X).ιChainComplex s)
  (singularSubdivideGenerator R s)).symm

/-- Iterated barycentric subdivision, as a chain map. -/
noncomputable def singularSubdivisionIterate
  (R : C) (X : TopCat.{0}) (N : ℕ) :
  ((singularChainComplexFunctor C).obj R).obj X →
  ((singularChainComplexFunctor C).obj R).obj X :=
  ((fun f => f >> singularSubdivision R X)^[N]) (|< _ )

lemma singularSubdivisionIterate_zero
  (R : C) (X : TopCat.{0}) :
  singularSubdivisionIterate R X 0 = |< _ :=
  rfl

lemma singularSubdivisionIterate_succ
  (R : C) (X : TopCat.{0}) (n : ℕ) :
  singularSubdivisionIterate R X (n + 1) =
  singularSubdivisionIterate R X n >> singularSubdivision R X :=
  Function.iterate_succ_apply' _ n _

lemma singularSubdivisionIterate_f_succ
  (R : C) (X : TopCat.{0}) (N n : ℕ) :
  (singularSubdivisionIterate R X (N + 1)).f n =
  (singularSubdivisionIterate R X N).f n >>
  singularSubdivisionHom R X n := by
  rw [singularSubdivisionIterate_succ]
  rfl

end

end PlatonicSolids.SingularExcision

```

A.14 PlatonicSolids/SingularExcision/ModuleSupport.lean

Finite coproduct support in ModuleCat (46 lines).

```

/-
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-/
import Mathlib.Algebra.Category.ModuleCat.Products
import Mathlib.Algebra.Category.ModuleCat.EpiMono
import Mathlib.Algebra.DirectSum.Module

/-!

```

Finite-support access for categorical coproducts in `ModuleCat`.

This is why the elementwise small-simplices proof is specialized to module coefficients: every element of the categorical coproduct is, through `ModuleCat.coprodIsoDirectSum`, a genuinely finite direct sum.

-/

open CategoryTheory Limits

universe u v w

namespace ModuleCat

variable {R : Type u} [Ring R] {ι : Type v}

variable (Z : ι → ModuleCat.{max v w} R) [DecidableEq ι] [HasCoproduct Z]

/-- The finite set of summands on which an element of a categorical coproduct of modules is nonzero. -/

noncomputable def coprodSupport

(x : Limits.sigmaObj (C := ModuleCat.{max v w} R) Z) : Finset ι :=

by

classical

exact (coprodIsoDirectSum Z).hom x |>.support

/-- Every element of a categorical coproduct of modules is the finite sum of its components included into the coproduct. -/

lemma eq_sum_ι_apply_coprodSupport

(x : Limits.sigmaObj (C := ModuleCat.{max v w} R) Z) :

x = $\sum i \in \text{coprodSupport } Z \ x,$

Sigma.ι Z i ((coprodIsoDirectSum Z).hom x i) := by

classical

apply_fun (coprodIsoDirectSum Z).hom using

(ModuleCat.mono_iff_injective (coprodIsoDirectSum Z).hom).mp inferInstance

simp only [map_sum, ι_coprodIsoDirectSum_hom_apply]

exact (DirectSum.sum_support_of ((coprodIsoDirectSum Z).hom x)).symm

end ModuleCat

A.15 PlatonicSolids/SingularExcision/Subcomplexes.lean

Lands-in and small singular subcomplexes (174 lines).

/-

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-/

import Mathlib.AlgebraicTopology.SimplicialSet.SubcomplexColimits

import Mathlib.AlgebraicTopology.SingularHomology.Basic

import PlatonicSolids.RelativeHomology

/-!

Singular subcomplexes subordinate to a pair of subsets

The singular simplices which land in a subset form the range subcomplex of the singular set of the corresponding subtype. The simplices small with respect to a two-set cover form the supremum of the two ranges.

-/

open AlgebraicTopology CategoryTheory Limits Set

open scoped Topology

namespace SingularExcision

```

variable {Y : Type} [TopologicalSpace Y]

/-- The map of singular sets induced by the inclusion of a subtype. -/
noncomputable def subtypeSingularSetMap (A : Set Y) :
  TopCat.toSSet.obj (.of ↑A) → TopCat.toSSet.obj (.of Y) :=
  TopCat.toSSet.map (subtypeIncl Y A)

/-- The singular subcomplex consisting of simplices which factor through `A`. -/
noncomputable abbrev landsIn (A : Set Y) :
  (TopCat.toSSet.obj (.of Y)).Subcomplex :=
  SSet.Subcomplex.range (subtypeSingularSetMap A)

/-- A simplex is small for the pair `(A, B)` if it lands in one member. -/
noncomputable abbrev small (A B : Set Y) :
  (TopCat.toSSet.obj (.of Y)).Subcomplex :=
  landsIn A ⊔ landsIn B

/-- The singular set of a subtype maps onto the corresponding range subcomplex. -/
noncomputable def subtypeToLandsIn (A : Set Y) :
  TopCat.toSSet.obj (.of ↑A) → (landsIn A : SSet) :=
  SSet.Subcomplex.toRange (subtypeSingularSetMap A)

instance subtypeSingularSetMap_mono (A : Set Y) :
  Mono (subtypeSingularSetMap A) := by
  exact TopCat.toSSet.map_mono (subtypeIncl Y A)

instance subtypeToLandsIn_isIso (A : Set Y) :
  IsIso (subtypeToLandsIn A) := by
  change IsIso (SSet.Subcomplex.toRange (subtypeSingularSetMap A))
  infer_instance

/-- The singular set of `A` is canonically isomorphic to the subcomplex
of ambient singular simplices which land in `A`. -/
noncomputable def subtypeSingularSetIso (A : Set Y) :
  TopCat.toSSet.obj (.of ↑A) ≅ (landsIn A : SSet) :=
  asIso (subtypeToLandsIn A)

@[simp, reassoc]
lemma subtypeToLandsIn_ℓ (A : Set Y) :
  subtypeToLandsIn A ≫ (landsIn A).ℓ = subtypeSingularSetMap A :=
  SSet.Subcomplex.toRange_ℓ _

/-- Under the concrete description of singular simplices, the singular-set
map of a subtype is postcomposition with `Subtype.val`. -/
lemma subtypeSingularSetMap_toSSetObjEquiv (A : Set Y)
  (n : SimplexCategoryop)
  (σ : (TopCat.toSSet.obj (.of ↑A)).obj n) :
  (TopCat.of Y).toSSetObjEquiv n
    ((subtypeSingularSetMap A).app n σ) =
  (subtypeIncl Y A).hom.comp
    ((TopCat.of ↑A).toSSetObjEquiv n σ) :=
  rfl

/-- The two evident maps out of an intersection subtype. -/
def interToLeft (A B : Set Y) : TopCat.of ↑(A ∩ B) → TopCat.of ↑A :=
  TopCat.ofHom (ContinuousMap.inclusion fun _ h => h.1)

def interToRight (A B : Set Y) : TopCat.of ↑(A ∩ B) → TopCat.of ↑B :=
  TopCat.ofHom (ContinuousMap.inclusion fun _ h => h.2)

lemma interToLeft_comp_subtypeIncl (A B : Set Y) :
  interToLeft A B ≫ subtypeIncl Y A = subtypeIncl Y (A ∩ B) := by
  rfl

```

```

lemma interToRight_comp_subtypeIncl (A B : Set Y) :
  interToRight A B  $\gg$  subtypeIncl Y B = subtypeIncl Y (A  $\cap$  B) := by
  rfl

/-- Factoring through the intersection is equivalent to factoring through
both subsets. -/
lemma landsIn_inter (A B : Set Y) :
  landsIn (A  $\cap$  B) = landsIn A  $\sqcap$  landsIn B := by
  ext n  $\sigma$ 
  constructor
  · rintro  $\langle \tau, \text{rfl} \rangle$ 
  constructor
  · refine  $\langle (\text{TopCat.toSSet.map (interToLeft A B)).app n } \tau, ?_ \rangle$ 
    simp [subtypeSingularSetMap,  $\leftarrow$  Functor.map_comp] using
    congrArg (fun q  $\Rightarrow$  q.app n  $\tau$ )
    (congrArg TopCat.toSSet.map (interToLeft_comp_subtypeIncl A B))
  · refine  $\langle (\text{TopCat.toSSet.map (interToRight A B)).app n } \tau, ?_ \rangle$ 
    simp [subtypeSingularSetMap,  $\leftarrow$  Functor.map_comp] using
    congrArg (fun q  $\Rightarrow$  q.app n  $\tau$ )
    (congrArg TopCat.toSSet.map (interToRight_comp_subtypeIncl A B))
  · rintro  $\langle \langle \tau A, hA \rangle, \langle \tau B, hB \rangle \rangle$ 
    let fA := (TopCat.of  $\uparrow A$ ).toSSetObjEquiv n  $\tau A$ 
    let fB := (TopCat.of  $\uparrow B$ ).toSSetObjEquiv n  $\tau B$ 
    have hval :  $\forall x, (fA x : Y) = fB x$  := by
      intro x
      have h := congrArg
        (fun q  $\Rightarrow$  (TopCat.of Y).toSSetObjEquiv n q x)
        (hA.trans hB.symm)
      simp [fA, fB, subtypeSingularSetMap_toSSetObjEquiv] using h
    let fI : C(TopCat.of (stdSimplex  $\mathbb{R}$  (Fin (n.unop.len + 1))),
      TopCat.of  $\uparrow (A \cap B)$ ) :=
       $\langle$ fun x  $\Rightarrow$   $\langle fA x, (fA x).property$ , by
        rw [hval x]
        exact (fB x).property $\rangle$ ,
      (continuous_subtype_val.comp fA.continuous).subtype_mk _ $\rangle$ 
    refine  $\langle ((\text{TopCat.of } \uparrow (A \cap B)).\text{toSSetObjEquiv n}).\text{symm fI}, ?_ \rangle$ 
    apply (TopCat.of Y).toSSetObjEquiv n |>.injective
  ext x
  simp [fI, fA, subtypeSingularSetMap_toSSetObjEquiv] using
  congrArg (fun q  $\Rightarrow$  (TopCat.of Y).toSSetObjEquiv n q x) hA

/-- The corresponding simplicial sets are canonically isomorphic. -/
noncomputable def landsInInterIso (A B : Set Y) :
  (landsIn (A  $\cap$  B) : SSet)  $\cong$ 
  ((landsIn A  $\sqcap$  landsIn B :
    (TopCat.toSSet.obj (.of Y)).Subcomplex) : SSet) :=
  SSet.Subcomplex.eqToIso (landsIn_inter A B)

@[simp, reassoc]
lemma landsInInterIso_hom_ $\iota$  (A B : Set Y) :
  (landsInInterIso A B).hom  $\gg$  (landsIn A  $\sqcap$  landsIn B). $\iota$  =
  (landsIn (A  $\cap$  B)). $\iota$  :=
  SSet.Subcomplex.homOfLE_ $\iota$  (landsIn_inter A B).le

/-- The two lands-in subcomplexes and their infimum/supremum form a
pushout square of simplicial sets. -/
noncomputable def smallIsPushout (A B : Set Y) :
  IsPushout
  (SSet.Subcomplex.homOfLE (inf_le_left :
    landsIn A  $\sqcap$  landsIn B  $\leq$  landsIn A))
  (SSet.Subcomplex.homOfLE (inf_le_right :
    landsIn A  $\sqcap$  landsIn B  $\leq$  landsIn B))

```

```

      (SSet.Subcomplex.homOfLE (le_sup_left :
        landsIn A ≤ small A B))
      (SSet.Subcomplex.homOfLE (le_sup_right :
        landsIn B ≤ small A B)) :=
(show SSet.Subcomplex.BicartSq
  (landsIn A ⊔ landsIn B) (landsIn A) (landsIn B) (small A B) from
  { sup_eq := rfl
    inf_eq := rfl }).isPushout

/-- Degreewise membership in `landsIn A` is exactly factorization through
the singular set of the subtype. -/
lemma mem_landsIn_iff {A : Set Y} {n : SimplexCategoryop}
  (σ : (TopCat.toSSet.obj (.of Y)).obj n) :
  σ ∈ (landsIn A).obj n ↔
  ∃ τ : (TopCat.toSSet.obj (.of ↑A)).obj n,
    (subtypeSingularSetMap A).app n τ = σ :=
Iff.rfl

/-- Degreewise membership in the small subcomplex has the expected
disjunctive description. -/
lemma mem_small_iff {A B : Set Y} {n : SimplexCategoryop}
  (σ : (TopCat.toSSet.obj (.of Y)).obj n) :
  σ ∈ (small A B).obj n ↔
  σ ∈ (landsIn A).obj n ∨ σ ∈ (landsIn B).obj n :=
Iff.rfl

end SingularExcision

```

A.16 PlatonicSolids/SingularExcision/DegreewisePushout.lean

Degreewise pushout of small chains (109 lines).

```

/-
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-/
import Mathlib.CategoryTheory.Limits.Preserves.SigmaConst
import PlatonicSolids.SingularExcision.Subcomplexes

/-!
# Degreewise pushouts for small singular chains

The lattice pushout of the two lands-in subcomplexes remains a pushout
after evaluation in each simplicial degree and after applying the free
`R`-object functor `sigmaConst.obj R`.
-/

open AlgebraicTopology CategoryTheory Limits
open scoped Simplicial

namespace SingularExcision

variable {Y : Type} [TopologicalSpace Y]
variable {C : Type*} [Category C] [Preadditive C] [HasCoproducts.{0} C]

/-- The degreewise pushout square of sets of small singular simplices. -/
noncomputable def smallSimplexSetIsPushout (A B : Set Y)
  (n : SimplexCategoryop) :
  IsPushout
  ((SSet.Subcomplex.homOfLE (inf_le_left :
    landsIn A ⊔ landsIn B ≤ landsIn A)).app n)
  ((SSet.Subcomplex.homOfLE (inf_le_right :
    landsIn A ⊔ landsIn B ≤ landsIn B)).app n)

```

```

    ((SSet.Subcomplex.homOfLE (le_sup_left :
      landsIn A ≤ small A B)).app n)
    ((SSet.Subcomplex.homOfLE (le_sup_right :
      landsIn B ≤ small A B)).app n) :=
  (smallIsPushout A B).map ((evaluation SimplexCategoryop (Type)).obj n)

/-- Applying `sigmaConst.obj R` to the degreewise set pushout gives a
pushout in any coefficient category with the required coproducts. -/
noncomputable def smallSigmaConstIsPushout (R : C) (A B : Set Y)
  (n : SimplexCategoryop) :
  IsPushout
    ((sigmaConst.obj R).map
      ((SSet.Subcomplex.homOfLE (inf_le_left :
        landsIn A ⊔ landsIn B ≤ landsIn A)).app n))
    ((sigmaConst.obj R).map
      ((SSet.Subcomplex.homOfLE (inf_le_right :
        landsIn A ⊔ landsIn B ≤ landsIn B)).app n))
    ((sigmaConst.obj R).map
      ((SSet.Subcomplex.homOfLE (le_sup_left :
        landsIn A ≤ small A B)).app n))
    ((sigmaConst.obj R).map
      ((SSet.Subcomplex.homOfLE (le_sup_right :
        landsIn B ≤ small A B)).app n)) :=
  (smallSimplexSetIsPushout A B n).map (sigmaConst.obj R)

/-- The chain complex of the singular set of a subtype is isomorphic to
the chain complex on the corresponding lands-in subcomplex. -/
noncomputable def subtypeChainComplexIso (R : C) (A : Set Y) :
  (TopCat.toSSet.obj (.of ↑A)).chainComplex R ≅
  (landsIn A : SSet).chainComplex R :=
  ((SSet.chainComplexFunctor C).obj R).mapIso (subtypeSingularSetIso A)

/-- In each degree, the maps between chain objects of the two lands-in
subcomplexes and their supremum form a pushout square. -/
noncomputable def smallChainDegreeIsPushout (R : C) (A B : Set Y) (n : ℕ) :
  IsPushout
    ((SSet.chainComplexMap
      (SSet.Subcomplex.homOfLE (inf_le_left :
        landsIn A ⊔ landsIn B ≤ landsIn A)) R).f n)
    ((SSet.chainComplexMap
      (SSet.Subcomplex.homOfLE (inf_le_right :
        landsIn A ⊔ landsIn B ≤ landsIn B)) R).f n)
    ((SSet.chainComplexMap
      (SSet.Subcomplex.homOfLE (le_sup_left :
        landsIn A ≤ small A B)) R).f n)
    ((SSet.chainComplexMap
      (SSet.Subcomplex.homOfLE (le_sup_right :
        landsIn B ≤ small A B)) R).f n) := by
  simpa [SSet.chainComplexMap, SSet.chainComplexFunctor,
    AlgebraicTopology.alternatingFaceMapComplex] using
    smallSigmaConstIsPushout R A B (Opposite.op [n])

/-- The degreewise pushout assembles to a pushout of chain complexes. -/
noncomputable def smallChainIsPushout (R : C) (A B : Set Y) :
  IsPushout
    (SSet.chainComplexMap
      (SSet.Subcomplex.homOfLE (inf_le_left :
        landsIn A ⊔ landsIn B ≤ landsIn A)) R)
    (SSet.chainComplexMap
      (SSet.Subcomplex.homOfLE (inf_le_right :
        landsIn A ⊔ landsIn B ≤ landsIn B)) R)
    (SSet.chainComplexMap
      (SSet.Subcomplex.homOfLE (le_sup_left :

```

```

        landsIn A ≤ small A B)) R)
(SSet.chainComplexMap
  (SSet.Subcomplex.homOfLE (le_sup_right :
    landsIn B ≤ small A B)) R) := by
refine
  { w := by
    rw [← Functor.map_comp, ← Functor.map_comp]
    rfl
    isColimit' := ⟨?_⟩ }
apply HomologicalComplex.isColimitOfEval
intro n
exact (PushoutCocone.isColimitMapCoconeEquiv _ _).symm
  (smallChainDegreeIsPushout R A B n).isColimit

end SingularExcision

```

A.17 PlatonicSolids/SingularExcision/CokernelPushout.lean

Cokernel model of a preadditive pushout (111 lines).

```

/-
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-/
import Mathlib.CategoryTheory.Limits.Shapes.Pullback.IsPullback.Basic
import Mathlib.CategoryTheory.Limits.Shapes.Kernels
import PlatonicSolids.SingularExcision.DegreewisePushout

/-!
# A cokernel model for a preadditive pushout

The pushout of  $f : I \rightarrow A$  and  $g : I \rightarrow B$  is the cokernel of
 $(f, -g) : I \rightarrow A \boxplus B$ .
-/

open CategoryTheory Limits

universe v u

namespace SingularExcision

variable {C : Type u} [Category.{v} C] [Preadditive C]
  [HasBinaryBiprod C] [HasCokernels C]
variable {I A B P : C} (f : I → A) (g : I → B)

/-- The relation map  $(f, -g)$  presenting a pushout as a cokernel. -/
noncomputable def pushoutRelation : I → A ⊕ B :=
  biprod.lift f (-g)

omit [HasCokernels C] in
lemma pushoutRelation_eq :
  pushoutRelation f g = f >> biprod.inl + (-g) >> biprod.inr := by
  ext
  · simp [pushoutRelation, Preadditive.add_comp]
  · simp [pushoutRelation, Preadditive.add_comp]

/-- The two canonical maps into the cokernel of  $(f, -g)$ . -/
noncomputable def cokernelPushoutInl :
  A → cokernel (pushoutRelation f g) :=
  biprod.inl >> cokernel.π (pushoutRelation f g)

noncomputable def cokernelPushoutInr :
  B → cokernel (pushoutRelation f g) :=

```

```

biprod.inr >> cokernel.π (pushoutRelation f g)

lemma cokernelPushout_comm :
  f >> cokernelPushoutInl f g = g >> cokernelPushoutInr f g := by
  have h := cokernel.condition (pushoutRelation f g)
  rw [pushoutRelation_eq, Preadditive.add_comp] at h
  rw [← pushoutRelation_eq] at h
  apply sub_eq_zero.mp
  simpa only [cokernelPushoutInl, cokernelPushoutInr, Category.assoc,
    Preadditive.neg_comp, sub_eq_add_neg] using h

/-- The cokernel of `(f, -g)` satisfies the pushout universal property. -/
noncomputable def cokernelIsPushout :
  IsPushout f g (cokernelPushoutInl f g) (cokernelPushoutInr f g) := by
  refine
    { w := cokernelPushout_comm f g
      isColimit' := (PushoutCocone.IsColimit.mk _ ?_ ?_ ?_ ?_) }
  · intro s
    refine cokernel.desc (pushoutRelation f g) (biprod.desc s.inl s.inr) ?_
    rw [pushoutRelation_eq, Preadditive.add_comp]
    simp only [Category.assoc, biprod.inl_desc, biprod.inr_desc]
    simpa only [Preadditive.neg_comp, sub_eq_add_neg] using
      sub_eq_zero.mpr s.condition
  · intro s
    simp [cokernelPushoutInl]
  · intro s
    simp [cokernelPushoutInr]
  · intro s m hmA hmB
    apply (cancel_epi (cokernel.π (pushoutRelation f g))).1
    rw [cokernel.π_desc]
  ext
  · simpa [cokernelPushoutInl, Category.assoc] using hmA
  · simpa [cokernelPushoutInr, Category.assoc] using hmB

/-- Any chosen pushout is canonically isomorphic to the cokernel
presentation. -/
noncomputable def pushoutIsoCokernel {h : A → P} {k : B → P}
  (sq : IsPushout f g h k) :
  P ≅ cokernel (pushoutRelation f g) :=
  sq.isoIsPushout _ _ (cokernelIsPushout f g)

section SmallChains

open AlgebraicTopology
open scoped Simplicial

variable {Y : Type} [TopologicalSpace Y]
variable {D : Type*} [Category D] [Abelian D] [HasCoproducts.{0} D]

/-- In every degree, the small-chain object is the cokernel of the
signed map from the infimum of the two lands-in subcomplexes. -/
noncomputable def smallChainDegreeIsoCokernel (R : D)
  (A B : Set Y) (n : ℕ) :
  ((small A B : SSet).chainComplex R).X n ≅
  cokernel
    (pushoutRelation
      ((SSet.chainComplexMap
        (SSet.Subcomplex.homOfLE (inf_le_left :
          landsIn A ⊔ landsIn B ≤ landsIn A)) R).f n)
      ((SSet.chainComplexMap
        (SSet.Subcomplex.homOfLE (inf_le_right :
          landsIn A ⊔ landsIn B ≤ landsIn B)) R).f n)) :=
  pushoutIsoCokernel _ _

```

```

    (smallChainDegreeIsPushout R A B n)

end SmallChains

end SingularExcision

```

A.18 PlatonicSolids/SingularExcision/CoverChains.lean

Cover chains isomorphic to small singular chains (133 lines).

```

/-
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-/
import PlatonicSolids.MayerVietoris
import PlatonicSolids.SingularExcision.CokernelPushout

/-!
# Cover chains as small singular chains

The singular sets of the three subtypes in a two-set cover are transported
to the two lands-in subcomplexes and their infimum. The resulting pushout
identifies the existing `coverChains` cokernel with the chain complex on
small singular simplices.
-/

open AlgebraicTopology CategoryTheory Limits HomologicalComplex
open scoped Topology

universe v u

namespace SingularExcision

variable {Y : Type} [TopologicalSpace Y]

/-- The singular set of the intersection, expressed as the infimum of the
two ambient lands-in subcomplexes. -/
noncomputable def intersectionSingularSetIso (A B : Set Y) :
  TopCat.toSSet.obj (.of  $\uparrow(A \cap B)$ )  $\cong$ 
  ((landsIn A  $\sqcap$  landsIn B :
    (TopCat.toSSet.obj (.of Y)).Subcomplex) : SSet) :=
  subtypeSingularSetIso (A  $\cap$  B)  $\ll$  landsInInterIso A B

lemma intersectionSingularSetIso_hom_comp_left (A B : Set Y) :
  (intersectionSingularSetIso A B).hom  $\gg$ 
  SSet.Subcomplex.homOfLE (inf_le_left :
    landsIn A  $\sqcap$  landsIn B  $\leq$  landsIn A) =
  TopCat.toSSet.map (interInclLeft A B)  $\gg$ 
  (subtypeSingularSetIso A).hom := by
  rw [← cancel_mono (landsIn A). $\iota$ ]
  simp [intersectionSingularSetIso, subtypeSingularSetIso,
    subtypeSingularSetMap, interInclLeft,
    ← Functor.map_comp]
  simpa [interInclLeft, interToLeft] using
    congrArg TopCat.toSSet.map (interToLeft_comp_subtypeIncl A B).symm

lemma intersectionSingularSetIso_hom_comp_right (A B : Set Y) :
  (intersectionSingularSetIso A B).hom  $\gg$ 
  SSet.Subcomplex.homOfLE (inf_le_right :
    landsIn A  $\sqcap$  landsIn B  $\leq$  landsIn B) =
  TopCat.toSSet.map (interInclRight A B)  $\gg$ 
  (subtypeSingularSetIso B).hom := by
  rw [← cancel_mono (landsIn B). $\iota$ ]

```

```

simp [intersectionSingularSetIso, subtypeSingularSetIso,
      subtypeSingularSetMap, interInclRight,
      ← Functor.map_comp]
simpa [interInclRight, interToRight] using
      congrArg TopCat.toSSet.map (interToRight_comp_subtypeIncl A B).symm

variable {C : Type u} [Category.{v} C] [Abelian C]
[HasCoproducts.{0} C] [HasPullbacks C]

/-- Chain-level form of `intersectionSingularSetIso`. -/
noncomputable def intersectionChainComplexIso (R : C) (A B : Set Y) :
  ((singularChainComplexFunctor C).obj R).obj (.of ↑(A ∩ B)) ≅
  ((landsIn A ⊔ landsIn B :
    (TopCat.toSSet.obj (.of Y)).Subcomplex) : SSet).chainComplex R :=
  ((SSet.chainComplexFunctor C).obj R).mapIso
  (intersectionSingularSetIso A B)

omit [HasPullbacks C] in
lemma intersectionChainComplexIso_hom_comp_left (R : C) (A B : Set Y) :
  (intersectionChainComplexIso R A B).hom >>
  SSet.chainComplexMap
  (SSet.Subcomplex.homOfLE (inf_le_left :
    landsIn A ⊔ landsIn B ≤ landsIn A)) R =
  ((singularChainComplexFunctor C).obj R).map (interInclLeft A B) >>
  (subtypeChainComplexIso R A).hom := by
simpa [intersectionChainComplexIso, subtypeChainComplexIso,
      singularChainComplexFunctor, SSet.chainComplexMap, Functor.map_comp] using
      congrArg (fun q => ((SSet.chainComplexFunctor C).obj R).map q)
      (intersectionSingularSetIso_hom_comp_left A B)

omit [HasPullbacks C] in
lemma intersectionChainComplexIso_hom_comp_right (R : C) (A B : Set Y) :
  (intersectionChainComplexIso R A B).hom >>
  SSet.chainComplexMap
  (SSet.Subcomplex.homOfLE (inf_le_right :
    landsIn A ⊔ landsIn B ≤ landsIn B)) R =
  ((singularChainComplexFunctor C).obj R).map (interInclRight A B) >>
  (subtypeChainComplexIso R B).hom := by
simpa [intersectionChainComplexIso, subtypeChainComplexIso,
      singularChainComplexFunctor, SSet.chainComplexMap, Functor.map_comp] using
      congrArg (fun q => ((SSet.chainComplexFunctor C).obj R).map q)
      (intersectionSingularSetIso_hom_comp_right A B)

/-- The square of actual subtype singular chains is a pushout onto the
small singular chain complex. -/
noncomputable def subtypeChainsSmallIsPushout (R : C) (A B : Set Y) :
  IsPushout
  (((singularChainComplexFunctor C).obj R).map (interInclLeft A B))
  (((singularChainComplexFunctor C).obj R).map (interInclRight A B))
  ((subtypeChainComplexIso R A).hom >>
  SSet.chainComplexMap
  (SSet.Subcomplex.homOfLE (le_sup_left :
    landsIn A ≤ small A B)) R)
  ((subtypeChainComplexIso R B).hom >>
  SSet.chainComplexMap
  (SSet.Subcomplex.homOfLE (le_sup_right :
    landsIn B ≤ small A B)) R) := by
apply (smallChainIsPushout R A B).of_iso'
(intersectionChainComplexIso R A B)
(subtypeChainComplexIso R A)
(subtypeChainComplexIso R B)
(Iso.refl _)
· exact intersectionChainComplexIso_hom_comp_left R A B

```

```

· exact intersectionChainComplexIso_hom_comp_right R A B
· simp only [Iso.refl_hom, Category.comp_id]
  congr
· simp only [Iso.refl_hom, Category.comp_id]
  congr

/-- The existing Mayer--Vietoris cover-chain cokernel is canonically
isomorphic to the chain complex generated by small singular simplices. -/
noncomputable def coverChainsIsoSmall (R : C) (A B : Set Y) :
  coverChains (C := C) R (interInclLeft A B) (interInclRight A B)  $\cong$ 
    (small A B : SSet).chainComplex R :=
  (pushoutIsoCokernel
    (((singularChainComplexFunctor C).obj R).map (interInclLeft A B))
    (((singularChainComplexFunctor C).obj R).map (interInclRight A B))
    (subtypeChainsSmallIsPushout R A B)).symm

end SingularExcision

```

A.19 PlatonicSolids/SingularExcision/SmallChains.lean

Small-chain subgroup, invariance, and the retract (1078 lines).

```

/-
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-/
import Mathlib.Algebra.Category.ModuleCat.Colimits
import Mathlib.Algebra.DirectSum.Basic
import PlatonicSolids.SingularExcision.Algebra
import PlatonicSolids.SingularExcision.BarycentricGeometry
import PlatonicSolids.SingularExcision.CoverChains
import PlatonicSolids.SingularExcision.ModuleSupport
import PlatonicSolids.SingularExcision.SingularChains

/-!
Small singular chains for a two-set open cover.

Coefficients stay in `ModuleCat` so that finite coproduct support is
available; the geometric smallness lemmas are coefficient-independent.
-/

open AlgebraicTopology CategoryTheory Limits HomologicalComplex Set
open scoped DirectSum Simplicial Topology

namespace SingularExcision

noncomputable section

variable {Y : Type} [TopologicalSpace Y]

/-- If a singular simplex has image in `A`, it is a simplex of `landsIn A`. -/
lemma mem_landsIn_of_image_subset {n : SimplexCategoryop}
  (s : (TopCat.toSSet.obj (.of Y)).obj n) {A : Set Y}
  (h : Set.range ((TopCat.of Y).toSSetObjEquiv n s)  $\subseteq$  A) :
   $\exists \tau$ , (subtypeSingularSetMap A).app n  $\tau$  = s := by
  let fY := (TopCat.of Y).toSSetObjEquiv n s
  let f : C(TopCat.of (stdSimplex  $\mathbb{R}$  (Fin (n.unop.len + 1))),
    TopCat.of  $\uparrow$ A) :=
     $\langle$ fun x =>  $\langle$ fY x, h  $\langle$ x, rfl $\rangle$  $\rangle$ , fY.continuous.subtype_mk  $\_$  $\rangle$ 
  refine  $\langle$ ((TopCat.of  $\uparrow$ A).toSSetObjEquiv n).symm f, ? $\_$  $\rangle$ 
  apply ((TopCat.of Y).toSSetObjEquiv n).injective
  ext x
  rw [subtypeSingularSetMap_toSSetObjEquiv, Equiv.apply_symm_apply]

```

```

rfl

/-- A simplex of `landsIn A` has image contained in `A`. -/
lemma image_subset_of_mem_landsIn {n : SimplexCategoryop}
  {s : (TopCat.toSSet.obj (.of Y)).obj n} {A : Set Y}
  (h : s ∈ (landsIn A).obj n) :
  Set.range ((TopCat.of Y).toSSetObjEquiv n s) ⊆ A := by
  obtain ⟨τ, rfl⟩ := h
  intro y hy
  obtain ⟨x, rfl⟩ := hy
  rw [subtypeSingularSetMap_toSSetObjEquiv]
  exact ((TopCat.of ↑A).toSSetObjEquiv n τ x).property

/-- Image in `A` or `B` means the simplex is small for the pair. -/
lemma mem_small_of_image_subset_left_or_right {n : SimplexCategoryop}
  (s : (TopCat.toSSet.obj (.of Y)).obj n) {A B : Set Y}
  (h : Set.range ((TopCat.of Y).toSSetObjEquiv n s) ⊆ A ∨
    Set.range ((TopCat.of Y).toSSetObjEquiv n s) ⊆ B) :
  s ∈ (small A B).obj n := by
  change s ∈ (landsIn A).obj n ∪ (landsIn B).obj n
  rcases h with hA | hB
  · obtain ⟨τ, hτ⟩ := mem_landsIn_of_image_subset s hA
    exact Or.inl ⟨τ, hτ⟩
  · obtain ⟨τ, hτ⟩ := mem_landsIn_of_image_subset s hB
    exact Or.inr ⟨τ, hτ⟩

/-- A small simplex has image in `A` or in `B`. -/
lemma image_subset_left_or_right_of_mem_small {n : SimplexCategoryop}
  {s : (TopCat.toSSet.obj (.of Y)).obj n} {A B : Set Y}
  (h : s ∈ (small A B).obj n) :
  Set.range ((TopCat.of Y).toSSetObjEquiv n s) ⊆ A ∨
  Set.range ((TopCat.of Y).toSSetObjEquiv n s) ⊆ B := by
  rcases (show s ∈ (landsIn A).obj n ∪ (landsIn B).obj n by
    simpa [small, Subfunctor.max_obj] using h) with hA | hB
  · exact Or.inl (image_subset_of_mem_landsIn hA)
  · exact Or.inr (image_subset_of_mem_landsIn hB)

/-- After enough barycentric subdivisions, every affine piece of a
singular simplex is small for a two-set open cover. -/
lemma eventual_small_pieces_of_openCover
  {n : ℕ} (σ : C(stdSimplex ℝ (Fin (n + 1)), Y))
  {A B : Set Y} (hA : IsOpen A) (hB : IsOpen B)
  (hAB : A ∪ B = univ) :
  ∃ N : ℕ, ∀ τ ∈ ((PlatonicSolids.SingularExcision.subdivideStd
    (q := n) n)[N]
    (PlatonicSolids.SingularExcision.simplex
      (PlatonicSolids.SingularExcision.idSimplex n))).support,
  σ '' Set.range τ.realize ⊆ A ∨
  σ '' Set.range τ.realize ⊆ B :=
  PlatonicSolids.SingularExcision.exists_subdivideStd_iterate_image_subset_left_or_right
  σ hA hB hAB

/-- Reparametrized pieces of a sufficiently subdivided singular simplex
are small simplices. -/
lemma eventual_small_reparametrize_of_openCover
  {n : ℕ} (s : (TopCat.toSSet.obj (.of Y)) _[n])
  {A B : Set Y} (hA : IsOpen A) (hB : IsOpen B)
  (hAB : A ∪ B = univ) :
  ∃ N : ℕ, ∀ τ ∈ ((PlatonicSolids.SingularExcision.subdivideStd
    (q := n) n)[N]
    (PlatonicSolids.SingularExcision.simplex
      (PlatonicSolids.SingularExcision.idSimplex n))).support,
  PlatonicSolids.SingularExcision.reparametrizeSingularSimplex s τ ∈

```

```

      (small A B).obj (.op (.mk n)) := by
obtain ⟨N, hN⟩ :=
  eventual_small_pieces_of_openCover
  ((TopCat.of Y).toSSetObjEquiv _ s) hA hB hAB
refine ⟨N, fun τ hτ => ?_⟩
refine mem_small_of_image_subset_left_or_right _ ?_
have himg :
  Set.range ((TopCat.of Y).toSSetObjEquiv _
    (PlatonicSolids.SingularExcision.reparametrizeSingularSimplex s τ)) =
    ((TopCat.of Y).toSSetObjEquiv _ s) '' Set.range τ.realize := by
  rw [PlatonicSolids.SingularExcision.reparametrizeSingularSimplex_toContinuousMap]
  simp [ContinuousMap.coe_comp, Set.range_comp]
  simpa [himg] using hN τ hτ

/-- Reparametrizing a small simplex keeps it small: the image of a piece
is contained in the image of the parent. -/
lemma reparametrize_mem_small_of_mem_small {p q : ℕ}
  {A B : Set Y}
  (s : (TopCat.toSSet.obj (.of Y)) _[q])
  (σ : PlatonicSolids.SingularExcision.AffineSimplex
    (stdSimplex ℝ (Fin (q + 1))) p)
  (hs : s ∈ (small A B).obj (.op (.mk q))) :
  PlatonicSolids.SingularExcision.reparametrizeSingularSimplex s σ ∈
    (small A B).obj (.op (.mk p)) := by
refine mem_small_of_image_subset_left_or_right _ ?_
have himg :
  Set.range ((TopCat.of Y).toSSetObjEquiv _
    (PlatonicSolids.SingularExcision.reparametrizeSingularSimplex s σ)) =
    ((TopCat.of Y).toSSetObjEquiv _ s) '' Set.range σ.realize := by
  rw [PlatonicSolids.SingularExcision.reparametrizeSingularSimplex_toContinuousMap]
  simp [ContinuousMap.coe_comp, Set.range_comp]
  rw [himg]
  rcases image_subset_left_or_right_of_mem_small hs with hA | hB
  · exact Or.inl ((Set.image_subset_range _ _).trans hA)
  · exact Or.inr ((Set.image_subset_range _ _).trans hB)

end

/--! ## Module coefficients: total chains and the small retract -/

noncomputable section

open Classical

variable {k : Type} [Ring k] (R : ModuleCat.{0} k)
variable [HasCoproducts.{0} (ModuleCat.{0} k)]

/-- Underlying abelian group of singular `n`-chains. -/
abbrev singularChainType (X : TopCat.{0}) (n : ℕ) : Type :=
  (((singularChainComplexFunctor (ModuleCat.{0} k)).obj R).obj X).X n : Type)

instance (X : TopCat.{0}) (n : ℕ) : AddCommGroup (singularChainType R X n) :=
  inferInstance

/-- Direct sum of all singular chain groups. -/
abbrev TotalSingularChains (X : TopCat.{0}) : Type :=
  DirectSum ℕ (singularChainType R X)

instance (X : TopCat.{0}) : AddCommGroup (TotalSingularChains R X) :=
  inferInstance

noncomputable def dApply (X : TopCat.{0}) (n : ℕ) :
  singularChainType R X (n + 1) →+ singularChainType R X n :=

```

```

AddMonoidHom.mk'
  (fun x =>
    (((singularChainComplexFunctor (ModuleCat.{0} k)).obj R).obj X).d
      (n + 1) n).hom x)
  (by intro x y; simp [map_add])

noncomputable def SApply (X : TopCat.{0}) (n : ℕ) :
  singularChainType R X n →+ singularChainType R X n :=
  AddMonoidHom.mk'
  (fun x =>
    (PlatonicSolids.SingularExcision.singularSubdivisionHom R X n).hom x)
  (by intro x y; simp [map_add])

noncomputable def HApply (X : TopCat.{0}) (n : ℕ) :
  singularChainType R X n →+ singularChainType R X (n + 1) :=
  AddMonoidHom.mk'
  (fun x =>
    (PlatonicSolids.SingularExcision.singularPrismHom R X n).hom x)
  (by intro x y; simp [map_add])

noncomputable def dTotal (X : TopCat.{0}) :
  TotalSingularChains R X →+ TotalSingularChains R X :=
  DirectSum.toAddMonoid fun n =>
    match n with
    | 0 => 0
    | n + 1 => (DirectSum.of (singularChainType R X) n).comp (dApply R X n)

noncomputable def STotal (X : TopCat.{0}) :
  TotalSingularChains R X →+ TotalSingularChains R X :=
  DirectSum.toAddMonoid fun n =>
    (DirectSum.of (singularChainType R X) n).comp (SApply R X n)

noncomputable def HTotal (X : TopCat.{0}) :
  TotalSingularChains R X →+ TotalSingularChains R X :=
  DirectSum.toAddMonoid fun n =>
    (DirectSum.of (singularChainType R X) (n + 1)).comp (HApply R X n)

lemma dTotal_of_zero (X : TopCat.{0})
  (y : singularChainType R X 0) :
  dTotal R X (DirectSum.of (singularChainType R X) 0 y) = 0 :=
  DirectSum.toAddMonoid_of _ 0 y

lemma dTotal_of_succ (X : TopCat.{0}) (n : ℕ)
  (y : singularChainType R X (n + 1)) :
  dTotal R X (DirectSum.of (singularChainType R X) (n + 1) y) =
  DirectSum.of (singularChainType R X) n (dApply R X n y) :=
  DirectSum.toAddMonoid_of _ (n + 1) y

lemma STotal_of (X : TopCat.{0}) (n : ℕ)
  (y : singularChainType R X n) :
  STotal R X (DirectSum.of (singularChainType R X) n y) =
  DirectSum.of (singularChainType R X) n (SApply R X n y) :=
  DirectSum.toAddMonoid_of _ n y

lemma HTotal_of (X : TopCat.{0}) (n : ℕ)
  (y : singularChainType R X n) :
  HTotal R X (DirectSum.of (singularChainType R X) n y) =
  DirectSum.of (singularChainType R X) (n + 1) (HApply R X n y) :=
  DirectSum.toAddMonoid_of _ n y

lemma SApply_dApply (X : TopCat.{0}) (n : ℕ)
  (y : singularChainType R X (n + 1)) :
  SApply R X n (dApply R X n y) = dApply R X n (SApply R X (n + 1) y) := by

```

```

have h :=
  (PlatonicSolids.SingularExcision.singularSubdivision R X).comm (n + 1) n
calc
  SApply R X n (dApply R X n y) =
    ModuleCat.Hom.hom
      ((PlatonicSolids.SingularExcision.singularSubdivision R X).f n)
      (ModuleCat.Hom.hom
        (((singularChainComplexFunctor (ModuleCat.{0} k)).obj R).obj X).d
        (n + 1) n) y) :=
    rfl
  _ = ModuleCat.Hom.hom
    (((singularChainComplexFunctor (ModuleCat.{0} k)).obj R).obj X).d
    (n + 1) n >>
    (PlatonicSolids.SingularExcision.singularSubdivision R X).f n) y :=
  (ModuleCat.comp_apply _ _ _).symm
  _ = ModuleCat.Hom.hom
    ((PlatonicSolids.SingularExcision.singularSubdivision R X).f (n + 1) >>
    (((singularChainComplexFunctor (ModuleCat.{0} k)).obj R).obj X).d
    (n + 1) n)) y := by
    rw [← h]
  _ = dApply R X n (SApply R X (n + 1) y) := by
    rw [ModuleCat.comp_apply]; rfl

lemma homotopyApply_zero (X : TopCat.{0})
  (y : singularChainType R X 0) :
  y - SApply R X 0 y = dApply R X 0 (HApply R X 0 y) := by
  have hS := PlatonicSolids.SingularExcision.singularSubdivisionHom_zero R X
  have hT := PlatonicSolids.SingularExcision.singularPrismHom_zero R X
  simp [SApply, HApply, dApply, hS, hT]

lemma homotopyApply_succ (X : TopCat.{0}) (n : ℕ)
  (y : singularChainType R X (n + 1)) :
  y - SApply R X (n + 1) y =
    HApply R X n (dApply R X n y) +
    dApply R X (n + 1) (HApply R X (n + 1) y) := by
  have hcomm :=
    (PlatonicSolids.SingularExcision.singularPrismHomotopy R X).comm (n + 1)
  have hdNext :
    dNext (n + 1)
    (PlatonicSolids.SingularExcision.singularPrismHomotopy R X).hom =
    (((singularChainComplexFunctor (ModuleCat.{0} k)).obj R).obj X).d
    (n + 1) n) >>
    PlatonicSolids.SingularExcision.singularPrismHom R X n := by
    rw [Homotopy.dNext_succ_chainComplex]
  simp [PlatonicSolids.SingularExcision.singularPrismHomotopy]
  have hprevD :
    prevD (n + 1)
    (PlatonicSolids.SingularExcision.singularPrismHomotopy R X).hom =
    PlatonicSolids.SingularExcision.singularPrismHom R X (n + 1) >>
    (((singularChainComplexFunctor (ModuleCat.{0} k)).obj R).obj X).d
    (n + 2) (n + 1)) := by
    rw [Homotopy.prevD_chainComplex]
  simp [PlatonicSolids.SingularExcision.singularPrismHomotopy]
  have hfun :=
    congrArg (fun f : _ → _ => ModuleCat.Hom.hom f y) hcomm
  rw [hdNext, hprevD] at hfun
  simp [SApply, HApply, dApply, ModuleCat.hom_add, ModuleCat.comp_apply] at hfun
  rw [sub_eq_iff_eq_add]
  simpa [add_assoc, add_left_comm, add_comm] using hfun

lemma homotopyApply (X : TopCat.{0}) (n : ℕ)
  (y : singularChainType R X n) :
  y - SApply R X n y =

```

```

      (match n with
      | 0 => 0
      | m + 1 => HApply R X m (dApply R X m y)) +
      dApply R X n (HApply R X n y) := by
cases n with
| zero =>
  simp using homotopyApply_zero R X y
| succ m =>
  simp using homotopyApply_succ R X m y

/-- Ungraded subdivision data on the total singular chain group. -/
noncomputable def subdivisionData_singular (X : TopCat.{0}) :
  SubdivisionData (TotalSingularChains R X) where
d := dTotal R X
S := STotal R X
H := HTotal R X
S_comm := by
  intro x
  refine DirectSum.induction_on x ?_ ?_ ?_
  · simp
  · intro n y
    match n with
    | 0 =>
      simp [dTotal_of_zero, STotal_of]
    | n + 1 =>
      simp [dTotal_of_succ, STotal_of, SApply_dApply]
  · intro a b ha hb
    simp [map_add, ha, hb]
homotopy := by
  intro x
  refine DirectSum.induction_on x ?_ ?_ ?_
  · simp
  · intro n y
    have h := homotopyApply R X n y
    match n with
    | 0 =>
      simp [dTotal_of_zero, STotal_of, HTotal_of,
        PlatonicSolids.SingularExcision.singularPrismHom_zero,
        PlatonicSolids.SingularExcision.singularSubdivisionHom_zero,
        HApply, SApply]
    | n + 1 =>
      simp [dTotal_of_succ, STotal_of, HTotal_of]
      have h' := congrArg (DirectSum.of (singularChainType R X) (n + 1)) h
      simp [map_add, map_sub, add_comm] using h'
  · intro a b ha hb
    simp only [map_add]
    have hsum : a + b - (STotal R X a + STotal R X b) =
      (a - STotal R X a) + (b - STotal R X b) := by abel
    rw [hsum, ha, hb]
    abel

variable {Y : Type} [TopologicalSpace Y]

/-- A degree-`n` chain is small when it is in the image of the
small-subcomplex inclusion. -/
def isSmallChain {A B : Set Y} {n : ℕ}
  (x : singularChainType R (.of Y) n) : Prop :=
  ∃ y : ((small A B : SSet).chainComplex R).X n,
  ModuleCat.Hom.hom ((SSet.chainComplexMap (small A B).ι R).f n) y = x

lemma isSmallChain_zero {A B : Set Y} {n : ℕ} :
  isSmallChain (R := R) (A := A) (B := B) (n := n) 0 :=
  ⟨0, map_zero _⟩

```

```

lemma isSmallChain_add {A B : Set Y} {n : ℕ}
  {x₁ x₂ : singularChainType R (.of Y) n}
  (h₁ : isSmallChain (R := R) (A := A) (B := B) x₁)
  (h₂ : isSmallChain (R := R) (A := A) (B := B) x₂) :
  isSmallChain (R := R) (A := A) (B := B) (x₁ + x₂) := by
  obtain ⟨y₁, hy₁⟩ := h₁
  obtain ⟨y₂, hy₂⟩ := h₂
  refine ⟨y₁ + y₂, ?_⟩
  rw [map_add, hy₁, hy₂]
  exact rfl

lemma isSmallChain_neg {A B : Set Y} {n : ℕ}
  {x : singularChainType R (.of Y) n}
  (h : isSmallChain (R := R) (A := A) (B := B) x) :
  isSmallChain (R := R) (A := A) (B := B) (-x) := by
  obtain ⟨y, hy⟩ := h
  refine ⟨-y, ?_⟩
  rw [map_neg, hy]
  exact rfl

/-- The small-chain subgroup of the total singular chain group. -/
def smallChainsSubgroup (A B : Set Y) :
  AddSubgroup (TotalSingularChains R (.of Y)) where
  carrier := { x | ∀ n, isSmallChain (R := R) (A := A) (B := B) (x n) }
  zero_mem' := fun n => by
    convert isSmallChain_zero (R := R) (A := A) (B := B) (n := n)
  add_mem' hx hy n := isSmallChain_add (R := R) (hx n) (hy n)
  neg_mem' hx n := isSmallChain_neg (R := R) (hx n)

lemma isSmallChain_d {A B : Set Y} {n : ℕ}
  {x : singularChainType R (.of Y) (n + 1)}
  (hx : isSmallChain (R := R) (A := A) (B := B) x) :
  isSmallChain (R := R) (A := A) (B := B) (dApply R (.of Y) n x) := by
  obtain ⟨y, hy⟩ := hx
  refine ⟨((small A B : SSet).chainComplex R).d (n + 1) n y, ?_⟩
  have hcomm := (SSet.chainComplexMap (small A B).ι R).comm (n + 1) n
  calc
    ModuleCat.Hom.hom ((SSet.chainComplexMap (small A B).ι R).f n)
      (((small A B : SSet).chainComplex R).d (n + 1) n y) =
      ModuleCat.Hom.hom
        (((small A B : SSet).chainComplex R).d (n + 1) n >>
          (SSet.chainComplexMap (small A B).ι R).f n) y :=
      (ModuleCat.comp_apply _ _ _).symm
  _ = ModuleCat.Hom.hom
    ((SSet.chainComplexMap (small A B).ι R).f (n + 1) >>
      ((TopCat.toSSet.obj (.of Y)).chainComplex R).d (n + 1) n) y := by
  rw [← hcomm]
  _ = dApply R (.of Y) n x := by
  rw [ModuleCat.comp_apply, hy]
  rfl

/-- The inclusion of a small simplex, as an ambient chain, is small. -/
lemma isSmallChain_ι {A B : Set Y} {n : ℕ}
  (s : (TopCat.toSSet.obj (.of Y)) _[n])
  (hs : s ∈ (small A B).obj (.op (.mk n))) (r : R) :
  isSmallChain (R := R) (A := A) (B := B)
    (ModuleCat.Hom.hom
      ((TopCat.toSSet.obj (.of Y)).ιChainComplex (R := R) s) r) := by
  let s' : (small A B : SSet) _[n] := ⟨s, hs⟩
  refine ⟨ModuleCat.Hom.hom ((small A B : SSet).ιChainComplex (R := R) s') r, ?_⟩
  have hι :=
    SSet.ι_chainComplexMap_f (X := (small A B : SSet))

```

```

(Y := TopCat.toSSet.obj (.of Y)) (f := (small A B).ι) (R := R) s'
have happ := congrArg (fun f : _ → _ => ModuleCat.Hom.hom f r) hι
have hcomp :
  ModuleCat.Hom.hom
    (((small A B : SSet).ιChainComplex (R := R) s') >>
      (SSet.chainComplexMap (small A B).ι R).f n) r =
  ModuleCat.Hom.hom ((SSet.chainComplexMap (small A B).ι R).f n)
    (ModuleCat.Hom.hom ((small A B : SSet).ιChainComplex (R := R) s') r) := by
  rw [ModuleCat.hom_comp, LinearMap.comp_apply]
calc
  ModuleCat.Hom.hom ((SSet.chainComplexMap (small A B).ι R).f n)
    (ModuleCat.Hom.hom ((small A B : SSet).ιChainComplex (R := R) s') r) =
  ModuleCat.Hom.hom
    (((small A B : SSet).ιChainComplex (R := R) s') >>
      (SSet.chainComplexMap (small A B).ι R).f n) r :=
  hcomp.symm
_ = ModuleCat.Hom.hom
  ((TopCat.toSSet.obj (.of Y)).ιChainComplex (R := R)
    ((small A B).ι.app (.op (.mk n)) s')) r :=
  happ
_ = ModuleCat.Hom.hom
  ((TopCat.toSSet.obj (.of Y)).ιChainComplex (R := R) s) r := by
  congr 2

lemma SApply_ι {X : TopCat.{0}} {n : ℕ}
  (s : (TopCat.toSSet.obj X) _[n]) (r : R) :
  SApply R X n
    (ModuleCat.Hom.hom ((TopCat.toSSet.obj X).ιChainComplex (R := R) s) r) =
  ModuleCat.Hom.hom
    (PlatonicSolids.SingularExcision.singularSubdivideGenerator R s) r := by
  have h :=
    PlatonicSolids.SingularExcision.ι_singularSubdivisionHom (R := R) (X := X) s
  have happ := congrArg (fun f : _ → _ => ModuleCat.Hom.hom f r) h
  have hcomp :
    ModuleCat.Hom.hom
      (((TopCat.toSSet.obj X).ιChainComplex (R := R) s) >>
        PlatonicSolids.SingularExcision.singularSubdivisionHom R X n) r =
    SApply R X n
      (ModuleCat.Hom.hom
        ((TopCat.toSSet.obj X).ιChainComplex (R := R) s) r) := by
  rw [ModuleCat.hom_comp, LinearMap.comp_apply]; rfl
  exact hcomp.symm.trans happ

lemma HApply_ι {X : TopCat.{0}} {n : ℕ}
  (s : (TopCat.toSSet.obj X) _[n]) (r : R) :
  HApply R X n
    (ModuleCat.Hom.hom ((TopCat.toSSet.obj X).ιChainComplex (R := R) s) r) =
  ModuleCat.Hom.hom
    (PlatonicSolids.SingularExcision.singularPrismGenerator R s) r := by
  have h :=
    PlatonicSolids.SingularExcision.ι_singularPrismHom (R := R) (X := X) s
  have happ := congrArg (fun f : _ → _ => ModuleCat.Hom.hom f r) h
  have hcomp :
    ModuleCat.Hom.hom
      (((TopCat.toSSet.obj X).ιChainComplex (R := R) s) >>
        PlatonicSolids.SingularExcision.singularPrismHom R X n) r =
    HApply R X n
      (ModuleCat.Hom.hom
        ((TopCat.toSSet.obj X).ιChainComplex (R := R) s) r) := by
  rw [ModuleCat.hom_comp, LinearMap.comp_apply]; rfl
  exact hcomp.symm.trans happ

lemma isSmallChain_nsmul {A B : Set Y} {n : ℕ}

```

```

{x : singularChainType R (.of Y) n} (m : ℕ)
(hx : isSmallChain (R := R) (A := A) (B := B) x) :
isSmallChain (R := R) (A := A) (B := B) (m • x) := by
induction m with
| zero =>
  simpa using isSmallChain_zero (R := R) (A := A) (B := B) (n := n)
| succ m ih =>
  rw [succ_nsmul]
  exact isSmallChain_add (R := R) ih hx

lemma isSmallChain_zsmul {A B : Set Y} {n : ℕ}
{x : singularChainType R (.of Y) n} (z : ℤ)
(hx : isSmallChain (R := R) (A := A) (B := B) x) :
isSmallChain (R := R) (A := A) (B := B) (z • x) := by
cases z with
| ofNat m =>
  simpa using isSmallChain_nsmul (R := R) m hx
| negSucc m =>
  simpa [negSucc_zsmul] using
    isSmallChain_neg (R := R)
    (isSmallChain_nsmul (R := R) (m + 1) hx)

lemma isSmallChain_realizeAffine {A B : Set Y} {p q : ℕ}
(s : (TopCat.toSSet.obj (.of Y)) _[q])
(c : PlatonicSolids.SingularExcision.AffineChain
  (stdSimplex ℝ (Fin (q + 1))) p)
(hc : ∀ τ ∈ c.support,
  PlatonicSolids.SingularExcision.reparametrizeSingularSimplex s τ ∈
  (small A B).obj (.op (.mk p)))
(r : R) :
isSmallChain (R := R) (A := A) (B := B)
  (ModuleCat.Hom.hom
    (PlatonicSolids.SingularExcision.realizeAffineChain R s c) r) := by
induction c using Finsupp.induction with
| zero =>
  simpa [PlatonicSolids.SingularExcision.realizeAffineChain_zero] using
    isSmallChain_zero (R := R) (A := A) (B := B) (n := p)
| single_add τ z c hτ hz ih =>
  have hx :
    ModuleCat.Hom.hom
      (PlatonicSolids.SingularExcision.realizeAffineChain R s
        (Finsupp.single τ z + c)) r =
    z • ModuleCat.Hom.hom
      (PlatonicSolids.SingularExcision.realizeAffineSimplex R s τ) r +
    ModuleCat.Hom.hom
      (PlatonicSolids.SingularExcision.realizeAffineChain R s c) r := by
  rw [map_add, PlatonicSolids.SingularExcision.realizeAffineChain_single]
  simp [ModuleCat.hom_add, LinearMap.map_smul]
  rw [hx]
  refine isSmallChain_add (R := R) ?_
  (ih fun ρ hρ => hc ρ (by
    have hne : ρ ≠ τ := fun e => hτ (e ► hρ)
    simp [Finsupp.mem_support_iff, Finsupp.add_apply, Finsupp.single_apply, hne] at hρ ⊢
    exact hρ))
  have hτs := hc τ (by
    simpa [Finsupp.mem_support_iff, Finsupp.add_apply,
      Finsupp.notMem_support_iff.mp hτ] using hz)
  have hgen := isSmallChain_ι (R := R) (A := A) (B := B)
    (PlatonicSolids.SingularExcision.reparametrizeSingularSimplex s τ)
    hτs r
  simpa [PlatonicSolids.SingularExcision.realizeAffineSimplex] using
    isSmallChain_zsmul (R := R) z hgen

```

```

lemma isSmallChain_S_generator {A B : Set Y} {n : ℕ}
  {s : (TopCat.toSSet.obj (.of Y)) _[n]}
  (hs : s ∈ (small A B).obj (.op (.mk n))) (r : R) :
  isSmallChain (R := R) (A := A) (B := B)
    (SApply R (.of Y) n
      (ModuleCat.Hom.hom
        ((TopCat.toSSet.obj (.of Y)).ιChainComplex (R := R) s) r)) := by
  rw [SApply_ι]
  exact isSmallChain_realizeAffine (R := R) (A := A) (B := B) s
    (PlatonicSolids.SingularExcision.standardSubdivision n)
    (fun τ _ => reparametrize_mem_small_of_mem_small s τ hs) r

lemma isSmallChain_H_generator {A B : Set Y} {n : ℕ}
  {s : (TopCat.toSSet.obj (.of Y)) _[n]}
  (hs : s ∈ (small A B).obj (.op (.mk n))) (r : R) :
  isSmallChain (R := R) (A := A) (B := B)
    (HApply R (.of Y) n
      (ModuleCat.Hom.hom
        ((TopCat.toSSet.obj (.of Y)).ιChainComplex (R := R) s) r)) := by
  rw [HApply_ι]
  exact isSmallChain_realizeAffine (R := R) (A := A) (B := B) s
    (PlatonicSolids.SingularExcision.standardPrism n)
    (fun τ _ => reparametrize_mem_small_of_mem_small s τ hs) r

lemma realizeAffineChain_comp_singularSubdivisionHom
  {X : TopCat.{0}} {p q : ℕ}
  (s : (TopCat.toSSet.obj X) _[q])
  (c : PlatonicSolids.SingularExcision.AffineChain
    (stdSimplex ℝ (Fin (q + 1))) p) :
  PlatonicSolids.SingularExcision.realizeAffineChain R s c >>
    PlatonicSolids.SingularExcision.singularSubdivisionHom R X p =
    PlatonicSolids.SingularExcision.realizeAffineChain R s
      (PlatonicSolids.SingularExcision.subdivideStd (q := q) p c) := by
  induction c using Finsupp.induction with
  | zero =>
    simp [PlatonicSolids.SingularExcision.realizeAffineChain_zero]
  | single_add τ z c hτ hz ih =>
    simp [map_add, Preadditive.add_comp, ih,
      PlatonicSolids.SingularExcision.realizeAffineChain_single,
      Preadditive.zsmul_comp]
    have hσ :
      PlatonicSolids.SingularExcision.realizeAffineSimplex R s τ >>
        PlatonicSolids.SingularExcision.singularSubdivisionHom R X p =
        PlatonicSolids.SingularExcision.realizeAffineChain R s
          (PlatonicSolids.SingularExcision.subdivideStd (q := q) p
            (PlatonicSolids.SingularExcision.simplex τ)) := by
      rw [PlatonicSolids.SingularExcision.realizeAffineSimplex]
      erw [PlatonicSolids.SingularExcision.ι_singularSubdivisionHom]
      rw [PlatonicSolids.SingularExcision.singularSubdivideGenerator,
        ← PlatonicSolids.SingularExcision.realizeAffineChain_subdivideStd_simplex]
      rw [hσ]
      rw [show Finsupp.single τ z = z • PlatonicSolids.SingularExcision.simplex τ
        by simp [PlatonicSolids.SingularExcision.simplex]]
      simp [map_zsmul]

lemma ι_singularSubdivisionIterate
  {X : TopCat.{0}} {n : ℕ}
  (s : (TopCat.toSSet.obj X) _[n]) (N : ℕ) :
  (TopCat.toSSet.obj X).ιChainComplex (R := R) s >>
    (PlatonicSolids.SingularExcision.singularSubdivisionIterate R X N).f n =
    PlatonicSolids.SingularExcision.realizeAffineChain R s
      ((PlatonicSolids.SingularExcision.subdivideStd (q := n) n)^[N]
        (PlatonicSolids.SingularExcision.simplex

```

```

        (PlatonicSolids.SingularExcision.idSimplex n))) := by
induction N with
| zero =>
  simp [PlatonicSolids.SingularExcision.singularSubdivisionIterate_zero,
    PlatonicSolids.SingularExcision.realizeAffineChain_simplex,
    PlatonicSolids.SingularExcision.realizeAffineSimplex_id]
  exact Category.comp_id _
| succ N ih =>
  rw [PlatonicSolids.SingularExcision.singularSubdivisionIterate_f_succ]
  erw [← Category.assoc, ih, realizeAffineChain_comp_singularSubdivisionHom]
  rw [Function.iterate_succ_apply']

lemma eventual_small_generator_of_openCover
  {n : ℕ} (s : (TopCat.toSSet.obj (.of Y)) _[n])
  {A B : Set Y} (hA : IsOpen A) (hB : IsOpen B)
  (hAB : A ∪ B = univ) (r : R) :
  ∃ N : ℕ, isSmallChain (R := R) (A := A) (B := B)
  (ModuleCat.Hom.hom
    ((PlatonicSolids.SingularExcision.singularSubdivisionIterate
      R (.of Y) N).f n)
    (ModuleCat.Hom.hom
      ((TopCat.toSSet.obj (.of Y)).ιChainComplex (R := R) s) r)) := by
obtain ⟨N, hN⟩ :=
  eventual_small_reparametrize_of_openCover s hA hB hAB
refine ⟨N, ?_⟩
have hiter := ι_singularSubdivisionIterate (R := R) (X := .of Y) s N
have hcomp :
  ModuleCat.Hom.hom
    (((TopCat.toSSet.obj (.of Y)).ιChainComplex (R := R) s) >>
      (PlatonicSolids.SingularExcision.singularSubdivisionIterate
        R (.of Y) N).f n) r =
  ModuleCat.Hom.hom
    ((PlatonicSolids.SingularExcision.singularSubdivisionIterate
      R (.of Y) N).f n)
  (ModuleCat.Hom.hom
    ((TopCat.toSSet.obj (.of Y)).ιChainComplex (R := R) s) r) := by
  rw [ModuleCat.hom_comp, LinearMap.comp_apply]
  rfl
have happ := congrArg (fun f : _ → _ => ModuleCat.Hom.hom f r) hiter
have hval :
  ModuleCat.Hom.hom
    ((PlatonicSolids.SingularExcision.singularSubdivisionIterate
      R (.of Y) N).f n)
  (ModuleCat.Hom.hom
    ((TopCat.toSSet.obj (.of Y)).ιChainComplex (R := R) s) r) =
  ModuleCat.Hom.hom
    (PlatonicSolids.SingularExcision.realizeAffineChain R s
      ((PlatonicSolids.SingularExcision.subdivideStd (q := n) n)^[N]
        (PlatonicSolids.SingularExcision.simplex
          (PlatonicSolids.SingularExcision.idSimplex n)))) r :=
  hcomp.symm.trans happ
  rw [hval]
  exact isSmallChain_realizeAffine (R := R) (A := A) (B := B) s _ hN r

/-- Index type of singular `n`-simplices. -/
abbrev SingularSimplex (X : TopCat.{0}) (n : ℕ) : Type :=
  (TopCat.toSSet.obj X) _[n]

/-- Degree-`n` singular chains are the coproduct of copies of `R`. -/
lemma singularChain_X_eq (X : TopCat.{0}) (n : ℕ) :
  (((singularChainComplexFunctor (ModuleCat.{0} k)).obj R).obj X).X n =
  ∐ fun _ : SingularSimplex X n ↦ R :=
  rfl

```

```

/-- Small `n`-chains are the coproduct of copies of `R` indexed by small
simplices. -/
lemma smallChain_X_eq (A B : Set Y) (n : ℕ) :
  (((small A B : SSet).chainComplex R).X n) =
    ∐ fun _ : (small A B : SSet) _[n] ↦ R :=
  rfl

/-- Every singular `n`-chain is a finite linear combination of generators. -/
lemma singularChain_eq_sum {X : TopCat.{0}} {n : ℕ}
  (x : singularChainType R X n) :
  x = ∑ s ∈ ModuleCat.coprodSupport (fun _ : SingularSimplex X n ↦ R) x,
    ModuleCat.Hom.hom
      ((TopCat.toSSet.obj X).ιChainComplex (R := R) s)
      ((ModuleCat.coprodIsoDirectSum
        (fun _ : SingularSimplex X n ↦ R)).hom x s) := by
  classical
  simp [SSet.ιChainComplex] using
    ModuleCat.eq_sum_ι_apply_coprodSupport
    (fun _ : SingularSimplex X n ↦ R) x

/-- Every small `n`-chain is a finite linear combination of small generators. -/
lemma smallChain_eq_sum {A B : Set Y} {n : ℕ}
  (y : ((small A B : SSet).chainComplex R).X n) :
  y = ∑ s ∈ ModuleCat.coprodSupport
    (fun _ : (small A B : SSet) _[n] ↦ R) y,
    ModuleCat.Hom.hom
      ((small A B : SSet).ιChainComplex (R := R) s)
      ((ModuleCat.coprodIsoDirectSum
        (fun _ : (small A B : SSet) _[n] ↦ R)).hom y s) := by
  classical
  simp [SSet.ιChainComplex] using
    ModuleCat.eq_sum_ι_apply_coprodSupport
    (fun _ : (small A B : SSet) _[n] ↦ R) y

lemma SApply_sum {X : TopCat.{0}} {n : ℕ} {ι : Type}
  (s : Finset ι) (f : ι → singularChainType R X n) :
  SApply R X n (∑ i ∈ s, f i) = ∑ i ∈ s, SApply R X n (f i) :=
  map_sum (SApply R X n) f s

lemma HApply_sum {X : TopCat.{0}} {n : ℕ} {ι : Type}
  (s : Finset ι) (f : ι → singularChainType R X n) :
  HApply R X n (∑ i ∈ s, f i) = ∑ i ∈ s, HApply R X n (f i) :=
  map_sum (HApply R X n) f s

lemma SApply_iterate_sum {X : TopCat.{0}} {n N : ℕ} {ι : Type}
  (s : Finset ι) (f : ι → singularChainType R X n) :
  (SApply R X n)^[N] (∑ i ∈ s, f i) = ∑ i ∈ s, (SApply R X n)^[N] (f i) := by
  induction N with
  | zero => simp
  | succ N ih =>
    rw [Function.iterate_succ_apply', ih, SApply_sum]
    simp [Function.iterate_succ_apply']

lemma isSmallChain_sum {A B : Set Y} {n : ℕ} {ι : Type}
  (s : Finset ι) (f : ι → singularChainType R (.of Y) n)
  (hf : ∀ i ∈ s, isSmallChain (R := R) (A := A) (B := B) (f i)) :
  isSmallChain (R := R) (A := A) (B := B) (∑ i ∈ s, f i) := by
  classical
  revert hf
  refine Finset.induction_on s ?_ ?_
  · intro _
    simp using isSmallChain_zero (R := R) (A := A) (B := B) (n := n)

```

```

· intro i s his ih hf
  rw [Finset.sum_insert his]
  exact isSmallChain_add (R := R)
    (hf i (Finset.mem_insert_self i s))
    (ih fun j hj => hf j (Finset.mem_insert_of_mem hj))

lemma SApply_iterate {X : TopCat.{0}} (n N : ℕ)
  (y : singularChainType R X n) :
  (SApply R X n)^[N] y =
    ModuleCat.Hom.hom
      ((PlatonicSolids.SingularExcision.singularSubdivisionIterate R X N).f n)
    y := by
induction N with
| zero =>
  simp [PlatonicSolids.SingularExcision.singularSubdivisionIterate_zero]
| succ N ih =>
  rw [Function.iterate_succ_apply', ih,
    PlatonicSolids.SingularExcision.singularSubdivisionIterate_f_succ]
  rfl

lemma STotal_apply (X : TopCat.{0}) (n : ℕ)
  (x : TotalSingularChains R X) :
  STotal R X x n = SApply R X n (x n) := by
  refine DirectSum.induction_on x ?_ ?_ ?_
· simp
· intro m y
  rw [STotal_of]
  by_cases h : m = n
· subst h
  simp [DirectSum.of_eq_same]
· rw [DirectSum.of_eq_of_ne m n _ (Ne.symm h),
  DirectSum.of_eq_of_ne m n y (Ne.symm h)]
  simp
· intro a b ha hb
  simp [map_add, ha, hb]

lemma STotal_iterate_apply (X : TopCat.{0}) (n N : ℕ)
  (x : TotalSingularChains R X) :
  ((STotal R X)^[N] x) n = (SApply R X n)^[N] (x n) := by
  induction N with
| zero => simp
| succ N ih =>
  rw [Function.iterate_succ_apply', STotal_apply, ih,
    Function.iterate_succ_apply']

lemma dTotal_apply (X : TopCat.{0}) (n : ℕ)
  (x : TotalSingularChains R X) :
  dTotal R X x n = dApply R X n (x (n + 1)) := by
  refine DirectSum.induction_on x ?_ ?_ ?_
· simp
· intro m y
  match m with
| 0 =>
  rw [dTotal_of_zero,
    DirectSum.of_eq_of_ne 0 (n + 1) y (Nat.succ_ne_zero n)]
  simp
| m + 1 =>
  rw [dTotal_of_succ]
  by_cases h : m = n
· subst h
  simp [DirectSum.of_eq_same]
· rw [DirectSum.of_eq_of_ne m n _ (Ne.symm h),
  DirectSum.of_eq_of_ne (m + 1) (n + 1) y (mt Nat.succ.inj (Ne.symm h))]
```

```

      simp
    · intro a b ha hb
      simp [map_add, ha, hb]

lemma HTotal_apply_zero (X : TopCat.{0})
  (x : TotalSingularChains R X) :
  HTotal R X x 0 = 0 := by
  refine DirectSum.induction_on x ?_ ?_ ?_
  · simp
  · intro m y
    rw [HTotal_of]
    exact DirectSum.of_eq_of_ne (m + 1) 0 _ (Nat.succ_ne_zero m).symm
  · intro a b ha hb
    simp [map_add, ha, hb]

lemma HTotal_apply_succ (X : TopCat.{0}) (n : ℕ)
  (x : TotalSingularChains R X) :
  HTotal R X x (n + 1) = HApply R X n (x n) := by
  refine DirectSum.induction_on x ?_ ?_ ?_
  · simp
  · intro m y
    rw [HTotal_of]
    by_cases h : m = n
    · subst h
      simp [DirectSum.of_eq_same]
    · rw [DirectSum.of_eq_of_ne (m + 1) (n + 1) _ (mt Nat.succ.inj (Ne.symm h)),
        DirectSum.of_eq_of_ne m n y (Ne.symm h)]
      simp
  · intro a b ha hb
    simp [map_add, ha, hb]

/-- Subdivision preserves smallness of an arbitrary small chain. -/
lemma isSmallChain_S {A B : Set Y} {n : ℕ}
  {x : singularChainType R (.of Y) n}
  (hx : isSmallChain (R := R) (A := A) (B := B) x) :
  isSmallChain (R := R) (A := A) (B := B) (SApply R (.of Y) n x) := by
  classical
  obtain ⟨y, hy⟩ := hx
  have hy_sum := smallChain_eq_sum (R := R) (A := A) (B := B) y
  have hx_sum :
    x = ∑ s ∈ ModuleCat.coprodSupport
      (fun _ : (small A B : SSet) _[n] ↦ R) y,
      ModuleCat.Hom.hom
        ((TopCat.toSSet.obj (.of Y)).ιChainComplex (R := R)
          ((small A B).ι.app _ s))
        ((ModuleCat.coprodIsoDirectSum
          (fun _ : (small A B : SSet) _[n] ↦ R)).hom y s) := by
  have hmap := congrArg
    (ModuleCat.Hom.hom ((SSet.chainComplexMap (small A B).ι R).f n)) hy_sum
  rw [hy] at hmap
  rw [hmap, map_sum]
  refine Finset.sum_congr rfl fun s _ => ?_
  have hι :=
    SSet.ι_chainComplexMap_f (X := (small A B : SSet))
    (Y := TopCat.toSSet.obj (.of Y)) (f := (small A B).ι) (R := R) s
  have happ :=
    congrArg (fun f : _ → _ =>
      ModuleCat.Hom.hom f
        ((ModuleCat.coprodIsoDirectSum
          (fun _ : (small A B : SSet) _[n] ↦ R)).hom y s)) hι
  have hcomp :
    ModuleCat.Hom.hom
      (((small A B : SSet).ιChainComplex (R := R) s) >>

```

```

      (SSet.chainComplexMap (small A B).ι R).f n)
    ((ModuleCat.coprodIsoDirectSum
      (fun _ : (small A B : SSet) _[n] ↦ R)).hom y s) =
    ModuleCat.Hom.hom ((SSet.chainComplexMap (small A B).ι R).f n)
      (ModuleCat.Hom.hom ((small A B : SSet).ι ChainComplex (R := R) s)
        ((ModuleCat.coprodIsoDirectSum
          (fun _ : (small A B : SSet) _[n] ↦ R)).hom y s))) := by
    rw [ModuleCat.hom_comp, LinearMap.comp_apply]
  exact hcomp.symm.trans happ
rw [hx_sum]
refine (SApply_sum (R := R) (X := .of Y)
  (ModuleCat.coprodSupport (fun _ : (small A B : SSet) _[n] ↦ R) y)
  (fun s => ModuleCat.Hom.hom
    ((TopCat.toSSet.obj (.of Y)).ι ChainComplex (R := R)
      ((small A B).ι.app _ s))
    ((ModuleCat.coprodIsoDirectSum
      (fun _ : (small A B : SSet) _[n] ↦ R)).hom y s))).symm ▸ ?_
refine isSmallChain_sum (R := R) _ _ fun s _ => ?_
simp [SApply_ι] using
  isSmallChain_S_generator (R := R) (A := A) (B := B) s.property
  ((ModuleCat.coprodIsoDirectSum
    (fun _ : (small A B : SSet) _[n] ↦ R)).hom y s)

/-- The prism operator preserves smallness of an arbitrary small chain. -/
lemma isSmallChain_H {A B : Set Y} {n : ℕ}
  {x : singularChainType R (.of Y) n}
  (hx : isSmallChain (R := R) (A := A) (B := B) x) :
  isSmallChain (R := R) (A := A) (B := B) (HApply R (.of Y) n x) := by
classical
obtain ⟨y, hy⟩ := hx
have hy_sum : smallChain_eq_sum (R := R) (A := A) (B := B) y
have hx_sum :
  x = ∑ s ∈ ModuleCat.coprodSupport
    (fun _ : (small A B : SSet) _[n] ↦ R) y,
    ModuleCat.Hom.hom
      ((TopCat.toSSet.obj (.of Y)).ι ChainComplex (R := R)
        ((small A B).ι.app _ s))
      ((ModuleCat.coprodIsoDirectSum
        (fun _ : (small A B : SSet) _[n] ↦ R)).hom y s) := by
have hmap := congrArg
  (ModuleCat.Hom.hom ((SSet.chainComplexMap (small A B).ι R).f n)) hy_sum
rw [hy] at hmap
rw [hmap, map_sum]
refine Finset.sum_congr rfl fun s _ => ?_
have hι :=
  SSet.ι_chainComplexMap_f (X := (small A B : SSet))
    (Y := TopCat.toSSet.obj (.of Y)) (f := (small A B).ι) (R := R) s
have happ :=
  congrArg (fun f : _ → _ =>
    ModuleCat.Hom.hom f
      ((ModuleCat.coprodIsoDirectSum
        (fun _ : (small A B : SSet) _[n] ↦ R)).hom y s)) hι
have hcomp :
  ModuleCat.Hom.hom
    (((small A B : SSet).ι ChainComplex (R := R) s) >>
      (SSet.chainComplexMap (small A B).ι R).f n)
    ((ModuleCat.coprodIsoDirectSum
      (fun _ : (small A B : SSet) _[n] ↦ R)).hom y s) =
  ModuleCat.Hom.hom ((SSet.chainComplexMap (small A B).ι R).f n)
    (ModuleCat.Hom.hom ((small A B : SSet).ι ChainComplex (R := R) s)
      ((ModuleCat.coprodIsoDirectSum
        (fun _ : (small A B : SSet) _[n] ↦ R)).hom y s))) := by
  rw [ModuleCat.hom_comp, LinearMap.comp_apply]

```

```

    exact hcomp.symm.trans happ
  rw [hx_sum]
  refine (HApply_sum (R := R) (X := .of Y)
    (ModuleCat.coprodSupport (fun _ : (small A B : SSet) _[n] ↦ R) y)
    (fun s => ModuleCat.Hom.hom
      ((TopCat.toSSet.obj (.of Y)).ιChainComplex (R := R)
        ((small A B).ι.app _ s))
      ((ModuleCat.coprodIsoDirectSum
        (fun _ : (small A B : SSet) _[n] ↦ R)).hom y s))).symm ▶ ?_
  refine isSmallChain_sum (R := R) _ _ fun s _ => ?_
  simp [HApply_ι] using
    isSmallChain_H_generator (R := R) (A := A) (B := B) s.property
    ((ModuleCat.coprodIsoDirectSum
      (fun _ : (small A B : SSet) _[n] ↦ R)).hom y s)

/-- The small-chain subgroup is invariant under `d`, `S`, and `H`. -/
lemma smallInvariant_of_openCover (A B : Set Y) :
  (subdivisionData_singular R (.of Y)).SmallInvariant
  (smallChainsSubgroup R A B) where
d_mem := by
  intro x hx n
  change isSmallChain (R := R) (A := A) (B := B) (dTotal R (.of Y) x n)
  rw [dTotal_apply]
  exact isSmallChain_d (R := R) (hx (n + 1))
S_mem := by
  intro x hx n
  change isSmallChain (R := R) (A := A) (B := B) (STotal R (.of Y) x n)
  rw [STotal_apply]
  exact isSmallChain_S (R := R) (hx n)
H_mem := by
  intro x hx n
  match n with
  | 0 =>
    change isSmallChain (R := R) (A := A) (B := B)
      (HTotal R (.of Y) x 0)
    rw [HTotal_apply_zero]
    exact isSmallChain_zero (R := R) (A := A) (B := B) (n := 0)
  | n + 1 =>
    change isSmallChain (R := R) (A := A) (B := B)
      (HTotal R (.of Y) x (n + 1))
    rw [HTotal_apply_succ]
    exact isSmallChain_H (R := R) (hx n)

lemma isSmallChain_S_iterate {A B : Set Y} {n : ℕ}
  {x : singularChainType R (.of Y) n} (N : ℕ)
  (hx : isSmallChain (R := R) (A := A) (B := B) x) :
  isSmallChain (R := R) (A := A) (B := B) ((SApply R (.of Y) n)^[N] x) := by
  induction N with
  | zero => simp
  | succ N ih =>
    rw [Function.iterate_succ_apply']
    exact isSmallChain_S (R := R) ih

lemma eventual_small_generator_iterate_of_openCover
  {n : ℕ} (s : (TopCat.toSSet.obj (.of Y)) _[n])
  {A B : Set Y} (hA : IsOpen A) (hB : IsOpen B)
  (hAB : A ∪ B = univ) (r : R) :
  ∃ N : ℕ, ∀ N' ≥ N,
    isSmallChain (R := R) (A := A) (B := B)
      ((SApply R (.of Y) n)^[N']
        (ModuleCat.Hom.hom
          ((TopCat.toSSet.obj (.of Y)).ιChainComplex (R := R) s) r)) := by
  obtain ⟨N, hN⟩ :=

```

```

    eventual_small_generator_of_openCover (R := R) s hA hB hAB r
  refine ⟨N, fun N' hN' => ?_⟩
  obtain ⟨k, hk⟩ := Nat.exists_eq_add_of_le hN'
  rw [hk, Nat.add_comm, Function.iterate_add_apply]
  exact isSmallChain_S_iterate (R := R) k (by
    simp [SApply_iterate] using hN)

/-- Every finite singular chain becomes small after enough subdivisions. -/
lemma eventual_small_of_degree
  {n : ℕ} (x : singularChainType R (.of Y) n)
  {A B : Set Y} (hA : IsOpen A) (hB : IsOpen B)
  (hAB : A ∪ B = univ) :
  ∃ N : ℕ, isSmallChain (R := R) (A := A) (B := B)
  ((SApply R (.of Y) n)^[N] x) := by
  classical
  have hx := singularChain_eq_sum (R := R) x
  let supp := ModuleCat.coprodSupport (fun _ : SingularSimplex (.of Y) n ↦ R) x
  have hgen : ∀ s ∈ supp, ∃ N : ℕ, ∀ N' ≥ N,
    isSmallChain (R := R) (A := A) (B := B)
    ((SApply R (.of Y) n)^[N']
      (ModuleCat.Hom.hom
        ((TopCat.toSSet.obj (.of Y)).ιChainComplex (R := R) s)
        ((ModuleCat.coprodIsoDirectSum
          (fun _ : SingularSimplex (.of Y) n ↦ R)).hom x s)))) :=
  fun s _ =>
    eventual_small_generator_iterate_of_openCover (R := R) s hA hB hAB _
  choose! N hN using hgen
  refine ⟨supp.sup N, ?_⟩
  rw [hx]
  refine (SApply_iterate_sum (R := R) (X := .of Y) supp
    (fun s => ModuleCat.Hom.hom
      ((TopCat.toSSet.obj (.of Y)).ιChainComplex (R := R) s)
      ((ModuleCat.coprodIsoDirectSum
        (fun _ : SingularSimplex (.of Y) n ↦ R)).hom x s)))) .symm ▸ ?_
  refine isSmallChain_sum (R := R) _ _ fun s hs =>
    hN s hs _ (Finset.le_sup hs)

lemma eventual_small_of_degree_le
  {n : ℕ} (x : singularChainType R (.of Y) n)
  {A B : Set Y} (hA : IsOpen A) (hB : IsOpen B)
  (hAB : A ∪ B = univ) :
  ∃ N : ℕ, ∀ N' ≥ N, isSmallChain (R := R) (A := A) (B := B)
  ((SApply R (.of Y) n)^[N'] x) := by
  obtain ⟨N, hN⟩ := eventual_small_of_degree (R := R) x hA hB hAB
  refine ⟨N, fun N' hN' => ?_⟩
  obtain ⟨k, hk⟩ := Nat.exists_eq_add_of_le hN'
  rw [hk, Nat.add_comm, Function.iterate_add_apply]
  exact isSmallChain_S_iterate (R := R) k hN

lemma eventual_small_of_chain
  (x : TotalSingularChains R (.of Y))
  {A B : Set Y} (hA : IsOpen A) (hB : IsOpen B)
  (hAB : A ∪ B = univ) :
  ∃ N : ℕ, (STotal R (.of Y))^[N] x ∈ smallChainsSubgroup R A B := by
  classical
  have hdeg : ∀ n ∈ x.support, ∃ N : ℕ, ∀ N' ≥ N,
    isSmallChain (R := R) (A := A) (B := B)
    ((SApply R (.of Y) n)^[N'] (x n)) :=
  fun n _ => eventual_small_of_degree_le (R := R) (x n) hA hB hAB
  choose! N hN using hdeg
  refine ⟨x.support.sup N, fun n => ?_⟩
  rw [STotal_iterate_apply]
  by_cases hn : n ∈ x.support

```

```

· exact hN n hn _ (Finset.le_sup hn)
· have hx0 : x n = 0 := DFinsupp.notMem_support_iff.mp hn
  simp [hx0] using isSmallChain_zero (R := R) (A := A) (B := B) (n := n)

/-- Every homology class of `Y` has a small cycle representative, and
every small boundary is the boundary of a small chain. -/
theorem small_cycles_and_boundaries_singular
  {A B : Set Y} (hA : IsOpen A) (hB : IsOpen B)
  (hAB : A ∪ B = univ) :
  (∀ z : TotalSingularChains R (.of Y),
    (subdivisionData_singular R (.of Y)).Cycle z →
    ∃ b ∈ smallChainsSubgroup R A B,
    (subdivisionData_singular R (.of Y)).Cycle b ∧
    (subdivisionData_singular R (.of Y)).Homologous z b) ∧
  (∀ b : TotalSingularChains R (.of Y),
    b ∈ smallChainsSubgroup R A B →
    (subdivisionData_singular R (.of Y)).Cycle b →
    (∃ x, b = (subdivisionData_singular R (.of Y)).d x) →
    ∃ y ∈ smallChainsSubgroup R A B,
    b = (subdivisionData_singular R (.of Y)).d y) :=
  (subdivisionData_singular R (.of Y)).small_cycles_and_boundaries
  (smallChainsSubgroup R A B)
  (smallInvariant_of_openCover (R := R) A B)
  (fun x => eventual_small_of_chain (R := R) x hA hB hAB)

end

end SingularExcision

```

A.20 PlatonicSolids/SingularExcision/OpenCoverQI.lean

Open-cover quasi-isomorphism and IsExcisiveSubspaces (523 lines).

```

/-
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-/
import Mathlib.Algebra.Category.ModuleCat.Abelian
import Mathlib.Algebra.Category.ModuleCat.EpiMono
import Mathlib.Algebra.Homology.ConcreteCategory
import Mathlib.Algebra.Homology.QuasiIso
import Mathlib.Algebra.Homology.ShortComplex.ModuleCat
import Mathlib.CategoryTheory.Limits.MonoCoproduct
import PlatonicSolids.SingularExcision.CoverChains
import PlatonicSolids.SingularExcision.SmallChains

/-!
Open-cover excision: the small-subcomplex inclusion is a
quasi-isomorphism, so an open cover is excisive. The geometric retract
is in `SmallChains`; the triangle `coverComparison = iso.hom >> small`
identifies the cover comparison with that inclusion.
-/

open AlgebraicTopology CategoryTheory Limits HomologicalComplex Set
open scoped Topology

set_option linter.unusedSectionVars false

namespace SingularExcision

noncomputable section

variable {k : Type} [Ring k] (R : ModuleCat.{0} k)

```

```

variable [HasCoproducts.{0} (ModuleCat.{0} k)]
variable [HasPullbacks (ModuleCat.{0} k)]
variable {Y : Type} [TopologicalSpace Y]

open Classical

/-- Ambient singular chains of `Y`. -/
abbrev ambientChains : ChainComplex (ModuleCat.{0} k)  $\mathbb{N}$  :=
  ((singularChainComplexFunctor (ModuleCat.{0} k)).obj R).obj (.of Y)

/-- Small-subcomplex inclusion on singular chains. -/
def small $\iota$  (A B : Set Y) :
  (small A B : SSet).chainComplex R  $\longrightarrow$ 
    ((singularChainComplexFunctor (ModuleCat.{0} k)).obj R).obj (.of Y) :=
  SSet.chainComplexMap (small A B). $\iota$  R

lemma toTotal_cycle_zero (z : singularChainType R (.of Y) 0) :
  (subdivisionData_singular R (.of Y)).Cycle
    (DirectSum.of (singularChainType R (.of Y)) 0 z) :=
  dTotal_of_zero R (.of Y) z

lemma toTotal_cycle_succ {n :  $\mathbb{N}$ } {z : singularChainType R (.of Y) (n + 1)}
  (hz : dApply R (.of Y) n z = 0) :
  (subdivisionData_singular R (.of Y)).Cycle
    (DirectSum.of (singularChainType R (.of Y)) (n + 1) z) := by
  change dTotal R (.of Y) (DirectSum.of _ (n + 1) z) = 0
  rw [dTotal_of_succ, hz, map_zero]

lemma inclChains_eq_subtypeIso_comp_landsIn $\iota$  (A : Set Y) :
  ((singularChainComplexFunctor (ModuleCat.{0} k)).obj R).map
    (subtypeIncl Y A) =
  (subtypeChainComplexIso (C := ModuleCat.{0} k) R A).hom  $\gg$ 
  SSet.chainComplexMap (landsIn A). $\iota$  R := by
  dsimp [subtypeChainComplexIso, singularChainComplexFunctor,
    SSet.chainComplexMap, subtypeSingularSetIso]
  rw [← Functor.map_comp, subtypeToLandsIn_ $\iota$ , subtypeSingularSetMap]

lemma landsIn $\iota$ _eq_to_small $\iota$  (A B : Set Y) :
  SSet.chainComplexMap (landsIn A). $\iota$  R =
  SSet.chainComplexMap
    (SSet.Subcomplex.homOfLE (le_sup_left : landsIn A  $\leq$  small A B)) R  $\gg$ 
  small $\iota$  R A B := by
  change
    ((SSet.chainComplexFunctor (ModuleCat.{0} k)).obj R).map (landsIn A). $\iota$  =
    ((SSet.chainComplexFunctor (ModuleCat.{0} k)).obj R).map
      (SSet.Subcomplex.homOfLE (le_sup_left : landsIn A  $\leq$  small A B))  $\gg$ 
    ((SSet.chainComplexFunctor (ModuleCat.{0} k)).obj R).map (small A B). $\iota$ 
  rw [← Functor.map_comp]
  rfl

lemma landsIn $\iota$ _eq_to_small $\iota$ _right (A B : Set Y) :
  SSet.chainComplexMap (landsIn B). $\iota$  R =
  SSet.chainComplexMap
    (SSet.Subcomplex.homOfLE (le_sup_right : landsIn B  $\leq$  small A B)) R  $\gg$ 
  small $\iota$  R A B := by
  change
    ((SSet.chainComplexFunctor (ModuleCat.{0} k)).obj R).map (landsIn B). $\iota$  =
    ((SSet.chainComplexFunctor (ModuleCat.{0} k)).obj R).map
      (SSet.Subcomplex.homOfLE (le_sup_right : landsIn B  $\leq$  small A B))  $\gg$ 
    ((SSet.chainComplexFunctor (ModuleCat.{0} k)).obj R).map (small A B). $\iota$ 
  rw [← Functor.map_comp]
  rfl

```

```

/-- The two inclusions of `A` and `B` into `Y`, after transport to
lands-in subcomplexes, factor through `small`. -/
lemma coverToX_eq_desc_comp_small (A B : Set Y) :
  coverToX (C := ModuleCat.{0} k) R
    (interInclLeft A B) (interInclRight A B)
    (subtypeIncl Y A) (subtypeIncl Y B) (subset_square_comm A B) =
  biprod.desc
    ((subtypeChainComplexIso (C := ModuleCat.{0} k) R A).hom >>
     SSet.chainComplexMap
       (SSet.Subcomplex.homOfLE (le_sup_left :
         landsIn A ≤ small A B)) R)
    ((subtypeChainComplexIso (C := ModuleCat.{0} k) R B).hom >>
     SSet.chainComplexMap
       (SSet.Subcomplex.homOfLE (le_sup_right :
         landsIn B ≤ small A B)) R) >>
  small R A B := by
  rw [coverToX]
  rw [inclChains_eq_subtypeIso_comp_landsIn (R := R) A]
  rw [inclChains_eq_subtypeIso_comp_landsIn (R := R) B]
  rw [landsIn_eq_to_small (R := R) A B]
  rw [landsIn_eq_to_small_right (R := R) A B]
  refine biprod.hom_ext' _ _ ?_ ?_
  · rw [biprod.inl_desc]
    exact (biprod.inl_desc_assoc
      ((subtypeChainComplexIso (C := ModuleCat.{0} k) R A).hom >>
       SSet.chainComplexMap
         (SSet.Subcomplex.homOfLE (le_sup_left : landsIn A ≤ small A B)) R)
      ((subtypeChainComplexIso (C := ModuleCat.{0} k) R B).hom >>
       SSet.chainComplexMap
         (SSet.Subcomplex.homOfLE (le_sup_right : landsIn B ≤ small A B)) R)
      (small R A B)).symm
  · rw [biprod.inr_desc]
    exact (biprod.inr_desc_assoc
      ((subtypeChainComplexIso (C := ModuleCat.{0} k) R A).hom >>
       SSet.chainComplexMap
         (SSet.Subcomplex.homOfLE (le_sup_left : landsIn A ≤ small A B)) R)
      ((subtypeChainComplexIso (C := ModuleCat.{0} k) R B).hom >>
       SSet.chainComplexMap
         (SSet.Subcomplex.homOfLE (le_sup_right : landsIn B ≤ small A B)) R)
      (small R A B)).symm

/-- Abbreviation for the small-pushout legs. -/
abbrev smallInl (A B : Set Y) :=
  (subtypeChainComplexIso (C := ModuleCat.{0} k) R A).hom >>
  SSet.chainComplexMap
    (SSet.Subcomplex.homOfLE (le_sup_left : landsIn A ≤ small A B)) R

abbrev smallInr (A B : Set Y) :=
  (subtypeChainComplexIso (C := ModuleCat.{0} k) R B).hom >>
  SSet.chainComplexMap
    (SSet.Subcomplex.homOfLE (le_sup_right : landsIn B ≤ small A B)) R

lemma mvIncl_eq_pushoutRelation (A B : Set Y) :
  mvIncl (C := ModuleCat.{0} k) R (interInclLeft A B) (interInclRight A B) =
  pushoutRelation
    (((singularChainComplexFunctor (ModuleCat.{0} k)).obj R).map
     (interInclLeft A B))
    (((singularChainComplexFunctor (ModuleCat.{0} k)).obj R).map
     (interInclRight A B)) :=
  rfl

lemma π_comp_coverChainsIsoSmall_hom (A B : Set Y) :
  cokernel.π (mvIncl (C := ModuleCat.{0} k) R)

```

```

    (interInclLeft A B) (interInclRight A B)) >>
    (coverChainsIsoSmall (C := ModuleCat.{0} k) R A B).hom =
    biprod.desc (smallInl (R := R) A B) (smallInr (R := R) A B) := by
    let f := ((singularChainComplexFunctor (ModuleCat.{0} k)).obj R).map
    (interInclLeft A B)
    let g := ((singularChainComplexFunctor (ModuleCat.{0} k)).obj R).map
    (interInclRight A B)
    let sq := subtypeChainsSmallIsPushout (C := ModuleCat.{0} k) R A B
    have hinl := sq.inl_isoIsPushout_inv _ _ (cokernelIsPushout f g)
    have hinr := sq.inr_isoIsPushout_inv _ _ (cokernelIsPushout f g)
    refine biprod.hom_ext' _ _ ?_ ?_
    · refine Eq.trans (Eq.trans ?_ hinl)
      (biprod.inl_desc (smallInl (R := R) A B)
      (smallInr (R := R) A B)).symm
    exact (Category.assoc
    (biprod.inl : _ →
      ((singularChainComplexFunctor (ModuleCat.{0} k)).obj R).obj (.of ↑A) ⊞
      ((singularChainComplexFunctor (ModuleCat.{0} k)).obj R).obj (.of ↑B))
      (cokernel.π (mvIncl (C := ModuleCat.{0} k) R
      (interInclLeft A B) (interInclRight A B)))
      (coverChainsIsoSmall (C := ModuleCat.{0} k) R A B).hom).symm
    · refine Eq.trans (Eq.trans ?_ hinr)
      (biprod.inr_desc (smallInl (R := R) A B)
      (smallInr (R := R) A B)).symm
    exact (Category.assoc
    (biprod.inr : _ →
      ((singularChainComplexFunctor (ModuleCat.{0} k)).obj R).obj (.of ↑A) ⊞
      ((singularChainComplexFunctor (ModuleCat.{0} k)).obj R).obj (.of ↑B))
      (cokernel.π (mvIncl (C := ModuleCat.{0} k) R
      (interInclLeft A B) (interInclRight A B)))
      (coverChainsIsoSmall (C := ModuleCat.{0} k) R A B).hom).symm

lemma coverComparison_eq_iso_comp_smallℓ (A B : Set Y) :
  coverComparison (C := ModuleCat.{0} k) R
    (interInclLeft A B) (interInclRight A B)
    (subtypeIncl Y A) (subtypeIncl Y B) (subset_square_comm A B) =
    (coverChainsIsoSmall (C := ModuleCat.{0} k) R A B).hom >>
    smallℓ R A B := by
  refine (cancel_epi (cokernel.π
    (mvIncl (C := ModuleCat.{0} k) R (interInclLeft A B)
    (interInclRight A B)))) .1 ?_
  simp only [coverComparison, cokernel.π_desc]
  rw [coverToX_eq_desc_comp_smallℓ, ← π_comp_coverChainsIsoSmall_hom]
  exact Category.assoc
    (cokernel.π (mvIncl (C := ModuleCat.{0} k) R
    (interInclLeft A B) (interInclRight A B)))
    (coverChainsIsoSmall (C := ModuleCat.{0} k) R A B).hom
    (smallℓ R A B)

instance smallℓ_mono_f (A B : Set Y) (n : ℕ) : Mono ((smallℓ R A B).f n) := by
  dsimp [smallℓ, SSet.chainComplexMap, SSet.chainComplexFunctor]
  exact MonoCoproduct.mono_map'_of_injective
    (fun _ : (TopCat.toSSet.obj (.of Y)).obj (.op (.mk n)) → R)
    (fun s : (small A B).obj (.op (.mk n)) → (s : _))
  Subtype.val_injective

instance smallℓ_mono (A B : Set Y) : Mono (smallℓ R A B) :=
  HomologicalComplex.mono_of_mono_f _ fun n => smallℓ_mono_f (R := R) A B n

lemma smallℓ_hom_injective (A B : Set Y) (n : ℕ) :
  Function.Injective (ModuleCat.Hom.hom ((smallℓ R A B).f n)) :=
  (ModuleCat.mono_iff_injective _).1 inferInstance

```

```

lemma ambient_d_apply {n : ℕ} (z : singularChainType R (.of Y) (n + 1)) :
  ModuleCat.Hom.hom
    (((singularChainComplexFunctor (ModuleCat.{0} k)).obj R).obj (.of Y)).d
      (n + 1) n) z =
    dApply R (.of Y) n z :=
  rfl

lemma of_mem_smallChainsSubgroup {A B : Set Y} {n : ℕ}
  {z : singularChainType R (.of Y) n}
  (hz : isSmallChain (R := R) (A := A) (B := B) z) :
  DirectSum.of (singularChainType R (.of Y)) n z ∈ smallChainsSubgroup R A B := by
  intro m
  by_cases h : m = n
  · subst h
    simp [DirectSum.of_eq_same] using hz
  · have hm : (DirectSum.of (singularChainType R (.of Y)) n z) m = 0 :=
      DirectSum.of_eq_of_ne n m z h
    simp [hm] using isSmallChain_zero (R := R) (A := A) (B := B) (n := m)

lemma cycle_component {n : ℕ} {z : TotalSingularChains R (.of Y)}
  (hz : (subdivisionData_singular R (.of Y)).Cycle z) :
  dApply R (.of Y) n (z (n + 1)) = 0 := by
  have h : dTotal R (.of Y) z = 0 := hz
  rw [← dTotal_apply, h]
  simp

lemma homologyπ_eq_zero_iff_boundary
  (K : ChainComplex (ModuleCat.{0} k) ℕ) (n : ℕ)
  (z : K.cycles n) :
  K.homologyπ n z = 0 ↔
  ∃ x : K.X (n + 1), K.toCycles (n + 1) n x = z := by
  have hex : (ShortComplex.mk (K.toCycles (n + 1) n) (K.homologyπ n)
    (K.toCycles_comp_homologyπ (n + 1) n)).Exact :=
    ShortComplex.exact_of_g_is_cokernel _
    (K.homologyIsCokernel (i := n + 1) (j := n) (ChainComplex.prev ℕ n))
  rw [ShortComplex.moduleCat_exact_iff] at hex
  constructor
  · exact hex z
  · rintro ⟨x, rfl⟩
    exact congrArg (fun f => f x) (K.toCycles_comp_homologyπ (n + 1) n)

lemma iCycles_apply_toCycles
  (K : ChainComplex (ModuleCat.{0} k) ℕ) (n : ℕ) (x : K.X (n + 1)) :
  ModuleCat.Hom.hom (K.iCycles n) (K.toCycles (n + 1) n x) =
  ModuleCat.Hom.hom (K.d (n + 1) n) x := by
  have hcomp : K.toCycles (n + 1) n >> K.iCycles n = K.d (n + 1) n :=
    K.liftCycles_i (K.d (n + 1) n) ((ComplexShape.down ℕ).next n) rfl
    (K.d_comp_d _ _ _)
  rw [← LinearMap.comp_apply, ← ModuleCat.hom_comp, hcomp]

lemma iCycles_apply_cyclesMap (A B : Set Y) (n : ℕ)
  (za : ((small A B : SSet).chainComplex R).cycles n) :
  ModuleCat.Hom.hom ((ambientChains (R := R)).iCycles n)
    (cyclesMap (smallℓ R A B) n za) =
  ModuleCat.Hom.hom ((smallℓ R A B).f n)
    (ModuleCat.Hom.hom
      (((small A B : SSet).chainComplex R).iCycles n) za) := by
  have hcomp := cyclesMap_i (smallℓ R A B) n
  rw [← LinearMap.comp_apply, ← ModuleCat.hom_comp, hcomp,
    ModuleCat.hom_comp, LinearMap.comp_apply]

lemma iCycles_d_apply (K : ChainComplex (ModuleCat.{0} k) ℕ) (n : ℕ)
  (z : K.cycles n) :

```

```

ModuleCat.Hom.hom (K.d n ((ComplexShape.down ℕ).next n))
  (ModuleCat.Hom.hom (K.iCycles n) z) = 0 := by
have hcomp := K.iCycles_d n ((ComplexShape.down ℕ).next n)
rw [← LinearMap.comp_apply, ← ModuleCat.hom_comp, hcomp]
simp

lemma iCycles_hom_injective (K : ChainComplex (ModuleCat.{0} k) ℕ) (n : ℕ) :
  Function.Injective (ModuleCat.Hom.hom (K.iCycles n)) :=
  (ModuleCat.mono_iff_injective _).1 inferInstance

lemma homologous_degree {z b w : TotalSingularChains R (.of Y)} {n : ℕ}
  (h : z - b = dTotal R (.of Y) w) :
  z n - b n = dApply R (.of Y) n (w (n + 1)) := by
have hn := congrArg (fun x : TotalSingularChains R (.of Y) => x n) h
change (z - b) n = dTotal R (.of Y) w n at hn
rw [dTotal_apply] at hn
exact hn

lemma cycle_of_iCycles {n : ℕ}
  (z : ((ambientChains (R := R)).cycles n)) :
  (subdivisionData_singular R (.of Y)).Cycle
    (DirectSum.of (singularChainType R (.of Y)) n
      (ModuleCat.Hom.hom ((ambientChains (R := R)).iCycles n) z)) := by
cases n with
| zero => exact toTotal_cycle_zero (R := R) _
| succ n =>
  refine toTotal_cycle_succ (R := R) ?_
  have hz := iCycles_d_apply (ambientChains (R := R)) (n + 1) z
  rw [ChainComplex.next_nat_succ] at hz
  simpa [dApply] using hz

lemma small_cycle_of_iCycles {A B : Set Y} {n : ℕ}
  (z : ((small A B : SSet).chainComplex R).cycles n) :
  (subdivisionData_singular R (.of Y)).Cycle
    (DirectSum.of (singularChainType R (.of Y)) n
      (ModuleCat.Hom.hom ((smallℓ R A B).f n)
        (ModuleCat.Hom.hom
          (((small A B : SSet).chainComplex R).iCycles n) z))) := by
cases n with
| zero => exact toTotal_cycle_zero (R := R) _
| succ n =>
  refine toTotal_cycle_succ (R := R) ?_
  have hz := iCycles_d_apply ((small A B : SSet).chainComplex R) (n + 1) z
  rw [ChainComplex.next_nat_succ] at hz
  have hcomp := (smallℓ R A B).comm (n + 1) n
  change ModuleCat.Hom.hom
    (((singularChainComplexFunctor (ModuleCat.{0} k)).obj R).obj (.of Y)).d
      (n + 1) n)
    (ModuleCat.Hom.hom ((smallℓ R A B).f (n + 1))
      (ModuleCat.Hom.hom
        (((small A B : SSet).chainComplex R).iCycles (n + 1)) z)) = 0
  rw [← LinearMap.comp_apply, ← ModuleCat.hom_comp, hcomp,
    ModuleCat.hom_comp, LinearMap.comp_apply, hz, map_zero]

lemma homologyMap_apply_homologyπ (A B : Set Y) (n : ℕ)
  (za : ((small A B : SSet).chainComplex R).cycles n) :
  ModuleCat.Hom.hom (homologyMap (smallℓ R A B) n)
    (ModuleCat.Hom.hom
      (((small A B : SSet).chainComplex R).homologyπ n) za) =
  ModuleCat.Hom.hom ((ambientChains (R := R)).homologyπ n)
    (cyclesMap (smallℓ R A B) n za) := by
have h := congrArg (fun f => ModuleCat.Hom.hom f za)
  (homologyMap_naturality (smallℓ R A B) (i := n))

```

```

convert h using 1 <=> rw [ModuleCat.hom_comp, LinearMap.comp_apply]

lemma d_small_of_d_image {A B : Set Y} {n : ℕ}
  (y : ((small A B : SSet).chainComplex R).X (n + 1))
  {z : ((small A B : SSet).chainComplex R).X n}
  (h : ModuleCat.Hom.hom ((smallℓ R A B).f n)
    (((small A B : SSet).chainComplex R).d (n + 1) n y) =
    ModuleCat.Hom.hom ((smallℓ R A B).f n) z) :
  ((small A B : SSet).chainComplex R).d (n + 1) n y = z :=
smallℓ_hom_injective (R := R) A B n h

lemma smallℓ_quasiIso {A B : Set Y}
  (hA : IsOpen A) (hB : IsOpen B) (hAB : A ∪ B = univ) :
  QuasiIso (smallℓ R A B) := by
  refine (quasiIso_iff (smallℓ R A B)).2 fun n =>
    (quasiIsoAt_iff_isIso_homologyMap (smallℓ R A B) n).2 ?_
  refine (ConcreteCategory.isIso_iff_bijective
    (homologyMap (smallℓ R A B) n)).2 ⟨?inj, ?surj⟩
  · intro a b hab
    apply sub_eq_zero.mp
    set c := a - b with hcdef
    have hc : ModuleCat.Hom.hom (homologyMap (smallℓ R A B) n) c = 0 := by
      simp [c, hab]
    obtain ⟨za, hza⟩ := (ModuleCat.epi_iff_surjective
      (((small A B : SSet).chainComplex R).homologyπ n)).1 inferInstance c
    have hbound :
      ModuleCat.Hom.hom ((ambientChains (R := R)).homologyπ n)
        (cyclesMap (smallℓ R A B) n za) = 0 := by
      rw [← homologyMap_apply_homologyπ, hza, hc]
    obtain ⟨x, hx⟩ :=
      (homologyπ_eq_zero_iff_boundary (ambientChains (R := R)) n _).1 hbound
    set zsmall :=
      ModuleCat.Hom.hom
        (((small A B : SSet).chainComplex R).iCycles n) za
    set btot := DirectSum.of (singularChainType R (.of Y)) n
      (ModuleCat.Hom.hom ((smallℓ R A B).f n) zsmall)
    have hbmemb : btot ∈ smallChainsSubgroup R A B :=
      of_mem_smallChainsSubgroup (R := R) ⟨zsmall, rfl⟩
    have hbcycle := small_cycle_of_iCycles (R := R) (A := A) (B := B) za
    have hdx :
      ModuleCat.Hom.hom ((ambientChains (R := R)).d (n + 1) n) x =
      ModuleCat.Hom.hom ((smallℓ R A B).f n) zsmall := by
      have h1 := iCycles_apply_toCycles (ambientChains (R := R)) n x
      have h2 := iCycles_apply_cyclesMap (R := R) A B n za
      rw [hx] at h1
      exact h1.symm.trans h2
    have hbound : ∃ w, btot = (subdivisionData_singular R (.of Y)).d w :=
      (DirectSum.of (singularChainType R (.of Y)) (n + 1) x, by
        change btot = dTotal R (.of Y)
        (DirectSum.of (singularChainType R (.of Y)) (n + 1) x)
        rw [dTotal_of_succ]
        simp [btot, dApply, hdx])
    obtain ⟨ytot, hymemb, hyeq⟩ :=
      (small_cycles_and_boundaries_singular (R := R) hA hB hAB).2
      btot hbmemb hbcycle hbound
    obtain ⟨ypred, hypred⟩ := hymemb (n + 1)
    have hdy :
      ModuleCat.Hom.hom ((smallℓ R A B).f n)
        (((small A B : SSet).chainComplex R).d (n + 1) n ypred) =
      ModuleCat.Hom.hom ((smallℓ R A B).f n) zsmall := by
      have hb_n : btot n = ModuleCat.Hom.hom ((smallℓ R A B).f n) zsmall := by
        simp [btot, DirectSum.of_eq_same]
      have hdeg := congrArg (fun w : TotalSingularChains R (.of Y) => w n) hyeq

```

```

have hytot : dApply R (.of Y) n (ytot (n + 1)) =
  ModuleCat.Hom.hom ((smallℓ R A B).f n) zsmall := by
  have : btot n =
    (subdivisionData_singular R (.of Y)).d ytot n := hdeg
  rw [show (subdivisionData_singular R (.of Y)).d ytot =
    dTotal R (.of Y) ytot from rfl] at this
  rw [dTotal_apply, hb_n] at this
  exact this.symm
have hcomm := (smallℓ R A B).comm (n + 1) n
rw [← LinearMap.comp_apply, ← ModuleCat.hom_comp, ← hcomm,
  ModuleCat.hom_comp, LinearMap.comp_apply]
have hypred' :
  ModuleCat.Hom.hom ((smallℓ R A B).f (n + 1)) ypred = ytot (n + 1) :=
  hypred
rw [hypred']
exact hytot
have hza_eq :
  za = ((small A B : SSet).chainComplex R).toCycles (n + 1) n ypred := by
  refine (iCycles_hom_injective ((small A B : SSet).chainComplex R) n) ?_
  have := d_small_of_d_image (R := R) (A := A) (B := B) ypred hdy
  rw [iCycles_apply_toCycles, this]
rw [← hza, hza_eq]
exact congrArg (fun f => ModuleCat.Hom.hom f ypred)
(((small A B : SSet).chainComplex R).toCycles_comp_homology $\pi$  (n + 1) n)
intro h
obtain ⟨zcyc, hzcyc⟩ := (ModuleCat.epi_iff_surjective
  ((ambientChains (R := R)).homology $\pi$  n)).1 inferInstance h
set z := ModuleCat.Hom.hom ((ambientChains (R := R)).iCycles n) zcyc
have hzcyc := cycle_of_iCycles (R := R) zcyc
obtain ⟨btot, hbmem, hbcycle, hhom⟩ :=
  (small_cycles_and_boundaries_singular (R := R) hA hB hAB).1 _ hzcyc
obtain ⟨w, hw⟩ := hhom
obtain ⟨yb, hyb⟩ := hbmem n
have hycycle :
  ModuleCat.Hom.hom
    (((small A B : SSet).chainComplex R).d n
      ((ComplexShape.down  $\mathbb{N}$ ).next n)) yb = 0 := by
  refine smallℓ_hom_injective (R := R) A B
    ((ComplexShape.down  $\mathbb{N}$ ).next n) ?_
  rw [map_zero]
  cases n with
  | zero =>
    rw [ChainComplex.next_nat_zero]
    have hcomm := (smallℓ R A B).comm 0 0
    rw [← LinearMap.comp_apply, ← ModuleCat.hom_comp, ← hcomm,
      ModuleCat.hom_comp, LinearMap.comp_apply]
    simp
  | succ m =>
    rw [ChainComplex.next_nat_succ]
    have hcomm := (smallℓ R A B).comm (m + 1) m
    rw [← LinearMap.comp_apply, ← ModuleCat.hom_comp, ← hcomm,
      ModuleCat.hom_comp, LinearMap.comp_apply]
    have hyb' :
      ModuleCat.Hom.hom ((smallℓ R A B).f (m + 1)) yb = btot (m + 1) :=
      hyb
    rw [hyb']
    exact cycle_component (R := R) (n := m) hbcycle
let ycyc := ((small A B : SSet).chainComplex R).cyclesMk
  (x := yb) (j := (ComplexShape.down  $\mathbb{N}$ ).next n) (hj := rfl)
  (hx := by
    change (forget2 (ModuleCat.{0} k) Ab).map _ yb = 0
    simp using hycycle)
refine ⟨((small A B : SSet).chainComplex R).homology $\pi$  n ycyc, ?_⟩

```

```

rw [← hzcyc, homologyMap_apply_homologyπ]
have hiycyc :
  ModuleCat.Hom.hom ((ambientChains (R := R)).iCycles n)
    (cyclesMap (smallℓ R A B) n ycyc) =
  ModuleCat.Hom.hom ((smallℓ R A B).f n) yb := by
have hmap := iCycles_apply_cyclesMap (R := R) A B n ycyc
have hy : ModuleCat.Hom.hom
  (((small A B : SSet).chainComplex R).iCycles n) ycyc = yb := by
  simpa using i_cyclesMk ((small A B : SSet).chainComplex R) yb
  ((ComplexShape.down ℕ).next n) rfl (by
    change (forget₂ (ModuleCat.{0} k) Ab).map _ yb = 0
    simpa using hycycle)
rw [hmap, hy]
have hex :
  (ambientChains (R := R)).toCycles (n + 1) n (w (n + 1)) =
  zcyc - cyclesMap (smallℓ R A B) n ycyc := by
  refine (iCycles_hom_injective (ambientChains (R := R)) n) ?_
have hdeg := homologous_degree (R := R) (n := n) hw
rw [iCycles_apply_toCycles, map_sub, hiycyc]
have hyb' : ModuleCat.Hom.hom ((smallℓ R A B).f n) yb = btot n := hyb
rw [hyb']
simpa [z, DirectSum.of_eq_same, dApply] using hdeg.symm
have hπ0 :
  (ambientChains (R := R)).homologyπ n
  ((ambientChains (R := R)).toCycles (n + 1) n (w (n + 1))) = 0 :=
  (homologyπ_eq_zero_iff_boundary (ambientChains (R := R)) n _).2
  ⟨w (n + 1), rfl⟩
have hzero :
  (ambientChains (R := R)).homologyπ n
  (zcyc - cyclesMap (smallℓ R A B) n ycyc) = 0 := by
  rw [← hex]
  exact hπ0
rw [map_sub] at hzero
exact (eq_of_sub_eq_zero hzero).symm

lemma IsExcisiveSubspaces_of_openCover {A B : Set Y}
  (hA : IsOpen A) (hB : IsOpen B) (hAB : A ∪ B = univ) :
  IsExcisiveSubspaces (C := ModuleCat.{0} k) R A B := by
  dsimp [IsExcisiveSubspaces, IsExcisive]
  rw [coverComparison_eq_iso_comp_smallℓ]
  haveI : QuasiIso (smallℓ R A B) := smallℓ_quasiIso (R := R) hA hB hAB
  infer_instance

lemma IsExcisiveSubspaces_puncturedSphere (n : ℕ) :
  IsExcisiveSubspaces (C := ModuleCat.{0} k) R
  (puncturedNorth n) (puncturedSouth n) :=
  IsExcisiveSubspaces_of_openCover (R := R)
  (isOpen_puncturedNorth n) (isOpen_puncturedSouth n) (punctured_cover n)

end

end SingularExcision

```

A.21 PlatonicSolids/SphereSingularHomology.lean

Stereographic $U \cap V \simeq_h S^{n-1}$ and $H_{n+1}(S^{n+1}) \cong H_n(S^n)$ (893 lines).

/-

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-/

import Mathlib.Algebra.Category.ModuleCat.Abelian

import Mathlib.Algebra.Homology.HomologicalComplexBiprod

```

import Mathlib.Analysis.Convex.Contractible
import Mathlib.Analysis.Normed.Module.Ball.RadialEquiv
import PlatonicSolids.Incidence
import PlatonicSolids.SingularExcision.OpenCoverQI

/-!
Singular  $H_n(S^n) \cong \mathbb{R}$  for  $n > 0$  from the stereographic
Mayer-Vietoris cover, after the open-cover quasi-isomorphism. The
intersection of the charts is homotopy-equivalent to  $S^{n-1}$ ; the
connecting map is an isomorphism in positive degrees;  $H_1(S^1)$  is
identified with  $\ker \epsilon$  of the two-point space. The optional Euler
bridge platonic_solids_3d_of_sphere feeds those Betti numbers into
the existing 3D incidence theorem without changing Palomar compared
types.
-/

open AlgebraicTopology CategoryTheory Limits HomologicalComplex Metric Set ContinuousMap
open scoped Topology InnerProductSpace

set_option linter.unusedSectionVars false

variable {k : Type} [Ring k] (R : ModuleCat.{0} k)
variable [HasCoproducts.{0} (ModuleCat.{0} k)]
variable [HasPullbacks (ModuleCat.{0} k)]

noncomputable section

/-! ## Stereographic intersection homotopy-equivalent to a sphere -/

lemma southPole_eq_neg_northPole (n : ℕ) :
  (southPole n : EuclideanSpace ℝ (Fin (n + 1))) =
    (-1 : ℝ) • (northPole n : EuclideanSpace ℝ (Fin (n + 1))) := by
  simp [southPole, northPole]

lemma stereoToFun_north_south (n : ℕ) :
  stereoToFun (northPole n : EuclideanSpace ℝ (Fin (n + 1)))
    (southPole n : EuclideanSpace ℝ (Fin (n + 1))) = 0 := by
  have hz :
    ((ℝ • (northPole n : EuclideanSpace ℝ (Fin (n + 1)))).orthogonal).orthogonalProjection
      (southPole n : EuclideanSpace ℝ (Fin (n + 1))) = 0 := by
  rw [southPole_eq_neg_northPole, map_smul,
      Submodule.orthogonalProjection_orthogonalComplement_singleton_eq_zero]
  simp
  rw [stereoToFun_apply, hz, smul_zero]

lemma homeoPuncturedNorth_apply (n : ℕ) (x : ↑(puncturedNorth n)) :
  homeoPuncturedNorth n x =
    stereoToFun (northPole n : EuclideanSpace ℝ (Fin (n + 1))) (x : _) := by
  apply Subtype.ext
  dsimp [homeoPuncturedNorth]
  rfl

lemma homeoPuncturedNorth_south (n : ℕ) :
  homeoPuncturedNorth n ⟨southPole n, by
    simpa [puncturedNorth] using (northPole_ne_southPole n).symm⟩ = 0 := by
  rw [homeoPuncturedNorth_apply, stereoToFun_north_south]

lemma homeoPuncturedNorth_symm_zero (n : ℕ) :
  (homeoPuncturedNorth n).symm 0 =
    ⟨southPole n, by simpa [puncturedNorth] using (northPole_ne_southPole n).symm⟩ := by
  apply (homeoPuncturedNorth n).injective
  simp [homeoPuncturedNorth_south]

```

```

lemma homeoPuncturedNorth_eq_zero_iff (n : ℕ) (x : ↑(puncturedNorth n)) :
  homeoPuncturedNorth n x = 0 ↔ (x : MetricSphere n) = southPole n := by
  constructor
  · intro hx
    have hx' : x = (homeoPuncturedNorth n).symm 0 := by
      rw [← hx, Homeomorph.symm_apply_apply]
      rw [hx', homeoPuncturedNorth_symm_zero]
  · rintro h
    have hx : x = ⟨southPole n, by
      simp [puncturedNorth] using (northPole_ne_southPole n).symm⟩ :=
      Subtype.ext h
    rw [hx, homeoPuncturedNorth_south]

/-- Stereographic projection from the north pole identifies `U ∩ V` with
the punctured hyperplane. -/
def homeoPuncturedInter (n : ℕ) :
  ↑(puncturedNorth n ∩ puncturedSouth n) ≃ₜ
  ({0}ᶜ : Set ((ℝ · (northPole n : EuclideanSpace ℝ (Fin (n + 1)))).orthogonal)) where
toFun p :=
  ⟨homeoPuncturedNorth n ⟨p.1, p.2.1⟩,
  mt (homeoPuncturedNorth_eq_zero_iff n ⟨p.1, p.2.1⟩).1 p.2.2⟩
invFun w :=
  ⟨(homeoPuncturedNorth n).symm w.1, by
  refine ⟨((homeoPuncturedNorth n).symm w.1).2, ?_⟩
  exact mt (homeoPuncturedNorth_eq_zero_iff n ((homeoPuncturedNorth n).symm w.1)).2
  (by rw [(homeoPuncturedNorth n).apply_symm_apply]; exact w.2)⟩
left_inv p := by
  ext
  simp
right_inv w := by
  ext
  simp
continuous_toFun := by
  rw [continuous_induced_rng]
  exact (homeoPuncturedNorth n).continuous.comp
  (Continuous.subtype_mk continuous_subtype_val _)
continuous_invFun := by
  rw [continuous_induced_rng]
  convert (continuous_subtype_val.comp
  ((homeoPuncturedNorth n).symm.continuous.comp continuous_subtype_val)) using 1

instance contractibleSpace_Ioi_zero : ContractibleSpace (Ioi (0 : ℝ)) :=
  (convex_Ioi (0 : ℝ)).contractibleSpace nonempty_Ioi

/-- A product with a contractible space is homotopy-equivalent to the first factor. -/
def homotopyEquiv_prod_contractible
  {X Y : Type} [TopologicalSpace X] [TopologicalSpace Y] [ContractibleSpace Y] :
  (X × Y) ≃ₕ X :=
  (ContinuousMap.HomotopyEquiv.prodCongr (ContinuousMap.HomotopyEquiv.refl X)
  (Classical.choice (ContractibleSpace.hequiv_unit Y))).trans
  (Homeomorph.prodUnique X Unit).toHomotopyEquiv

/-- Polar coordinates: the punctured hyperplane is homotopy-equivalent to
its unit sphere. -/
def homotopyEquiv_puncturedHyperplane_sphere (n : ℕ) :
  ({0}ᶜ : Set ((ℝ · (northPole n : EuclideanSpace ℝ (Fin (n + 1)))).orthogonal)) ≃ₕ
  ↑(sphere (0 : ((ℝ · (northPole n : EuclideanSpace ℝ (Fin (n + 1)))).orthogonal)) 1) :=
  (homeomorphUnitSphereProd
  ((ℝ · (northPole n : EuclideanSpace ℝ (Fin (n + 1)))).orthogonal)).toHomotopyEquiv.trans
  homotopyEquiv_prod_contractible

/-- The unit sphere in the polar hyperplane of `S^{n+1}` is homeomorphic to `S^n`. -/
def homeoHyperplaneSphere (n : ℕ) :

```

```

    ↑(sphere (0 : ((ℝ · (northPole (n + 1) : EuclideanSpace ℝ (Fin (n + 2)))).orthogonal)) 1)
    ≃t
    MetricSphere n :=
haveI : Fact (Module.finrank ℝ (EuclideanSpace ℝ (Fin (n + 2))) = n + 1 + 1) :=
  (by simp)
have hv : (northPole (n + 1) : EuclideanSpace ℝ (Fin (n + 2))) ≠ 0 := by
  simp [northPole]
let b := OrthonormalBasis.fromOrthogonalSpanSingleton (n + 1) hv
have himg : b.repr.toHomeomorph '' sphere 0 1 =
  sphere (0 : EuclideanSpace ℝ (Fin (n + 1))) 1 := by
  rw [LinearIsometryEquiv.coe_toHomeomorph, LinearIsometryEquiv.image_sphere]
  simp
(b.repr.toHomeomorph.image (sphere 0 1)).trans (Homeomorph.setCongr himg)

/-- The stereographic intersection of `S^{n+1}` is homotopy-equivalent to `S^n`. -/
def homotopyEquiv_puncturedInter_sphere (n : ℕ) :
  ↑(puncturedNorth (n + 1) ∩ puncturedSouth (n + 1)) ≃h MetricSphere n :=
  (homeoPuncturedInter (n + 1)).toHomotopyEquiv.trans
  ((homotopyEquiv_puncturedHyperplane_sphere (n + 1)).trans
   (homeoHyperplaneSphere n).toHomotopyEquiv)

/-! ## Mayer-Vietoris connecting map, hex discharged -/

/-- Connecting map of the stereographic cover, with excision supplied by
the open-cover quasi-isomorphism. -/
def singularHomology_sphere_mayerVietorisδ (n : ℕ) :
  (((singularChainComplexFunctor (ModuleCat.{0} k)).obj R).obj
   (.of (MetricSphere (n + 1)))) .homology (n + 1) →
  (((singularChainComplexFunctor (ModuleCat.{0} k)).obj R).obj
   (.of ↑(puncturedNorth (n + 1) ∩ puncturedSouth (n + 1)))).homology n :=
  mayerVietorisδ (C := ModuleCat.{0} k) R
  (interInclLeft (puncturedNorth (n + 1)) (puncturedSouth (n + 1)))
  (interInclRight (puncturedNorth (n + 1)) (puncturedSouth (n + 1)))
  (subtypeIncl (MetricSphere (n + 1)) (puncturedNorth (n + 1)))
  (subtypeIncl (MetricSphere (n + 1)) (puncturedSouth (n + 1)))
  (subset_square_comm _ _)
  (SingularExcision.IsExcisiveSubspaces_puncturedSphere (R := R) (n + 1)) n

lemma isZero_homology_chainBiprod
  {U V : TopCat.{0}} {n : ℕ}
  (hU : IsZero (((singularHomologyFunctor (ModuleCat.{0} k) n).obj R).obj U))
  (hV : IsZero (((singularHomologyFunctor (ModuleCat.{0} k) n).obj R).obj V)) :
  IsZero (((singularChainComplexFunctor (ModuleCat.{0} k)).obj R).obj U ⊕
  ((singularChainComplexFunctor (ModuleCat.{0} k)).obj R).obj V).homology n) := by
  let KU := ((singularChainComplexFunctor (ModuleCat.{0} k)).obj R).obj U
  let KV := ((singularChainComplexFunctor (ModuleCat.{0} k)).obj R).obj V
  change IsZero ((KU ⊕ KV).homology n)
  rw [IsZero.iff_id_eq_zero, ← homologyMap_id (KU ⊕ KV) n, ← biprod.total]
  rw [homologyMap_add, homologyMap_comp, homologyMap_comp]
  have hf : homologyMap (biprod.fst : KU ⊕ KV → KU) n = 0 := hU.eq_of_tgt _ _
  have hs : homologyMap (biprod.snd : KU ⊕ KV → KV) n = 0 := hV.eq_of_tgt _ _
  simp [hf, hs]

lemma isZero_homology_chartSum (n m : ℕ) (hm : m ≠ 0) :
  IsZero (((singularChainComplexFunctor (ModuleCat.{0} k)).obj R).obj
  (.of ↑(puncturedNorth n)) ⊕
  ((singularChainComplexFunctor (ModuleCat.{0} k)).obj R).obj
  (.of ↑(puncturedSouth n)))).homology m :=
  isZero_homology_chainBiprod (R := R)
  (isZero_singularHomology_puncturedNorth (C := ModuleCat.{0} k) R n hm)
  (isZero_singularHomology_puncturedSouth (C := ModuleCat.{0} k) R n hm)

lemma triadδ_sphere_isIso {n : ℕ} (hn : 0 < n) :

```

```

IsIso (triad $\delta$  (C := ModuleCat.{0} k) R
  (interInclLeft (puncturedNorth (n + 1)) (puncturedSouth (n + 1)))
  (interInclRight (puncturedNorth (n + 1)) (puncturedSouth (n + 1))) n) :=
(triad_shortExact (C := ModuleCat.{0} k) R
  (interInclLeft (puncturedNorth (n + 1)) (puncturedSouth (n + 1)))
  (interInclRight (puncturedNorth (n + 1)) (puncturedSouth (n + 1)))) .isIso_ $\delta$ 
(n + 1) n (by simp)
(isZero_homology_chartSum (R := R) (n + 1) (n + 1) (Nat.succ_ne_zero n))
(isZero_homology_chartSum (R := R) (n + 1) n hn.ne')

lemma singularHomology_sphere_mayerVietoris $\delta$ _isIso {n :  $\mathbb{N}$ } (hn : 0 < n) :
  IsIso (singularHomology_sphere_mayerVietoris $\delta$  (R := R) n) := by
  dsimp [singularHomology_sphere_mayerVietoris $\delta$ , mayerVietoris $\delta$ ]
  haveI := triad $\delta$ _sphere_isIso (R := R) hn
  infer_instance

/-- In positive degrees the stereographic connecting map identifies
`H_{n+1}(S^{n+1})` with `H_n` of the chart intersection. -/
def singularHomology_sphere_mayerVietoris $\delta$ _iso {n :  $\mathbb{N}$ } (hn : 0 < n) :
  (((singularChainComplexFunctor (ModuleCat.{0} k)).obj R).obj
    (.of (MetricSphere (n + 1)))) .homology (n + 1)  $\cong$ 
    (((singularChainComplexFunctor (ModuleCat.{0} k)).obj R).obj
      (.of ( $\uparrow$ (puncturedNorth (n + 1)  $\cap$  puncturedSouth (n + 1))))) .homology n :=
  haveI := singularHomology_sphere_mayerVietoris $\delta$ _isIso (R := R) hn
  asIso (singularHomology_sphere_mayerVietoris $\delta$  (R := R) n)

/-- For `n  $\geq$  1`, `H_{n+1}(S^{n+1})` is isomorphic to `H_n(S^n)`. -/
def singularHomology_sphere_succ_iso {n :  $\mathbb{N}$ } (hn : 0 < n) :
  (((singularHomologyFunctor (ModuleCat.{0} k) (n + 1)).obj R).obj
    (.of (MetricSphere (n + 1))))  $\cong$ 
    (((singularHomologyFunctor (ModuleCat.{0} k) n).obj R).obj
      (.of (MetricSphere n)))) :=
  (singularHomology_sphere_mayerVietoris $\delta$ _iso (R := R) hn).trans
  (singularHomologyIso_of_homotopyEquiv
    (homotopyEquiv_puncturedInter_sphere n) R n)

/-- Iterated connecting isomorphism: for `n  $\geq$  1`,
`H_{n+j}(S^{n+j})` is isomorphic to `H_n(S^n)`. -/
def singularHomology_sphere_iterate_iso
  {n :  $\mathbb{N}$ } (hn : 0 < n) : (j :  $\mathbb{N}$ )  $\rightarrow$ 
  (((singularHomologyFunctor (ModuleCat.{0} k) (n + j)).obj R).obj
    (.of (MetricSphere (n + j))))  $\cong$ 
    (((singularHomologyFunctor (ModuleCat.{0} k) n).obj R).obj
      (.of (MetricSphere n))))
  | 0 => Iso.refl _
  | j + 1 =>
    (singularHomology_sphere_succ_iso (R := R) (n := n + j) (by omega)).trans
    (singularHomology_sphere_iterate_iso hn j)

/-! ## `H_*(S^n)` in all degrees -/

/-- Connecting map of the stereographic cover of `S^{n+1}` in any degree. -/
def singularHomology_sphere_mayerVietoris $\delta$ _deg (n m :  $\mathbb{N}$ ) :
  (((singularChainComplexFunctor (ModuleCat.{0} k)).obj R).obj
    (.of (MetricSphere (n + 1)))) .homology (m + 1)  $\rightarrow$ 
    (((singularChainComplexFunctor (ModuleCat.{0} k)).obj R).obj
      (.of ( $\uparrow$ (puncturedNorth (n + 1)  $\cap$  puncturedSouth (n + 1))))) .homology m :=
  mayerVietoris $\delta$  (C := ModuleCat.{0} k) R
  (interInclLeft (puncturedNorth (n + 1)) (puncturedSouth (n + 1)))
  (interInclRight (puncturedNorth (n + 1)) (puncturedSouth (n + 1)))
  (subtypeIncl (MetricSphere (n + 1)) (puncturedNorth (n + 1)))
  (subtypeIncl (MetricSphere (n + 1)) (puncturedSouth (n + 1)))
  (subset_square_comm _ _)

```

```

(SingularExcision.IsExcisiveSubspaces_puncturedSphere (R := R) (n + 1)) m

lemma triadδ_sphere_isIso_of_pos (n m : ℕ) (hm : 0 < m) :
  IsIso (triadδ (C := ModuleCat.{0} k) R
    (interInclLeft (puncturedNorth (n + 1)) (puncturedSouth (n + 1)))
    (interInclRight (puncturedNorth (n + 1)) (puncturedSouth (n + 1))) m) :=
(triad_shortExact (C := ModuleCat.{0} k) R
  (interInclLeft (puncturedNorth (n + 1)) (puncturedSouth (n + 1)))
  (interInclRight (puncturedNorth (n + 1)) (puncturedSouth (n + 1)))) .isIso_δ
(m + 1) m (by simp)
(isZero_homology_chartSum (R := R) (n + 1) (m + 1) (Nat.succ_ne_zero m))
(isZero_homology_chartSum (R := R) (n + 1) m hm.ne')

lemma singularHomology_sphere_mayerVietorisδ_deg_isIso
  (n : ℕ) {m : ℕ} (hm : 0 < m) :
  IsIso (singularHomology_sphere_mayerVietorisδ_deg (R := R) n m) := by
dsimp [singularHomology_sphere_mayerVietorisδ_deg, mayerVietorisδ]
haveI := triadδ_sphere_isIso_of_pos (R := R) n m hm
infer_instance

/-- For  $m \geq 1$ ,  $H_{m+1}(S^{n+1})$  is isomorphic to  $H_m(S^n)$ . -/
def singularHomology_sphere_succ_iso_of_pos {m : ℕ} (hm : 0 < m) (n : ℕ) :
  (((singularHomologyFunctor (ModuleCat.{0} k) (m + 1)).obj R).obj
    (.of (MetricSphere (n + 1)))) ≅
  (((singularHomologyFunctor (ModuleCat.{0} k) m).obj R).obj
    (.of (MetricSphere n))) :=
haveI := singularHomology_sphere_mayerVietorisδ_deg_isIso (R := R) n hm
(asIso (singularHomology_sphere_mayerVietorisδ_deg (R := R) n m)).trans
  (singularHomologyIso_of_homotopyEquiv
    (homotopyEquiv_puncturedInter_sphere n) R m)

/-- The 0-sphere in EuclideanSpace  $\mathbb{R}$  (Fin 1) is homeomorphic to  $\{\pm 1\} \subset \mathbb{R}$ . -/
def homeoMetricSphere0 :
  MetricSphere 0  $\simeq_t$   $\uparrow$ (Metric.sphere (0 : ℝ) 1) :=
let e : EuclideanSpace ℝ (Fin 1)  $\simeq_t$  ℝ :=
  (PiLp.homeomorph 2 fun _ : Fin 1  $\mapsto$  ℝ).trans (Homeomorph.funUnique (Fin 1) ℝ)
have himg : e '' sphere 0 1 = sphere (0 : ℝ) 1 := by
  ext x
  constructor
  · rintro ⟨y, hy, rfl⟩
    have hy1 : ||y|| = 1 := by simp [mem_sphere_iff_norm] using hy
    have : |e y| = ||y|| := by
      simp [e, EuclideanSpace.norm_eq, Real.norm_eq_abs, Finset.univ_unique,
        Fin.default_eq_zero, Real.sqrt_sq_eq_abs, PiLp.homeomorph]
    simp [mem_sphere_iff_norm, Real.norm_eq_abs, this] using hy1
  · intro hx
    refine ⟨e.symm x, ?_, e.apply_symm_apply x⟩
    have hx1 : |x| = 1 := by simp [mem_sphere_iff_norm, Real.norm_eq_abs] using hx
    have : ||e.symm x|| = |x| := by
      simp [e, EuclideanSpace.norm_eq, Real.norm_eq_abs, Finset.univ_unique,
        Fin.default_eq_zero, Real.sqrt_sq_eq_abs, PiLp.homeomorph]
    simp [mem_sphere_iff_norm, this] using hx1
  (e.image (sphere 0 1)).trans (Homeomorph.setCongr himg)

lemma isZero_singularHomology_metricSphere0 {m : ℕ} (hm : m ≠ 0) :
  IsZero (((singularHomologyFunctor (ModuleCat.{0} k) m).obj R).obj
    (.of (MetricSphere 0))) :=
(isZero_singularHomology_sphere0 (C := ModuleCat.{0} k) R hm).of_iso
  (singularHomologyIso_of_homeomorph (homeoMetricSphere0) R m)

/--  $H_m(S^1) = 0$  for  $m \geq 2$ . -/
lemma isZero_singularHomology_sphere1_of_ge_two {m : ℕ} (hm : 2 ≤ m) :
  IsZero (((singularHomologyFunctor (ModuleCat.{0} k) m).obj R).obj

```

```

      (.of (MetricSphere 1))) := by
have hm0 : 0 < m - 1 := Nat.sub_pos_of_lt hm
convert (isZero_singularHomology_metricSphere0 (R := R) hm0.ne').of_iso
  (singularHomology_sphere_succ_iso_of_pos (R := R) hm0 0)
exact (Nat.sub_add_cancel (Nat.le_trans (by norm_num : 1 ≤ 2) hm)).symm

lemma triadδ_sphere_mono_zero (n : ℕ) :
  Mono (triadδ (C := ModuleCat.{0} k) R
    (interInclLeft (puncturedNorth n) (puncturedSouth n))
    (interInclRight (puncturedNorth n) (puncturedSouth n)) 0) :=
  (triad_shortExact (C := ModuleCat.{0} k) R
    (interInclLeft (puncturedNorth n) (puncturedSouth n))
    (interInclRight (puncturedNorth n) (puncturedSouth n))) .mono_δ
  1 0 (by simp) (isZero_homology_chartSum (R := R) n 1 one_ne_zero)

instance pathConnectedSpace_puncturedInter {n : ℕ} [NeZero n] :
  PathConnectedSpace ↑(puncturedNorth (n + 1) ∩ puncturedSouth (n + 1)) :=
  pathConnectedSpace_of_homotopyEquiv (homotopyEquiv_puncturedInter_sphere n)

lemma homologyMap_mvIncl_zero_mono
  {n : ℕ} [NeZero n] :
  Mono (homologyMap
    (mvIncl (C := ModuleCat.{0} k) R
      (interInclLeft (puncturedNorth (n + 1)) (puncturedSouth (n + 1)))
      (interInclRight (puncturedNorth (n + 1)) (puncturedSouth (n + 1)))) 0) := by
have : IsIso (homologyMap
  (((singularChainComplexFunctor (ModuleCat.{0} k)).obj R).map
    (interInclLeft (puncturedNorth (n + 1)) (puncturedSouth (n + 1)))) 0) :=
  inferInstance
have hfst :
  homologyMap
    (mvIncl (C := ModuleCat.{0} k) R
      (interInclLeft (puncturedNorth (n + 1)) (puncturedSouth (n + 1)))
      (interInclRight (puncturedNorth (n + 1)) (puncturedSouth (n + 1)))) 0 >>
  homologyMap
    (biprod.fst :
      ((singularChainComplexFunctor (ModuleCat.{0} k)).obj R).obj
        (.of ↑(puncturedNorth (n + 1))) ⊕
      ((singularChainComplexFunctor (ModuleCat.{0} k)).obj R).obj
        (.of ↑(puncturedSouth (n + 1))) →
      ((singularChainComplexFunctor (ModuleCat.{0} k)).obj R).obj
        (.of ↑(puncturedNorth (n + 1)))) 0 =
  homologyMap
    (((singularChainComplexFunctor (ModuleCat.{0} k)).obj R).map
      (interInclLeft (puncturedNorth (n + 1)) (puncturedSouth (n + 1)))) 0 := by
rw [← homologyMap_comp]
congr 1
exact biprod.lift_fst _ _
exact mono_of_mono_fac hfst

/-- `H_1(S^n) = 0` for `n ≥ 2`. -/
lemma isZero_singularHomology_sphere_one {n : ℕ} (hn : 2 ≤ n) :
  IsZero (((singularHomologyFunctor (ModuleCat.{0} k) 1).obj R).obj
    (.of (MetricSphere n))) := by
have hn1 : n - 1 + 1 = n := Nat.sub_add_cancel (by omega)
haveI : NeZero (n - 1) := ⟨by omega⟩
haveI : PathConnectedSpace ↑(puncturedNorth n ∩ puncturedSouth n) := by
  rw [show n = n - 1 + 1 from hn1.symm]
  infer_instance
have hmono : Mono (homologyMap
  (mvIncl (C := ModuleCat.{0} k) R
    (interInclLeft (puncturedNorth n) (puncturedSouth n))
    (interInclRight (puncturedNorth n) (puncturedSouth n)))) 0) := by

```

```

have hfst :
  homologyMap
    (mvIncl (C := ModuleCat.{0} k) R
      (interInclLeft (puncturedNorth n) (puncturedSouth n))
      (interInclRight (puncturedNorth n) (puncturedSouth n))) 0 >>
  homologyMap
    (biprod.fst :
      ((singularChainComplexFunctor (ModuleCat.{0} k)).obj R).obj
        (.of ↑(puncturedNorth n)) ⊕
        ((singularChainComplexFunctor (ModuleCat.{0} k)).obj R).obj
          (.of ↑(puncturedSouth n)) →
        ((singularChainComplexFunctor (ModuleCat.{0} k)).obj R).obj
          (.of ↑(puncturedNorth n))) 0 =
    homologyMap
      (((singularChainComplexFunctor (ModuleCat.{0} k)).obj R).map
        (interInclLeft (puncturedNorth n) (puncturedSouth n))) 0 := by
  rw [← homologyMap_comp]
  congr 1
  exact biprod.lift_fst _ _
  exact mono_of_mono_fac hfst
have hz := (triad_shortExact (C := ModuleCat.{0} k) R
  (interInclLeft (puncturedNorth n) (puncturedSouth n))
  (interInclRight (puncturedNorth n) (puncturedSouth n))).δ_comp
1 0 (by simp)
have hδ0 : triadδ (C := ModuleCat.{0} k) R
  (interInclLeft (puncturedNorth n) (puncturedSouth n))
  (interInclRight (puncturedNorth n) (puncturedSouth n)) 0 = 0 := by
  refine (cancel_mono (homologyMap
    (mvIncl (C := ModuleCat.{0} k) R
      (interInclLeft (puncturedNorth n) (puncturedSouth n))
      (interInclRight (puncturedNorth n) (puncturedSouth n)))) 0)).1 ?_
  erw [hz]
  exact (Limits.zero_comp (f := homologyMap
    (mvIncl (C := ModuleCat.{0} k) R
      (interInclLeft (puncturedNorth n) (puncturedSouth n))
      (interInclRight (puncturedNorth n) (puncturedSouth n)))) 0)).symm
haveI := triadδ_sphere_mono_zero (R := R) n
refine IsZero.of_iso ?_
  (excisiveHomologyIso (C := ModuleCat.{0} k) R
    (interInclLeft (puncturedNorth n) (puncturedSouth n))
    (interInclRight (puncturedNorth n) (puncturedSouth n))
    (subtypeIncl (MetricSphere n) (puncturedNorth n))
    (subtypeIncl (MetricSphere n) (puncturedSouth n))
    (subset_square_comm _ _))
  (SingularExcision.IsExcisiveSubspaces_puncturedSphere (R := R) n) 1).symm
rw [IsZero.iff_id_eq_zero]
refine (cancel_mono (triadδ (C := ModuleCat.{0} k) R
  (interInclLeft (puncturedNorth n) (puncturedSouth n))
  (interInclRight (puncturedNorth n) (puncturedSouth n)) 0)).1 ?_
  rw [hδ0]
  simp

/-- Drop one dimension:  $H_a(S^b) \cong H_{a-1}(S^{b-1})$  when  $a \geq 2$  and  $b \geq 1$ . -/
def singularHomology_sphere_down_iso
  {a b : ℕ} (ha : 2 ≤ a) (hb : 1 ≤ b) :
  (((singularHomologyFunctor (ModuleCat.{0} k) a).obj R).obj
    (.of (MetricSphere b))) ≅
  (((singularHomologyFunctor (ModuleCat.{0} k) (a - 1)).obj R).obj
    (.of (MetricSphere (b - 1))))) :=
  by
    have hm : 0 < a - 1 := Nat.sub_pos_of_lt ha
    have ha' : a - 1 + 1 = a := Nat.sub_add_cancel (by omega)
    have hb' : b - 1 + 1 = b := Nat.sub_add_cancel hb

```

```

have type_eq :
  (((singularHomologyFunctor (ModuleCat.{0} k) a).obj R).obj
    (.of (MetricSphere b))) =
    (((singularHomologyFunctor (ModuleCat.{0} k) (a - 1 + 1)).obj R).obj
      (.of (MetricSphere (b - 1 + 1)))) := by
  rw [ha', hb']
exact (eqToIso type_eq).trans
  (singularHomology_sphere_succ_iso_of_pos (R := R) hm (b - 1))

/-- For  $n \geq 1$ ,  $H_n(S^n)$  is isomorphic to  $H_1(S^1)$ . -/
def singularHomology_sphere_iso_sphere1 {n : ℕ} (hn : 0 < n) :
  (((singularHomologyFunctor (ModuleCat.{0} k) n).obj R).obj
    (.of (MetricSphere n))) ≅
    (((singularHomologyFunctor (ModuleCat.{0} k) 1).obj R).obj
      (.of (MetricSphere 1)))) :=
  by
    have hn' : 1 + (n - 1) = n := Nat.add_sub_cancel' hn
    have type_eq :
      (((singularHomologyFunctor (ModuleCat.{0} k) n).obj R).obj
        (.of (MetricSphere n))) =
        (((singularHomologyFunctor (ModuleCat.{0} k) (1 + (n - 1))).obj R).obj
          (.of (MetricSphere (1 + (n - 1)))))) := by
      rw [hn']
    exact (eqToIso type_eq).trans
      (singularHomology_sphere_iterate_iso (R := R) (n := 1)
        (by norm_num : 0 < 1) (n - 1))

/-- Positive-degree singular homology of  $S^n$  vanishes off the top degree. -/
lemma isZero_singularHomology_sphere
  {n m : ℕ} (hn : 0 < n) (hm : m ≠ 0) (hne : m ≠ n) :
  IsZero (((singularHomologyFunctor (ModuleCat.{0} k) m).obj R).obj
    (.of (MetricSphere n))) := by
  rcases lt_or_gt_of_ne hne with hlt | hgt
  · --  $0 < m < n$ : induct on  $n$ , reducing to  $H_1$  of a sphere of dimension  $\geq 2$ .
    induction n generalizing m with
    | zero => exact (hn.ne rfl).elim
    | succ n ih =>
      have hn0 : n ≠ 0 := by omega
      cases m with
      | zero => exact (hm rfl).elim
      | succ m =>
        cases m with
        | zero =>
          exact isZero_singularHomology_sphere_one (R := R) (by omega)
        | succ m =>
          refine (ih (m := m + 1) (by omega) (by omega) (by omega)
            (by omega)).of_iso
            (singularHomology_sphere_down_iso (R := R) (a := m + 2) (b := n + 1)
              (by omega) (by omega))
  · --  $m > n \geq 1$ : induct on  $n$ , reducing to  $H_*(S^1)$  with degree  $\geq 2$ .
    induction n generalizing m with
    | zero => exact (hn.ne rfl).elim
    | succ n ih =>
      by_cases hn0 : n = 0
      · subst hn0
        exact isZero_singularHomology_sphere1_of_ge_two (R := R) (by omega)
      · refine (ih (m := m - 1) (by omega) (by omega) (by omega)
          (by omega)).of_iso
          (singularHomology_sphere_down_iso (R := R) (a := m) (b := n + 1)
            (by omega) (by omega))

/-! ##  $H_1(S^1) \cong \mathbb{R}$  and the full calculation of  $H_*(S^n)$  -/

```

```

instance preservesBinaryBiproduct_homologyFunctor (n : ℕ)
  (K L : ChainComplex (ModuleCat.{0} k) ℕ) :
  PreservesBinaryBiproduct K L
  (homologyFunctor (ModuleCat.{0} k) (ComplexShape.down ℕ) n) :=
  preservesBinaryBiproduct_of_preservesBiproduct _ K L

lemma homologyMap_mvIncl
  {W U V : TopCat.{0}} (iU : W → U) (iV : W → V) (n : ℕ) :
  homologyMap (mvIncl (C := ModuleCat.{0} k) R iU iV) n >>
  ((homologyFunctor (ModuleCat.{0} k) (ComplexShape.down ℕ) n).mapBiprod
    (((singularChainComplexFunctor (ModuleCat.{0} k)).obj R).obj U)
    (((singularChainComplexFunctor (ModuleCat.{0} k)).obj R).obj V)).hom =
  biprod.lift
    (homologyMap (((singularChainComplexFunctor (ModuleCat.{0} k)).obj R).map iU) n)
    (-homologyMap (((singularChainComplexFunctor (ModuleCat.{0} k)).obj R).map iV) n) := by
  let F := homologyFunctor (ModuleCat.{0} k) (ComplexShape.down ℕ) n
  let KU := ((singularChainComplexFunctor (ModuleCat.{0} k)).obj R).obj U
  let KV := ((singularChainComplexFunctor (ModuleCat.{0} k)).obj R).obj V
  have h := biprod.map_lift_mapBiprod (F := F) (X := KU) (Y := KV)
    (((singularChainComplexFunctor (ModuleCat.{0} k)).obj R).map iU)
    (-(((singularChainComplexFunctor (ModuleCat.{0} k)).obj R).map iV))
  simpa [mvIncl, homologyMap_neg] using h

lemma biprod_lift_id_neg_id_mono (X : ModuleCat.{0} k) :
  Mono (biprod.lift (⊣ X) (⊣ X)) :=
  mono_of_mono_fac (biprod.lift_fst (⊣ X) (⊣ X))

lemma biprod_lift_comp_map {W X Y X' Y' : ModuleCat.{0} k}
  (f : W → X) (g : W → Y) (a : X → X') (b : Y → Y') :
  biprod.lift f g >> biprod.map a b = biprod.lift (f >> a) (g >> b) := by
  refine biprod.hom_ext _ _ ?_ ?_
  · simp
  · simp

lemma biprod_comp_lift {W' W X Y : ModuleCat.{0} k}
  (h : W' → W) (f : W → X) (g : W → Y) :
  h >> biprod.lift f g = biprod.lift (h >> f) (h >> g) := by
  refine biprod.hom_ext _ _ ?_ ?_
  · simp
  · simp

lemma homologyMap_mvIncl_comp_aug
  {W U V : TopCat.{0}} [PathConnectedSpace U] [PathConnectedSpace V]
  (iU : W → U) (iV : W → V) :
  homologyMap (mvIncl (C := ModuleCat.{0} k) R iU iV) 0 >>
  ((homologyFunctor (ModuleCat.{0} k) (ComplexShape.down ℕ) 0).mapBiprod
    (((singularChainComplexFunctor (ModuleCat.{0} k)).obj R).obj U)
    (((singularChainComplexFunctor (ModuleCat.{0} k)).obj R).obj V)).hom >>
  biprod.map (U.singularHomology0 R) (V.singularHomology0 R) =
  W.singularHomology0 R >> biprod.lift (⊣ R) (⊣ R) := by
  have hmv := homologyMap_mvIncl (R := R) iU iV 0
  have hU := singularHomology0_natural iU R
  have hV := singularHomology0_natural iV R
  have hneg :
    (-homologyMap (((singularChainComplexFunctor (ModuleCat.{0} k)).obj R).map iV) 0) >>
    V.singularHomology0 R =
    -W.singularHomology0 R := by
    rw [Preadditive.neg_comp, hV]; rfl
  calc
    homologyMap (mvIncl (C := ModuleCat.{0} k) R iU iV) 0 >>
    ((homologyFunctor (ModuleCat.{0} k) (ComplexShape.down ℕ) 0).mapBiprod
      (((singularChainComplexFunctor (ModuleCat.{0} k)).obj R).obj U)
      (((singularChainComplexFunctor (ModuleCat.{0} k)).obj R).obj V)).hom >>

```

```

    biprod.map (U.singularHomology0 R) (V.singularHomology0 R) =
    biprod.lift
      (homologyMap (((singularChainComplexFunctor (ModuleCat.{0} k)).obj R).map iU) 0)
      (-homologyMap (((singularChainComplexFunctor (ModuleCat.{0} k)).obj R).map iV) 0) >>
    biprod.map (U.singularHomology0 R) (V.singularHomology0 R) := by
    rw [← Category.assoc, hmv]; rfl
  _ = biprod.lift (W.singularHomology0 R) (-W.singularHomology0 R) := by
    rw [biprod_lift_comp_map, hU, hneg]; rfl
  _ = W.singularHomology0 R >> biprod.lift (⊘ R) (⊘ R) := by
    rw [biprod_comp_lift, Category.comp_id, Preadditive.comp_neg, Category.comp_id]

/-- Connecting map of the triad is a kernel of `mvIncl_*` when it is mono. -/
noncomputable def triadδ_isoKernel
  {W U V : TopCat.{0}} (iU : W → U) (iV : W → V) [Mono iU]
  [Mono (triadδ (C := ModuleCat.{0} k) R iU iV 0)] :
  (coverChains (C := ModuleCat.{0} k) R iU iV).homology 1 ≅
  kernel (homologyMap (mvIncl (C := ModuleCat.{0} k) R iU iV) 0) :=
  (limit.isoLimitCone
    (KernelFork.of
      (triadδ (C := ModuleCat.{0} k) R iU iV 0)
      (by
        have h := (triad_shortExact (C := ModuleCat.{0} k) R iU iV).δ_comp
          1 0 (by simp)
        exact h),
      (triad_exact_at_W (C := ModuleCat.{0} k) R iU iV 0).fIsKernel)).symm

def sphere0_pos : ↑(Metric.sphere (0 : ℝ) 1) :=
  ⟨1, by simp [mem_sphere_iff_norm]⟩

def sphere0_neg : ↑(Metric.sphere (0 : ℝ) 1) :=
  ⟨-1, by simp [mem_sphere_iff_norm]⟩

lemma sphere0_eq_pos_or_neg (x : ↑(Metric.sphere (0 : ℝ) 1)) :
  x = sphere0_pos ∨ x = sphere0_neg := by
  have hx : |(x : ℝ)| = 1 := by
    have hmem := x.2
    rw [mem_sphere_iff_norm, Real.norm_eq_abs, sub_zero] at hmem
    exact hmem
  rcases (abs_eq (show (0 : ℝ) ≤ 1 from by norm_num)).mp hx with h | h
  · exact Or.inl (Subtype.ext h)
  · exact Or.inr (Subtype.ext h)

lemma not_joined_sphere0_pos_neg :
  ¬ Joined sphere0_pos sphere0_neg := by
  intro h
  have h0 : h.somePath 0 = sphere0_pos := Path.source _
  have h1 : h.somePath 1 = sphere0_neg := Path.target _
  have heq := TotallyDisconnectedSpace.eq_of_continuous
    (h.somePath : unitInterval → ↑(Metric.sphere (0 : ℝ) 1))
    h.somePath.continuous (0 : unitInterval) (1 : unitInterval)
  rw [h0, h1] at heq
  have : (1 : ℝ) = -1 := congrArg Subtype.val heq
  norm_num at this

lemma zerothHomotopy_sphere0_pos_ne_neg :
  ZerothHomotopy.mk sphere0_pos ≠ ZerothHomotopy.mk sphere0_neg := by
  intro h
  exact not_joined_sphere0_pos_neg (Quotient.exact h)

lemma zerothHomotopy_sphere0_eq (c : ZerothHomotopy ↑(Metric.sphere (0 : ℝ) 1)) :
  c = ZerothHomotopy.mk sphere0_pos ∨ c = ZerothHomotopy.mk sphere0_neg := by
  obtain ⟨x, rfl⟩ := ZerothHomotopy.mk_surjective c
  rcases sphere0_eq_pos_or_neg x with hx | hx

```

```

· exact Or.inl (congrArg _ hx)
· exact Or.inr (congrArg _ hx)

noncomputable def equivZerothHomotopy_sphere0 :
  ZerothHomotopy  $\uparrow$  (Metric.sphere (0 :  $\mathbb{R}$ ) 1)  $\simeq$  Bool := by
classical
refine
{ toFun := fun c => decide (c = ZerothHomotopy.mk sphere0_pos)
  invFun := fun b => bif b then ZerothHomotopy.mk sphere0_pos
    else ZerothHomotopy.mk sphere0_neg
  left_inv := ?_
  right_inv := ?_ }
· intro c
rcases zerothHomotopy_sphere0_eq c with hc | hc
· simp [hc]
· have hne :  $\neg$  ZerothHomotopy.mk sphere0_neg = ZerothHomotopy.mk sphere0_pos :=
  zerothHomotopy_sphere0_pos_ne_neg.symm
  simp [hc, hne]
· intro b
cases b with
| true => simp
| false =>
  have hne :  $\neg$  ZerothHomotopy.mk sphere0_neg = ZerothHomotopy.mk sphere0_pos :=
    zerothHomotopy_sphere0_pos_ne_neg.symm
    simp [hne]

/-- Coproduct of two copies of `R`, indexed by `Bool`, is the biproduct. -/
noncomputable def sigmaBoolIsoBiprod :
  ( $\coprod$  fun _ : Bool  $\mapsto$  R)  $\cong$  R  $\boxplus$  R where
hom := Sigma.desc fun b => bif b then biprod.inl else biprod.inr
inv := biprod.desc (Sigma. $\iota$  (fun _ : Bool  $\mapsto$  R) true) (Sigma. $\iota$  (fun _ : Bool  $\mapsto$  R) false)
hom_inv_id := by
  refine Sigma.hom_ext _ _ ?_
  intro b
  cases b with
  | false =>
    rw [Sigma. $\iota$ _desc_assoc]
    change biprod.inr  $\gg$  biprod.desc _ _ = _
    rw [biprod.inr_desc, Category.comp_id]
  | true =>
    rw [Sigma. $\iota$ _desc_assoc]
    change biprod.inl  $\gg$  biprod.desc _ _ = _
    rw [biprod.inl_desc, Category.comp_id]
inv_hom_id := by
  refine biprod.hom_ext' _ _ ?_ ?_
  · rw [biprod.inl_desc_assoc, Sigma. $\iota$ _desc, Category.comp_id]
    rfl
  · rw [biprod.inr_desc_assoc, Sigma. $\iota$ _desc, Category.comp_id]
    rfl

/-- Shear `(x, y)  $\mapsto$  (x, x+y)` on a biproduct. -/
noncomputable def biprodShearIso (X : ModuleCat.{0} k) : X  $\boxplus$  X  $\cong$  X  $\boxplus$  X where
hom := biprod.lift biprod.fst (biprod.fst + biprod.snd)
inv := biprod.lift biprod.fst (biprod.snd + (-biprod.fst))
hom_inv_id := by
  refine biprod.hom_ext _ _ ?_ ?_
  · simp
  · simp [sub_eq_add_neg, add_assoc, add_neg_cancel_comm_assoc, add_neg_cancel]
inv_hom_id := by
  refine biprod.hom_ext _ _ ?_ ?_
  · simp
  · simp [sub_eq_add_neg, add_assoc, add_neg_cancel_comm_assoc, add_neg_cancel]

```

```

lemma biprodShearIso_hom_snd (X : ModuleCat.{0} k) :
  (biprodShearIso X).hom >> biprod.snd = biprod.fst + biprod.snd :=
  biprod.lift_snd _ _

lemma biprod_fst_add_snd_eq_desc (X : ModuleCat.{0} k) :
  (biprod.fst : X  $\boxplus$  X  $\longrightarrow$  X) + biprod.snd = biprod.desc ( $\mathbb{K}$  X) ( $\mathbb{K}$  X) := by
  refine biprod.hom_ext' _ _ ?_ ?_
  · simp
  · simp

lemma sigmaBoolIsoBiprod_hom_desc :
  (sigmaBoolIsoBiprod (R := R)).hom >> biprod.desc ( $\mathbb{K}$  R) ( $\mathbb{K}$  R) =
    Sigma.desc fun _ : Bool  $\mapsto$   $\mathbb{K}$  R := by
  refine Sigma.hom_ext _ _ ?_
  intro b
  cases b with
  | false =>
    dsimp [sigmaBoolIsoBiprod]
    rw [Sigma.l_desc_assoc]
    change biprod.inr >> biprod.desc ( $\mathbb{K}$  R) ( $\mathbb{K}$  R) = _
    rw [biprod.inr_desc, Sigma.l_desc]
  | true =>
    dsimp [sigmaBoolIsoBiprod]
    rw [Sigma.l_desc_assoc]
    change biprod.inl >> biprod.desc ( $\mathbb{K}$  R) ( $\mathbb{K}$  R) = _
    rw [biprod.inl_desc, Sigma.l_desc]

lemma sigma_desc_id_sphere0 :
  (Sigma.reindex equivZerothHomotopy_sphere0
    (fun _ : Bool  $\mapsto$  R)).hom >>
    Sigma.desc (fun _ : Bool  $\mapsto$   $\mathbb{K}$  R) =
    Sigma.desc fun _ : ZerothHomotopy  $\uparrow$ (Metric.sphere (0 :  $\mathbb{R}$ ) 1)  $\mapsto$   $\mathbb{K}$  R := by
  refine Sigma.hom_ext _ _ ?_
  intro c
  rw [Sigma.l_reindex_hom_assoc]
  refine Eq.trans (Sigma.l_desc (fun _ : Bool  $\mapsto$   $\mathbb{K}$  R)
    (equivZerothHomotopy_sphere0 c)) ?_
  exact (Sigma.l_desc (fun _ : ZerothHomotopy  $\uparrow$ (Metric.sphere (0 :  $\mathbb{R}$ ) 1)  $\mapsto$   $\mathbb{K}$  R) c).symm

/-- `ker(epsilon : H_0(S^0)  $\rightarrow$  R)  $\cong$  R`. -/
noncomputable def kernel_singularHomology_0E_sphere0 :
  kernel ((TopCat.of  $\uparrow$ (Metric.sphere (0 :  $\mathbb{R}$ ) 1)).singularHomology_0E R)  $\cong$  R :=
  let e0 := TopCat.singularHomology_0Iso (.of  $\uparrow$ (Metric.sphere (0 :  $\mathbb{R}$ ) 1)) R
  let ere : ( $\coprod$  fun _ : ZerothHomotopy  $\uparrow$ (Metric.sphere (0 :  $\mathbb{R}$ ) 1)  $\mapsto$  R)  $\cong$ 
    ( $\coprod$  fun _ : Bool  $\mapsto$  R) :=
    Sigma.reindex equivZerothHomotopy_sphere0 (fun _ : Bool  $\mapsto$  R)
  have hEps : (TopCat.of  $\uparrow$ (Metric.sphere (0 :  $\mathbb{R}$ ) 1)).singularHomology_0E R =
    e0.hom >> ere.hom >> (sigmaBoolIsoBiprod (R := R)).hom >>
      (biprodShearIso R).hom >> biprod.snd := by
  have h1 := TopCat.singularHomology_0Iso_sigma_desc_id
    (.of  $\uparrow$ (Metric.sphere (0 :  $\mathbb{R}$ ) 1)) R
  have h2 := sigma_desc_id_sphere0 (R := R)
  have h3 := sigmaBoolIsoBiprod_hom_desc (R := R)
  have h4 := biprodShearIso_hom_snd (X := R)
  have h5 := biprod_fst_add_snd_eq_desc (X := R)
  calc
    (TopCat.of  $\uparrow$ (Metric.sphere (0 :  $\mathbb{R}$ ) 1)).singularHomology_0E R =
      e0.hom >> Sigma.desc (fun _  $\mapsto$   $\mathbb{K}$  R) := h1.symm
    _ = e0.hom >> ere.hom >> Sigma.desc fun _ : Bool  $\mapsto$   $\mathbb{K}$  R := by
      rw [← h2]; rfl
    _ = e0.hom >> ere.hom >> (sigmaBoolIsoBiprod (R := R)).hom >>
      biprod.desc ( $\mathbb{K}$  R) ( $\mathbb{K}$  R) := by
      rw [← h3]

```

```

_ = e0.hom >> ere.hom >> (sigmaBoolIsoBiprod (R := R)).hom >>
  (biprod.fst + biprod.snd) := by
  rw [h5]
_ = e0.hom >> ere.hom >> (sigmaBoolIsoBiprod (R := R)).hom >>
  (biprodShearIso R).hom >> biprod.snd := by
  rw [← h4]
(kernel.congr _ _ hEps).trans <|
(kernelIsIsoComp (e0.hom >> ere.hom >> (sigmaBoolIsoBiprod (R := R)).hom >>
  (biprodShearIso R).hom) biprod.snd).trans
kernelBiprodSndIso

/-- Stereographic intersection of  $S^1$  is homotopy-equivalent to  $S^0 \subset \mathbb{R}$ . -/
def homotopyEquiv_puncturedInter_sphere0 :
  ↑(puncturedNorth 1 ∩ puncturedSouth 1)  $\simeq_h$  ↑(Metric.sphere (0 : ℝ) 1) :=
  (homotopyEquiv_puncturedInter_sphere 0).trans homeoMetricSphere0.toHomotopyEquiv

/--  $H_1(S^1; R) \cong R$ . -/
noncomputable def singularHomology_sphere1 :
  (((singularHomologyFunctor (ModuleCat.{0} k) 1).obj R).obj
    (.of (MetricSphere 1)))  $\cong R :=$ 
  let iU := interInclLeft (puncturedNorth 1) (puncturedSouth 1)
  let iV := interInclRight (puncturedNorth 1) (puncturedSouth 1)
  haveI : Mono (triadδ (C := ModuleCat.{0} k) R iU iV 0) :=
    triadδ_sphere_mono_zero (R := R) 1
  have hcmp := homologyMap_mvIncl_comp_aug (R := R) iU iV
  let F := homologyFunctor (ModuleCat.{0} k) (ComplexShape.down ℕ) 0
  let KU := ((singularChainComplexFunctor (ModuleCat.{0} k)).obj R).obj
    (.of ↑(puncturedNorth 1))
  let KV := ((singularChainComplexFunctor (ModuleCat.{0} k)).obj R).obj
    (.of ↑(puncturedSouth 1))
  let epsU := (TopCat.of ↑(puncturedNorth 1)).singularHomology0 R
  let epsV := (TopCat.of ↑(puncturedSouth 1)).singularHomology0 R
  haveI : IsIso epsU := inferInstance
  haveI : IsIso epsV := inferInstance
  let thetaIso :
    (((singularHomologyFunctor (ModuleCat.{0} k) 0).obj R).obj
      (.of ↑(puncturedNorth 1)))  $\boxplus$ 
      (((singularHomologyFunctor (ModuleCat.{0} k) 0).obj R).obj
        (.of ↑(puncturedSouth 1)))  $\cong R \boxplus R :=$ 
  { hom := biprod.map epsU epsV
    inv := biprod.map (inv epsU) (inv epsV)
    hom_inv_id := by
      refine biprod.hom_ext _ _ ?_ ?_
      · simp
      · simp
    inv_hom_id := by
      refine biprod.hom_ext _ _ ?_ ?_
      · simp
      · simp }
  let q := (F.mapBiprod KU KV).trans thetaIso
  haveI : Mono (biprod.lift (⊗ R) (⊗ R)) := biprod_lift_id_neg_id_mono R
  let eW := singularHomologyIso_of_homotopyEquiv
    (homotopyEquiv_puncturedInter_sphere0) R 0
  have hnata : eW.hom >>
    (TopCat.of ↑(Metric.sphere (0 : ℝ) 1)).singularHomology0 R =
    (TopCat.of ↑(puncturedNorth 1 ∩ puncturedSouth 1)).singularHomology0 R :=
    singularHomology0_natural (TopCat.ofHom homotopyEquiv_puncturedInter_sphere0.toFun) R
  (excisiveHomologyIso (C := ModuleCat.{0} k) R iU iV
    (subtypeIncl (MetricSphere 1) (puncturedNorth 1))
    (subtypeIncl (MetricSphere 1) (puncturedSouth 1))
    (subset_square_comm _ _))
    (SingularExcision.IsExcisiveSubspaces_puncturedSphere (R := R) 1) 1).symm.trans <|
    (triadδ_isoKernel (R := R) iU iV).trans <|

```

```

(kernel.mapIso
  (f := homologyMap (mvIncl (C := ModuleCat.{0} k) R iU iV) 0)
  ((TopCat.of  $\uparrow$ (puncturedNorth 1  $\cap$  puncturedSouth 1)).singularHomology $_{0\in R} \gg$ 
    biprod.lift ( $\not\equiv$  R) ( $\not\equiv$  R))
  (Iso.refl _) q (by
    simp using hcmp)).trans <|
(kernelCompMono
  ((TopCat.of  $\uparrow$ (puncturedNorth 1  $\cap$  puncturedSouth 1)).singularHomology $_{0\in R}$ 
    (biprod.lift ( $\not\equiv$  R) ( $\not\equiv$  R))).trans <|
  (kernel.congr _ _ hnat.symm)).trans <|
  (kernelIsIsoComp eW.hom
    ((TopCat.of  $\uparrow$ (Metric.sphere (0 :  $\mathbb{R}$ ) 1)).singularHomology $_{0\in R}$ )).trans
    (kernel_singularHomology $_{0\in \text{sphere0}}$  (R := R))

/-- For `n > 0`, `H_n(S^n; R)  $\cong$  R`. -/
noncomputable def singularHomology_sphere {n :  $\mathbb{N}$ } (hn : 0 < n) :
  (((singularHomologyFunctor (ModuleCat.{0} k) n).obj R).obj
    (.of (MetricSphere n)))  $\cong$  R :=
  (singularHomology_sphere_iso_sphere1 (R := R) hn).trans
  (singularHomology_sphere1 (R := R))

/-- `H_0(S^2; R)  $\cong$  R` by path-connectedness. -/
noncomputable def singularHomology_sphere2_zero :
  (((singularHomologyFunctor (ModuleCat.{0} k) 0).obj R).obj
    (.of (MetricSphere 2)))  $\cong$  R :=
  haveI : NeZero (2 :  $\mathbb{N}$ ) := (by decide)
  singularHomology_0Iso_pathConnected _ R

/-- `H_1(S^2; R) = 0`. -/
lemma isZero_singularHomology_sphere2_one :
  IsZero (((singularHomologyFunctor (ModuleCat.{0} k) 1).obj R).obj
    (.of (MetricSphere 2))) :=
  isZero_singularHomology_sphere (R := R) (n := 2) (m := 1)
  (by decide) (by decide) (by decide)

/-- `H_2(S^2; R)  $\cong$  R`. -/
noncomputable def singularHomology_sphere2 :
  (((singularHomologyFunctor (ModuleCat.{0} k) 2).obj R).obj
    (.of (MetricSphere 2)))  $\cong$  R :=
  singularHomology_sphere (R := R) (n := 2) (by decide)

/-- The 3D classification from a cellular 2-complex whose Betti numbers
  equal the singular Betti numbers of `Metric.sphere` in dimension 2
  (`singularHomology_sphere2_zero`, `isZero_singularHomology_sphere2_one`,
  `singularHomology_sphere2`). Euler-Poincaré supplies `V+F=E+2`; regularity
  then forces a Platonic pair. Palomar compared types are unchanged. -/
theorem platonic_solids_3d_of_sphere
  {K : Type*} [Field K]
  {C2 C1 C0 : Type*}
  [AddCommGroup C2] [Module K C2] [FiniteDimensional K C2]
  [AddCommGroup C1] [Module K C1] [FiniteDimensional K C1]
  [AddCommGroup C0] [Module K C0] [FiniteDimensional K C0]
  (d2 : C2  $\rightarrow_1$  [K] C1) (d1 : C1  $\rightarrow_1$  [K] C0)
  (V E F p q :  $\mathbb{N}$ )
  (hp : 3  $\leq$  p) (hq : 3  $\leq$  q)
  (h_edges_faces : p * F = 2 * E)
  (h_edges_verts : q * V = 2 * E)
  (hV : Module.finrank K C0 = V)
  (hE : Module.finrank K C1 = E)
  (hF : Module.finrank K C2 = F)
  (hcomp : d1.comp d2 = 0)
  (hs0 : betti0 d1 = 1)
  (hs1 : betti1 d2 d1 = 0)

```

```

    (hs2 : betti2 d2 = 1) :
    IsPlatonicPair p q :=
    platonic_solids_3d_of_homology d2 d1 V E F p q hp hq
    h_edges_faces h_edges_verts hV hE hF hcomp hs0 hs1 hs2

end

```

A.22 PlatonicSolids/Sphere2Homology.lean

Cellular $H_*(S^2; \mathbb{Q})$ of the tetrahedron surface (129 lines).

```

/-
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-/
import Mathlib.LinearAlgebra.Matrix.Notation
import Mathlib.LinearAlgebra.Matrix.Rank
import Mathlib.LinearAlgebra.Matrix.ToLin
import Mathlib.Tactic
import PlatonicSolids.EulerPoincare
import PlatonicSolids.Incidence

/-!
Paper §2.1. Cellular homology of the standard simplicial  $(S^2)$ :
the boundary of a tetrahedron (4 vertices, 6 edges, 4 triangles).

This is the usual CW / simplicial calculation
`H0 ≅ ℚ`, `H1 = 0`, `H2 ≅ ℚ`. Singular `H0` of `Metric.sphere`
and `H_*(S^0)` are in `PlatonicSolids/SingularHomology.lean`;
the  $S^3$ -sphere calculation is `PlatonicSolids/Sphere3Homology.lean`.
-/

open scoped Matrix
open Matrix Module

/-- Oriented incidence vertices ← edges: columns `01,02,03,12,13,23`. -/
def tetra_d1 : Matrix (Fin 4) (Fin 6) ℚ :=
  !![ -1, -1, -1, 0, 0, 0;
      1, 0, 0, -1, -1, 0;
      0, 1, 0, 1, 0, -1;
      0, 0, 1, 0, 1, 1]

/-- Oriented incidence edges ← faces: columns `012,013,023,123`. -/
def tetra_d2 : Matrix (Fin 6) (Fin 4) ℚ :=
  !![ 1, 1, 0, 0;
      -1, 0, 1, 0;
      0, -1, -1, 0;
      1, 0, 0, 1;
      0, 1, 0, -1;
      0, 0, 1, 1]

lemma tetra_comp : tetra_d1 * tetra_d2 = 0 := by
  ext i j
  fin_cases i <|> fin_cases j <|> simp [tetra_d1, tetra_d2]

noncomputable def tetra_del1 : (Fin 6 → ℚ) →1 [ℚ] (Fin 4 → ℚ) := tetra_d1.toLin'
noncomputable def tetra_del2 : (Fin 4 → ℚ) →1 [ℚ] (Fin 6 → ℚ) := tetra_d2.toLin'

lemma tetra_del_comp : tetra_del1.comp tetra_del2 = 0 := by
  rw [tetra_del1, tetra_del2, ← toLin'_mul, tetra_comp]
  exact map_zero _

lemma tetra_d1_rank_ge : 3 ≤ tetra_d1.rank := by

```

```

have hI : ((1 : Matrix (Fin 3) (Fin 3) ℚ).rank) = 3 := rank_one
have hsub :
  tetra_d1.submatrix ![1, 2, 3] ![0, 1, 2] = (1 : Matrix (Fin 3) (Fin 3) ℚ) := by
  ext i j
  fin_cases i <=> fin_cases j <=> simp [tetra_d1]
have : (tetra_d1.submatrix ![1, 2, 3] ![0, 1, 2]).rank = 3 := by
  rw [hsub]; exact hI
exact this.symm.trans_le (rank_submatrix_le _ _ _)

lemma tetra_d2_rank_ge : 3 ≤ tetra_d2.rank := by
  have hI : ((1 : Matrix (Fin 3) (Fin 3) ℚ).rank) = 3 := rank_one
  have hsub :
    tetra_d2.submatrix ![3, 4, 5] ![0, 1, 2] = (1 : Matrix (Fin 3) (Fin 3) ℚ) := by
    ext i j
    fin_cases i <=> fin_cases j <=> simp [tetra_d2]
  have : (tetra_d2.submatrix ![3, 4, 5] ![0, 1, 2]).rank = 3 := by
    rw [hsub]; exact hI
  exact this.symm.trans_le (rank_submatrix_le _ _ _)

lemma tetra_d1_rank : tetra_d1.rank = 3 := by
  have hsum := rank_add_rank_le_card_of_mul_eq_zero tetra_comp
  have h1 := tetra_d1_rank_ge
  have h2 := tetra_d2_rank_ge
  simp only [Fintype.card_fin] at hsum
  omega

lemma tetra_d2_rank : tetra_d2.rank = 3 := by
  have hsum := rank_add_rank_le_card_of_mul_eq_zero tetra_comp
  have h1 := tetra_d1_rank_ge
  have h2 := tetra_d2_rank_ge
  simp only [Fintype.card_fin] at hsum
  omega

lemma tetra_del1_range : finrank ℚ (LinearMap.range tetra_del1) = 3 := by
  simpa [tetra_del1, toLin'_apply'] using tetra_d1_rank

lemma tetra_del2_range : finrank ℚ (LinearMap.range tetra_del2) = 3 := by
  simpa [tetra_del2, toLin'_apply'] using tetra_d2_rank

/--  $H_0(S^2; \mathbb{Q}) \cong \mathbb{Q}$ . -/
theorem sphere2_betti0 : betti0 tetra_del1 = 1 := by
  simp [betti0, tetra_del1_range, finrank_fintype_fun_eq_card]

/--  $H_1(S^2; \mathbb{Q}) = 0$ . -/
theorem sphere2_betti1 : betti1 tetra_del2 tetra_del1 = 0 := by
  have hr := tetra_del1.finrank_range_add_finrank_ker
  simp [betti1, tetra_del2_range, tetra_del1_range, finrank_fintype_fun_eq_card] at hr ⊢
  omega

/--  $H_2(S^2; \mathbb{Q}) \cong \mathbb{Q}$ . -/
theorem sphere2_betti2 : betti2 tetra_del2 = 1 := by
  have hr := tetra_del2.finrank_range_add_finrank_ker
  simp [betti2, tetra_del2_range, finrank_fintype_fun_eq_card] at hr ⊢
  omega

/-- Cellular homology of the standard simplicial 2-sphere. -/
theorem homology_sphere2 :
  betti0 tetra_del1 = 1 ∧ betti1 tetra_del2 tetra_del1 = 0 ∧ betti2 tetra_del2 = 1 :=
  ⟨sphere2_betti0, sphere2_betti1, sphere2_betti2⟩

/-- Euler's formula on the tetrahedron, from its computed Betti numbers. -/
theorem euler_tetrahedron : (4 : ℕ) + 4 = 6 + 2 :=
  euler_formula_of_sphere2 tetra_del2 tetra_del1

```

```

    (by simp [finrank_fintype_fun_eq_card])
    (by simp [finrank_fintype_fun_eq_card])
    (by simp [finrank_fintype_fun_eq_card])
    tetra_del_comp sphere2_betti0 sphere2_betti1 sphere2_betti2

/-- The tetrahedron incidence structure is Platonic, using homology not a
raw Euler hypothesis. -/
theorem tetrahedron_platonic : IsPlatonicPair 3 3 :=
  platonic_solids_3d_of_homology tetra_del2 tetra_del1 4 6 4 3 3
  (by decide) (by decide) (by decide) (by decide)
  (by simp [finrank_fintype_fun_eq_card])
  (by simp [finrank_fintype_fun_eq_card])
  (by simp [finrank_fintype_fun_eq_card])
  tetra_del_comp sphere2_betti0 sphere2_betti1 sphere2_betti2

```

A.23 PlatonicSolids/Sphere3Homology.lean

Cellular $H_*(S^3; \mathbb{Q})$ of the 4-simplex boundary (195 lines).

```

/-
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-/
import Mathlib.LinearAlgebra.Matrix.Notation
import Mathlib.LinearAlgebra.Matrix.Rank
import Mathlib.LinearAlgebra.Matrix.ToLin
import Mathlib.Tactic
import PlatonicSolids.EulerPoincare

/-!
Paper §3.1. Cellular homology of the standard simplicial  $(S^3)$ :
the boundary of a 4-simplex (5 vertices, 10 edges, 10 triangles,
5 tetrahedra).

Ranks are  $\langle 4, 6, 4 \rangle$ , so  $\langle b_0 = 1, b_1 = 0, b_2 = 0, b_3 = 1 \rangle$  and
 $\chi = 0$ . This is the combinatorial 3-sphere the paper uses; Mathlib
does not compute singular homology of Metric.sphere in positive
degree.
-/

open scoped Matrix
open Matrix Module

/-- Vertices  $\langle 0..4 \rangle \leftarrow$  edges  $\langle 01,02,03,04,12,13,14,23,24,34 \rangle$ . -/
def pentatope_d1 : Matrix (Fin 5) (Fin 10)  $\mathbb{Q}$  :=
  !![ -1, -1, -1, -1, 0, 0, 0, 0, 0, 0;
      1, 0, 0, 0, -1, -1, -1, 0, 0, 0;
      0, 1, 0, 0, 1, 0, 0, -1, -1, 0;
      0, 0, 1, 0, 0, 1, 0, 1, 0, -1;
      0, 0, 0, 1, 0, 0, 1, 0, 1, 1]

/-- Edges  $\leftarrow$  triangles
 $\langle 012,013,014,023,024,034,123,124,134,234 \rangle$ . -/
def pentatope_d2 : Matrix (Fin 10) (Fin 10)  $\mathbb{Q}$  :=
  !![ 1, 1, 1, 0, 0, 0, 0, 0, 0, 0;
      -1, 0, 0, 1, 1, 0, 0, 0, 0, 0;
      0, -1, 0, -1, 0, 1, 0, 0, 0, 0;
      0, 0, -1, 0, -1, -1, 0, 0, 0, 0;
      1, 0, 0, 0, 0, 0, 1, 1, 0, 0;
      0, 1, 0, 0, 0, 0, -1, 0, 1, 0;
      0, 0, 1, 0, 0, 0, 0, -1, -1, 0;
      0, 0, 0, 1, 0, 0, 1, 0, 0, 1;
      0, 0, 0, 0, 1, 0, 0, 1, 0, -1;

```

```

0, 0, 0, 0, 0, 1, 0, 0, 1, 1]

/-- Triangles ← tetrahedra `0123,0124,0134,0234,1234`. -/
def pentatope_d3 : Matrix (Fin 10) (Fin 5) ℚ :=
  !![ -1, -1, 0, 0, 0;
      1, 0, -1, 0, 0;
      0, 1, 1, 0, 0;
      -1, 0, 0, -1, 0;
      0, -1, 0, 1, 0;
      0, 0, -1, -1, 0;
      1, 0, 0, 0, -1;
      0, 1, 0, 0, 1;
      0, 0, 1, 0, -1;
      0, 0, 0, 1, 1]

lemma pentatope_comp21 : pentatope_d1 * pentatope_d2 = 0 := by
  native_decide

lemma pentatope_comp32 : pentatope_d2 * pentatope_d3 = 0 := by
  native_decide

noncomputable def pentatope_del1 : (Fin 10 → ℚ) →1[ℚ] (Fin 5 → ℚ) :=
  pentatope_d1.toLin'
noncomputable def pentatope_del2 : (Fin 10 → ℚ) →1[ℚ] (Fin 10 → ℚ) :=
  pentatope_d2.toLin'
noncomputable def pentatope_del3 : (Fin 5 → ℚ) →1[ℚ] (Fin 10 → ℚ) :=
  pentatope_d3.toLin'

lemma pentatope_del_comp21 : pentatope_del1.comp pentatope_del2 = 0 := by
  rw [pentatope_del1, pentatope_del2, ← toLin'_mul, pentatope_comp21]
  exact map_zero _

lemma pentatope_del_comp32 : pentatope_del2.comp pentatope_del3 = 0 := by
  rw [pentatope_del2, pentatope_del3, ← toLin'_mul, pentatope_comp32]
  exact map_zero _

lemma pentatope_d1_rank_ge : 4 ≤ pentatope_d1.rank := by
  have hI : ((1 : Matrix (Fin 4) (Fin 4) ℚ).rank) = 4 := rank_one
  have hsub :
    pentatope_d1.submatrix ![1, 2, 3, 4] ![0, 1, 2, 3] =
      (1 : Matrix (Fin 4) (Fin 4) ℚ) := by
    ext i j
    fin_cases i <;> fin_cases j <;> simp [pentatope_d1]
  have : (pentatope_d1.submatrix ![1, 2, 3, 4] ![0, 1, 2, 3]).rank = 4 := by
    rw [hsub]; exact hI
  exact this.symm.trans_le (rank_submatrix_le _ _ _)

lemma pentatope_d3_rank_ge : 4 ≤ pentatope_d3.rank := by
  have hI : ((1 : Matrix (Fin 4) (Fin 4) ℚ).rank) = 4 := rank_one
  have hsub :
    pentatope_d3.submatrix ![6, 7, 8, 9] ![0, 1, 2, 3] =
      (1 : Matrix (Fin 4) (Fin 4) ℚ) := by
    ext i j
    fin_cases i <;> fin_cases j <;> simp [pentatope_d3]
  have : (pentatope_d3.submatrix ![6, 7, 8, 9] ![0, 1, 2, 3]).rank = 4 := by
    rw [hsub]; exact hI
  exact this.symm.trans_le (rank_submatrix_le _ _ _)

/-- A `6 × 6` minor of `d2` with determinant `1`. -/
def pentatope_d2_minor : Matrix (Fin 6) (Fin 6) ℚ :=
  !![ 1, 1, 1, 0, 0, 0;
      -1, 0, 0, 1, 1, 0;
      0, -1, 0, -1, 0, 1;

```

```

1, 0, 0, 0, 0, 0;
0, 1, 0, 0, 0, 0;
0, 0, 0, 1, 0, 0]

lemma pentatope_d2_minor_eq :
  pentatope_d2.submatrix ![0, 1, 2, 4, 5, 7] ![0, 1, 2, 3, 4, 5] =
    pentatope_d2_minor := by
  ext i j
  fin_cases i <;> fin_cases j <;> simp [pentatope_d2, pentatope_d2_minor]

lemma pentatope_d2_minor_det : pentatope_d2_minor.det = 1 := by
  native_decide

lemma pentatope_d2_rank_ge : 6 ≤ pentatope_d2.rank := by
  have hU : IsUnit pentatope_d2_minor :=
    (Matrix.isUnit_iff_isUnit_det _).2 (by simp [pentatope_d2_minor_det])
  have : pentatope_d2_minor.rank = 6 := by
    simp using rank_of_isUnit pentatope_d2_minor hU
  rw [← pentatope_d2_minor_eq] at this
  exact this.symm.trans_le (rank_submatrix_le _ _ _)

lemma pentatope_d1_rank : pentatope_d1.rank = 4 := by
  have hsum := rank_add_rank_le_card_of_mul_eq_zero pentatope_comp21
  have h1 := pentatope_d1_rank_ge
  have h2 := pentatope_d2_rank_ge
  simp only [Fintype.card_fin] at hsum
  omega

lemma pentatope_d2_rank : pentatope_d2.rank = 6 := by
  have hsum := rank_add_rank_le_card_of_mul_eq_zero pentatope_comp21
  have h1 := pentatope_d1_rank_ge
  have h2 := pentatope_d2_rank_ge
  simp only [Fintype.card_fin] at hsum
  omega

lemma pentatope_d3_rank : pentatope_d3.rank = 4 := by
  have hsum := rank_add_rank_le_card_of_mul_eq_zero pentatope_comp32
  have h2 := pentatope_d2_rank
  have h3 := pentatope_d3_rank_ge
  simp only [Fintype.card_fin] at hsum
  omega

lemma pentatope_del1_range : finrank ℚ (LinearMap.range pentatope_del1) = 4 := by
  simp [pentatope_del1, toLin'_apply'] using pentatope_d1_rank

lemma pentatope_del2_range : finrank ℚ (LinearMap.range pentatope_del2) = 6 := by
  simp [pentatope_del2, toLin'_apply'] using pentatope_d2_rank

lemma pentatope_del3_range : finrank ℚ (LinearMap.range pentatope_del3) = 4 := by
  simp [pentatope_del3, toLin'_apply'] using pentatope_d3_rank

theorem sphere3_betti0 : betti0 pentatope_del1 = 1 := by
  simp [betti0, pentatope_del1_range, finrank_fintype_fun_eq_card]

theorem sphere3_betti1 : betti1 pentatope_del2 pentatope_del1 = 0 := by
  have hr := pentatope_del1.finrank_range_add_finrank_ker
  simp [betti1, pentatope_del2_range, pentatope_del1_range,
    finrank_fintype_fun_eq_card] at hr ⊢
  omega

theorem sphere3_betti2 : betti2Rel pentatope_del3 pentatope_del2 = 0 := by
  have hr := pentatope_del2.finrank_range_add_finrank_ker
  simp [betti2Rel, pentatope_del3_range, pentatope_del2_range,

```

```

    finrank_fintype_fun_eq_card] at hr ⊢
  omega

theorem sphere3_betti3 : betti3 pentatope_del3 = 1 := by
  have hr := pentatope_del3.finrank_range_add_finrank_ker
  simp [betti3, pentatope_del3_range, finrank_fintype_fun_eq_card] at hr ⊢
  omega

/-- Cellular homology of the standard simplicial 3-sphere. -/
theorem homology_sphere3 :
  betti0 pentatope_del1 = 1 ∧
  betti1 pentatope_del2 pentatope_del1 = 0 ∧
  betti2Rel pentatope_del3 pentatope_del2 = 0 ∧
  betti3 pentatope_del3 = 1 :=
  ⟨sphere3_betti0, sphere3_betti1, sphere3_betti2, sphere3_betti3⟩

/--  $\chi(S^3) = 0$  from the computed Betti numbers of the 4-simplex boundary. -/
theorem eulerChar_sphere3 : ((5 : ℕ) : ℤ) - 10 + 10 - 5 = 0 :=
  euler_formula_of_sphere3 pentatope_del3 pentatope_del2 pentatope_del1
  (by simp [finrank_fintype_fun_eq_card])
  (by simp [finrank_fintype_fun_eq_card])
  (by simp [finrank_fintype_fun_eq_card])
  (by simp [finrank_fintype_fun_eq_card])
  pentatope_del_comp32 pentatope_del_comp21
  sphere3_betti0 sphere3_betti1 sphere3_betti2 sphere3_betti3

```

A.24 PlatonicSolids/EulerBetti.lean

Betti arithmetic of $\chi(S^2) = 2$ and $\chi(S^3) = 0$ (24 lines).

```

/-
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-/
import Mathlib.Tactic

/-!
Paper §2.1 / §3.1. Betti arithmetic of  $\chi(S^2)$  and  $\chi(S^3)$ .

The algebraic Euler-Poincaré identities are `euler_poincare` and
`euler_poincare3` in `PlatonicSolids/EulerPoincare.lean`. The numbers
`b_0 = 1`, `b_1 = 0`, `b_2 = 1` for the standard simplicial `S^2` are
`homology_sphere2`; the numbers `b_0 = 1`, `b_1 = 0`, `b_2 = 0`,
`b_3 = 1` for the standard simplicial `S^3` are `homology_sphere3`.
Singular `H_0` of `Metric.sphere` in dimension `n ≥ 1` and the full
calculation of `H_*(S^0)` are in `PlatonicSolids/SingularHomology.lean`.
These lemmas record the resulting ` $\chi(S^2) = 2$ ` and ` $\chi(S^3) = 0$ `.
-/

/--  $\chi(S^2) = b_0 - b_1 + b_2 = 1 - 0 + 1$ . -/
lemma eulerChar_sphere2_betti : (1 : ℤ) - 0 + 1 = 2 := by norm_num

/--  $\chi(S^3) = b_0 - b_1 + b_2 - b_3 = 1 - 0 + 0 - 1$ . -/
lemma eulerChar_sphere3_betti : (1 : ℤ) - 0 + 0 - 1 = 0 := by norm_num

```

A.25 PlatonicSolids/PlatonicPair.lean

The five solutions of the Platonic Diophantine inequality (38 lines).

```

/-
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```

```

-/
import Mathlib.Tactic

/-!
Paper §2.3. The five solutions of  $(p-2)(q-2) < 4$  for  $p, q \geq 3$ .
-/

/-- Schläfli pairs of the five Platonic solids. -/
def IsPlatonicPair (p q : ℕ) : Prop :=
  (p = 3 ∧ q = 3) ∨
  (p = 3 ∧ q = 4) ∨
  (p = 4 ∧ q = 3) ∨
  (p = 3 ∧ q = 5) ∨
  (p = 5 ∧ q = 3)

lemma pq_ge_two_p_add_two_q {p q : ℕ} (hp : 4 ≤ p) (hq : 4 ≤ q) :
  2 * p + 2 * q ≤ p * q := by
  zify
  nlinarith

/-- The inequality  $pq < 2p + 2q$  for  $p, q \geq 3$  has exactly the five Platonic solutions. -/
theorem platonic_pairs_of_inequality {p q : ℕ}
  (hp : 3 ≤ p) (hq : 3 ≤ q) (h : p * q < 2 * p + 2 * q) :
  IsPlatonicPair p q := by
  rcases show p = 3 ∨ q = 3 ∨ (4 ≤ p ∧ 4 ≤ q) by omega with rfl | rfl | ⟨hp4, hq4⟩
  · rcases show q = 3 ∨ q = 4 ∨ q = 5 by omega with rfl | rfl | rfl <;> simp [IsPlatonicPair]
  · rcases show p = 3 ∨ p = 4 ∨ p = 5 by omega with rfl | rfl | rfl <;> simp [IsPlatonicPair]
  · exact (lt_irrefl _ (h.trans_le (pq_ge_two_p_add_two_q hp4 hq4))).elim

theorem IsPlatonicPair.ge_three {p q : ℕ} (h : IsPlatonicPair p q) : 3 ≤ p ∧ 3 ≤ q := by
  rcases h with ⟨rfl, rfl⟩ | ⟨rfl, rfl⟩ | ⟨rfl, rfl⟩ | ⟨rfl, rfl⟩ | ⟨rfl, rfl⟩ <;> decide

theorem IsPlatonicPair.inequality {p q : ℕ} (h : IsPlatonicPair p q) :
  p * q < 2 * p + 2 * q := by
  rcases h with ⟨rfl, rfl⟩ | ⟨rfl, rfl⟩ | ⟨rfl, rfl⟩ | ⟨rfl, rfl⟩ | ⟨rfl, rfl⟩ <;> decide

```

A.26 PlatonicSolids/Reciprocal.lean

$1/p + 1/q > 1/2$ iff the integer inequality (44 lines).

```

/-
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-/
import Mathlib.Data.Real.Basic
import PlatonicSolids.PlatonicPair

/-!
Paper §2.3 / §3.3.  $1/p + 1/q > 1/2$  is the reciprocal form of the
Platonic Diophantine inequality, and is the condition  $D_3 > 0$ .
-/

```

```

lemma reciprocal_iff {p q : ℕ} (hp : 0 < p) (hq : 0 < q) :
  (1 / (p : ℝ) + 1 / q > 1 / 2) ↔ p * q < 2 * p + 2 * q := by
  have : (0 : ℝ) < p := Nat.cast_pos.mpr hp
  have : (0 : ℝ) < q := Nat.cast_pos.mpr hq
  constructor
  · intro h
    have h' : (p + q : ℝ) / (p * q) > 1 / 2 := by
      convert h using 1
      field_simp
      ring
    rw [gt_iff_lt, lt_div_iff₀ (by positivity)] at h'

```

```

    exact_mod_cast (by linarith : (p * q : ℝ) < 2 * p + 2 * q)
  · intro h
    have hR : (p * q : ℝ) < 2 * p + 2 * q := mod_cast h
    have h' : (p + q : ℝ) / (p * q) > 1 / 2 := by
      rw [gt_iff_lt, lt_div_iff₀ (by positivity)]
      linarith
    convert h' using 1
    field_simp
    ring

theorem platonic_pair_iff {p q : ℕ} :
  IsPlatonicPair p q ↔ 3 ≤ p ∧ 3 ≤ q ∧ p * q < 2 * p + 2 * q := by
  constructor
  · intro h
    exact ⟨h.ge_three.1, h.ge_three.2, h.inequality⟩
  · exact fun ⟨hp, hq, h⟩ => platonic_pairs_of_inequality hp hq h

theorem platonic_pair_iff_reciprocal {p q : ℕ} (hp : 3 ≤ p) (hq : 3 ≤ q) :
  IsPlatonicPair p q ↔ 1 / (p : ℝ) + 1 / q > 1 / 2 := by
  rw [platonic_pair_iff, reciprocal_iff (by omega) (by omega)]
  simp [hp, hq]

```

A.27 PlatonicSolids/Incidence.lean

Regularity incidence plus Euler force a Platonic pair and $E > 0$ (83 lines).

```

/-
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-/
import PlatonicSolids.EulerPoincare
import PlatonicSolids.PlatonicPair

/-!
Paper §2.2-2.4. Regularity incidence `pF = 2E`, `qV = 2E` and Euler
`V + F = E + 2` force a Platonic pair, and force `E > 0` with no extra
non-degeneracy axiom.
-/

/-- The Euler + regularity identities algebraically prohibit `E = 0`. -/
theorem edges_pos_of_regular
  (V E F p q : ℕ)
  (hp : 3 ≤ p) (hq : 3 ≤ q)
  (h_edges_faces : p * F = 2 * E)
  (h_edges_verts : q * V = 2 * E)
  (h_euler : V + F = E + 2) :
  0 < E := by
  have h_lhs : p * q * (V + F) = (2 * p + 2 * q) * E := calc
    p * q * (V + F) = p * (q * V) + q * (p * F) := by ring
    _ = p * (2 * E) + q * (2 * E) := by rw [h_edges_verts, h_edges_faces]
    _ = (2 * p + 2 * q) * E := by ring
  have h_rhs : p * q * (V + F) = p * q * E + 2 * p * q := calc
    p * q * (V + F) = p * q * (E + 2) := by rw [h_euler]
    _ = p * q * E + 2 * p * q := by ring
  have h_eq : (2 * p + 2 * q) * E = p * q * E + 2 * p * q := by rw [← h_lhs, h_rhs]
  by_contra hz
  have hE0 : E = 0 := Nat.eq_zero_of_not_pos hz
  subst hE0
  simp only [mul_zero, zero_add] at h_eq
  exact (Nat.pos_iff_ne_zero.mp (by positivity : 0 < 2 * p * q)) h_eq.symm

/-- Any regular incidence structure with  $\chi = 2$  is a Platonic pair. -/
theorem platonic_solids_3d

```

```

(V E F p q : ℕ)
(hp : 3 ≤ p) (hq : 3 ≤ q)
(h_edges_faces : p * F = 2 * E)
(h_edges_verts : q * V = 2 * E)
(h_euler : V + F = E + 2) :
  IsPlatonicPair p q := by
  have h_lhs : p * q * (V + F) = (2 * p + 2 * q) * E := calc
    p * q * (V + F) = p * (q * V) + q * (p * F) := by ring
    _ = p * (2 * E) + q * (2 * E) := by rw [h_edges_verts, h_edges_faces]
    _ = (2 * p + 2 * q) * E := by ring
  have h_eq : (2 * p + 2 * q) * E = p * q * E + 2 * p * q := by
    calc (2 * p + 2 * q) * E = p * q * (V + F) := h_lhs.symm
    _ = p * q * (E + 2) := by rw [h_euler]
    _ = p * q * E + 2 * p * q := by ring
  have h_ineq : p * q < 2 * p + 2 * q := by
    obtain hlt | hge := lt_or_ge (p * q) (2 * p + 2 * q)
    · exact hlt
    · have : (2 * p + 2 * q) * E ≤ p * q * E := Nat.mul_le_mul_right E hge
      have : 0 < 2 * p * q := by positivity
      omega
  exact platonic_pairs_of_inequality hp hq h_ineq

/-- The 3D classification from a cellular 2-complex with the Betti
numbers of  $S^2$ . Euler-Poincaré (euler_formula_of_sphere2) supplies
 $V + F = E + 2$ ; regularity incidence then forces a Platonic pair.
The optional singular-homology form is platonic_solids_3d_of_sphere
in SphereSingularHomology.lean. -/
theorem platonic_solids_3d_of_homology
  {K : Type*} [Field K]
  {C₂ C₁ C₀ : Type*}
  [AddCommGroup C₂] [Module K C₂] [FiniteDimensional K C₂]
  [AddCommGroup C₁] [Module K C₁] [FiniteDimensional K C₁]
  [AddCommGroup C₀] [Module K C₀] [FiniteDimensional K C₀]
  (d₂ : C₂ →₁[K] C₁) (d₁ : C₁ →₁[K] C₀)
  (V E F p q : ℕ)
  (hp : 3 ≤ p) (hq : 3 ≤ q)
  (h_edges_faces : p * F = 2 * E)
  (h_edges_verts : q * V = 2 * E)
  (hV : Module.finrank K C₀ = V)
  (hE : Module.finrank K C₁ = E)
  (hF : Module.finrank K C₂ = F)
  (hcomp : d₁.comp d₂ = 0)
  (hb0 : betti0 d₁ = 1) (hb1 : betti1 d₂ d₁ = 0) (hb2 : betti2 d₂ = 1) :
  IsPlatonicPair p q :=
  platonic_solids_3d V E F p q hp hq h_edges_faces h_edges_verts
  (euler_formula_of_sphere2 d₂ d₁ hV hE hF hcomp hb0 hb1 hb2)

```

A.28 PlatonicSolids/Angles.lean

$\pi/n \in (0, \pi/2)$ so sine and cosine are positive (28 lines).

```

/-
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-/
import Mathlib.Analysis.SpecialFunctions.Trigonometric.Basic

/-!
Paper §3.3. For  $n \geq 3$ , the Schläfli angle  $\pi/n$  lies in  $(0, \pi/2)$ ,
so sin and cos are positive.
-/
open Real

```

```

lemma angle_pos {n : ℕ} (hn : 1 ≤ n) : 0 < π / n := by
  have : (0 : ℝ) < n := Nat.cast_pos.mpr hn
  positivity

lemma angle_lt_pi_div_two {n : ℕ} (hn : 3 ≤ n) : π / n < π / 2 :=
  div_lt_div_of_pos_left pi_pos (by positivity : (0 : ℝ) < 2)
  (by exact_mod_cast (lt_of_lt_of_le (by decide : (2 : ℕ) < 3) hn))

lemma sin_angle_pos {n : ℕ} (hn : 3 ≤ n) : 0 < sin (π / n) :=
  sin_pos_of_pos_of_lt_pi (angle_pos (Nat.succ_le_of_lt (Nat.zero_lt_of_lt hn)))
  (angle_lt_pi_div_two hn |>.trans (by linarith [pi_pos]))

lemma cos_angle_pos {n : ℕ} (hn : 3 ≤ n) : 0 < cos (π / n) :=
  cos_pos_of_mem_Ioo (by linarith [angle_pos (Nat.succ_le_of_lt (Nat.zero_lt_of_lt hn)), pi_pos
  ],
  angle_lt_pi_div_two hn)

```

A.29 PlatonicSolids/GramMatrix.lean

Schläfli Gram matrix and the scalar Δ (30 lines).

```

/-
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-/
import Mathlib.Analysis.SpecialFunctions.Trigonometric.Basic
import Mathlib.LinearAlgebra.Matrix.Hermitian
import Mathlib.LinearAlgebra.Matrix.Notation

/-!
Paper §3.2. The Schläfli Gram matrix of  $\{p, q, r\}$  and the scalar
 $\Delta(p, q, r)$  that will equal its determinant.
-/

open scoped Matrix
open Real Matrix

/-- Canonical bilinear form of the rank-4 Schläfli Coxeter system  $\{p, q, r\}$ . -/
noncomputable def schlaflGram (p q r : ℕ) : Matrix (Fin 4) (Fin 4) ℝ :=
  !![1, -cos (π / p), 0, 0;
    -cos (π / p), 1, -cos (π / q), 0;
    0, -cos (π / q), 1, -cos (π / r);
    0, 0, -cos (π / r), 1]

/--  $\Delta(p, q, r) = \sin^2(\pi/p) \sin^2(\pi/r) - \cos^2(\pi/q)$ . -/
noncomputable def schlaflDet (p q r : ℕ) : ℝ :=
  sin (π / p) ^ 2 * sin (π / r) ^ 2 - cos (π / q) ^ 2

lemma schlaflGram_isHermitian (p q r : ℕ) : (schlaflGram p q r).IsHermitian := by
  ext i j
  fin_cases i <;> fin_cases j <;> simp [schlaflGram, conjTranspose]

```

A.30 PlatonicSolids/LeadingMinors.lean

Closed forms of the leading minors D_1, D_2, D_3 (34 lines).

```

/-
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-/
import Mathlib.LinearAlgebra.Matrix.Determinant.Basic

```

```

import PlatonicSolids.GramMatrix

/-!
Paper §3.3-3.4. Leading principal minors `D1`, `D2`, `D3` of the Gram
matrix, matching the paper's closed forms.
-/

open scoped Matrix
open Real Matrix

def leadingSubmatrix (k : ℕ) (hk : k ≤ 4) (A : Matrix (Fin 4) (Fin 4) ℝ) :
  Matrix (Fin k) (Fin k) ℝ :=
  A.submatrix (Fin.castLE hk) (Fin.castLE hk)

noncomputable def leadingMinor (k : ℕ) (hk : k ≤ 4) (p q r : ℕ) : ℝ :=
  (leadingSubmatrix k hk (schlafliGram p q r)).det

lemma leadingMinor_one (p q r : ℕ) : leadingMinor 1 (by decide) p q r = 1 := by
  simp [leadingMinor, leadingSubmatrix, schlafliGram]

lemma leadingMinor_two (p q r : ℕ) :
  leadingMinor 2 (by decide) p q r = sin (π / p) ^ 2 := by
  simp [leadingMinor, leadingSubmatrix, schlafliGram, det_fin_two]
  linarith [sin_sq_add_cos_sq (π / p)]

lemma leadingMinor_three (p q r : ℕ) :
  leadingMinor 3 (by decide) p q r = sin (π / p) ^ 2 - cos (π / q) ^ 2 := by
  simp [leadingMinor, leadingSubmatrix, schlafliGram, det_fin_three]
  linarith [sin_sq_add_cos_sq (π / p)]

```

A.31 PlatonicSolids/TrailingMinor.lean

Trailing 3×3 minor is the vertex-figure cell minor (22 lines).

```

/-
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-/

import Mathlib.LinearAlgebra.Matrix.Determinant.Basic
import PlatonicSolids.GramMatrix

/-!
Paper §3.3. The trailing  $3 \times 3$  principal minor is the cell minor of the
vertex figure `{q, r}`.
-/

open scoped Matrix
open Real Matrix

noncomputable def trailing3 (p q r : ℕ) : Matrix (Fin 3) (Fin 3) ℝ :=
  (schlafliGram p q r).submatrix (fun i : Fin 3 => i.succ) (fun i => i.succ)

lemma trailing3_det (p q r : ℕ) :
  (trailing3 p q r).det = sin (π / q) ^ 2 - cos (π / r) ^ 2 := by
  simp [trailing3, schlafliGram, det_fin_three]
  linarith [sin_sq_add_cos_sq (π / q)]

```

A.32 PlatonicSolids/D3IffPlatonic.lean

$D_3 > 0$ iff the cell is Platonic (58 lines).

```

/-

```

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```

-/
import PlatonicSolids.Angles
import PlatonicSolids.LeadingMinors
import PlatonicSolids.Reciprocal
import PlatonicSolids.TrailingMinor

/-!
Paper §3.3.  $D_3 > 0$  iff the cell is Platonic; the trailing  $3 \times 3$  minor
is the same statement for the vertex figure.
-/

open Real

lemma angle_mem_Icc {n : ℕ} (hn : 3 ≤ n) :
   $\pi / n \in \text{Set.Icc } (-(\pi / 2)) (\pi / 2) :=$ 
  (by linarith [angle_pos (Nat.succ_le_of_lt (Nat.zero_lt_of_lt hn)), pi_pos],
   (angle_lt_pi_div_two hn).le)

lemma complement_mem_Icc {n : ℕ} (hn : 3 ≤ n) :
   $\pi / 2 - \pi / n \in \text{Set.Icc } (-(\pi / 2)) (\pi / 2) :=$ 
  (by linarith [angle_lt_pi_div_two hn, pi_pos],
   by linarith [angle_pos (Nat.succ_le_of_lt (Nat.zero_lt_of_lt hn))])

/--  $\sin(\pi/p) > \cos(\pi/q)$  iff  $1/p + 1/q > 1/2$ . -/
lemma sin_gt_cos_iff_reciprocal {p q : ℕ} (hp : 3 ≤ p) (hq : 3 ≤ q) :
   $\sin(\pi / p) > \cos(\pi / q) \leftrightarrow 1 / (p : \mathbb{R}) + 1 / q > 1 / 2 :=$  by
  have hp0 :  $(0 : \mathbb{R}) < p := \text{Nat.cast_pos.mpr } (\text{Nat.zero_lt_of_lt hp})$ 
  have hq0 :  $(0 : \mathbb{R}) < q := \text{Nat.cast_pos.mpr } (\text{Nat.zero_lt_of_lt hq})$ 
  rw [← sin_pi_div_two_sub ( $\pi / q$ ), gt_iff_lt, gt_iff_lt]
  rw [strictMonoOn_sin.lt_iff_lt (complement_mem_Icc hq) (angle_mem_Icc hp)]
  constructor
  · intro h
    have hsum :  $\pi / p + \pi / q > \pi / 2 :=$  by linarith
    have hmul :  $\pi * (1 / (p : \mathbb{R}) + 1 / q) > \pi * (1 / 2) :=$  by
      convert hsum using 1 <=> field_simp
    exact lt_of_mul_lt_mul_left hmul pi_pos.le
  · intro h
    have hmul :  $\pi * (1 / (p : \mathbb{R}) + 1 / q) > \pi * (1 / 2) :=$ 
      mul_lt_mul_of_pos_left h pi_pos
    have hsum :  $\pi / p + \pi / q > \pi / 2 :=$  by
      convert hmul using 1 <=> field_simp
    linarith

/-- Leading  $3 \times 3$  minor  $D_3 > 0$  iff  $\{p, q\}$  is Platonic. -/
theorem leadingMinor_three_pos_iff {p q r : ℕ} (hp : 3 ≤ p) (hq : 3 ≤ q) :
   $0 < \text{leadingMinor } 3 \text{ (by decide) } p \ q \ r \leftrightarrow \text{IsPlatonicPair } p \ q :=$  by
  rw [leadingMinor_three, platonic_pair_iff_reciprocal hp hq, sub_pos,
      sq_lt_sq0 (cos_angle_pos hq).le (sin_angle_pos hp).le]
  exact sin_gt_cos_iff_reciprocal hp hq

/-- Trailing  $3 \times 3$  minor  $> 0$  iff the vertex figure  $\{q, r\}$  is Platonic. -/
theorem trailing3_pos_iff {p q r : ℕ} (hq : 3 ≤ q) (hr : 3 ≤ r) :
   $0 < (\text{trailing3 } p \ q \ r).\text{det} \leftrightarrow \text{IsPlatonicPair } q \ r :=$  by
  rw [trailing3_det, ← leadingMinor_three (p := q) (q := r) (r := p)]
  exact leadingMinor_three_pos_iff hq hr

```

A.33 PlatonicSolids/QuadForm.lean

Quadratic form of the Gram matrix (26 lines).

/-

```

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-/
import PlatonicSolids.GramMatrix

/-!
Paper §3.3. The quadratic form of the Schläfli Gram matrix.
-/

open scoped Matrix
open Real Matrix

private lemma sum_fin4 (f : Fin 4 → ℝ) :  $\sum i, f i = f 0 + f 1 + f 2 + f 3$  := by
  rw [Fin.sum_univ_succ, Fin.sum_univ_succ, Fin.sum_univ_succ]
  simp
  abel

lemma schlaefliGram_quad (p q r : ℕ) (x : Fin 4 → ℝ) :
  star x ·v (schlaefliGram p q r ·v x) =
    x 0 ^ 2 + x 1 ^ 2 + x 2 ^ 2 + x 3 ^ 2
    - 2 * cos (π / p) * x 0 * x 1
    - 2 * cos (π / q) * x 1 * x 2
    - 2 * cos (π / r) * x 2 * x 3 := by
  simp [schlaefliGram, dotProduct, mulVec, sum_fin4]
  ring

```

A.34 PlatonicSolids/CompletingSquare.lean

LDL / completing-the-square identity (55 lines).

```

/-
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-/
import Mathlib.Tactic

/-!
Paper §3.3. Completing the square (LDL) for the rank-4 tridiagonal
quadratic form that the Schläfli Gram matrix induces.
-/

/-- Weighted sum of squares after the triangular change of variables. -/
lemma band_quad_squares (α β γ : ℝ) (x0 x1 x2 x3 : ℝ)
  (h2 : 1 - α ^ 2 ≠ 0) (h3 : 1 - α ^ 2 - β ^ 2 ≠ 0) :
  x0 ^ 2 + x1 ^ 2 + x2 ^ 2 + x3 ^ 2 + 2 * α * x0 * x1 + 2 * β * x1 * x2 +
    2 * γ * x2 * x3 =
    (x0 + α * x1) ^ 2
    + (1 - α ^ 2) * (x1 + β / (1 - α ^ 2) * x2) ^ 2
    + ((1 - α ^ 2 - β ^ 2) / (1 - α ^ 2)) *
      (x2 + γ * (1 - α ^ 2) / (1 - α ^ 2 - β ^ 2) * x3) ^ 2
    + ((1 - α ^ 2 - β ^ 2 - γ ^ 2 * (1 - α ^ 2)) / (1 - α ^ 2 - β ^ 2)) *
      x3 ^ 2 := by
  field_simp [h2, h3]
  ring

lemma weighted_squares_pos {a b c d c1 c2 c3 : ℝ}
  (hc1 : 0 < c1) (hc2 : 0 < c2) (hc3 : 0 < c3)
  (h : a ≠ 0 ∨ b ≠ 0 ∨ c ≠ 0 ∨ d ≠ 0) :
  0 < a ^ 2 + c1 * b ^ 2 + c2 * c ^ 2 + c3 * d ^ 2 := by
  rcases h with ha | hb | hc | hd
  · nlinarith [sq_pos_of_ne_zero ha, sq_nonneg b, sq_nonneg c, sq_nonneg d]
  · nlinarith [sq_pos_of_ne_zero hb, sq_nonneg a, sq_nonneg c, sq_nonneg d]
  · nlinarith [sq_pos_of_ne_zero hc, sq_nonneg a, sq_nonneg b, sq_nonneg d]

```

```

· llinarith [sq_pos_of_ne_zero hd, sq_nonneg a, sq_nonneg b, sq_nonneg c]

/-- The triangular change of variables is invertible. -/
lemma ldl_coords_eq_zero {α β γ x0 x1 x2 x3 : ℝ}
  (_h2 : 1 - α ^ 2 ≠ 0) (_h3 : 1 - α ^ 2 - β ^ 2 ≠ 0)
  (h01 : x0 + α * x1 = 0)
  (h12 : x1 + β / (1 - α ^ 2) * x2 = 0)
  (h23 : x2 + γ * (1 - α ^ 2) / (1 - α ^ 2 - β ^ 2) * x3 = 0)
  (h3z : x3 = 0) :
  x0 = 0 ∧ x1 = 0 ∧ x2 = 0 ∧ x3 = 0 := by
  have hx3 : x3 = 0 := h3z
  have hx2 : x2 = 0 := by
    rw [hx3] at h23
    simp using h23
  have hx1 : x1 = 0 := by
    rw [hx2] at h12
    simp at h12
    exact h12
  have hx0 : x0 = 0 := by
    rw [hx1] at h01
    simp using h01
  exact ⟨hx0, hx1, hx2, hx3⟩

```

A.35 PlatonicSolids/DetFormula.lean

$\det M = \Delta$ (43 lines).

```

/-
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-/
import Mathlib.LinearAlgebra.Matrix.Determinant.Basic
import PlatonicSolids.GramMatrix

/-!
Paper §3.4. The determinant of the rank-4 Schläfli Gram matrix is
 $\Delta = \sin^2(\pi/p) \sin^2(\pi/r) - \cos^2(\pi/q)$ .
-/

open scoped Matrix
open Real Matrix

private lemma sum_fin4 (f : Fin 4 → ℝ) :  $\sum i, f i = f 0 + f 1 + f 2 + f 3$  := by
  rw [Fin.sum_univ_succ, Fin.sum_univ_succ, Fin.sum_univ_succ]
  simp
  abel

/-- Determinant of the tridiagonal Gram pattern. -/
lemma det_band (a b c : ℝ) :
  (!! [1, a, 0, 0;
    a, 1, b, 0;
    0, b, 1, c;
    0, 0, c, 1] : Matrix (Fin 4) (Fin 4) ℝ).det =
  (1 - a ^ 2) * (1 - c ^ 2) - b ^ 2 := by
  rw [det_succ_row_zero, sum_fin4]
  simp [det_fin_three, Fin.succAbove]
  ring

lemma schlaefliGram_det (p q r : ℕ) :
  (schlaefliGram p q r).det = schlaefliDet p q r := by
  unfold schlaefliGram schlaefliDet
  rw [det_band]
  have hp : 1 - (-cos (π / p)) ^ 2 = sin (π / p) ^ 2 := by llinarith [sin_sq_add_cos_sq (π / p)]

```

```

have hr : 1 - (-cos (π / r)) ^ 2 = sin (π / r) ^ 2 := by linarith [sin_sq_add_cos_sq (π / r)]
rw [hp, hr]
ring

lemma leadingMinor_four (p q r : ℕ) :
  (schlaflGram p q r).det = schlaflDet p q r :=
  schlaflGram_det p q r

```

A.36 PlatonicSolids/Sylvester.lean

Sylvester: positive definite iff leading minors are positive (100 lines).

```

/-
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-/
import Mathlib.Analysis.Matrix.PosDef
import PlatonicSolids.CompletingSquare
import PlatonicSolids.DetFormula
import PlatonicSolids.LleadingMinors
import PlatonicSolids.QuadForm

/-!
Paper §3.3. Sylvester's criterion for the Schläfli Gram matrix:
positive-definite iff the four leading principal minors are positive.
`PosDef →` minors uses Mathlib (`PosDef.submatrix`, `PosDef.det_pos`);
the converse is the completing-the-square identity.
-/

open scoped Matrix
open Real Matrix

lemma leadingMinors_pos_of_posDef {p q r : ℕ} (h : (schlaflGram p q r).PosDef) :
  0 < leadingMinor 1 (by decide) p q r ∧
  0 < leadingMinor 2 (by decide) p q r ∧
  0 < leadingMinor 3 (by decide) p q r ∧
  0 < (schlaflGram p q r).det := by
  have h1 := h.submatrix (Fin.castLE_injective (by decide : 1 ≤ 4))
  have h2 := h.submatrix (Fin.castLE_injective (by decide : 2 ≤ 4))
  have h3 := h.submatrix (Fin.castLE_injective (by decide : 3 ≤ 4))
  refine ⟨?, ?, ?, PosDef.det_pos h⟩
  · simp [leadingMinor, leadingSubmatrix] using PosDef.det_pos h1
  · simp [leadingMinor, leadingSubmatrix] using PosDef.det_pos h2
  · simp [leadingMinor, leadingSubmatrix] using PosDef.det_pos h3

/-- Converse Sylvester for this Gram family: positive leading minors ⇒ `PosDef`. -/
theorem schlaflGram_posDef_of_minors {p q r : ℕ}
  (h1 : 0 < leadingMinor 1 (by decide) p q r)
  (h2 : 0 < leadingMinor 2 (by decide) p q r)
  (h3 : 0 < leadingMinor 3 (by decide) p q r)
  (h4 : 0 < (schlaflGram p q r).det) :
  (schlaflGram p q r).PosDef := by
  have hD1 : leadingMinor 1 (by decide) p q r = 1 := leadingMinor_one p q r
  have : (0 : ℝ) < 1 := hD1 ▸ h1
  clear this
  refine posDef_iff_dotProduct_mulVec.mpr ⟨schlaflGram_isHermitian p q r, ?_⟩
  intro x hx
  set α := -cos (π / p)
  set β := -cos (π / q)
  set γ := -cos (π / r)
  have hα : 1 - α ^ 2 = sin (π / p) ^ 2 := by
    simp [α]; linarith [sin_sq_add_cos_sq (π / p)]
  have hβ : 1 - α ^ 2 - β ^ 2 = sin (π / p) ^ 2 - cos (π / q) ^ 2 := by

```

```

    simp [α, β]; linarith [sin_sq_add_cos_sq (π / p)]
have hD2 : leadingMinor 2 (by decide) p q r = sin (π / p) ^ 2 := leadingMinor_two p q r
have hD3 : leadingMinor 3 (by decide) p q r =
    sin (π / p) ^ 2 - cos (π / q) ^ 2 := leadingMinor_three p q r
have h2ne : 1 - α ^ 2 ≠ 0 := by
    rw [hα]; exact ne_of_gt (hD2 ▸ h2)
have h3ne : 1 - α ^ 2 - β ^ 2 ≠ 0 := by
    rw [hβ]; exact ne_of_gt (hD3 ▸ h3)
have hc1 : 0 < 1 - α ^ 2 := by rw [hα]; exact hD2 ▸ h2
have hc2 : 0 < 1 - α ^ 2 - β ^ 2 := by rw [hβ]; exact hD3 ▸ h3
have hγ : 1 - γ ^ 2 = sin (π / r) ^ 2 := by
    simp [γ]; linarith [sin_sq_add_cos_sq (π / r)]
have hc3 : 0 < 1 - α ^ 2 - β ^ 2 - γ ^ 2 * (1 - α ^ 2) := by
    have : 1 - α ^ 2 - β ^ 2 - γ ^ 2 * (1 - α ^ 2) =
        (1 - α ^ 2) * (1 - γ ^ 2) - β ^ 2 := by ring
    rw [this, hα, hγ]
have hβsq : β ^ 2 = cos (π / q) ^ 2 := by simp [β]
rw [hβsq]
have hdet : sin (π / p) ^ 2 * sin (π / r) ^ 2 - cos (π / q) ^ 2 =
    (schlafligram p q r).det := by
    rw [schlafligram_det, schlafliDet]
    rw [hdet]
    exact h4
have hQ : star x ·v (schlafligram p q r *v x) =
    (x 0 + α * x 1) ^ 2
    + (1 - α ^ 2) * (x 1 + β / (1 - α ^ 2) * x 2) ^ 2
    + ((1 - α ^ 2 - β ^ 2) / (1 - α ^ 2)) *
        (x 2 + γ * (1 - α ^ 2) / (1 - α ^ 2 - β ^ 2) * x 3) ^ 2
    + ((1 - α ^ 2 - β ^ 2 - γ ^ 2 * (1 - α ^ 2)) / (1 - α ^ 2 - β ^ 2)) *
        x 3 ^ 2 := by
    calc star x ·v (schlafligram p q r *v x)
      = x 0 ^ 2 + x 1 ^ 2 + x 2 ^ 2 + x 3 ^ 2
        - 2 * cos (π / p) * x 0 * x 1
        - 2 * cos (π / q) * x 1 * x 2
        - 2 * cos (π / r) * x 2 * x 3 := schlafligram_quad p q r x
    _ = x 0 ^ 2 + x 1 ^ 2 + x 2 ^ 2 + x 3 ^ 2
        + 2 * α * x 0 * x 1 + 2 * β * x 1 * x 2 + 2 * γ * x 2 * x 3 := by
        simp [α, β, γ]; ring
    _ = _ := band_quad_squares α β γ (x 0) (x 1) (x 2) (x 3) h2ne h3ne
rw [hQ]
refine weighted_squares_pos hc1 (div_pos hc2 hc1) (div_pos hc3 hc2) ?_
by_contra hforms
push Not at hforms
obtain ⟨h01, h12, h23, h3z⟩ := hforms
obtain ⟨hx0, hx1, hx2, hx3⟩ := ldl_coords_eq_zero h2ne h3ne h01 h12 h23 h3z
have : x = 0 := by
    ext i
    fin_cases i <|> simp [hx0, hx1, hx2, hx3]
exact hx this

```

A.37 PlatonicSolids/TrigValues.lean

Exact
 $\sin^2(\pi/n)$ and
 $\cos^2(\pi/n)$ for n
in
3, 4, 5 (63 lines).
/-

```

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-/
import Mathlib.Analysis.SpecialFunctions.Trigonometric.Basic

/-!
Paper §4. Exact  $\sin^2(\pi/n)$  and  $\cos^2(\pi/n)$  for  $n \in \{3,4,5\}$ , from
Mathlib's sin_pi_div_* / cos_pi_div_*.
-/

open Real

lemma sq_cos_pi_div_three :  $\cos(\pi / 3)^2 = 1 / 4$  := by
  rw [cos_pi_div_three]; norm_num

lemma sq_sin_pi_div_four :  $\sin(\pi / 4)^2 = 1 / 2$  := by
  rw [sin_pi_div_four, div_pow, sq_sqrt (by norm_num : (0 : ℝ) ≤ 2)]; norm_num

lemma sq_cos_pi_div_four :  $\cos(\pi / 4)^2 = 1 / 2$  := by
  rw [cos_pi_div_four, div_pow, sq_sqrt (by norm_num : (0 : ℝ) ≤ 2)]; norm_num

lemma sq_cos_pi_div_five :  $\cos(\pi / 5)^2 = (3 + \sqrt{5}) / 8$  := by
  rw [cos_pi_div_five]
  have :  $(1 + \sqrt{5})^2 = 6 + 2 * \sqrt{5}$  := by
    ring_nf
    rw [sq_sqrt (by norm_num : (0 : ℝ) ≤ 5)]
    ring
  field_simp
  rw [this]
  ring

lemma sq_sin_pi_div_five :  $\sin(\pi / 5)^2 = (5 - \sqrt{5}) / 8$  := by
  have h :  $\sin(\pi / 5)^2 = 1 - \cos(\pi / 5)^2$  := by
    linarith [sin_sq_add_cos_sq (pi / 5)]
  rw [h, sq_cos_pi_div_five]
  field_simp
  ring

lemma sqrt5_bounds :  $2 < \sqrt{5} \wedge \sqrt{5} < 3$  := by
  constructor
  · rw [lt_sqrt (by norm_num)]; norm_num
  · rw [sqrt_lt' (by norm_num)]; norm_num

lemma three_mul_sqrt5_lt_seven :  $3 * \sqrt{5} < 7$  := by
  have h :  $(3 * \sqrt{5})^2 < (7 : ℝ)^2$  := by
    calc (3 *  $\sqrt{5}$ )^2 = 9 * ( $\sqrt{5}$ ^2) := by ring
    _ = 9 * 5 := by rw [sq_sqrt (by norm_num)]
    _ = 45 := by norm_num
    _ < 49 := by norm_num
    _ = (7 : ℝ)^2 := by norm_num
  have :  $0 \leq 3 * \sqrt{5}$  := by positivity
  nlinarith

lemma two_mul_sqrt5_gt_three :  $3 < 2 * \sqrt{5}$  := by
  have h :  $(3 : ℝ)^2 < (2 * \sqrt{5})^2$  := by
    calc (3 : ℝ)^2 = 9 := by norm_num
    _ < 20 := by norm_num
    _ = 4 * 5 := by norm_num
    _ = 4 * ( $\sqrt{5}$ ^2) := by rw [sq_sqrt (by norm_num)]
    _ = (2 *  $\sqrt{5}$ )^2 := by ring
  have :  $0 \leq 2 * \sqrt{5}$  := by positivity

```

A.38 PlatonicSolids/CompatibleTriples.lean

The eleven Platonic-compatible Schläfli triples (64 lines).

```

/-
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-/
import PlatonicSolids.PlatonicPair

/-!
Paper §4. The eleven Schläfli triples whose cell and vertex figure
are both Platonic.
-/

/-- Compatible middle index: 11 of the 25 Platonic-pair products survive. -/
theorem platonic_compatible_eleven {p q r : ℕ}
  (hc : IsPlatonicPair p q) (hv : IsPlatonicPair q r) :
  p = 3 ∧ q = 3 ∧ r = 3 ∨
  p = 3 ∧ q = 3 ∧ r = 4 ∨
  p = 3 ∧ q = 3 ∧ r = 5 ∨
  p = 3 ∧ q = 4 ∧ r = 3 ∨
  p = 4 ∧ q = 3 ∧ r = 3 ∨
  p = 4 ∧ q = 3 ∧ r = 4 ∨
  p = 4 ∧ q = 3 ∧ r = 5 ∨
  p = 3 ∧ q = 5 ∧ r = 3 ∨
  p = 5 ∧ q = 3 ∧ r = 3 ∨
  p = 5 ∧ q = 3 ∧ r = 4 ∨
  p = 5 ∧ q = 3 ∧ r = 5 := by
  rcases hc with ⟨rfl, rfl⟩ | ⟨rfl, rfl⟩ | ⟨rfl, rfl⟩ | ⟨rfl, rfl⟩ | ⟨rfl, rfl⟩
  · -- cell {3,3}
    rcases hv with ⟨_, rfl⟩ | ⟨_, rfl⟩ | ⟨h, _⟩ | ⟨_, rfl⟩ | ⟨h, _⟩
    · exact .inl ⟨rfl, rfl, rfl⟩
    · exact .inr <| .inl ⟨rfl, rfl, rfl⟩
    · cases h
    · exact .inr <| .inr <| .inl ⟨rfl, rfl, rfl⟩
    · cases h
  · -- cell {3,4}
    rcases hv with ⟨h, _⟩ | ⟨h, _⟩ | ⟨_, rfl⟩ | ⟨h, _⟩ | ⟨h, _⟩
    · cases h
    · cases h
    · exact .inr <| .inr <| .inr <| .inl ⟨rfl, rfl, rfl⟩
    · cases h
    · cases h
  · -- cell {4,3}
    rcases hv with ⟨_, rfl⟩ | ⟨_, rfl⟩ | ⟨h, _⟩ | ⟨_, rfl⟩ | ⟨h, _⟩
    · exact .inr <| .inr <| .inr <| .inr <| .inl ⟨rfl, rfl, rfl⟩
    · exact .inr <| .inr <| .inr <| .inr <| .inr <| .inl ⟨rfl, rfl, rfl⟩
    · cases h
    · exact .inr <| .inr <| .inr <| .inr <| .inr <| .inr <| .inl ⟨rfl, rfl, rfl⟩
    · cases h
  · -- cell {3,5}
    rcases hv with ⟨h, _⟩ | ⟨h, _⟩ | ⟨h, _⟩ | ⟨h, _⟩ | ⟨_, rfl⟩
    · cases h
    · cases h
    · cases h
    · cases h
    · exact .inr <| .inr <| .inr <| .inr <| .inr <| .inr <| .inr <| .inl ⟨rfl, rfl, rfl⟩
  · -- cell {5,3}
    rcases hv with ⟨_, rfl⟩ | ⟨_, rfl⟩ | ⟨h, _⟩ | ⟨_, rfl⟩ | ⟨h, _⟩
    · exact .inr <| .inr <| .inr <| .inr <| .inr <| .inr <| .inr <| .inr <|

```

```

    .inl ⟨rfl, rfl, rfl⟩
  · exact .inr <| .inr <| .inr <| .inr <| .inr <| .inr <| .inr <|
    .inr <| .inl ⟨rfl, rfl, rfl⟩
  · cases h
  · exact .inr <| .inr <| .inr <| .inr <| .inr <| .inr <| .inr <|
    .inr <| .inr ⟨rfl, rfl, rfl⟩
  · cases h

```

A.39 PlatonicSolids/SignTable.lean

Sign of Δ on those eleven triples (121 lines).

```

/-
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-/
import PlatonicSolids.GramMatrix
import PlatonicSolids.TrigValues

/-!
Paper §4. Sign of `Δ` on the eleven Platonic-compatible triples.
-/

open Real

private lemma unfold_det (p q r : ℕ) :
  schlafliDet p q r = sin (π / p) ^ 2 * sin (π / r) ^ 2 - cos (π / q) ^ 2 :=
  rfl

private lemma cast_angles :
  sin (π / (3 : ℕ)) ^ 2 = sin (π / 3) ^ 2 ∧
  sin (π / (4 : ℕ)) ^ 2 = sin (π / 4) ^ 2 ∧
  sin (π / (5 : ℕ)) ^ 2 = sin (π / 5) ^ 2 ∧
  cos (π / (3 : ℕ)) ^ 2 = cos (π / 3) ^ 2 ∧
  cos (π / (4 : ℕ)) ^ 2 = cos (π / 4) ^ 2 ∧
  cos (π / (5 : ℕ)) ^ 2 = cos (π / 5) ^ 2 := by
  simp [Nat.cast_ofNat]

lemma schlafliDet_333 : schlafliDet 3 3 3 = 5 / 16 := by
  rw [unfold_det, cast_angles.1, cast_angles.2.2.2.1, sq_sin_pi_div_three, sq_cos_pi_div_three]
  norm_num

lemma schlafliDet_334 : schlafliDet 3 3 4 = 1 / 8 := by
  rw [unfold_det, cast_angles.1, cast_angles.2.1, cast_angles.2.2.2.1,
    sq_sin_pi_div_three, sq_sin_pi_div_four, sq_cos_pi_div_three]
  norm_num

lemma schlafliDet_433 : schlafliDet 4 3 3 = 1 / 8 := by
  rw [unfold_det, cast_angles.2.1, cast_angles.1, cast_angles.2.2.2.1,
    sq_sin_pi_div_four, sq_sin_pi_div_three, sq_cos_pi_div_three]
  norm_num

lemma schlafliDet_343 : schlafliDet 3 4 3 = 1 / 16 := by
  rw [unfold_det, cast_angles.1, cast_angles.2.2.2.2.1, sq_sin_pi_div_three, sq_cos_pi_div_four]
  norm_num

lemma schlafliDet_335 : schlafliDet 3 3 5 = (7 - 3 * √5) / 32 := by
  rw [unfold_det, cast_angles.1, cast_angles.2.2.1, cast_angles.2.2.2.1,
    sq_sin_pi_div_three, sq_sin_pi_div_five, sq_cos_pi_div_three]
  ring

lemma schlafliDet_533 : schlafliDet 5 3 3 = (7 - 3 * √5) / 32 := by

```

```

    rw [unfold_det, cast_angles.2.2.1, cast_angles.1, cast_angles.2.2.2.1,
        sq_sin_pi_div_five, sq_sin_pi_div_three, sq_cos_pi_div_three]
    ring

lemma schlaflidDet_434 : schlaflidDet 4 3 4 = 0 := by
  rw [unfold_det, cast_angles.2.1, cast_angles.2.2.2.1, sq_sin_pi_div_four, sq_cos_pi_div_three]
  ]
  norm_num

lemma schlaflidDet_353 : schlaflidDet 3 5 3 = (3 - 2 *  $\sqrt{5}$ ) / 16 := by
  rw [unfold_det, cast_angles.1, cast_angles.2.2.2.2.2, sq_sin_pi_div_three, sq_cos_pi_div_five]
  ]
  ring

lemma schlaflidDet_435 : schlaflidDet 4 3 5 = (1 -  $\sqrt{5}$ ) / 16 := by
  rw [unfold_det, cast_angles.2.1, cast_angles.2.2.1, cast_angles.2.2.2.1,
      sq_sin_pi_div_four, sq_sin_pi_div_five, sq_cos_pi_div_three]
  ring

lemma schlaflidDet_534 : schlaflidDet 5 3 4 = (1 -  $\sqrt{5}$ ) / 16 := by
  rw [unfold_det, cast_angles.2.2.1, cast_angles.2.1, cast_angles.2.2.2.1,
      sq_sin_pi_div_five, sq_sin_pi_div_four, sq_cos_pi_div_three]
  ring

lemma schlaflidDet_535 : schlaflidDet 5 3 5 = (14 - 10 *  $\sqrt{5}$ ) / 64 := by
  rw [unfold_det, cast_angles.2.2.1, cast_angles.2.2.2.1, sq_sin_pi_div_five,
      sq_cos_pi_div_three]
  have : ((5 -  $\sqrt{5}$ ) / 8) ^ 2 = (30 - 10 *  $\sqrt{5}$ ) / 64 := by
    field_simp
    ring_nf
    rw [sq_sqrt (by norm_num : (0 :  $\mathbb{R}$ ) ≤ 5)]
    ring
  rw [← pow_two, this]
  ring

lemma schlaflidDet_333_pos : 0 < schlaflidDet 3 3 3 := by rw [schlaflidDet_333]; norm_num
lemma schlaflidDet_334_pos : 0 < schlaflidDet 3 3 4 := by rw [schlaflidDet_334]; norm_num
lemma schlaflidDet_433_pos : 0 < schlaflidDet 4 3 3 := by rw [schlaflidDet_433]; norm_num
lemma schlaflidDet_343_pos : 0 < schlaflidDet 3 4 3 := by rw [schlaflidDet_343]; norm_num

lemma schlaflidDet_335_pos : 0 < schlaflidDet 3 3 5 := by
  rw [schlaflidDet_335]
  have : 0 < 7 - 3 *  $\sqrt{5}$  := by linarith [three_mul_sqrt5_lt_seven]
  positivity

lemma schlaflidDet_533_pos : 0 < schlaflidDet 5 3 3 := by
  rw [schlaflidDet_533]
  have : 0 < 7 - 3 *  $\sqrt{5}$  := by linarith [three_mul_sqrt5_lt_seven]
  positivity

lemma schlaflidDet_434_nonpos : schlaflidDet 4 3 4 ≤ 0 := by
  rw [schlaflidDet_434]

lemma schlaflidDet_353_neg : schlaflidDet 3 5 3 < 0 := by
  rw [schlaflidDet_353]
  have : 3 - 2 *  $\sqrt{5}$  < 0 := by linarith [two_mul_sqrt5_gt_three]
  exact div_neg_of_neg_of_pos this (by positivity)

lemma schlaflidDet_435_neg : schlaflidDet 4 3 5 < 0 := by
  rw [schlaflidDet_435]
  have : 1 -  $\sqrt{5}$  < 0 := by linarith [sqrt5_bounds.1]
  exact div_neg_of_neg_of_pos this (by positivity)

```

```

lemma schlaflDet_534_neg : schlaflDet 5 3 4 < 0 := by
  rw [schlaflDet_534]
  have : 1 -  $\sqrt{5}$  < 0 := by linarith [sqrt5_bounds.1]
  exact div_neg_of_neg_of_pos this (by positivity)

lemma schlaflDet_535_neg : schlaflDet 5 3 5 < 0 := by
  rw [schlaflDet_535]
  have : 14 - 10 *  $\sqrt{5}$  < 0 := by
    have : 14 < 10 *  $\sqrt{5}$  := by nlinarith [sqrt5_bounds.1]
    linarith
  exact div_neg_of_neg_of_pos this (by positivity)

```

A.40 PlatonicSolids/Regular4.lean

The six regular convex 4-polytopes (28 lines).

```

/-
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-/
import PlatonicSolids.PlatonicPair

/-!
Paper §4, Group I. The six regular convex 4-polytopes.
-/

/-- Schläfli triples of the six regular convex 4-polytopes. -/
def IsRegular4Polytope (p q r : ℕ) : Prop :=
  (p = 3 ∧ q = 3 ∧ r = 3) ∨
  (p = 4 ∧ q = 3 ∧ r = 3) ∨
  (p = 3 ∧ q = 3 ∧ r = 4) ∨
  (p = 3 ∧ q = 4 ∧ r = 3) ∨
  (p = 5 ∧ q = 3 ∧ r = 3) ∨
  (p = 3 ∧ q = 3 ∧ r = 5)

theorem IsRegular4Polytope.ge_three {p q r : ℕ} (h : IsRegular4Polytope p q r) :
  3 ≤ p ∧ 3 ≤ q ∧ 3 ≤ r := by
  rcases h with ⟨rfl, rfl, rfl⟩ | ⟨rfl, rfl, rfl⟩ | ⟨rfl, rfl, rfl⟩
  | ⟨rfl, rfl, rfl⟩ | ⟨rfl, rfl, rfl⟩ | ⟨rfl, rfl, rfl⟩ <=> decide

theorem IsRegular4Polytope.platonic {p q r : ℕ} (h : IsRegular4Polytope p q r) :
  IsPlatonicPair p q ∧ IsPlatonicPair q r := by
  rcases h with ⟨rfl, rfl, rfl⟩ | ⟨rfl, rfl, rfl⟩ | ⟨rfl, rfl, rfl⟩
  | ⟨rfl, rfl, rfl⟩ | ⟨rfl, rfl, rfl⟩ | ⟨rfl, rfl, rfl⟩ <=> simp [IsPlatonicPair]

```

A.41 PlatonicSolids/Classification.lean

$\Delta > 0$ (equivalently M positive definite) iff one of the six (84 lines).

```

/-
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-/
import PlatonicSolids.Angles
import PlatonicSolids.CompatibleTriples
import PlatonicSolids.D3IffPlatonic
import PlatonicSolids.Regular4
import PlatonicSolids.SignTable
import PlatonicSolids.Sylvester
import PlatonicSolids.TrailingMinor

```

```

/-!
Paper §4. The six regular convex 4-polytopes are exactly the
Platonic-compatible triples with  $\Delta > 0$ , equivalently those whose
Schläfli Gram matrix is positive definite.
-/

open Real Matrix

/-- Palomar compared 4D theorem: Platonic cell and vertex figure with
 $\Delta > 0$  force one of the six regular convex 4-polytopes. -/
theorem regular_polychora_classification (p q r : ℕ)
  (h_cell : IsPlatonicPair p q)
  (h_vf : IsPlatonicPair q r)
  (h_det : 0 < schlaflidDet p q r) :
  IsRegular4Polytope p q r := by
  rcases platonic_compatible_eleven h_cell h_vf with
  | ⟨rfl, rfl, rfl⟩ | ⟨rfl, rfl, rfl⟩ | ⟨rfl, rfl, rfl⟩ | ⟨rfl, rfl, rfl⟩
  | ⟨rfl, rfl, rfl⟩ | ⟨rfl, rfl, rfl⟩ | ⟨rfl, rfl, rfl⟩ | ⟨rfl, rfl, rfl⟩
  | ⟨rfl, rfl, rfl⟩ | ⟨rfl, rfl, rfl⟩ | ⟨rfl, rfl, rfl⟩
  · exact .inl ⟨rfl, rfl, rfl⟩
  · exact .inr <| .inr <| .inl ⟨rfl, rfl, rfl⟩
  · exact .inr <| .inr <| .inr <| .inr <| .inr ⟨rfl, rfl, rfl⟩
  · exact .inr <| .inr <| .inr <| .inl ⟨rfl, rfl, rfl⟩
  · exact .inr <| .inl ⟨rfl, rfl, rfl⟩
  · exact (not_lt_of_ge schlaflidDet_434_nonpos h_det).elim
  · exact (lt_asymm schlaflidDet_435_neg h_det).elim
  · exact (lt_asymm schlaflidDet_353_neg h_det).elim
  · exact .inr <| .inr <| .inr <| .inr <| .inl ⟨rfl, rfl, rfl⟩
  · exact (lt_asymm schlaflidDet_534_neg h_det).elim
  · exact (lt_asymm schlaflidDet_535_neg h_det).elim

theorem schlaflidDet_pos_of_regular {p q r : ℕ} (h : IsRegular4Polytope p q r) :
  0 < schlaflidDet p q r := by
  rcases h with ⟨rfl, rfl, rfl⟩ | ⟨rfl, rfl, rfl⟩ | ⟨rfl, rfl, rfl⟩
  | ⟨rfl, rfl, rfl⟩ | ⟨rfl, rfl, rfl⟩ | ⟨rfl, rfl, rfl⟩
  · exact schlaflidDet_333_pos
  · exact schlaflidDet_433_pos
  · exact schlaflidDet_334_pos
  · exact schlaflidDet_343_pos
  · exact schlaflidDet_533_pos
  · exact schlaflidDet_335_pos

theorem regular_of_posDef {p q r : ℕ} (hp : 3 ≤ p) (hq : 3 ≤ q) (hr : 3 ≤ r)
  (h : (schlaflidGram p q r).PosDef) :
  IsRegular4Polytope p q r := by
  obtain ⟨_, _, h3, h4⟩ := leadingMinors_pos_of_posDef h
  have hcell : IsPlatonicPair p q := (leadingMinor_three_pos_iff hp hq).mp h3
  have hvf : IsPlatonicPair q r := by
    have hs := h.submatrix (Fin.succ_injective (n := 3))
    have : 0 < (trailing3 p q r).det := by
      simpa [trailing3] using PosDef.det_pos hs
    exact (trailing3_pos_iff hq hr).mp this
  have hdet : 0 < schlaflidDet p q r := by
    rw [← schlaflidGram_det]
    exact PosDef.det_pos h
  exact regular_polychora_classification p q r hcell hvf hdet

theorem posDef_of_regular {p q r : ℕ} (h : IsRegular4Polytope p q r) :
  (schlaflidGram p q r).PosDef := by
  obtain ⟨hp, hq, _hr⟩ := h.ge_three
  obtain ⟨hcell, _hvf⟩ := h.platonic
  refine schlaflidGram_posDef_of_minors ?_ ?_ ?_ ?_
  · rw [leadingMinor_one]; norm_num

```

```

· rw [leadingMinor_two]; exact sq_pos_of_pos (sin_angle_pos hp)
· exact (leadingMinor_three_pos_iff hp hq).mpr hcell
· rw [schlafligram_det]; exact schlafliDet_pos_of_regular h

/-- Finite regular convex 4-polytopes are exactly the triples `{p,q,r}`
with `{p,q,r} ≥ 3` whose Schläfli Gram matrix is positive definite. -/
theorem regular_polychora_iff {p q r : ℕ} (hp : 3 ≤ p) (hq : 3 ≤ q) (hr : 3 ≤ r) :
  (schlafliGram p q r).PosDef ↔ IsRegular4Polytope p q r :=
  ⟨regular_of_posDef hp hq hr, posDef_of_regular⟩

```

A.42 Challenge.lean

Palomar statement of record (deliberate sorries) (113 lines).

```

/-
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Authors: Lars Warren Ericson.
-/
import Mathlib.Analysis.SpecialFunctions.Trigonometric.Basic
import Mathlib.LinearAlgebra.Matrix.Notation
import Mathlib.Tactic

/-!
# Palomar statement of record (Platonic solids and regular polychora)

This module states the compared Palomar family and the Mathlib type
surface it depends on. Challenge may use `sorry`; Solution supplies the
proofs.

The selected theorems are:

1. `platonic_solids_3d` - Euler  $(V-E+F=2)$  plus regularity incidence
 $(pF=2E)$ ,  $(qV=2E)$  force a Platonic pair  $((p,q))$ .
2. `edges_pos_of_regular` - the same identities algebraically force
 $(E>0)$ ; non-degeneracy is a theorem, not an axiom.
3. `regular_polychora_classification` - Platonic cells and vertex figures
with positive Schläfli Gram determinant are exactly the six regular
convex 4-polytopes.

## How to read this file

The definitions below are the vocabulary of the claim. A reader who wants
to check what has been proved should read this file and need not read
anything else. `Solution.lean` imports the sorry-free development in
`PlatonicSolids.lean`. Each supporting lemma lives in a section-sized
module under `PlatonicSolids/`, meant to be read next to the matching
paragraph of `PlatonicSolids.md`.
-/

open scoped Matrix
open Real Matrix

/-- The 5 Platonic pairs (p, q):
(3, 3): Tetrahedron
(3, 4): Octahedron
(4, 3): Cube
(3, 5): Icosahedron
(5, 3): Dodecahedron -/
def IsPlatonicPair (p q : ℕ) : Prop :=
  (p = 3 ∧ q = 3) ∨
  (p = 3 ∧ q = 4) ∨
  (p = 4 ∧ q = 3) ∨

```

```

(p = 3 ∧ q = 5) ∨
(p = 5 ∧ q = 3)

/-- Canonical bilinear form of the rank-4 Schläfli Coxeter system {p, q, r}. -/
noncomputable def schlaflGram (p q r : ℕ) : Matrix (Fin 4) (Fin 4) ℝ :=
  !![1, -cos (π / p), 0, 0;
    -cos (π / p), 1, -cos (π / q), 0;
    0, -cos (π / q), 1, -cos (π / r);
    0, 0, -cos (π / r), 1]

/-- Δ(p, q, r) = sin²(π/p) sin²(π/r) - cos²(π/q). -/
noncomputable def schlaflDet (p q r : ℕ) : ℝ :=
  sin (π / p) ^ 2 * sin (π / r) ^ 2 - cos (π / q) ^ 2

/-- The 6 regular convex 4-polytopes (Schläfli symbols {p, q, r}):
  {3, 3, 3}: 5-cell (4D Simplex)
  {4, 3, 3}: 8-cell (Tesseract / Hypercube)
  {3, 3, 4}: 16-cell (Cross-polytope)
  {3, 4, 3}: 24-cell (Exceptional 4D regular polytope)
  {5, 3, 3}: 120-cell
  {3, 3, 5}: 600-cell -/
def IsRegular4Polytope (p q r : ℕ) : Prop :=
  (p = 3 ∧ q = 3 ∧ r = 3) ∨
  (p = 4 ∧ q = 3 ∧ r = 3) ∨
  (p = 3 ∧ q = 3 ∧ r = 4) ∨
  (p = 3 ∧ q = 4 ∧ r = 3) ∨
  (p = 5 ∧ q = 3 ∧ r = 3) ∨
  (p = 3 ∧ q = 3 ∧ r = 5)

/-- Theorem: A Platonic configuration cannot have zero edges.
  The topological Euler characteristic (χ = 2) and regularity conditions (p, q ≥ 3)
  algebraically force E ≥ 1 without requiring any geometric non-degeneracy axiom. -/
theorem edges_pos_of_regular
  (V E F p q : ℕ)
  (hp : 3 ≤ p) (hq : 3 ≤ q)
  (h_edges_faces : p * F = 2 * E)
  (h_edges_verts : q * V = 2 * E)
  (h_euler : V + F = E + 2) :
  0 < E := by
  sorry

/-- The 3D Main Theorem:
  Any polyhedron satisfying Euler's formula (V - E + F = 2)
  and regularity incidence relations (pF = 2E, qV = 2E)
  must have (p, q) belonging to the 5 Platonic solids. -/
theorem platonic_solids_3d
  (V E F p q : ℕ)
  (hp : 3 ≤ p) (hq : 3 ≤ q)
  (h_edges_faces : p * F = 2 * E)
  (h_edges_verts : q * V = 2 * E)
  (h_euler : V + F = E + 2) :
  IsPlatonicPair p q := by
  sorry

/-- The Main Classification Theorem for 4D Regular Polychora:
  Any triple {p, q, r} whose 3D cells {p, q} and vertex figures {q, r} are
  Platonic solids, and whose Coxeter Gram matrix is positive-definite (Δ > 0),
  is strictly one of the 6 regular convex 4-polytopes. -/
theorem regular_polychora_classification (p q r : ℕ)
  (h_cell : IsPlatonicPair p q)
  (h_vf : IsPlatonicPair q r)
  (h_det : 0 < schlaflDet p q r) :
  IsRegular4Polytope p q r := by

```

sorry

A.43 Solution.lean

Palomar solution module: imports PlatonicSolids (20 lines).

```
/-
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-/
import PlatonicSolids

/-!
# Solutions to the Challenge

The declarations of `Challenge.lean`, proved. Importing `PlatonicSolids`
supplies `platonic_solids_3d`, `edges_pos_of_regular`,
`regular_polychora_classification` and the definitions they depend on
(`IsPlatonicPair`, `IsRegular4Polytope`, `schlafliGram`, `schlafliDet`),
with the same names and types as in the Challenge module.
Comparator compares those declarations.

The proofs are split across section-sized modules under `PlatonicSolids/`.
There is no `sorry` in those files.
-/
```